Table 10 of AIS-007 (Revision 5)

Criteria for extension of approval

Continuation document ALP-829383b7-T10-R3-C · TEST – Bada Dost rear axle 5.13

Engineering draft. Manufacturer and test-agency acceptance / signatures remain pending.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
CMVR 96; applicable IS 11852 / IS 15986 route Service braking performance	Name and address of engine manufacturing plant (In case of imported vehicles, above details shall be supplied for importer also): Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX → Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Axle spacing in case of multi axle vehicles.: NA → NA Other axle: NA → NA Gross combination weight kg (applicable for articulated / Tractor Trailer combination vehicles): NA → NA Front axle: NA → NA Rear axle: NA → NA Trailer axle / other axles: NA → NA If applicable, maximum mass of trailer whic5h may be coupled: NA → NA Unbraked trailer: NA → NA Reference mass (for vehicles with GVW less than or equal to 3.5 ton) (kg.): Refer Annexure T7-B → Refer Annexure T7-B Make: APOLLO → APOLLO	CEA / report needs review	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade description(s): CEAT WINMILE X3-R → CEAT WINMILE		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 Tyre rolling Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing:		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Refer - TC7719XX000 003XXXF → Refer - TC7719XX000 003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022-2023/300002XXX	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	X/TA/XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L- 756XXXX → CM/L- 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E- Marking: NA → NA Others (For articulated / combination vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	rating, Speed rating, Load rating or Load		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00 R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 325 mm → 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26 kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	& 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Final Drive (4x2 or 4x4 or 6x2 or 6x4 or others) Front / Rear / Both: (4 x 2) Rear → (4 x 2) Rear Rear axle make: Single Reduction, Hypoid, Semi-floating Banjo type → Single Reduction, Hypoid, Semi-floating Banjo type Rear axle ratio: 4.56 → 5.13 Differential type: Single Reduction, Hypoid, Semi-floating Banjo type → Single Reduction, Hypoid, Semi-floating Banjo type Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2.		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	YYYY – 18.2 :1 Engine coolant temperature: Red Color – Thermometer Symbol → Red Color – Thermometer Symbol Engine low oil pressure: Red Color – Oil	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Tank Symbol → Red Color – Oil Tank Symbol Brake failure: Red Color – Parking and Oil symbol together → Red Color – Parking and Oil symbol together Brake failure: → Installation drawing showing rear extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size, make, unladen pressure) etc., distance between tyre & RUPD RH & LH, frame rear width, P1, P2 points.: - → - Parking brake: Hand operated [LHS] → Hand operated [LHS] Service Brake: Foot operated → Foot operated Service Brake: → Type of coupling device for brake: NA → NA Detailed engineering drawing as specified in AIS – 013 or photographs showing the mounting details with dimensions.: NA → NA Distance between the contour points of the orthogonal projections on to plane “P” of the accelerator pedal and service brake pedal bearing surfaces, “E” in		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	mm.: -- → -- Distance between the projection of the service brake pedal on to the reference plane “P”, to the right, “H” in mm.: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Distance between the projection of the service brake pedal on to the0 reference plane “P”, to the left, “J” in mm.: -- → -- In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → -- Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side, distance between front extremity & FUPD, width of FUPD, ID marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Drawing / Images of position & arrangement of Engine: -- → -- Drawing/ Images of Under bonnet (Engine house) Components.: -- → -- 1st · Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	· Overall Ratio / AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69 2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC: 12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56 4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 → 20.44 Make: M/s Brakes India Private Limited → M/s Brakes India Private Limited Type (Mechanical/hydraulic/air/air assisted/vacuum assisted/others):		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Vacuum assisted hydraulic → Vacuum assisted hydraulic Control system & braking wheel: Hydraulic acting on all wheels → Hydraulic acting on all wheels Schematic layout indicating method of split of brake system, location of valves, reservoirs, ABS components etc. (Attach	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	drawing and indicate the drawing number): AJ1XXX00 → AJ1XXX00 Anti-Lock braking system Provided (Yes / No / Optional): NA → NA If yes, details of ABS: - → - Make: - → - Category of ABS: - → - Nos. of directly controlled wheel(s): - → - Brief description of failure warning tell–tale: - → - Wheel Speed Sensors: - → - No. of sensors: - → - Make of sensors: - → - Type of sensors: - → - Modulator: - → - Nos. of Modulators: - → - Make of Modulators: - → - Identification No. / Part No. of Modulator: - → - Brief description and features: - → - Controller: - → - Nos. of Controller: - → - Make of Controller: - → - Identification No. / Part No. of Controller: - → - Brief description and features: - → - Height of Center of Gravity		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(mm): - → - Un laden condition: - → - Laden condition: - → - Slack adjuster (only applicable for M2, M3, N and T category of vehicles.): - → - Front (Automatic / Manual): - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make of slack adjuster: - → - Model and Type: - → - Rear (Automatic / Manual): - → - Make of slack adjuster: - → - Model and Type: - → - Front wheel: 132x47.3x10.35 → 132x47.3x10.35 Rear wheel: LEADING - 314x51.8x5.6 TRAILING – 246x51.8x5.6 → LEADING - 314x51.8x5.6 TRAILING – 246x51.8x5.6 Others: NA → NA Type of liner wear indicator (window/ acoustic/ optical/ any other): Window → Window Front: 250 sq.cm → 250 sq.cm Rear: 570 sq.cm → 570 sq.cm Others: NA → NA Front wheel / axle: Option 1: SYY 6XXXYA → Option 1: SYY 6XXXYA Rear wheel / axle: APACHE 4059M → APACHE 4059M Others: NA → NA Whether asbestos or asbestos-free: Asbestos Free → Asbestos Free Front axle (Disc / Drum): DISC → DISC Effective Diameter (mm): 212 → 212 Rear axle (Disc / drum): DRUM → DRUM Effective Diameter (mm): 282 → 282 Other axle (		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Disc / Drum): NA → NA Effective diameter (mm): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Material (if the braking surface is not ferrous): NA → NA Front wheel: -- → -- Rear wheel: -- → -- Others: -- → -- Make: M/s XXXX YYYY → M/s XXXX YYYY Type (only applicable for M2, M3, N and T category of vehicles.): N Category → N Category Inner diameter of the master cylinder (mm): 22.81 → 22.81 Operating stroke (mm): 41 → 41 Type of supply tank (only applicable for M2, M3, N and T category of vehicles.): → Brake Chamber: Rear → Rear Brake Chamber: Parking → Parking Make and type: NA → NA Make and type: NA → NA Size mm: NA → NA Size mm: NA → NA Inner diameter mm: NA → NA Inner diameter mm: NA → NA Stroke mm: NA → NA Stroke mm: NA → NA Make and Identification No.: MAKE: XXX GIRLING TAC No: AB2XX1. CoP No: SYL/29/2020-2021/300000YY60/ADDM/XXXX → MAKE: XXX GIRLING TAC No: AB2XX1. CoP No: SYL/29/2020-2021/300000YY60/ADDM/XXXX		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Type of expansion (HL / HR): HL → HL Free Length of hoses: FOR NON ABS. FRONT : L1 – 470, L2 -417.7 REAR :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	L1 – 405, L2 – 360.9 → FOR NON ABS. FRONT : L1 – 470, L2 -417.7 REAR : L1 – 405, L2 – 360.9 Thickness of lining (mm): 0.6 → 0.6 Nominal bore dia. (mm): 3.1 → 3.1 Type (Visual display/ audible/others): Visual display → Visual display Operation pressure (kg/cm2 / bar / kPa): NA → NA Type of safety device: NA → NA Make: M/s XXXXXYY → M/s XXXXXYY Type (mechanical/spring brake): MECHANICAL → MECHANICAL Acting on Transmission/wheel: WHEEL → WHEEL Control System & Braking wheel: Hand operated & Rear wheel → Hand operated & Rear wheel Lining / pad: Lining → Lining Name of producer: M/s XXXXYYY → M/s XXXXYYY Dimension (mm) (D x W x T): LEADING - 305x48.8x5.6 TRAILING – 256x48.8x5.6 → LEADING - 305x48.8x5.6 TRAILING – 256x48.8x5.6 Area (cm 2): 279 sq.cm → 279 sq.cm Material: XYZ 420 → XYZ 420	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Diameter of brake drum/disc (mm): 272 → 272 Type: UNFAILED PART OF THE SERVICE BRAKE → UNFAILED PART OF THE SERVICE BRAKE Make: Ms. GULF Oil → Ms. GULF Oil Trade name: DOT 4 → DOT 4 Specification/ grade as		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	per Indian standard: DOT 4 → DOT 4 Brake fluid level indicator (Yes/No): Yes → Yes Type (Manual/Auto) Front and Rear: Manual → Manual Lever length: → Pneumatic Brake Actuator: NA → NA Type (as per IS): NA → NA Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No Brake Assist System (Fitted : Yes/No): No → No Category of Brake Assist System A / B: -- → -- For category A systems, define the force threshold at which the ratio between pedal force and brake pressure increases: -- → -- For category B systems, define the brake pedal speed which must be achieved in	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	order to activate the Brake Assist System (e.g. pedal stroke speed (mm/s) during a given time interval): -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Motor Vehicle stability function information document: -- → -- System: -- → -- System variants: -- → -- System options: -- → -- Control function (directional/roll-over/both) including an explanation of the basic function and/or philosophy of the control: -- → -- System configurations (where appropriate): -- → -- System identification including software level identifier: -- → -- List of motor vehicles by description and configuration that are covered by the information document: -- → -- Schematic diagrams of the respective configurations installed on the motor vehicles defined in item 2.1. above with consideration given to the following:: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Lift axles: -- → -- Steering axles: -- → -- Anti-lock braking configurations: -- → -- Wheelbase, track, centre of gravity height: -- → -- Wheel type (single or twin) and tyre type (e.g. structure, category of use, size): -- → --		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Management of the engine or any other source(s) of motive power e.g.Torque Management, control of fuel supply etc.: -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Pneumatic circuits System schematics including anti-lock braking configurations associated with the trailer types defined in paragraph 6.2.1. of this annex.: -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies that	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Vehicle is/ is not equipped to		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	tow a trailer with electric braking system (Yes/No): No → No Make: Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. → Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. Model name / identification: ALXYZABC0 → ALXYZABC0 Type: Compression Ignition → Compression Ignition Bore x stroke (mm): 80 X 98 → 80 X 98 No. of cylinders: 3 → 3 Displacement (cm^3): 1468 cc → 1468 cc Compression ratio: 16.5±1.0 : 1 → 16.5±1.0 : 1 Max. Engine output (KW @ rpm): 55.1 Kw @ 3300 rpm → 55.1 Kw @ 3300 rpm Max. Torque (Nm @ rpm): 170Nm @ 1600 to 2400 rpm → 170Nm @ 1600 to 2400 rpm Air cleaner type: Dry type → Dry type Type of Fuel: Diesel → Diesel Front axle ratio: NA → NA Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2. YYYY Pvt.Ltd Service brake Type (Mechanical/hydraulic/air/air assisted/vacuum assisted/others): Vacuum assisted hydraulic → Vacuum assisted		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	hydraulic Auto Slack Adjuster Fitted (Yes / No / Optional): Yes (Automatic wear adjustment) → Yes (Automatic wear adjustment) Anti-Lock braking system Provided(Yes/No/Optional): Optional → Optional Front (Disc / Drum): Disc → Disc Rear (Disc / Drum): Drum → Drum Total braking area (cm²): Front axle-254 sq.cm Rear axle-580 sq.cm → Front axle-254 sq.cm Rear axle-580 sq.cm Parking brake type: Mechanical → Mechanical Secondary brake type: Un failed part of the service brake → Un failed part of the service brake Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	12 PR Tyre size designation including ply rating: 215/75 R15 LT → 215/75 R15 LT Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	→ 3.26 kg/cm2 Wiper motor: XYZ Ltd. → XYZ Ltd. Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Maximum GVW kg (for rigid vehicles): Refer Annexure T7 – A Details for Recommended load body - Refer Annexure T7 - B → Refer Annexure T7 – A Details for Recommended load body - Refer Annexure T7 - B Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Engine Manufacturer::		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Ashok Leyland Limited → Ashok Leyland Limited Manufacturer's engine model code (as marked on the engine, or other means of identification): Model Code: ALXYZABC0 Enclosure – 4A → Model Code: ALXYZABC0 Enclosure – 4A Location of engine model code and engine serial number on block / crankcase (drawing to be attached): Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A → Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer Enclosure – 4A Working principle: positive ignition / compression- ignition, four-stroke / two-stroke / rotary cycle (DI/IDI) (NA/TC/TCIC/Any other): Compression Ignition, four-stroke, DI, TCIC → Compression Ignition, four-stroke, DI, TCIC Number, arrangement of cylinders:: 3, Inline → 3, Inline Bore----------(mm): 80 → 80 Stroke: ----------(mm): 98 → 98 Firing order:: 1-2-3 → 1-2-3 Engine		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	capacity:----------(cm3): 1468 → 1468 Volumetric compression ratio (specify tolerance): 17.5±1.0 : 1 → 17.5±1.0 : 1 Drawings of combustion chamber and piston crown and, in the case of positive ignition engine, piston rings:: Refer Enclosure – 4B → Refer Enclosure – 4B Vehicle with Defeat Device Yes / No: No → No If yes – Details of the Defeat device: NA → NA Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	equivalent) in PPM content by volume in the exhaust gas with the engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum net power:..... kW at min-1: 55.1 KW @ 3100 ± 50 rpm → 55.1 KW @ 3100 ± 50 rpm Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm Maximum net torque: .....Nm at: -------min-1 ...(manufacturer's		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	declared value): 170 Nm @ 1600 – 2400 rpm → 170 Nm @ 1600 – 2400 rpm Maximum rated speed (Specify tolerance): 3100 rpm ± 50 rpm → 3100 rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Type of fuel used : Mono Fuel / Bi-Fuel / Flex Fuel / Dual Fuel: Mono Fuel → Mono Fuel Fuel type: Diesel / Gasoline / LPG / CNG / Biomethane / Bio-H methane / HCNG / Biogas / LNG / Ethanol ((E85) / 2 (E100)) /ED (95) Flex fuel Methanol M15 / M100 / MD95 / Biodiesel up to 100% /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Hydrogen.: Diesel → Diesel Maximum amount of biofuel acceptable in fuel (manufacturer's declared value): per cent by volume: 7% → 7% By fuel injection (compression-ignition only): (Yes/No): Yes → Yes System description:: Unit Pump Common Rail Direct Injection system → Unit Pump Common Rail Direct Injection system Working principle: direct-injection/pre-chamber/swirl Chamber: Direct Injection → Direct Injection Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) &		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: Plunger Type & FHL0XYZ0 → Plunger Type & FHL0XYZ0 Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Injection advance curve (Specify the tolerances):: ECU Controlled → ECU Controlled Governor: ECU Controlled → ECU Controlled Cut-off point under load:...(min-1): 3300 → 3300 Cut-off point without load (Maximum	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	no-load speed)…(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Multi-Hole Injectors (7 holes) FKXXYYZZ → Multi-Hole Injectors (7 holes) FKXXYYZZ Opening pressure (kPa) (specify the tolerances) or characteristic diagram: 15000 to 23000 kPa → 15000 to 23000 kPa Length and internal diameter…(mm): Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) -		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	193.5± 3 Internal Diameter – 3mm → Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm Common rail make & type: M/s. XXXXYYY, Hot Forged Rail Type → M/s. XXXXYYY, Hot Forged Rail Type Cold start system: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description:: NA → NA Auxiliary starting aid: NA → NA Make(s):: NA → NA Type(s):: NA → NA System description:: NA → NA Electronic controlled injection: (Yes/No):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA → NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s. XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and Metering type Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make and type of air temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type By fuel injection (positive ignition only): (Yes/No): No → No Working principle: intake manifold (single/multi- point)/direct injection/other (specify): NA → NA Make(s) of Injection Pump :: NA → NA Type(s) & Identification		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number of Injection Pump:: NA → NA System description (in the case of systems other than continuous injection give equivalent details):: NA → NA Make and type of the control unit:: NA → NA Make and type of the fuel regulator:: NA → NA Make and type of the air-flow sensor:: NA → NA Make and type of the micro-switch:: NA → NA Make and type of the throttle housing:: NA → NA Make and type of the water temperature sensor:: NA → NA Make and type of the air temperature	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	sensor:: NA → NA Injectors: Opening pressure (Specify the tolerances):,..... (kPa) or characteristic diagram:: NA → NA Make(s):: NA → NA Type(s):: NA → NA Identification Number: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA Cut-off point under load (min-1): NA → NA Injection timing:: NA → NA Cold start system:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Operating principle(s):: NA → NA Operating limits/settings(Specify the tolerances): NA → NA Pressure, (Specify the tolerances).....(kPa) or characteristic diagram:: 320 ± 50 kPa → 320 ± 50 kPa Rated voltage:...V, positive/negative ground: 12V, Negative Ground → 12V, Negative Ground Generator/ Alternator: Alternator → Alternator Type:: DC – Alternator with Voltage regulator → DC – Alternator with Voltage regulator Nominal output: ...VA: 1205 VA (For Non AC Variants) 850 Amps	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	1745 VA (For AC Variants) 120 Amps → 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps Ignition: NA → NA Make(s):: NA → NA Type(s):: NA → NA Working principle:: NA → NA Ignition advance curve or map(Specify the tolerances):: NA → NA Static ignition timing (Specify the tolerances):.....degrees before TDC: NA → NA Spark plugs: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Gap setting----------(mm): NA → NA Nominal resistance (kilo ohm) (if resistive type): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Ignition coil(s): NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Ignition condenser: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA EMI suppressor cap / Device / Electronic unit: NA → NA Make: NA → NA Type (Resistive/Capacitive): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Identification number: NA → NA Nominal resistance (kilo ohm): NA → NA Terminology and Drawing of interference Suppression equipment: NA → NA H.T. Cable: NA → NA Make and Place: NA → NA Type (Resistive/Non-resistive): NA → NA Length (mm) (if resistive type): NA → NA Outside dia. (mm) (if resistive type): NA → NA Nominal resistance kilo ohm, (if resistive type): NA → NA Radiator drawing(s): Refer enclosure 4C2 & 4C4 → Refer enclosure 4C2 & 4C4 Radiator drawing(s): Refer enclosurev 4C1 & 4C3 → Refer enclosurev 4C1 & 4C3 Make: Mahle Behr → Mahle Behr Make: Tata Toyo → Tata Toyo Type (s): Cross flow → Cross flow Relief valve pressure setting: 0.9 kg/cm2 → 0.9 kg/cm2 Fan characteristics (Fan power, kW) Enclose the fan		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	power curve corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part No: 9 → 9	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Material of blades (metal / plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ± 150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150 (AC) Radiator core open area (cm²): 1906 cm2 → 1906 cm2 Cooling system: liquid/air: Liquid → Liquid Nominal setting of the engine temperature control o mechanism (C)::: 85 deg C → 85 deg C Liquid: Yes → Yes Nature of liquid:: Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L → Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	coolant capacity: approx..8 L Circulating pump(s): (Yes/No): Yes → Yes Characteristics of Circulating pump:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm → Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm Make(s):: XXXXX Fasteners Ltd → XXXXX Fasteners Ltd Type(s):: Impeller Type → Impeller Type Drive ratio(s):: 1: 1.12 → 1: 1.12 Description of the fan and its drive mechanism:: Electric Fan , Motor Driven → Electric Fan , Motor Driven Blower: (Yes/No): No → No Characteristics of blower: NA → NA Make(s):: NA → NA Type(s):: NA → NA Drive ratio(s):: NA → NA Pressure charger: (Yes/No) Turbocharger / Supercharger: Yes, Turbo charger → Yes, Turbo charger Make(s):: M/s. XYZ, India → M/s. XYZ, India Type(s) & Identification Number:: Waste Gate with Vacuum Actuator, X3M1XXXX → Waste Gate with Vacuum Actuator, X3M1XXXX Description of the system (maximum charge pressure:.....(kPa), waste-gate, if applicable): Waste gate turbocharger with vacuum		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	actuator Maximum Charge pressure - 145 kPa → Waste gate turbocharger with vacuum	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	actuator Maximum Charge pressure - 145 kPa Inter-cooler: (Yes/No): Yes → Yes Type: air-air / air-water: Air – Air → Air – Air Make :: Mahle Behr → Mahle Behr Make :: Tata Toyo → Tata Toyo Identification Number:: B1V42XXX → B1V42XXX Identification Number:: B1V42YYY → B1V42YYY Air pressure drop across the intercooler: 15 kPa → 15 kPa Intake depression at rated engine speed and at 100 per cent load (Specify location of measurement) Minimum allowable:... (kPa) Maximum allowable:... (kPa): Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Description and drawings of inlet pipes and their accessories (plenum chamber, heating device, additional air intakes, etc.):: Intake manifold with Electronic throttle valve Refer Enclosure 4D → Intake manifold with Electronic throttle valve Refer Enclosure 4D Intake manifold description		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	including dimensional drawings and/or photographs):: Aluminium Intake manifold with pressure sensor Refer Enclosure 4D → Aluminium Intake manifold with pressure sensor Refer Enclosure 4D Air filter, Dimensional drawings: Refer Enclosure 4E → Refer Enclosure 4E Make(s):: M/s. Fleet guard → M/s. Fleet guard Type(s) & Identification Number:: Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 → Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 Intake silencer, Dimensional drawing: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description and Dimensional drawing of the exhaust manifold:: Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F → Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F Description and drawing with dimensions of the exhaust system:(Description with general arrangement of exhaust system along with its routing indicating the lengths of exhaust	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	pipe, tail pipe and: Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly → Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly Effective volume of exhaust-system (specify the tolerance & range) in liters (from exhaust manifold / TC outlet to tail pipe end), Enclose the exhaust system dimensional drawing and indicate the volume of each parts clearly: 13.05 ± 1 litre Refer Enclosure 4G → 13.05 ± 1 litre Refer Enclosure 4G Description and drawing of the exhaust Muffler/Silencer:: NA → NA Type(s) & Identification	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number:: NA → NA Make: NA → NA Number:.....: NA → NA Internal diameter of exhaust pipe (mm)(min): diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G → diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G Minimum distance between exhaust pipe(s) and the fuel line: >150 mm → >150 mm Auxiliary Noise shields for compliance to IS 3028 and / OR AIS-020 (If Provided): NA → NA Material: NA → NA Layout of noise shield / Photographs / Diagram Showing arrangements indicating fitment on vehicle.: NA → NA Maximum allowable exhaust back pressure at rated engine speed and at 100 per cent load (location of measurement): ... (kPa): 30.0 kpa (150 mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Minimum cross-sectional areas of inlet and outlet ports:: Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm → Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm Inlet (mm): 8.00 → 8.00 Exhaust (mm): 8.00 → 8.00 Opening: 17° bTDC → 17° bTDC Closing: 51° aBDC → 51° aBDC Opening: 66° bBDC → 66°	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	bBDC Closing: 25° aTDC → 25° aTDC Transfer: NA → NA Opening: NA → NA Closing: NA → NA Reference and/or setting ranges (Specify the tolerances):: On Cold Engine → On Cold Engine Inlet: 0.3 to 0.4 mm → 0.3 to 0.4 mm Exhaust: 0.3 to 0.4 mm → 0.3 to 0.4 mm Distribution by ports: NA → NA Volume of crank-case cavity with piston at TDC: NA → NA Reed valve fitted (Yes / No): No → No Description of inlet ports, scavenging and exhaust ports with corresponding timing.: NA → NA Device for recycling crankcase gases (description and drawings):: Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J → Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx Catalyst (if any, and if not covered by another heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	integrated and Passive Selective catalytic reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF / DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No Element – 1 No. → Catalytic Converter – 1No Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type for LNT (periodic / continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:……cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap Bulk Density (GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	and Characteristics:: Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY Type and Identification number:: Type - Passive NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Honey Comb type Material: Cordierite → Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make:: NA → NA Type and Idetification number :: NA → NA NOx Sensor		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Before /After of SCR):: No → No Make:: NA → NA Type and Identification number:: NA → NA Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems, description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent engine test bench cycles, between two cycles where regenerative	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	phases occur under the conditions equivalent to Type I test (Distance "D" in t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Distance “d” in AIS 137): 2 cycles → 2 cycles Description of method employed to determine the number of cycles between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter (DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	International standard (where appropriate):: NA → NA Frequency of reagent refill: continuous/maintenance (where		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	appropriate):: NA → NA Normal operating temperature range (°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY → YYYYYYY Air injection: (Yes/No): No → No Type (pulse air, air pump, etc.) & Identification Number:: NA → NA Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other): Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FEW0XXYY Characteristics (flow rate, etc.):: Flow rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes EGR		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Cooler Make and Identification Number: M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:.....g: NA → NA Canister working capacity:.....g: NA → NA Canister bed	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	volume:.....litre: NA → NA Canister Purge and vent strategy description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve : (Yes/No): NA → NA Make and Identification Number: NA → NA Operating pressure (kPa): NA → NA Liquid		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	fuel hoses material :: NA → NA Schematic drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or drawing (Periodic / Continuous): NA → NA Particulate trap: (Yes/No): Yes → Yes Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dimensions and shape of the particulate trap (capacity):: Dimension :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure 4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type Material - Silicon Carbide → Structure - Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI Electronic Control Unit (ECU): Yes → Yes Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number: P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Misfire detection:: NA → NA Oxygen sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K → Refer Enclosure 4K Particulate traps monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method)::	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer Enclosure 4K → Refer Enclosure 4K List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K → Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K A description of the type and number of the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	EUDC A description of the type of the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing all sensed components with the strategy for fault detection and MI activation (fixed number of driving cycles or statistical method), including a list of relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive explanation for the data	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	given in service $05 Test ID $21 to FF and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles –		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Diagnostics on Controller Area Network (CAN) – Part 4: Requirements for emissions- related systems", a comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137): Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA → NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Systems to ensure the correct operation of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	deactivation of the driver inducement, for use by the rescue services [Note: cross- reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling /disable after parking: NA → NA Lowest concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct operation of NOx control measures: NA → NA Details of any devices designed to influence fuel economy (if not covered by other items):: NA → NA LPG fuelling system: (Yes/No): No → No Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to LPG or back:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	System layout (electrical connections, vacuum connections, compensation hoses, etc.): NA → NA NG fuelling system: (Yes/No): No → No Approval number: NA → NA Electronic engine management control unit for NG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation:: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to NG or back: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.):: NA → NA Hydrogen fuelling system: (Yes/No): No → No Electronic engine management control unit for hydrogen fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Description of the safeguarding of the catalyst at switch- over from petrol to hydrogen or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA H2NG fuelling system: (Yes/No): No → No Percentage of hydrogen in the fuel (maximum specified by the manufacturer): NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to H2NG or back:: NA → NA System lay-out (electrical connections,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	vacuum connections compensation hoses, etc.):: NA → NA Electric motor(Traction): No → No Type (winding, excitation):: NA → NA Maximum hourly output: kW(manufacturer’s declared value): NA → NA Maximum net power:kW(manufacturer’s declared		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	value): NA → NA Maximum 30 minutes power: kW (manufacturer’s declared value): NA → NA Operating voltage: V: NA → NA Make and Identification Number:: → Battery (Traction): NA → NA Number of cells:: NA → NA Mass: ……(kg): NA → NA Capacity: Ah (Amp-hours): NA → NA Position:: NA → NA Engines or motor combinations: NA → NA Hybrid Electric Vehicle: (Yes/No): No → No Category of Hybrid Electric vehicle Off Vehicle Charging/Not Off Vehicle Charging9: NA → NA Operating mode switch: with/without: NA → NA Selectable modes: NA → NA Pure electric: (Yes/No): NA → NA Pure fuel consuming: (Yes/No): NA → NA Hybrid modes: (Yes/No) (if yes, short description): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Description of the energy storage device: (traction battery, capacitor, flywheel/generator...): NA → NA Make(s):: NA → NA Type(s) :: NA → NA Identification number:: NA → NA Kind of electrochemical couple:: NA → NA Energy: ..... (for battery: voltage and capacity Ah in 2 h, for capacitor: J): NA → NA Charger: on		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	board/external/without: NA → NA Electric machines (describe each type of electric machine separately): NA → NA Make:: NA → NA Type:: NA → NA Primary use: traction motor/generator: NA → NA When used as traction motor: mono motor / multi motors (number):: NA → NA Maximum power: kW: NA → NA Working principle:: NA → NA Direct current/alternating current/number of phases:: NA → NA Separate excitation/series/compound: NA → NA Synchronous/asynchronous: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA Identification number:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Power controller: NA → NA Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Vehicle electric range..... km (according to AIS 137):: NA → NA Manufacturer’s recommendation for preconditioning:: NA → NA CO2 (g/km) (Applicable for category M1 with GVW<3.5 T): NA → NA Declared : (rounded to 3 decimal places): NA → NA Fuel consumption (l/100 km) for Petrol, LPG or Diesel and (kg/100km) for CNG and (kWh/100 km) for Electric Driven		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Vehicles(Applicable for category M1 with GVW<3.5 T) (rounded to 3 decimal places)*: NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA *calculated on the declared CO2: NA → NA CO2 Reducing technologies Available (Yes/No): No → No Regenerative braking: NA → NA Start-Stop System: NA → NA Tyre pressure monitoring system: NA → NA 6 or more Speed Transmission: NA → NA Any other technology (manufacture to	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	enclose detail): NA → NA Electric energy consumption for electric vehicles Wh/km: NA → NA Electric energy consumption for pure electric vehicles Wh/km: NA → NA Electric energy consumption for externally chargeable hybrid electric vehicles: NA → NA Electric energy consumption (condition A, combined) Wh/km: NA → NA Electric energy consumption (condition B, combined) Wh/km: NA → NA Electric energy consumption (weighted combined) ….Wh/km: NA → NA Moment of inertia of engine flywheel:: 0.1212 kg/m2 → 0.1212 kg/m2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2 Make: XXYY System → XXYY System Type: 12V → 12V Identification mark: X5XYZ100 → X5XYZ100 Rated voltage: 12V → 12V Number of sweep Frequencies: 2 → 2 Highest sweep frequency (Cycles / min): 62±8 rpm → 62±8 rpm Lowest sweep frequency (Cycles / min): 40±5 rpm → 40±5 rpm Washer tank motor / Washer pump: ELECTRIC MOTOR WITH PUMP → ELECTRIC MOTOR WITH PUMP Make: XYZ → XYZ	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Model: PEAD64XYZS00 → PEAD64XYZS00 Schematic Diagram of Engine cooling system: - → - Maximum vehicle speed for continuous operation (km/h) (Only for AC horns): 80 Kmph → 80 Kmph A list, specifying the parts which constitute the headlamp cleaner and drawings thereof, (e.g. pumps, nozzles, valves, motors and wipers);: -- → -- Description and drawings/photographs of the shapes and constituent materials of the part of the body forming the engine compartment and the part of the passenger compartment nearest to it:: -- → -- Drawings or photographs of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	the position of the metal components housed in the engine compartment (e.g. heating appliances, spare wheel, air filter, steering mechanism, etc.):: -- → -- List of all Electrical components, which include Broadband EMI sources (like HAVC Motor, Wiper Motor, Window motor and Horn etc.) The list shall include Device name, Make/s and Part No. /Type ID/Drawing No: -- → -- System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	only in case of steering column locking systems: Locks the Steering → Locks the Steering Components placed in Engine compartment: - → - Frequency bands (Hz) · Max Output Power (W): Antenna Position at Vehicle, specific conditions for installation and/or use → Antenna Position at Vehicle, specific conditions for installation and/or use Example of Engine No: HJEZXYZ891 → HJEZXYZ891		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
CMVR 96; applicable IS 11852 / IS 15986 route Parking braking performance	Name and address of engine manufacturing plant (In case of imported vehicles, above details shall be supplied for importer also): Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX → Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Axle spacing in case of multi axle vehicles.: NA → NA Other axle: NA → NA Gross combination weight kg (applicable for articulated / Tractor Trailer combination vehicles): NA → NA Front axle: NA → NA	CEA / report needs review	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Rear axle: NA → NA Trailer axle / other axles: NA → NA If applicable, maximum mass of trailer whic5h may be coupled: NA → NA Unbraked trailer: NA → NA Reference mass (for vehicles with GVW less than or equal to 3.5 ton) (kg.): Refer Annexure T7-B → Refer Annexure T7-B Make: APOLLO → APOLLO Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade description(s): CEAT WINMILE X3-R → CEAT WINMILE		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	142 Tyre rolling Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing:		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Refer - TC7719XX000 003XXXF → Refer - TC7719XX000	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L-756XXXX → CM/L-756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Spare wheel: 1 → 1 Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E- Marking: NA → NA Others (For articulated / combination vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00 R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	M2, M3, N and T category of vehicles.): 325 mm → 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26 kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Final Drive (4x2 or 4x4 or 6x2 or 6x4 or others) Front / Rear / Both: (4 x 2) Rear → (4 x 2) Rear Rear axle make: Single Reduction, Hypoid, Semi-floating Banjo type → Single Reduction, Hypoid, Semi-floating Banjo type Rear axle ratio: 4.56 → 5.13 Differential type: Single Reduction, Hypoid, Semi-floating Banjo type → Single Reduction, Hypoid, Semi-floating Banjo type Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2.	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	YYYY – 18.2 :1 Engine coolant temperature: Red Color – Thermometer Symbol → Red Color – Thermometer Symbol Engine low oil pressure: Red Color – Oil Tank Symbol → Red Color – Oil Tank Symbol Brake failure: Red Color – Parking and Oil symbol together → Red Color – Parking and Oil symbol together Brake failure: → Installation drawing showing rear extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size, make, unladen pressure) etc., distance between tyre & RUPD RH & LH, frame rear width, P1, P2 points.: - → - Parking brake: Hand operated [LHS] → Hand operated [LHS] Service Brake: Foot operated → Foot operated Service Brake: → Type of coupling device for brake: NA → NA Detailed engineering drawing as specified in AIS – 013 or photographs showing the mounting details with dimensions.: NA → NA Distance between the contour points of the orthogonal projections on to plane “P” of the accelerator pedal and service brake pedal bearing surfaces, “E” in	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	mm.: -- → -- Distance between the projection of the service brake pedal on to the reference plane “P”, to the right, “H” in mm.: -- → -- Distance between the projection of the service brake pedal on to the0 reference plane “P”, to the left, “J” in mm.: -- → -- In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → -- Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side, distance between front extremity & FUPD, width of FUPD, ID marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Drawing / Images of position & arrangement of Engine: -- → -- Drawing/ Images of Under bonnet (Engine house) Components.: -- → -- 1st · Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	· Overall Ratio / AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC: 12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56 4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 → 20.44 Make: M/s Brakes India Private Limited → M/s Brakes India Private Limited Type (Mechanical/hydraulic/air/air assisted/vacuum assisted/others):		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Vacuum assisted hydraulic → Vacuum assisted hydraulic Control system & braking wheel: Hydraulic acting on all wheels →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Hydraulic acting on all wheels Schematic layout indicating method of split of brake system, location of valves, reservoirs, ABS components etc. (Attach drawing and indicate the drawing number): AJ1XXX00 → AJ1XXX00 Anti-Lock braking system Provided (Yes / No / Optional): NA → NA If yes, details of ABS: - → - Make: - → - Category of ABS: - → - Nos. of directly controlled wheel(s): - → - Brief description of failure warning tell–tale: - → - Wheel Speed Sensors: - → - No. of sensors: - → - Make of sensors: - → - Type of sensors: - → - Modulator: - → - Nos. of Modulators: - → - Make of Modulators: - → - Identification No. / Part No. of Modulator: - → - Brief description and features: - → - Controller: - → - Nos. of Controller: - → - Make of Controller: - → - Identification No. / Part No. of Controller: - → - Brief description and features: - → - Height of Center of Gravity		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(mm): - → - Un laden condition: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Laden condition: - → - Slack adjuster (only applicable for M2, M3, N and T category of vehicles.): - → - Front (Automatic / Manual): - → - Make of slack adjuster: - → - Model and Type: - → - Rear (Automatic / Manual): - → - Make of slack adjuster: - → - Model and Type: - → - Front wheel: 132x47.3x10.35 → 132x47.3x10.35 Rear wheel: LEADING - 314x51.8x5.6 TRAILING – 246x51.8x5.6 → LEADING - 314x51.8x5.6 TRAILING – 246x51.8x5.6 Others: NA → NA Type of liner wear indicator (window/ acoustic/ optical/ any other): Window → Window Front: 250 sq.cm → 250 sq.cm Rear: 570 sq.cm → 570 sq.cm Others: NA → NA Front wheel / axle: Option 1: SYY 6XXXYA → Option 1: SYY 6XXXYA Rear wheel / axle: APACHE 4059M → APACHE 4059M Others: NA → NA Whether asbestos or asbestos-free: Asbestos Free → Asbestos Free Front axle (Disc / Drum): DISC → DISC Effective Diameter (mm): 212 → 212 Rear axle (Disc / drum): DRUM → DRUM Effective Diameter (mm): 282 → 282 Other axle (		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Disc / Drum): NA → NA Effective diameter (mm): NA → NA Material (if the braking surface is not ferrous): NA → NA Front wheel: -- → -- Rear wheel: -- → -- Others: -- → -- Make: M/s XXXX YYYY → M/s XXXX YYYY Type (only applicable for M2, M3, N and T category of vehicles.): N Category → N Category Inner diameter of the master cylinder (mm): 22.81 → 22.81 Operating stroke (mm): 41 → 41 Type of supply tank (only applicable for M2, M3, N and T category of vehicles.): → Brake Chamber: Rear → Rear Brake Chamber: Parking → Parking Make and type: NA → NA Make and type: NA → NA Size mm: NA → NA Size mm: NA → NA Inner diameter mm: NA → NA Inner diameter mm: NA → NA Stroke mm: NA → NA Stroke mm: NA → NA Make and Identification No.: MAKE: XXX GIRLING TAC No: AB2XX1. CoP No: SYL/29/2020-2021/300000YY60/ADDM/XXXX → MAKE: XXX GIRLING TAC No: AB2XX1. CoP No: SYL/29/2020-2021/300000YY60/ADDM/X	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	XXX Type of expansion (HL / HR): HL → HL Free Length of hoses: FOR NON ABS. FRONT : L1 – 470, L2 -417.7 REAR :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	L1 – 405, L2 – 360.9 → FOR NON ABS. FRONT : L1 – 470, L2 -417.7 REAR : L1 – 405, L2 – 360.9 Thickness of lining (mm): 0.6 → 0.6 Nominal bore dia. (mm): 3.1 → 3.1 Type (Visual display/ audible/others): Visual display → Visual display Operation pressure (kg/cm2 / bar / kPa): NA → NA Type of safety device: NA → NA Make: M/s XXXXXYY → M/s XXXXXYY Type (mechanical/spring brake): MECHANICAL → MECHANICAL Acting on Transmission/wheel: WHEEL → WHEEL Control System & Braking wheel: Hand operated & Rear wheel → Hand operated & Rear wheel Lining / pad: Lining → Lining Name of producer: M/s XXXXYYY → M/s XXXXYYY Dimension (mm) (D x W x T): LEADING - 305x48.8x5.6 TRAILING – 256x48.8x5.6 → LEADING - 305x48.8x5.6 TRAILING – 256x48.8x5.6 Area (cm 2): 279 sq.cm → 279 sq.cm Material: XYZ 420 → XYZ 420	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Diameter of brake drum/disc (mm): 272 → 272 Type: UNFAILED PART OF THE SERVICE BRAKE → UNFAILED PART OF THE SERVICE BRAKE Make: Ms. GULF Oil → Ms. GULF Oil Trade name: DOT 4 → DOT 4 Specification/ grade as		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	per Indian standard: DOT 4 → DOT 4 Brake fluid level indicator (Yes/No): Yes → Yes Type (Manual/Auto) Front and Rear: Manual → Manual Lever length: → Pneumatic Brake Actuator: NA → NA Type (as per IS): NA → NA Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No Brake Assist System (Fitted : Yes/No): No → No Category of Brake Assist System A / B: -- → -- For category A systems, define the force threshold at which the ratio between pedal force and brake pressure increases: -- → -- For category B systems, define the brake pedal speed which must be achieved in	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	order to activate the Brake Assist System (e.g. pedal stroke speed (mm/s) during a given time interval): -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Motor Vehicle stability function information document: -- → -- System: -- → -- System variants: -- → -- System options: -- → -- Control function (directional/roll-over/both) including an explanation of the basic function and/or philosophy of the control: -- → -- System configurations (where appropriate): -- → -- System identification including software level identifier: -- → -- List of motor vehicles by description and configuration that are covered by the information document: -- → -- Schematic diagrams of the respective configurations installed on the motor vehicles defined in item 2.1. above with consideration given to the following:: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Lift axles: -- → -- Steering axles: -- → -- Anti-lock braking configurations: -- → -- Wheelbase, track, centre of gravity height: -- → -- Wheel type (single or twin) and tyre type (e.g. structure, category of use, size): -- → --		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Management of the engine or any other source(s) of motive power e.g.Torque Management, control of fuel supply etc.: -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Pneumatic circuits System schematics including anti-lock braking configurations associated with the trailer types defined in paragraph 6.2.1. of this annex.: -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies that	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Vehicle is/ is not equipped to		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	tow a trailer with electric braking system (Yes/No): No → No Make: Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. → Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. Model name / identification: ALXYZABC0 → ALXYZABC0 Type: Compression Ignition → Compression Ignition Bore x stroke (mm): 80 X 98 → 80 X 98 No. of cylinders: 3 → 3 Displacement (cm^3): 1468 cc → 1468 cc Compression ratio: 16.5±1.0 : 1 → 16.5±1.0 : 1 Max. Engine output (KW @ rpm): 55.1 Kw @ 3300 rpm → 55.1 Kw @ 3300 rpm Max. Torque (Nm @ rpm): 170Nm @ 1600 to 2400 rpm → 170Nm @ 1600 to 2400 rpm Air cleaner type: Dry type → Dry type Type of Fuel: Diesel → Diesel Front axle ratio: NA → NA Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2. YYYY Pvt.Ltd Service brake Type (Mechanical/hydraulic/air/air assisted/vacuum assisted/others): Vacuum assisted hydraulic → Vacuum assisted		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	hydraulic Auto Slack Adjuster Fitted (Yes / No / Optional): Yes (Automatic wear adjustment) → Yes (Automatic wear adjustment) Anti-Lock braking system Provided(Yes/No/Optional): Optional → Optional Front (Disc / Drum): Disc → Disc Rear (Disc / Drum): Drum → Drum Total braking area (cm²): Front axle-254 sq.cm Rear axle-580 sq.cm → Front axle-254 sq.cm Rear axle-580 sq.cm Parking brake type: Mechanical → Mechanical Secondary brake type: Un failed part of the service brake → Un failed part of the service brake Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	12 PR Tyre size designation including ply rating: 215/75 R15 LT → 215/75 R15 LT Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	→ 3.26 kg/cm2 Wiper motor: XYZ Ltd. → XYZ Ltd. Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Maximum GVW kg (for rigid vehicles): Refer Annexure T7 – A Details for Recommended load body - Refer Annexure T7 - B → Refer Annexure T7 – A Details for Recommended load body - Refer Annexure T7 - B Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Engine Manufacturer::		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Ashok Leyland Limited → Ashok Leyland Limited Manufacturer's engine model code (as marked on the engine, or other means of identification): Model Code: ALXYZABC0 Enclosure – 4A → Model Code: ALXYZABC0 Enclosure – 4A Location of engine model code and engine serial number on block / crankcase (drawing to be attached): Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A → Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer Enclosure – 4A Working principle: positive ignition / compression- ignition, four-stroke / two-stroke / rotary cycle (DI/IDI) (NA/TC/TCIC/Any other): Compression Ignition, four-stroke, DI, TCIC → Compression Ignition, four-stroke, DI, TCIC Number, arrangement of cylinders:: 3, Inline → 3, Inline Bore----------(mm): 80 → 80 Stroke: ----------(mm): 98 → 98 Firing order:: 1-2-3 → 1-2-3 Engine		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	capacity:----------(cm3): 1468 → 1468 Volumetric compression ratio (specify tolerance): 17.5±1.0 : 1 → 17.5±1.0 : 1 Drawings of combustion chamber and piston crown and, in the case of positive ignition engine, piston rings:: Refer Enclosure – 4B → Refer Enclosure – 4B Vehicle with Defeat Device Yes / No: No → No If yes – Details of the Defeat device: NA → NA Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	equivalent) in PPM content by volume in the exhaust gas with the engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum net power:..... kW at min-1: 55.1 KW @ 3100 ± 50 rpm → 55.1 KW @ 3100 ± 50 rpm Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm Maximum net torque: .....Nm at: -------min-1 ...(manufacturer's		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	declared value): 170 Nm @ 1600 – 2400 rpm → 170 Nm @ 1600 – 2400 rpm Maximum rated speed (Specify tolerance): 3100 rpm ± 50 rpm → 3100 rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Type of fuel used : Mono Fuel / Bi-Fuel / Flex Fuel / Dual Fuel: Mono Fuel → Mono Fuel Fuel type: Diesel / Gasoline / LPG / CNG / Biomethane / Bio-H methane / HCNG / Biogas / LNG / Ethanol ((E85) / 2 (E100)) /ED (95) Flex fuel Methanol M15 / M100 / MD95 / Biodiesel up to 100% /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Hydrogen.: Diesel → Diesel Maximum amount of biofuel acceptable in fuel (manufacturer's declared value): per cent by volume: 7% → 7% By fuel injection (compression-ignition only): (Yes/No): Yes → Yes System description:: Unit Pump Common Rail Direct Injection system → Unit Pump Common Rail Direct Injection system Working principle: direct-injection/pre-chamber/swirl Chamber: Direct Injection → Direct Injection Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) &		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: Plunger Type & FHL0XYZ0 → Plunger Type & FHL0XYZ0 Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Injection advance curve (Specify the tolerances):: ECU Controlled → ECU Controlled Governor: ECU Controlled → ECU Controlled Cut-off point under load:…(min-1): 3300 → 3300 Cut-off point without load (Maximum	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	no-load speed)…(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Multi-Hole Injectors (7 holes) FKXXYYZZ → Multi-Hole Injectors (7 holes) FKXXYYZZ Opening pressure (kPa) (specify the tolerances) or characteristic diagram: 15000 to 23000 kPa → 15000 to 23000 kPa Length and internal diameter…(mm): Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) -		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	193.5± 3 Internal Diameter – 3mm → Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm Common rail make & type: M/s. XXXXYYY, Hot Forged Rail Type → M/s. XXXXYYY, Hot Forged Rail Type Cold start system: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description:: NA → NA Auxiliary starting aid: NA → NA Make(s):: NA → NA Type(s):: NA → NA System description:: NA → NA Electronic controlled injection: (Yes/No):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA → NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s. XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and Metering type Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make and type of air temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type By fuel injection (positive ignition only): (Yes/No): No → No Working principle: intake manifold (single/multi- point)/direct injection/other (specify): NA → NA Make(s) of Injection Pump :: NA → NA Type(s) & Identification		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number of Injection Pump:: NA → NA System description (in the case of systems other than continuous injection give equivalent details):: NA → NA Make and type of the control unit:: NA → NA Make and type of the fuel regulator:: NA → NA Make and type of the air-flow sensor:: NA → NA Make and type of the micro-switch:: NA → NA Make and type of the throttle housing:: NA → NA Make and type of the water temperature sensor:: NA → NA Make and type of the air temperature	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	sensor:: NA → NA Injectors: Opening pressure (Specify the tolerances):,..... (kPa) or characteristic diagram:: NA → NA Make(s):: NA → NA Type(s):: NA → NA Identification Number: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA Cut-off point under load (min-1): NA → NA Injection timing:: NA → NA Cold start system:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Operating principle(s):: NA → NA Operating limits/settings(Specify the tolerances): NA → NA Pressure, (Specify the tolerances).....(kPa) or characteristic diagram:: 320 ± 50 kPa → 320 ± 50 kPa Rated voltage:...V, positive/negative ground: 12V, Negative Ground → 12V, Negative Ground Generator/ Alternator: Alternator → Alternator Type:: DC – Alternator with Voltage regulator → DC – Alternator with Voltage regulator Nominal output: ...VA: 1205 VA (For Non AC Variants) 850 Amps	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	1745 VA (For AC Variants) 120 Amps → 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps Ignition: NA → NA Make(s):: NA → NA Type(s):: NA → NA Working principle:: NA → NA Ignition advance curve or map(Specify the tolerances):: NA → NA Static ignition timing (Specify the tolerances):.....degrees before TDC: NA → NA Spark plugs: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Gap setting----------(mm): NA → NA Nominal resistance (kilo ohm) (if resistive type): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Ignition coil(s): NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Ignition condenser: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA EMI suppressor cap / Device / Electronic unit: NA → NA Make: NA → NA Type (Resistive/Capacitive): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Identification number: NA → NA Nominal resistance (kilo ohm): NA → NA Terminology and Drawing of interference Suppression equipment: NA → NA H.T. Cable: NA → NA Make and Place: NA → NA Type (Resistive/Non-resistive): NA → NA Length (mm) (if resistive type): NA → NA Outside dia. (mm) (if resistive type): NA → NA Nominal resistance kilo ohm, (if resistive type): NA → NA Radiator drawing(s): Refer enclosure 4C2 & 4C4 → Refer enclosure 4C2 & 4C4 Radiator drawing(s): Refer enclosurev 4C1 & 4C3 → Refer enclosurev 4C1 & 4C3 Make: Mahle Behr → Mahle Behr Make: Tata Toyo → Tata Toyo Type (s): Cross flow → Cross flow Relief valve pressure setting: 0.9 kg/cm2 → 0.9 kg/cm2 Fan characteristics (Fan power, kW) Enclose the fan		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	power curve corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part No: 9 → 9	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Material of blades (metal / plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ± 150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150 (AC) Radiator core open area (cm²): 1906 cm2 → 1906 cm2 Cooling system: liquid/air: Liquid → Liquid Nominal setting of the engine temperature control o mechanism (C)::: 85 deg C → 85 deg C Liquid: Yes → Yes Nature of liquid:: Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L → Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	coolant capacity: approx..8 L Circulating pump(s): (Yes/No): Yes → Yes Characteristics of Circulating pump:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm → Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm Make(s):: XXXXX Fasteners Ltd → XXXXX Fasteners Ltd Type(s):: Impeller Type → Impeller Type Drive ratio(s):: 1: 1.12 → 1: 1.12 Description of the fan and its drive mechanism:: Electric Fan , Motor Driven → Electric Fan , Motor Driven Blower: (Yes/No): No → No Characteristics of blower: NA → NA Make(s):: NA → NA Type(s):: NA → NA Drive ratio(s):: NA → NA Pressure charger: (Yes/No) Turbocharger / Supercharger: Yes, Turbo charger → Yes, Turbo charger Make(s):: M/s. XYZ, India → M/s. XYZ, India Type(s) & Identification Number:: Waste Gate with Vacuum Actuator, X3M1XXXX → Waste Gate with Vacuum Actuator, X3M1XXXX Description of the system (maximum charge pressure:.....(kPa), waste-gate, if applicable): Waste gate turbocharger with vacuum		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	actuator Maximum Charge pressure - 145 kPa → Waste gate turbocharger with vacuum actuator	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	actuator Maximum Charge pressure - 145 kPa Inter-cooler: (Yes/No): Yes → Yes Type: air-air / air-water: Air – Air → Air – Air Make :: Mahle Behr → Mahle Behr Make :: Tata Toyo → Tata Toyo Identification Number:: B1V42XXX → B1V42XXX Identification Number:: B1V42YYY → B1V42YYY Air pressure drop across the intercooler: 15 kPa → 15 kPa Intake depression at rated engine speed and at 100 per cent load (Specify location of measurement) Minimum allowable:... (kPa) Maximum allowable:... (kPa): Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Description and drawings of inlet pipes and their accessories (plenum chamber, heating device, additional air intakes, etc.):: Intake manifold with Electronic throttle valve Refer Enclosure 4D → Intake manifold with Electronic throttle valve Refer Enclosure 4D Intake manifold description		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	including dimensional drawings and/or photographs):: Aluminium Intake manifold with pressure sensor Refer Enclosure 4D → Aluminium Intake manifold with pressure sensor Refer Enclosure 4D Air filter, Dimensional drawings: Refer Enclosure 4E → Refer Enclosure 4E Make(s):: M/s. Fleet guard → M/s. Fleet guard Type(s) & Identification Number:: Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 → Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 Intake silencer, Dimensional drawing: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description and Dimensional drawing of the exhaust manifold:: Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F → Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F Description and drawing with dimensions of the exhaust system:(Description with general arrangement of exhaust system along with its routing indicating the lengths of exhaust	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	pipe, tail pipe and: Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly → Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly Effective volume of exhaust-system (specify the tolerance & range) in liters (from exhaust manifold / TC outlet to tail pipe end), Enclose the exhaust system dimensional drawing and indicate the volume of each parts clearly: 13.05 ± 1 litre Refer Enclosure 4G → 13.05 ± 1 litre Refer Enclosure 4G Description and drawing of the exhaust Muffler/Silencer:: NA → NA Type(s) & Identification	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number:: NA → NA Make: NA → NA Number:.....: NA → NA Internal diameter of exhaust pipe (mm)(min): diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G → diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G Minimum distance between exhaust pipe(s) and the fuel line: >150 mm → >150 mm Auxiliary Noise shields for compliance to IS 3028 and / OR AIS-020 (If Provided): NA → NA Material: NA → NA Layout of noise shield / Photographs / Diagram Showing arrangements indicating fitment on vehicle.: NA → NA Maximum allowable exhaust back pressure at rated engine speed and at 100 per cent load (location of measurement): ... (kPa): 30.0 kpa (150 mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Minimum cross-sectional areas of inlet and outlet ports:: Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm → Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm Inlet (mm): 8.00 → 8.00 Exhaust (mm): 8.00 → 8.00 Opening: 17° bTDC → 17° bTDC Closing: 51° aBDC → 51° aBDC Opening: 66° bBDC → 66°	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	bBDC Closing: 25° aTDC → 25° aTDC Transfer: NA → NA Opening: NA → NA Closing: NA → NA Reference and/or setting ranges (Specify the tolerances):: On Cold Engine → On Cold Engine Inlet: 0.3 to 0.4 mm → 0.3 to 0.4 mm Exhaust: 0.3 to 0.4 mm → 0.3 to 0.4 mm Distribution by ports: NA → NA Volume of crank-case cavity with piston at TDC: NA → NA Reed valve fitted (Yes / No): No → No Description of inlet ports, scavenging and exhaust ports with corresponding timing.: NA → NA Device for recycling crankcase gases (description and drawings):: Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J → Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx Catalyst (if any, and if not covered by another heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	integrated and Passive Selective catalytic reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF / DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No Element – 1 No. → Catalytic Converter – 1No Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type for LNT (periodic / continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:……cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap Bulk Density (GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	and Characteristics:: Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY Type and Identification number:: Type - Passive NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Honey Comb type Material: Cordierite → Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make:: NA → NA Type and Idetification number :: NA → NA NOx Sensor		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Before /After of SCR):: No → No Make:: NA → NA Type and Identification number:: NA → NA Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems, description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent engine test bench cycles, between two cycles where regenerative	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	phases occur under the conditions equivalent to Type I test (Distance "D" in t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Distance “d” in AIS 137): 2 cycles → 2 cycles Description of method employed to determine the number of cycles between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter (DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	International standard (where appropriate):: NA → NA Frequency of reagent refill: continuous/maintenance (where		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	appropriate):: NA → NA Normal operating temperature range (°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY → YYYYYYY Air injection: (Yes/No): No → No Type (pulse air, air pump, etc.) & Identification Number:: NA → NA Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other): Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FEW0XXYY Characteristics (flow rate, etc.):: Flow rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes EGR		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Cooler Make and Identification Number: M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:.....g: NA → NA Canister working capacity:.....g: NA → NA Canister bed	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	volume:.....litre: NA → NA Canister Purge and vent strategy description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve : (Yes/No): NA → NA Make and Identification Number: NA → NA Operating pressure (kPa): NA → NA Liquid		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	fuel hoses material :: NA → NA Schematic drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or drawing (Periodic / Continuous): NA → NA Particulate trap: (Yes/No): Yes → Yes Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dimensions and shape of the particulate trap (capacity):: Dimension :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure 4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type Material - Silicon Carbide → Structure - Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI Electronic Control Unit (ECU): Yes → Yes Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number: P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Misfire detection:: NA → NA Oxygen sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K → Refer Enclosure 4K Particulate traps monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method)::	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer Enclosure 4K → Refer Enclosure 4K List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K → Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K A description of the type and number of the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	EUDC A description of the type of the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing all sensed components with the strategy for fault detection and MI activation (fixed number of driving cycles or statistical method), including a list of relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive explanation for the data	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	given in service $05 Test ID $21 to FF and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles –		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Diagnostics on Controller Area Network (CAN) – Part 4: Requirements for emissions- related systems", a comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137): Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA → NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Systems to ensure the correct operation of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	deactivation of the driver inducement, for use by the rescue services [Note: cross- reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling /disable after parking: NA → NA Lowest concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct operation of NOx control measures: NA → NA Details of any devices designed to influence fuel economy (if not covered by other items):: NA → NA LPG fuelling system: (Yes/No): No → No Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to LPG or back:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	System layout (electrical connections, vacuum connections, compensation hoses, etc.): NA → NA NG fuelling system: (Yes/No): No → No Approval number: NA → NA Electronic engine management control unit for NG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation:: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to NG or back: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.):: NA → NA Hydrogen fuelling system: (Yes/No): No → No Electronic engine management control unit for hydrogen fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Description of the safeguarding of the catalyst at switch- over from petrol to hydrogen or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA H2NG fuelling system: (Yes/No): No → No Percentage of hydrogen in the fuel (maximum specified by the manufacturer): NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to H2NG or back:: NA → NA System lay-out (electrical connections,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	vacuum connections compensation hoses, etc.):: NA → NA Electric motor(Traction): No → No Type (winding, excitation):: NA → NA Maximum hourly output: kW(manufacturer’s declared value): NA → NA Maximum net power:kW(manufacturer’s declared		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	value): NA → NA Maximum 30 minutes power: kW (manufacturer’s declared value): NA → NA Operating voltage: V: NA → NA Make and Identification Number:: → Battery (Traction): NA → NA Number of cells:: NA → NA Mass: ……(kg): NA → NA Capacity: Ah (Amp-hours): NA → NA Position:: NA → NA Engines or motor combinations: NA → NA Hybrid Electric Vehicle: (Yes/No): No → No Category of Hybrid Electric vehicle Off Vehicle Charging/Not Off Vehicle Charging9: NA → NA Operating mode switch: with/without: NA → NA Selectable modes: NA → NA Pure electric: (Yes/No): NA → NA Pure fuel consuming: (Yes/No): NA → NA Hybrid modes: (Yes/No) (if yes, short description): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Description of the energy storage device: (traction battery, capacitor, flywheel/generator...): NA → NA Make(s):: NA → NA Type(s) :: NA → NA Identification number:: NA → NA Kind of electrochemical couple:: NA → NA Energy: ..... (for battery: voltage and capacity Ah in 2 h, for capacitor: J): NA → NA Charger: on		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	board/external/without: NA → NA Electric machines (describe each type of electric machine separately): NA → NA Make:: NA → NA Type:: NA → NA Primary use: traction motor/generator: NA → NA When used as traction motor: mono motor / multi motors (number):: NA → NA Maximum power: kW: NA → NA Working principle:: NA → NA Direct current/alternating current/number of phases:: NA → NA Separate excitation/series/compound: NA → NA Synchronous/asynchronous: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA Identification number:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Power controller: NA → NA Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Vehicle electric range..... km (according to AIS 137):: NA → NA Manufacturer’s recommendation for preconditioning:: NA → NA CO2 (g/km) (Applicable for category M1 with GVW<3.5 T): NA → NA Declared : (rounded to 3 decimal places): NA → NA Fuel consumption (l/100 km) for Petrol, LPG or Diesel and (kg/100km) for CNG and (kWh/100 km) for Electric Driven		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Vehicles(Applicable for category M1 with GVW<3.5 T) (rounded to 3 decimal places)*: NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA *calculated on the declared CO2: NA → NA CO2 Reducing technologies Available (Yes/No): No → No Regenerative braking: NA → NA Start-Stop System: NA → NA Tyre pressure monitoring system: NA → NA 6 or more Speed Transmission: NA → NA Any other technology (manufacture to	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	enclose detail): NA → NA Electric energy consumption for electric vehicles Wh/km: NA → NA Electric energy consumption for pure electric vehicles Wh/km: NA → NA Electric energy consumption for externally chargeable hybrid electric vehicles: NA → NA Electric energy consumption (condition A, combined) Wh/km: NA → NA Electric energy consumption (condition B, combined) Wh/km: NA → NA Electric energy consumption (weighted combined) ….Wh/km: NA → NA Moment of inertia of engine flywheel:: 0.1212 kg/m2 → 0.1212 kg/m2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2 Make: XXYY System → XXYY System Type: 12V → 12V Identification mark: X5XYZ100 → X5XYZ100 Rated voltage: 12V → 12V Number of sweep Frequencies: 2 → 2 Highest sweep frequency (Cycles / min): 62±8 rpm → 62±8 rpm Lowest sweep frequency (Cycles / min): 40±5 rpm → 40±5 rpm Washer tank motor / Washer pump: ELECTRIC MOTOR WITH PUMP → ELECTRIC MOTOR WITH PUMP Make: XYZ → XYZ	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Model: PEAD64XYZS00 → PEAD64XYZS00 Schematic Diagram of Engine cooling system: - → - Maximum vehicle speed for continuous operation (km/h) (Only for AC horns): 80 Kmph → 80 Kmph A list, specifying the parts which constitute the headlamp cleaner and drawings thereof, (e.g. pumps, nozzles, valves, motors and wipers);: -- → -- Description and drawings/photographs of the shapes and constituent materials of the part of the body forming the engine compartment and the part of the passenger compartment nearest to it:: -- → -- Drawings or photographs of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	the position of the metal components housed in the engine compartment (e.g. heating appliances, spare wheel, air filter, steering mechanism, etc.):: -- → -- List of all Electrical components, which include Broadband EMI sources (like HAVC Motor, Wiper Motor, Window motor and Horn etc.) The list shall include Device name, Make/s and Part No. /Type ID/Drawing No: -- → -- System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	only in case of steering column locking systems: Locks the Steering → Locks the Steering Components placed in Engine compartment: - → - Frequency bands (Hz) · Max Output Power (W): Antenna Position at Vehicle, specific conditions for installation and/or use → Antenna Position at Vehicle, specific conditions for installation and/or use Example of Engine No: HJEZXYZ891 → HJEZXYZ891		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
CMVR 96; applicable braking standard; N1 fitment / mandate review ABS / electronic braking applicability and performance	Name and address of engine manufacturing plant (In case of imported vehicles, above details shall be supplied for importer also): Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX → Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Axle spacing in case of multi axle vehicles.: NA → NA Other axle: NA → NA Gross combination weight kg (applicable for articulated / Tractor Trailer combination vehicles): NA → NA Front axle: NA → NA	CEA / report needs review	Confirm ABS fitted / applicable. Unknown fitment / scope is retained for review.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Rear axle: NA → NA Trailer axle / other axles: NA → NA If applicable, maximum mass of trailer whic5h may be coupled: NA → NA Unbraked trailer: NA → NA Reference mass (for vehicles with GVW less than or equal to 3.5 ton) (kg.): Refer Annexure T7-B → Refer Annexure T7-B Make: APOLLO → APOLLO Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade description(s): CEAT WINMILE X3-R → CEAT WINMILE		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	142 Tyre rolling Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing:		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Refer - TC7719XX000 003XXXF → Refer - TC7719XX000	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L-756XXXX → CM/L-756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Spare wheel: 1 → 1 Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E- Marking: NA → NA Others (For articulated / combination vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00 R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	M2, M3, N and T category of vehicles.): 325 mm → 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26 kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Final Drive (4x2 or 4x4 or 6x2 or 6x4 or others) Front / Rear / Both: (4 x 2) Rear → (4 x 2) Rear Rear axle make: Single Reduction, Hypoid, Semi-floating Banjo type → Single Reduction, Hypoid, Semi-floating Banjo type Rear axle ratio: 4.56 → 5.13 Differential type: Single Reduction, Hypoid, Semi-floating Banjo type → Single Reduction, Hypoid, Semi-floating Banjo type Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2.	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	YYYY – 18.2 :1 Engine coolant temperature: Red Color – Thermometer Symbol → Red Color – Thermometer Symbol Engine low oil pressure: Red Color – Oil Tank Symbol → Red Color – Oil Tank Symbol Brake failure: Red Color – Parking and Oil symbol together → Red Color – Parking and Oil symbol together Brake failure: → Installation drawing showing rear extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size, make, unladen pressure) etc., distance between tyre & RUPD RH & LH, frame rear width, P1, P2 points.: - → - Parking brake: Hand operated [LHS] → Hand operated [LHS] Service Brake: Foot operated → Foot operated Service Brake: → Type of coupling device for brake: NA → NA Detailed engineering drawing as specified in AIS – 013 or photographs showing the mounting details with dimensions.: NA → NA Distance between the contour points of the orthogonal projections on to plane “P” of the accelerator pedal and service brake pedal bearing surfaces, “E” in	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	mm.: -- → -- Distance between the projection of the service brake pedal on to the reference plane “P”, to the right, “H” in mm.: -- → -- Distance between the projection of the service brake pedal on to the0 reference plane “P”, to the left, “J” in mm.: -- → -- In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → -- Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side, distance between front extremity & FUPD, width of FUPD, ID marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Drawing / Images of position & arrangement of Engine: -- → -- Drawing/ Images of Under bonnet (Engine house) Components.: -- → -- 1st · Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	· Overall Ratio / AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC: 12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56 4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 → 20.44 Make: M/s Brakes India Private Limited → M/s Brakes India Private Limited Type (Mechanical/hydraulic/air/air assisted/vacuum assisted/others):		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Vacuum assisted hydraulic → Vacuum assisted hydraulic Control system & braking wheel: Hydraulic acting on all wheels →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Hydraulic acting on all wheels Schematic layout indicating method of split of brake system, location of valves, reservoirs, ABS components etc. (Attach drawing and indicate the drawing number): AJ1XXX00 → AJ1XXX00 Anti-Lock braking system Provided (Yes / No / Optional): NA → NA If yes, details of ABS: - → - Make: - → - Category of ABS: - → - Nos. of directly controlled wheel(s): - → - Brief description of failure warning tell–tale: - → - Wheel Speed Sensors: - → - No. of sensors: - → - Make of sensors: - → - Type of sensors: - → - Modulator: - → - Nos. of Modulators: - → - Make of Modulators: - → - Identification No. / Part No. of Modulator: - → - Brief description and features: - → - Controller: - → - Nos. of Controller: - → - Make of Controller: - → - Identification No. / Part No. of Controller: - → - Brief description and features: - → - Height of Center of Gravity		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(mm): - → - Un laden condition: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Laden condition: - → - Slack adjuster (only applicable for M2, M3, N and T category of vehicles.): - → - Front (Automatic / Manual): - → - Make of slack adjuster: - → - Model and Type: - → - Rear (Automatic / Manual): - → - Make of slack adjuster: - → - Model and Type: - → - Front wheel: 132x47.3x10.35 → 132x47.3x10.35 Rear wheel: LEADING - 314x51.8x5.6 TRAILING – 246x51.8x5.6 → LEADING - 314x51.8x5.6 TRAILING – 246x51.8x5.6 Others: NA → NA Type of liner wear indicator (window/ acoustic/ optical/ any other): Window → Window Front: 250 sq.cm → 250 sq.cm Rear: 570 sq.cm → 570 sq.cm Others: NA → NA Front wheel / axle: Option 1: SYY 6XXXYA → Option 1: SYY 6XXXYA Rear wheel / axle: APACHE 4059M → APACHE 4059M Others: NA → NA Whether asbestos or asbestos-free: Asbestos Free → Asbestos Free Front axle (Disc / Drum): DISC → DISC Effective Diameter (mm): 212 → 212 Rear axle (Disc / drum): DRUM → DRUM Effective Diameter (mm): 282 → 282 Other axle (		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Disc / Drum): NA → NA Effective diameter (mm): NA → NA Material (if the braking surface is not ferrous): NA → NA Front wheel: -- → -- Rear wheel: -- → -- Others: -- → -- Make: M/s XXXX YYYY → M/s XXXX YYYY Type (only applicable for M2, M3, N and T category of vehicles.): N Category → N Category Inner diameter of the master cylinder (mm): 22.81 → 22.81 Operating stroke (mm): 41 → 41 Type of supply tank (only applicable for M2, M3, N and T category of vehicles.): → Brake Chamber: Rear → Rear Brake Chamber: Parking → Parking Make and type: NA → NA Make and type: NA → NA Size mm: NA → NA Size mm: NA → NA Inner diameter mm: NA → NA Inner diameter mm: NA → NA Stroke mm: NA → NA Stroke mm: NA → NA Make and Identification No.: MAKE: XXX GIRLING TAC No: AB2XX1. CoP No: SYL/29/2020-2021/300000YY60/ADDM/XXXX → MAKE: XXX GIRLING TAC No: AB2XX1. CoP No: SYL/29/2020-2021/300000YY60/ADDM/X	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	XXX Type of expansion (HL / HR): HL → HL Free Length of hoses: FOR NON ABS. FRONT : L1 – 470, L2 -417.7 REAR :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	L1 – 405, L2 – 360.9 → FOR NON ABS. FRONT : L1 – 470, L2 -417.7 REAR : L1 – 405, L2 – 360.9 Thickness of lining (mm): 0.6 → 0.6 Nominal bore dia. (mm): 3.1 → 3.1 Type (Visual display/ audible/others): Visual display → Visual display Operation pressure (kg/cm2 / bar / kPa): NA → NA Type of safety device: NA → NA Make: M/s XXXXXYY → M/s XXXXXYY Type (mechanical/spring brake): MECHANICAL → MECHANICAL Acting on Transmission/wheel: WHEEL → WHEEL Control System & Braking wheel: Hand operated & Rear wheel → Hand operated & Rear wheel Lining / pad: Lining → Lining Name of producer: M/s XXXXYYY → M/s XXXXYYY Dimension (mm) (D x W x T): LEADING - 305x48.8x5.6 TRAILING – 256x48.8x5.6 → LEADING - 305x48.8x5.6 TRAILING – 256x48.8x5.6 Area (cm 2): 279 sq.cm → 279 sq.cm Material: XYZ 420 → XYZ 420	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Diameter of brake drum/disc (mm): 272 → 272 Type: UNFAILED PART OF THE SERVICE BRAKE → UNFAILED PART OF THE SERVICE BRAKE Make: Ms. GULF Oil → Ms. GULF Oil Trade name: DOT 4 → DOT 4 Specification/ grade as		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	per Indian standard: DOT 4 → DOT 4 Brake fluid level indicator (Yes/No): Yes → Yes Type (Manual/Auto) Front and Rear: Manual → Manual Lever length: → Pneumatic Brake Actuator: NA → NA Type (as per IS): NA → NA Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No Brake Assist System (Fitted : Yes/No): No → No Category of Brake Assist System A / B: -- → -- For category A systems, define the force threshold at which the ratio between pedal force and brake pressure increases: -- → -- For category B systems, define the brake pedal speed which must be achieved in	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	order to activate the Brake Assist System (e.g. pedal stroke speed (mm/s) during a given time interval): -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Motor Vehicle stability function information document: -- → -- System: -- → -- System variants: -- → -- System options: -- → -- Control function (directional/roll-over/both) including an explanation of the basic function and/or philosophy of the control: -- → -- System configurations (where appropriate): -- → -- System identification including software level identifier: -- → -- List of motor vehicles by description and configuration that are covered by the information document: -- → -- Schematic diagrams of the respective configurations installed on the motor vehicles defined in item 2.1. above with consideration given to the following:: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Lift axles: -- → -- Steering axles: -- → -- Anti-lock braking configurations: -- → -- Wheelbase, track, centre of gravity height: -- → -- Wheel type (single or twin) and tyre type (e.g. structure, category of use, size): -- → --		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Management of the engine or any other source(s) of motive power e.g.Torque Management, control of fuel supply etc.: -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Pneumatic circuits System schematics including anti-lock braking configurations associated with the trailer types defined in paragraph 6.2.1. of this annex.: -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies that	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Vehicle is/ is not equipped to		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	tow a trailer with electric braking system (Yes/No): No → No Make: Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. → Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. Model name / identification: ALXYZABC0 → ALXYZABC0 Type: Compression Ignition → Compression Ignition Bore x stroke (mm): 80 X 98 → 80 X 98 No. of cylinders: 3 → 3 Displacement (cm³): 1468 cc → 1468 cc Compression ratio: 16.5±1.0 : 1 → 16.5±1.0 : 1 Max. Engine output (KW @ rpm): 55.1 Kw @ 3300 rpm → 55.1 Kw @ 3300 rpm Max. Torque (Nm @ rpm): 170Nm @ 1600 to 2400 rpm → 170Nm @ 1600 to 2400 rpm Air cleaner type: Dry type → Dry type Type of Fuel: Diesel → Diesel Front axle ratio: NA → NA Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2. YYYY Pvt.Ltd Service brake Type (Mechanical/hydraulic/air/air assisted/vacuum assisted/others): Vacuum assisted hydraulic → Vacuum assisted		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	hydraulic Auto Slack Adjuster Fitted (Yes / No / Optional): Yes (Automatic wear adjustment) → Yes (Automatic wear adjustment) Anti-Lock braking system Provided(Yes/No/Optional): Optional → Optional Front (Disc / Drum): Disc → Disc Rear (Disc / Drum): Drum → Drum Total braking area (cm²): Front axle-254 sq.cm Rear axle-580 sq.cm → Front axle-254 sq.cm Rear axle-580 sq.cm Parking brake type: Mechanical → Mechanical Secondary brake type: Un failed part of the service brake → Un failed part of the service brake Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	12 PR Tyre size designation including ply rating: 215/75 R15 LT → 215/75 R15 LT Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	→ 3.26 kg/cm2 Wiper motor: XYZ Ltd. → XYZ Ltd. Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Maximum GVW kg (for rigid vehicles): Refer Annexure T7 – A Details for Recommended load body - Refer Annexure T7 - B → Refer Annexure T7 – A Details for Recommended load body - Refer Annexure T7 - B Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Engine Manufacturer::		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Ashok Leyland Limited → Ashok Leyland Limited Manufacturer's engine model code (as marked on the engine, or other means of identification): Model Code: ALXYZABC0 Enclosure – 4A → Model Code: ALXYZABC0 Enclosure – 4A Location of engine model code and engine serial number on block / crankcase (drawing to be attached): Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A → Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer Enclosure – 4A Working principle: positive ignition / compression- ignition, four-stroke / two-stroke / rotary cycle (DI/IDI) (NA/TC/TCIC/Any other): Compression Ignition, four-stroke, DI, TCIC → Compression Ignition, four-stroke, DI, TCIC Number, arrangement of cylinders:: 3, Inline → 3, Inline Bore----------(mm): 80 → 80 Stroke: ----------(mm): 98 → 98 Firing order:: 1-2-3 → 1-2-3 Engine		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	capacity:----------(cm3): 1468 → 1468 Volumetric compression ratio (specify tolerance): 17.5±1.0 : 1 → 17.5±1.0 : 1 Drawings of combustion chamber and piston crown and, in the case of positive ignition engine, piston rings:: Refer Enclosure – 4B → Refer Enclosure – 4B Vehicle with Defeat Device Yes / No: No → No If yes – Details of the Defeat device: NA → NA Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	equivalent) in PPM content by volume in the exhaust gas with the engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum net power:..... kW at min-1: 55.1 KW @ 3100 ± 50 rpm → 55.1 KW @ 3100 ± 50 rpm Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm Maximum net torque: .....Nm at: -------min-1 ...(manufacturer's		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	declared value): 170 Nm @ 1600 – 2400 rpm → 170 Nm @ 1600 – 2400 rpm Maximum rated speed (Specify tolerance): 3100 rpm ± 50 rpm → 3100 rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Type of fuel used : Mono Fuel / Bi-Fuel / Flex Fuel / Dual Fuel: Mono Fuel → Mono Fuel Fuel type: Diesel / Gasoline / LPG / CNG / Biomethane / Bio-H methane / HCNG / Biogas / LNG / Ethanol ((E85) / 2 (E100)) /ED (95) Flex fuel Methanol M15 / M100 / MD95 / Biodiesel up to 100% /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Hydrogen.: Diesel → Diesel Maximum amount of biofuel acceptable in fuel (manufacturer's declared value): per cent by volume: 7% → 7% By fuel injection (compression-ignition only): (Yes/No): Yes → Yes System description:: Unit Pump Common Rail Direct Injection system → Unit Pump Common Rail Direct Injection system Working principle: direct-injection/pre-chamber/swirl Chamber: Direct Injection → Direct Injection Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) &		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: Plunger Type & FHL0XYZ0 → Plunger Type & FHL0XYZ0 Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Injection advance curve (Specify the tolerances):: ECU Controlled → ECU Controlled Governor: ECU Controlled → ECU Controlled Cut-off point under load:...(min-1): 3300 → 3300 Cut-off point without load (Maximum	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	no-load speed)…(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Multi-Hole Injectors (7 holes) FKXXYYZZ → Multi-Hole Injectors (7 holes) FKXXYYZZ Opening pressure (kPa) (specify the tolerances) or characteristic diagram: 15000 to 23000 kPa → 15000 to 23000 kPa Length and internal diameter…(mm): Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) -		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	193.5± 3 Internal Diameter – 3mm → Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm Common rail make & type: M/s. XXXXYYY, Hot Forged Rail Type → M/s. XXXXYYY, Hot Forged Rail Type Cold start system: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description:: NA → NA Auxiliary starting aid: NA → NA Make(s):: NA → NA Type(s):: NA → NA System description:: NA → NA Electronic controlled injection: (Yes/No):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA → NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s. XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and Metering type Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make and type of air temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type By fuel injection (positive ignition only): (Yes/No): No → No Working principle: intake manifold (single/multi- point)/direct injection/other (specify): NA → NA Make(s) of Injection Pump :: NA → NA Type(s) & Identification		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number of Injection Pump:: NA → NA System description (in the case of systems other than continuous injection give equivalent details):: NA → NA Make and type of the control unit:: NA → NA Make and type of the fuel regulator:: NA → NA Make and type of the air-flow sensor:: NA → NA Make and type of the micro-switch:: NA → NA Make and type of the throttle housing:: NA → NA Make and type of the water temperature sensor:: NA → NA Make and type of the air temperature	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	sensor:: NA → NA Injectors: Opening pressure (Specify the tolerances):,..... (kPa) or characteristic diagram:: NA → NA Make(s):: NA → NA Type(s):: NA → NA Identification Number: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA Cut-off point under load (min-1): NA → NA Injection timing:: NA → NA Cold start system:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Operating principle(s):: NA → NA Operating limits/settings(Specify the tolerances): NA → NA Pressure, (Specify the tolerances).....(kPa) or characteristic diagram:: 320 ± 50 kPa → 320 ± 50 kPa Rated voltage:...V, positive/negative ground: 12V, Negative Ground → 12V, Negative Ground Generator/ Alternator: Alternator → Alternator Type:: DC – Alternator with Voltage regulator → DC – Alternator with Voltage regulator Nominal output: ...VA: 1205 VA (For Non AC Variants) 850 Amps	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	1745 VA (For AC Variants) 120 Amps → 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps Ignition: NA → NA Make(s):: NA → NA Type(s):: NA → NA Working principle:: NA → NA Ignition advance curve or map(Specify the tolerances):: NA → NA Static ignition timing (Specify the tolerances):.....degrees before TDC: NA → NA Spark plugs: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Gap setting----------(mm): NA → NA Nominal resistance (kilo ohm) (if resistive type): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Ignition coil(s): NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Ignition condenser: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA EMI suppressor cap / Device / Electronic unit: NA → NA Make: NA → NA Type (Resistive/Capacitive): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Identification number: NA → NA Nominal resistance (kilo ohm): NA → NA Terminology and Drawing of interference Suppression equipment: NA → NA H.T. Cable: NA → NA Make and Place: NA → NA Type (Resistive/Non-resistive): NA → NA Length (mm) (if resistive type): NA → NA Outside dia. (mm) (if resistive type): NA → NA Nominal resistance kilo ohm, (if resistive type): NA → NA Radiator drawing(s): Refer enclosure 4C2 & 4C4 → Refer enclosure 4C2 & 4C4 Radiator drawing(s): Refer enclosurev 4C1 & 4C3 → Refer enclosurev 4C1 & 4C3 Make: Mahle Behr → Mahle Behr Make: Tata Toyo → Tata Toyo Type (s): Cross flow → Cross flow Relief valve pressure setting: 0.9 kg/cm2 → 0.9 kg/cm2 Fan characteristics (Fan power, kW) Enclose the fan		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	power curve corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part No: 9 → 9	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Material of blades (metal / plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ± 150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150 (AC) Radiator core open area (cm²): 1906 cm2 → 1906 cm2 Cooling system: liquid/air: Liquid → Liquid Nominal setting of the engine temperature control o mechanism (C)::: 85 deg C → 85 deg C Liquid: Yes → Yes Nature of liquid:: Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L → Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	coolant capacity: approx..8 L Circulating pump(s): (Yes/No): Yes → Yes Characteristics of Circulating pump:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm → Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm Make(s):: XXXXX Fasteners Ltd → XXXXX Fasteners Ltd Type(s):: Impeller Type → Impeller Type Drive ratio(s):: 1: 1.12 → 1: 1.12 Description of the fan and its drive mechanism:: Electric Fan , Motor Driven → Electric Fan , Motor Driven Blower: (Yes/No): No → No Characteristics of blower: NA → NA Make(s):: NA → NA Type(s):: NA → NA Drive ratio(s):: NA → NA Pressure charger: (Yes/No) Turbocharger / Supercharger: Yes, Turbo charger → Yes, Turbo charger Make(s):: M/s. XYZ, India → M/s. XYZ, India Type(s) & Identification Number:: Waste Gate with Vacuum Actuator, X3M1XXXX → Waste Gate with Vacuum Actuator, X3M1XXXX Description of the system (maximum charge pressure:.....(kPa), waste-gate, if applicable): Waste gate turbocharger with vacuum		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	actuator Maximum Charge pressure - 145 kPa → Waste gate turbocharger with vacuum actuator	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	actuator Maximum Charge pressure - 145 kPa Inter-cooler: (Yes/No): Yes → Yes Type: air-air / air-water: Air – Air → Air – Air Make :: Mahle Behr → Mahle Behr Make :: Tata Toyo → Tata Toyo Identification Number:: B1V42XXX → B1V42XXX Identification Number:: B1V42YYY → B1V42YYY Air pressure drop across the intercooler: 15 kPa → 15 kPa Intake depression at rated engine speed and at 100 per cent load (Specify location of measurement) Minimum allowable:... (kPa) Maximum allowable:... (kPa): Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Description and drawings of inlet pipes and their accessories (plenum chamber, heating device, additional air intakes, etc.):: Intake manifold with Electronic throttle valve Refer Enclosure 4D → Intake manifold with Electronic throttle valve Refer Enclosure 4D Intake manifold description		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	including dimensional drawings and/or photographs):: Aluminium Intake manifold with pressure sensor Refer Enclosure 4D → Aluminium Intake manifold with pressure sensor Refer Enclosure 4D Air filter, Dimensional drawings: Refer Enclosure 4E → Refer Enclosure 4E Make(s):: M/s. Fleet guard → M/s. Fleet guard Type(s) & Identification Number:: Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 → Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 Intake silencer, Dimensional drawing: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description and Dimensional drawing of the exhaust manifold:: Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F → Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F Description and drawing with dimensions of the exhaust system:(Description with general arrangement of exhaust system along with its routing indicating the lengths of exhaust	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	pipe, tail pipe and: Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly → Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly Effective volume of exhaust-system (specify the tolerance & range) in liters (from exhaust manifold / TC outlet to tail pipe end), Enclose the exhaust system dimensional drawing and indicate the volume of each parts clearly: 13.05 ± 1 litre Refer Enclosure 4G → 13.05 ± 1 litre Refer Enclosure 4G Description and drawing of the exhaust Muffler/Silencer:: NA → NA Type(s) & Identification	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number:: NA → NA Make: NA → NA Number:.....: NA → NA Internal diameter of exhaust pipe (mm)(min): diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G → diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G Minimum distance between exhaust pipe(s) and the fuel line: >150 mm → >150 mm Auxiliary Noise shields for compliance to IS 3028 and / OR AIS-020 (If Provided): NA → NA Material: NA → NA Layout of noise shield / Photographs / Diagram Showing arrangements indicating fitment on vehicle.: NA → NA Maximum allowable exhaust back pressure at rated engine speed and at 100 per cent load (location of measurement): ... (kPa): 30.0 kpa (150 mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Minimum cross-sectional areas of inlet and outlet ports:: Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm → Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm Inlet (mm): 8.00 → 8.00 Exhaust (mm): 8.00 → 8.00 Opening: 17° bTDC → 17° bTDC Closing: 51° aBDC → 51° aBDC Opening: 66° bBDC → 66°	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	bBDC Closing: 25° aTDC → 25° aTDC Transfer: NA → NA Opening: NA → NA Closing: NA → NA Reference and/or setting ranges (Specify the tolerances):: On Cold Engine → On Cold Engine Inlet: 0.3 to 0.4 mm → 0.3 to 0.4 mm Exhaust: 0.3 to 0.4 mm → 0.3 to 0.4 mm Distribution by ports: NA → NA Volume of crank-case cavity with piston at TDC: NA → NA Reed valve fitted (Yes / No): No → No Description of inlet ports, scavenging and exhaust ports with corresponding timing.: NA → NA Device for recycling crankcase gases (description and drawings):: Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J → Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx Catalyst (if any, and if not covered by another heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	integrated and Passive Selective catalytic reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF / DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No Element – 1 No. → Catalytic Converter – 1No Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type for LNT (periodic / continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:……cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap Bulk Density (GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	and Characteristics:: Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY Type and Identification number:: Type - Passive NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Honey Comb type Material: Cordierite → Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make:: NA → NA Type and Idetification number :: NA → NA NOx Sensor		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Before /After of SCR):: No → No Make:: NA → NA Type and Identification number:: NA → NA Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems, description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent engine test bench cycles, between two cycles where regenerative	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	phases occur under the conditions equivalent to Type I test (Distance "D" in t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Distance “d” in AIS 137): 2 cycles → 2 cycles Description of method employed to determine the number of cycles between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter (DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	International standard (where appropriate):: NA → NA Frequency of reagent refill: continuous/maintenance (where		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	appropriate):: NA → NA Normal operating temperature range (°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY → YYYYYYY Air injection: (Yes/No): No → No Type (pulse air, air pump, etc.) & Identification Number:: NA → NA Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other): Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FEW0XXYY Characteristics (flow rate, etc.):: Flow rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes EGR		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Cooler Make and Identification Number: M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:.....g: NA → NA Canister working capacity:.....g: NA → NA Canister bed	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	volume:.....litre: NA → NA Canister Purge and vent strategy description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve : (Yes/No): NA → NA Make and Identification Number: NA → NA Operating pressure (kPa): NA → NA Liquid		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	fuel hoses material :: NA → NA Schematic drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or drawing (Periodic / Continuous): NA → NA Particulate trap: (Yes/No): Yes → Yes Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dimensions and shape of the particulate trap (capacity):: Dimension :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure 4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type Material - Silicon Carbide → Structure - Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI Electronic Control Unit (ECU): Yes → Yes Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number: P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Misfire detection:: NA → NA Oxygen sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K → Refer Enclosure 4K Particulate traps monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method)::	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer Enclosure 4K → Refer Enclosure 4K List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K → Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K A description of the type and number of the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	EUDC A description of the type of the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing all sensed components with the strategy for fault detection and MI activation (fixed number of driving cycles or statistical method), including a list of relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive explanation for the data	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	given in service $05 Test ID $21 to FF and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles –		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Diagnostics on Controller Area Network (CAN) – Part 4: Requirements for emissions- related systems", a comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137): Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA → NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Systems to ensure the correct operation of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	deactivation of the driver inducement, for use by the rescue services [Note: cross- reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling /disable after parking: NA → NA Lowest concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct operation of NOx control measures: NA → NA Details of any devices designed to influence fuel economy (if not covered by other items):: NA → NA LPG fuelling system: (Yes/No): No → No Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to LPG or back:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	System layout (electrical connections, vacuum connections, compensation hoses, etc.): NA → NA NG fuelling system: (Yes/No): No → No Approval number: NA → NA Electronic engine management control unit for NG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation:: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to NG or back: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.):: NA → NA Hydrogen fuelling system: (Yes/No): No → No Electronic engine management control unit for hydrogen fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Description of the safeguarding of the catalyst at switch- over from petrol to hydrogen or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA H2NG fuelling system: (Yes/No): No → No Percentage of hydrogen in the fuel (maximum specified by the manufacturer): NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to H2NG or back:: NA → NA System lay-out (electrical connections,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	vacuum connections compensation hoses, etc.):: NA → NA Electric motor(Traction): No → No Type (winding, excitation):: NA → NA Maximum hourly output: kW(manufacturer’s declared value): NA → NA Maximum net power:kW(manufacturer’s declared		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	value): NA → NA Maximum 30 minutes power: kW (manufacturer’s declared value): NA → NA Operating voltage: V: NA → NA Make and Identification Number:: → Battery (Traction): NA → NA Number of cells:: NA → NA Mass: ……(kg): NA → NA Capacity: Ah (Amp-hours): NA → NA Position:: NA → NA Engines or motor combinations: NA → NA Hybrid Electric Vehicle: (Yes/No): No → No Category of Hybrid Electric vehicle Off Vehicle Charging/Not Off Vehicle Charging9: NA → NA Operating mode switch: with/without: NA → NA Selectable modes: NA → NA Pure electric: (Yes/No): NA → NA Pure fuel consuming: (Yes/No): NA → NA Hybrid modes: (Yes/No) (if yes, short description): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Description of the energy storage device: (traction battery, capacitor, flywheel/generator...): NA → NA Make(s):: NA → NA Type(s) :: NA → NA Identification number:: NA → NA Kind of electrochemical couple:: NA → NA Energy: ..... (for battery: voltage and capacity Ah in 2 h, for capacitor: J): NA → NA Charger: on		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	board/external/without: NA → NA Electric machines (describe each type of electric machine separately): NA → NA Make:: NA → NA Type:: NA → NA Primary use: traction motor/generator: NA → NA When used as traction motor: mono motor / multi motors (number):: NA → NA Maximum power: kW: NA → NA Working principle:: NA → NA Direct current/alternating current/number of phases:: NA → NA Separate excitation/series/compound: NA → NA Synchronous/asynchronous: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA Identification number:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Power controller: NA → NA Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Vehicle electric range..... km (according to AIS 137):: NA → NA Manufacturer’s recommendation for preconditioning:: NA → NA CO2 (g/km) (Applicable for category M1 with GVW<3.5 T): NA → NA Declared : (rounded to 3 decimal places): NA → NA Fuel consumption (l/100 km) for Petrol, LPG or Diesel and (kg/100km) for CNG and (kWh/100 km) for Electric Driven		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Vehicles(Applicable for category M1 with GVW<3.5 T) (rounded to 3 decimal places)*: NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA *calculated on the declared CO2: NA → NA CO2 Reducing technologies Available (Yes/No): No → No Regenerative braking: NA → NA Start-Stop System: NA → NA Tyre pressure monitoring system: NA → NA 6 or more Speed Transmission: NA → NA Any other technology (manufacture to	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	enclose detail): NA → NA Electric energy consumption for electric vehicles Wh/km: NA → NA Electric energy consumption for pure electric vehicles Wh/km: NA → NA Electric energy consumption for externally chargeable hybrid electric vehicles: NA → NA Electric energy consumption (condition A, combined) Wh/km: NA → NA Electric energy consumption (condition B, combined) Wh/km: NA → NA Electric energy consumption (weighted combined) ….Wh/km: NA → NA Moment of inertia of engine flywheel:: 0.1212 kg/m2 → 0.1212 kg/m2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2 Make: XXYY System → XXYY System Type: 12V → 12V Identification mark: X5XYZ100 → X5XYZ100 Rated voltage: 12V → 12V Number of sweep Frequencies: 2 → 2 Highest sweep frequency (Cycles / min): 62±8 rpm → 62±8 rpm Lowest sweep frequency (Cycles / min): 40±5 rpm → 40±5 rpm Washer tank motor / Washer pump: ELECTRIC MOTOR WITH PUMP → ELECTRIC MOTOR WITH PUMP Make: XYZ → XYZ	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Model: PEAD64XYZS00 → PEAD64XYZS00 Schematic Diagram of Engine cooling system: - → - Maximum vehicle speed for continuous operation (km/h) (Only for AC horns): 80 Kmph → 80 Kmph A list, specifying the parts which constitute the headlamp cleaner and drawings thereof, (e.g. pumps, nozzles, valves, motors and wipers);: -- → -- Description and drawings/photographs of the shapes and constituent materials of the part of the body forming the engine compartment and the part of the passenger compartment nearest to it:: -- → -- Drawings or photographs of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	the position of the metal components housed in the engine compartment (e.g. heating appliances, spare wheel, air filter, steering mechanism, etc.):: -- → -- List of all Electrical components, which include Broadband EMI sources (like HAVC Motor, Wiper Motor, Window motor and Horn etc.) The list shall include Device name, Make/s and Part No. /Type ID/Drawing No: -- → -- System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	only in case of steering column locking systems: Locks the Steering → Locks the Steering Components placed in Engine compartment: - → - Frequency bands (Hz) · Max Output Power (W): Antenna Position at Vehicle, specific conditions for installation and/or use → Antenna Position at Vehicle, specific conditions for installation and/or use Example of Engine No: HJEZXYZ891 → HJEZXYZ891		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
CMVR 117; IS 11827; current edition review Speedometer indication and installation	Make: APOLLO → APOLLO Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade description(s): CEAT WINMILE X3-R → CEAT WINMILE X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling	CEA / report needs review	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Resistance as per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 Tyre rolling Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	– SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Tyre Tread Pattern along with Drawing: Refer - TC7719XX000 003XXXF → Refer - TC7719XX000 003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L-756XXXX → CM/L-756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification: TAC No. / BIS License No. / E- Marking: NA → NA Others (For articulated / combination vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00 R15 LT 12 PR	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other (for articulated/combination	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 325 mm → 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26 kg/cm2 Other: NA → NA Tyre		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 Speedometer:: 7.00 R15 LT → 7.00 R15 LT Speedometer:: 215/75 R15 LT → 215/75 R15 LT Type: Electronic →		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Electronic Make, and Identification No. / Part No./ Drawing No.: Pricol Limited → Pricol Limited Make, and Identification No. / Part No./ Drawing No.: Minda stoneridge → Minda stoneridge Make, and Identification No. / Part No./ Drawing No.: Pricol Limited → Pricol Limited Make, and Identification No. / Part No./ Drawing No.: Minda stoneri dge → Minda stoneri dge Make, and Identification No. / Part No./ Drawing No.: Pricol Limited → Pricol Limited Make, and Identification No. / Part No./ Drawing No.: Minda stoneridge → Minda stoneridge Make, and Identification No. / Part No./ Drawing No.: FL102XXX → FL102XXX Make, and Identification No. / Part No./ Drawing No.: FL103XXX → FL103XXX Make, and Identification No. / Part No./ Drawing No.: FL104XXX → FL104XXX Make, and Identification No. / Part No./ Drawing No.: FL103 XXX → FL103 XXX	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make, and Identification No. / Part No./ Drawing No.: FL102XXX → FL102XXX Make, and Identification No. / Part No./ Drawing No.: FL103XXX → FL103XXX Range: 0 -		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	130 km/h → 0 - 130 km/h Major marking: NA (Digital Indicator) → NA (Digital Indicator) Minor marking: NA (Digital Indicator) → NA (Digital Indicator) Speedometer ratio, Pulses / km RAR: 2440 → 2440 Speedometer ratio, Pulses / km RAR: 2424 → 2424 Speedometer ratio, Pulses / km RAR: 2590 → 2590 Ratio of Speedo drive: 0.60 → 0.60 Ratio of Speedo drive: 0.60 → 0.60 Ratio of Speedo drive: 0.64 → 0.64 Installation drawing showing rear extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size, make, unladen pressure) etc., distance between tyre & RUPD RH & LH, frame rear width, P1, P2 points.: - → - In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side, distance between front extremity &		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	FUPD, width of FUPD, ID marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- 1st · Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st · Overall Ratio / AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69 2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC: 12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56 4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC:		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 → 20.44 Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No Wheel type (single or twin) and tyre type (e.g. structure, category of use, size): -- → -- Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT 12 PR Tyre size designation including ply rating: 215/75 R15 LT → 215/75 R15 LT Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2 → 3.26 kg/cm2 Tyre pressure monitoring system: NA → NA		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
AIS-137 (Part 3); route / fuel review Type I tailpipe emissions	Base fuel type used for vehicle Type approval including blend level (Ethanol, Bio-diesel etc): Diesel (Maximum 7% bio-diesel blend) → Diesel (Maximum 7% bio-diesel blend) Name and address of engine manufacturing plant (In case of imported vehicles, above details shall be supplied for importer also): Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX → Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX Reference mass (for vehicles with GVW less than or equal to 3.5 ton) (kg.): Refer Annexure T7-B → Refer Annexure T7-B Make: APOLLO → APOLLO Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT	CEA / report needs review	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Brand(s) name(s) and/or Trade description(s): CEAT WINMILE X3-R → CEAT WINMILE X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 Tyre rolling Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer - TC7719XX000 003XXXF → Refer - TC7719XX000 003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022- 2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022- 2023/300002XXX X/TA/XXXX Identification: TAC No. / BIS License No. / E-		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Identification: TAC No. / BIS License No. / E- Marking: CM/L- 756XXXX → CM/L- 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E- Marking: NA → NA Others (For articulated / combination vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	114/109 Q Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00 R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other (for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 325 mm		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	→ 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26 kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Kmph, 60 Kmph & 40 Kmph with SLF Type: Fully synchronized 5 - forward gears with a Constant mesh reverse gear → Fully synchronized 5 - forward gears with a Constant mesh reverse gear Model name (if any): XXX-XXX → XXX-XXX Model name (if any): YYYY-YY → YYYY-YY Gear shifting control system (Sketch showing gear shifting arrangement): → No. of gears: Forward 5 + 1 Rev → Forward 5 + 1 Rev Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Type of steering gear (Re-circulating ball / Worm & Roller / Rack & Pinion / Others): Re-circulating ball type → Re-circulating ball type Make: 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd → 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2. YYYY – 18.2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	:1 Fuel Gauge: LED Bars → LED Bars Engine coolant temperature: Red Color – Thermometer Symbol → Red Color – Thermometer Symbol	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Engine low oil pressure: Red Color – Oil Tank Symbol → Red Color – Oil Tank Symbol Installation drawing showing rear extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size, make, unladen pressure) etc., distance between tyre & RUPD RH & LH, frame rear width, P1, P2 points.: - → - Make: M/s. XXXXYYYY → M/s. XXXXYYYY Material (Metallic / Plastic etc.): Metallic → Metallic Nominal thickness mm: 1.0mm → 1.0mm Capacity, Litre: 40 Lts, 30 Lts → 40 Lts, 30 Lts Detailed drawing of the fuel tank assembly with material specifications: FHSXXXXY FHSXXXXY → FHSXXXXY FHSXXXXY Detailed drawing indicating the position / location of the fuel tank(s) in the vehicle.: A5LXXX00 (50 LTS) A5LXXX00 (40 LTS) → A5LXXX00 (50 LTS) A5LXXX00 (40 LTS) Identification No. / Part No.: FHS00544 FfffXXXcdHXXS00744 → FHS00544 FfffXXXcdHXXS00744 Detailed		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	engineering drawing as specified in AIS – 013 or photographs showing the mounting details with dimensions.: NA → NA Cross sectional Drawing of Gear shift lever (Knob): FXA0XXYY (Non AC) / FXA0XXYY (AC) → FXA0XXYY (Non AC) / FXA0XXYY (AC) In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → -- Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side, distance between front extremity & FUPD, width of FUPD, ID marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Drawing / Images of position & arrangement of Engine: -- → -- Drawing/ Images of Under bonnet (Engine house) Components.: -- → -- 1st · Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st · Overall Ratio /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC: 12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56 4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 → 20.44 Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No Wheel type		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(single or twin) and tyre type (e.g. structure, category of use, size): -- → -- Gearbox type (e.g. manual, automated manual, semi-automatic, automatic): -- → -- Management of the engine or any other source(s) of motive power e.g.Torque Management, control of fuel supply etc.: -- → -- Make: Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. → Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. Model name / identification: ALXYZABC0 → ALXYZABC0 Type: Compression Ignition → Compression Ignition Bore x stroke (mm): 80 X 98 → 80 X 98 No. of cylinders: 3 → 3 Displacement (cm³): 1468 cc → 1468 cc Compression ratio: 16.5±1.0 : 1 → 16.5±1.0 : 1 Max. Engine output (KW @ rpm): 55.1 Kw @ 3300 rpm → 55.1 Kw @ 3300 rpm Max. Torque (Nm @ rpm): 170Nm @ 1600 to 2400 rpm → 170Nm @ 1600 to 2400 rpm Air cleaner type: Dry type → Dry type Type of Fuel: Diesel → Diesel Make and model: RSB	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Transmission (T210R) → RSB Transmission (T210R) Make and model: AVTEC (A5M210) → AVTEC (A5M210) Type: Manual gearbox with fully synchro mesh & Rev with sliding mesh → Manual gearbox with fully synchro mesh & Rev with sliding mesh No. of gears: 5 Forward and 1 Reverse → 5 Forward and 1 Reverse 1st: 4.62 → 4.62 1st: 4.60 → 4.60 1st: 21.53 → 21.53 1st: 21.44 → 21.44 2nd: 2.68 → 2.68 2nd: 2.69 → 2.69 2nd: 12.67 → 12.67 2nd: 12.71 → 12.71 3rd: 1.51 → 1.51 3rd: 1.49 → 1.49 3rd: 7.34 → 7.34 3rd: 7.25 → 7.25 4th: 1.0 → 1.0 4th: 1.0 → 1.0 4th: 4.56 → 4.56 4th: 4.56 → 4.56 5th: 0.70 → 0.70 5th: 0.71 → 0.71 5th: 3.64 → 3.64 5th: 3.67 → 3.67 Reverse: 4.39 → 4.39 Reverse: 20.36 → 20.36 Reverse: 4.38 → 4.38 Reverse: 20.34 → 20.34 Drive Axle (Front / Rear / All): Rear → Rear	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Front axle ratio: NA → NA Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres:		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT 12 PR Tyre size designation including ply rating: 215/75 R15 LT → 215/75 R15 LT Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2 → 3.26 kg/cm2 Material: Metallic → Metallic Capacity (l): 50 lts , 40 lts → 50 lts , 40 lts Dimensions: 2490WB → 2490WB Dimensions: 2640WB → 2640WB Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Overall width (mm): 1842 → 1842 Overall Length (mm): CC → CC Overall Length (mm): 4750 → 4750	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Overall Length (mm): 4988 → 4988 Overall Length (mm): FSD / HSD → FSD / HSD Overall Length (mm): 4925 → 4925 Overall Length (mm): 5225 → 5225 Overall height (mm): 1961 → 1961 Overall height (mm): 1948 → 1948 Maximum gradeability in 1st gear: GVW 2890 kg → GVW 2890 kg Maximum gradeability in 1st gear: GVW 3390 kg → GVW 3390 kg		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Maximum gradeability in 1st gear: 10 Deg → 10 Deg Maximum gradeability in 1st gear: 8.5 Deg → 8.5 Deg Maximum GVW kg (for rigid vehicles): Refer Annexure T7 – A Details for Recommended load body - Refer Annexure T7 - B → Refer Annexure T7 – A Details for Recommended load body - Refer Annexure T7 - B Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Declared (Rounded to 3 decimal places): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Description of energy converters and power plant. (In the case of a vehicle that can run either on petrol, diesel, etc., or also in combination with another fuel, items shall be repeated.1): Diesel → Diesel Engine Manufacturer:: Ashok Leyland Limited → Ashok Leyland Limited Manufacturer's engine model code (as marked on the engine, or other means of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	identification): Model Code: ALXYZABC0 Enclosure – 4A → Model Code: ALXYZABC0 Enclosure – 4A Location of engine model code and engine serial number on block / crankcase (drawing to be attached): Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A → Engine Model Code Location - On Engine Head Cover Engine Serial Number		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Location - On Engine Block Refer Enclosure – 4A Working principle: positive ignition / compression- ignition, four-stroke / two-stroke / rotary cycle (DI/IDI) (NA/TC/TCIC/Any other): Compression Ignition, four-stroke, DI, TCIC → Compression Ignition, four-stroke, DI, TCIC Number, arrangement of cylinders:: 3, Inline → 3, Inline Bore----------(mm): 80 → 80 Stroke: ----------(mm): 98 → 98 Firing order:: 1-2-3 → 1-2-3 Engine capacity:----------(cm3): 1468 → 1468 Volumetric compression ratio (specify	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	tolerance): 17.5±1.0 : 1 → 17.5±1.0 : 1 Drawings of combustion chamber and piston crown and, in the case of positive ignition engine, piston rings:: Refer Enclosure – 4B → Refer Enclosure – 4B Vehicle with Defeat Device Yes / No: No → No If yes – Details of the Defeat device: NA → NA Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane equivalent) in PPM content by volume in		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	the exhaust gas with the engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum net power:..... kW at min-1: 55.1 KW @ 3100 ± 50 rpm → 55.1 KW @ 3100 ± 50 rpm Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm Maximum net torque: .....Nm at: -------min-1 ...(manufacturer's declared value): 170 Nm @ 1600 – 2400 rpm → 170 Nm @ 1600 – 2400 rpm Maximum rated speed (Specify tolerance): 3100 rpm ± 50 rpm → 3100	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Type of fuel used : Mono Fuel / Bi-Fuel / Flex Fuel / Dual Fuel: Mono Fuel → Mono Fuel Fuel type: Diesel / Gasoline / LPG / CNG / Biomethane / Bio-H methane / HCNG / Biogas / LNG / Ethanol ((E85) / 2 (E100)) /ED (95) Flex fuel Methanol M15 / M100 / MD95 / Biodiesel up to 100% /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Hydrogen.: Diesel → Diesel Maximum amount of biofuel acceptable in fuel (manufacturer's declared value): per cent by volume: 7% → 7% By fuel injection (compression-ignition only): (Yes/No): Yes → Yes System description:: Unit Pump Common Rail Direct Injection system → Unit Pump Common Rail Direct Injection system Working principle: direct-injection/pre-chamber/swirl Chamber: Direct Injection → Direct Injection Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Plunger Type & FHL0XYZ0 → Plunger Type & FHL0XYZ0 Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net power engine speed (min-1) or	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Injection advance curve (Specify the tolerances):: ECU Controlled → ECU Controlled Governor: ECU Controlled → ECU Controlled Cut-off point under load:…(min-1): 3300 →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	3300 Cut-off point without load (Maximum no-load speed)…(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Multi-Hole Injectors (7 holes) FKXXYYZZ → Multi-Hole Injectors (7 holes) FKXXYYZZ Opening pressure (kPa) (specify the tolerances) or characteristic diagram: 15000 to 23000 kPa → 15000 to 23000 kPa Length and internal diameter…(mm): Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm → Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Common rail make & type: M/s. XXXXYYY, Hot Forged Rail Type → M/s. XXXXYYY, Hot Forged Rail Type Cold start system: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description:: NA → NA Auxiliary starting aid: NA → NA Make(s):: NA → NA Type(s):: NA → NA System description:: NA → NA Electronic controlled injection: (Yes/No): Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Rail, Direct Injection with Pressure Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA → NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s. XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and Metering type Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	of air temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type By fuel injection (positive ignition only): (Yes/No): No → No Working principle: intake manifold (single/multi- point)/direct injection/other (specify): NA → NA Make(s) of Injection Pump :: NA → NA Type(s) & Identification Number of Injection Pump:: NA → NA System description (in the case of systems other than continuous injection	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	give equivalent details):: NA → NA Make and type of the control unit:: NA → NA Make and type of the fuel regulator:: NA → NA Make and type of the air-flow sensor:: NA → NA Make and type of the micro-switch:: NA → NA Make and type of the throttle housing:: NA → NA Make and type of the water temperature sensor:: NA → NA Make and type of the air temperature sensor:: NA → NA Injectors: Opening pressure (Specify the		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	tolerances):,..... (kPa) or characteristic diagram:: NA → NA Make(s):: NA → NA Type(s):: NA → NA Identification Number: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA Cut-off point under load (min-1): NA → NA Injection timing:: NA → NA Cold start system:: NA → NA Operating principle(s):: NA → NA Operating limits/settings(Specify the	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	tolerances): NA → NA Pressure, (Specify the tolerances).....(kPa) or characteristic diagram:: 320 ± 50 kPa → 320 ± 50 kPa Rated voltage:...V, positive/negative ground: 12V, Negative Ground → 12V, Negative Ground Generator/ Alternator: Alternator → Alternator Type:: DC – Alternator with Voltage regulator → DC – Alternator with Voltage regulator Nominal output: ...VA: 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps → 1205 VA (For Non		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps Ignition: NA → NA Make(s):: NA → NA Type(s):: NA → NA Working principle:: NA → NA Ignition advance curve or map(Specify the tolerances):: NA → NA Static ignition timing (Specify the tolerances):.....degrees before TDC: NA → NA Spark plugs: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Gap setting----------(mm): NA → NA Nominal resistance (kilo ohm) (if	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	resistive type): NA → NA Ignition coil(s): NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Ignition condenser: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA EMI suppressor cap / Device / Electronic unit: NA → NA Make: NA → NA Type (Resistive/Capacitive): NA → NA Identification number: NA → NA Nominal resistance (kilo ohm): NA → NA Terminology and Drawing of interference Suppression equipment: NA → NA H.T. Cable: NA → NA Make and Place: NA → NA Type (Resistive/Non-resistive): NA → NA Length (mm) (if resistive type):		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	NA → NA Outside dia. (mm) (if resistive type): NA → NA Nominal resistance kilo ohm, (if resistive type): NA → NA Radiator drawing(s): Refer enclosure 4C2 & 4C4 → Refer enclosure 4C2 & 4C4 Radiator drawing(s): Refer enclosurev 4C1 & 4C3 → Refer enclosurev 4C1 & 4C3 Make: Mahle Behr → Mahle Behr Make: Tata Toyo → Tata Toyo	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Type (s): Cross flow → Cross flow Relief valve pressure setting: 0.9 kg/cm2 → 0.9 kg/cm2 Fan characteristics (Fan power, kW) Enclose the fan power curve corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part No: 9 → 9 Material of blades (metal / plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ± 150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(AC) Radiator core open area (cm²): 1906 cm2 → 1906 cm2 Cooling system: liquid/air: Liquid → Liquid Nominal setting of the engine temperature control o mechanism (C)::: 85 deg C → 85 deg C Liquid: Yes → Yes	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Nature of liquid:: Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L → Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L Circulating pump(s): (Yes/No): Yes → Yes Characteristics of Circulating pump: Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm → Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm Make(s):: XXXXX Fasteners Ltd → XXXXX Fasteners Ltd Type(s):: Impeller Type → Impeller Type Drive ratio(s):: 1: 1.12 → 1: 1.12 Description of the fan and its drive mechanism:: Electric Fan , Motor Driven → Electric Fan , Motor Driven Blower: (Yes/No): No → No Characteristics of blower:		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	NA → NA Make(s):: NA → NA Type(s):: NA → NA Drive ratio(s):: NA → NA Pressure charger: (Yes/No) Turbocharger	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	/ Supercharger: Yes, Turbo charger → Yes, Turbo charger Make(s):: M/s. XYZ, India → M/s. XYZ, India Type(s) & Identification Number:: Waste Gate with Vacuum Actuator, X3M1XXXX → Waste Gate with Vacuum Actuator, X3M1XXXX Description of the system (maximum charge pressure:.....(kPa), waste-gate, if applicable): Waste gate turbocharger with vacuum actuator Maximum Charge pressure - 145 kPa → Waste gate turbocharger with vacuum actuator Maximum Charge pressure - 145 kPa Inter-cooler: (Yes/No): Yes → Yes Type: air-air / air-water: Air – Air → Air – Air Make :: Mahle Behr → Mahle Behr Make :: Tata Toyo → Tata Toyo Identification Number:: B1V42XXX → B1V42XXX Identification Number:: B1V42YYY → B1V42YYY Air pressure drop across the intercooler: 15 kPa → 15 kPa Intake depression at rated engine speed and at 100 per cent load (Specify location of measurement) Minimum allowable:... (kPa)		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Maximum allowable:... (kPa): Minimum : 0 kPa	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Description and drawings of inlet pipes and their accessories (plenum chamber, heating device, additional air intakes, etc.):: Intake manifold with Electronic throttle valve Refer Enclosure 4D → Intake manifold with Electronic throttle valve Refer Enclosure 4D Intake manifold description including dimensional drawings and/or photographs):: Aluminium Intake manifold with pressure sensor Refer Enclosure 4D → Aluminium Intake manifold with pressure sensor Refer Enclosure 4D Air filter, Dimensional drawings: Refer Enclosure 4E → Refer Enclosure 4E Make(s):: M/s. Fleet guard → M/s. Fleet guard Type(s) & Identification Number:: Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 → Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 Intake silencer, Dimensional drawing: NA → NA Make(s)::		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	NA → NA Type(s) & Identification Number:: NA → NA Description and Dimensional drawing of the exhaust manifold:: Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F → Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F Description and drawing with dimensions of the exhaust system:(Description with general arrangement of exhaust system along with its routing indicating the lengths of exhaust pipe, tail pipe and: Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly → Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	provided across DPF and O2 sensors provided before and after LNT-DPF assembly Effective volume of exhaust-system (specify the tolerance & range) in liters (from exhaust manifold / TC outlet to tail pipe end), Enclose the exhaust system dimensional drawing and indicate the volume of each parts clearly: 13.05 ± 1 litre Refer Enclosure 4G → 13.05 ± 1 litre Refer Enclosure 4G Description and drawing of the exhaust Muffler/Silencer:: NA → NA Type(s) & Identification Number:: NA → NA Make: NA → NA Number:.....: NA → NA Internal diameter of exhaust pipe (mm)(min): diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G → diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G Minimum distance between exhaust pipe(s) and the fuel line: >150 mm → >150 mm Auxiliary Noise shields for compliance to IS 3028 and / OR AIS-020 (If Provided): NA → NA Material: NA → NA Layout of noise shield / Photographs / Diagram Showing arrangements indicating fitment on vehicle.: NA → NA Maximum allowable exhaust back	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	pressure at rated engine speed and at 100 per cent load (location of measurement): … (kPa): 30.0 kpa (150 mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Minimum cross-sectional areas of inlet and outlet ports:: Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm → Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm Inlet (mm): 8.00 → 8.00 Exhaust (mm): 8.00 → 8.00 Opening: 17° bTDC → 17° bTDC Closing: 51° aBDC → 51° aBDC Opening: 66° bBDC → 66° bBDC Closing: 25° aTDC → 25° aTDC Transfer: NA → NA Opening: NA → NA Closing: NA → NA Reference and/or setting ranges (Specify the tolerances):: On Cold Engine → On Cold Engine Inlet: 0.3 to 0.4 mm → 0.3 to 0.4 mm Exhaust: 0.3 to 0.4 mm → 0.3 to 0.4 mm Distribution by ports: NA → NA Volume of crank-case cavity with piston at TDC: NA → NA Reed valve fitted (Yes / No): No → No Description of inlet ports, scavenging and exhaust ports with corresponding timing.: NA → NA Device for recycling crankcase gases (description and	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	drawings):: Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J → Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx Catalyst (if any, and if not covered by another heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF / DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No Element – 1 No. → Catalytic Converter – 1No Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type for LNT (periodic / continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:……cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap Bulk Density (GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle and Characteristics:: Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY Type and Identification number:: Type - Passive NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite → Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA Make:: NA → NA Type and Idetification number :: NA → NA NOx Sensor (Before /After of SCR):: No → No Make:: NA → NA Type and Identification number:: NA → NA Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems, description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent engine test bench cycles, between two cycles where regenerative phases occur under the conditions equivalent to Type I test (Distance "D" in t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration (Distance “d” in AIS 137): 2 cycles → 2 cycles Description of method employed to determine the number of cycles between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter (DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA International standard (where appropriate):: NA → NA Frequency of reagent refill: continuous/maintenance (where appropriate):: NA → NA Normal operating temperature range (°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	YYYYYYY Air injection: (Yes/No): No → No Type (pulse air, air pump, etc.) & Identification Number:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other): Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY → FEW0XXYY Characteristics (flow rate, etc.):: Flow rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes EGR Cooler Make and Identification Number: M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:……..g: NA → NA Canister working capacity:……..g: NA → NA Canister bed volume:……..litre: NA → NA Canister Purge and vent strategy description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve : (Yes/No): NA → NA Make and Identification Number: NA → NA Operating pressure (kPa): NA → NA Liquid fuel hoses material :: NA → NA Schematic drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or drawing (Periodic / Continuous): NA → NA Particulate trap: (Yes/No): Yes → Yes	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dimensions and shape of the particulate trap (capacity):: Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic Regeneration Type	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh: 90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type Material - Silicon Carbide → Structure - Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI Electronic Control Unit (ECU): Yes → Yes Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA Misfire detection:: NA → NA Oxygen sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	→ Refer Enclosure 4K Particulate traps monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method):: Refer Enclosure 4K → Refer Enclosure 4K List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K → Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K A description of the type and number of the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3 EUDC A description of the type of the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing all sensed components with the strategy for fault detection and MI activation	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(fixed number of driving cycles or statistical method), including a list of relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output		
(continued)	codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive explanation for the data given in service $05 Test ID $21 to FF and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles – Diagnostics on Controller Area Network (CAN) – Part 4: Requirements for emissions- related systems", a comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137): Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA → NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA Systems to ensure the correct operation of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent deactivation of the driver inducement, for use by the rescue services [Note: cross- reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling /disable after parking: NA → NA Lowest concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	operation of NOx control measures: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Details of any devices designed to influence fuel economy (if not covered by other items):: NA → NA LPG fuelling system: (Yes/No): No → No Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to LPG or back:: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.): NA → NA NG fuelling system: (Yes/No): No → No Approval number: NA → NA Electronic engine management control unit for NG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation:: NA → NA Description of the safeguarding of the		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	catalyst at switch- over from petrol to NG or back: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.):: NA → NA Hydrogen fuelling system: (Yes/No): No → No Electronic engine management control unit for hydrogen fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to hydrogen or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA H2NG fuelling system: (Yes/No): No → No Percentage of hydrogen in the fuel (maximum specified by the manufacturer): NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to H2NG or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA Electric motor(Traction): No → No Type (winding, excitation):: NA → NA Maximum hourly output: kW(manufacturer’s declared value): NA → NA Maximum net power:kW(manufacturer’s declared value): NA → NA Maximum 30 minutes power: kW (manufacturer’s declared value): NA → NA Operating voltage: V: NA → NA Make and Identification Number:: → Battery (Traction): NA → NA Number of cells:: NA → NA Mass: ……(kg): NA → NA Capacity: Ah (Amp-hours): NA → NA Position:: NA → NA Engines or motor combinations: NA → NA Hybrid Electric Vehicle: (Yes/No): No → No Category of Hybrid Electric vehicle Off Vehicle Charging/Not Off Vehicle Charging9: NA → NA Operating mode switch: with/without: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Selectable modes: NA → NA Pure electric: (Yes/No): NA → NA Pure fuel consuming: (Yes/No): NA → NA Hybrid modes: (Yes/No) (if yes, short description): NA → NA Description of the energy storage device: (traction battery, capacitor, flywheel/generator...): NA → NA Make(s):: NA → NA Type(s) :: NA → NA Identification number:: NA → NA Kind of electrochemical couple:: NA → NA Energy: ..... (for battery: voltage and capacity Ah in 2 h, for capacitor: J): NA → NA Charger: on board/external/without: NA → NA Electric machines (describe each type of electric machine separately): NA → NA Make:: NA → NA Type:: NA → NA Primary use: traction motor/generator: NA → NA When used as traction motor: mono motor / multi motors (number):: NA → NA Maximum power: kW: NA → NA Working principle:: NA → NA Direct current/alternating current/number of phases:: NA → NA Separate excitation/series/compound: NA → NA Synchronous/asynchronous: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification number:: NA → NA Power controller: NA → NA Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Vehicle electric range..... km (according to AIS 137):: NA → NA Manufacturer’s recommendation for preconditioning:: NA → NA CO2 (g/km) (Applicable for category M1 with GVW<3.5 T): NA → NA Declared : (rounded to 3 decimal places): NA → NA Fuel consumption (l/100 km) for Petrol, LPG or Diesel and (kg/100km) for CNG and (kWh/100 km) for Electric Driven Vehicles(Applicable for category M1 with GVW<3.5 T) (rounded to 3 decimal places)*: NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA *calculated on the declared CO2: NA → NA CO2 Reducing technologies Available (Yes/No): No → No Regenerative braking: NA → NA Start-Stop System: NA → NA Tyre pressure monitoring system: NA → NA 6 or more Speed Transmission: NA → NA Any other technology (manufacture to	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	enclose detail): NA → NA Electric		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	energy consumption for electric vehicles Wh/km: NA → NA Electric energy consumption for pure electric vehicles Wh/km: NA → NA Electric energy consumption for externally chargeable hybrid electric vehicles: NA → NA Electric energy consumption (condition A, combined) Wh/km: NA → NA Electric energy consumption (condition B, combined) Wh/km: NA → NA Electric energy consumption (weighted combined) ….Wh/km: NA → NA Temperatures permitted by the manufacturer: NA → NA Cooling system: Yes → Yes Liquid cooling: Yes → Yes Maximum temperature at outlet: …….(°C): 100 °C → 100 °C Air cooling: NA → NA Reference point:: NA → NA Maximum temperature at reference point: …….(°C): NA → NA Maximum outlet temperature of the inlet intercooler: .. (°C): 45 °C → 45 °C Maximum exhaust temperature at the point in the exhaust pipe(s) adjacent to the outer flange(s) of the exhaust manifold: ...(°C): 740 °C → 740 °C Minimum: .. (°C): - 8 → - 8 Maximum: .. (°C): 42 → 42	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Minimum: ...(°C): - 5 → - 5 Maximum: ...(°C): 120 →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	120 Lubrication system: Yes → Yes Description of the system: Forced Lubrication system → Forced Lubrication system Position of the lubricant reservoir:: Sump beneath crankcase → Sump beneath crankcase Feed system (by pump/injection into intake/mixing with fuel, etc.): Pump → Pump Lubricating pump: Yes → Yes Make(s):: M/s. ZXWA → M/s. ZXWA Type(s):: XXYY Type → XXYY Type Mixture with fuel: No → No Percentage:: → Oil cooler: (Yes/No): Yes → Yes Drawing(s):..., or: Enclosure 4L → Enclosure 4L Make(s):: M/s. XXYY or Any other equivalent → M/s. XXYY or Any other equivalent Type(s):: DONUT → DONUT Moment of inertia of engine flywheel:: 0.1212 kg/m2 → 0.1212 kg/m2 Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2 Schematic Diagram of Engine cooling system: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Description and drawings/photographs of the shapes and constituent materials of the part of the body forming the engine compartment and the part of the passenger compartment nearest to it:: -- → --		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Drawings or photographs of the position of the metal components housed in the engine compartment (e.g. heating appliances, spare wheel, air filter, steering mechanism, etc.):: -- → -- Name of the manufacturer of the system (Steering column shaft / Gear shift control / Transmission) on which the Protective Device functions: XXYY Steering System → XXYY Steering System Type/Part no./Identification number(ID)/Drawing No.of the device on which the Protective Device functions: FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly → FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable only in case of steering column locking systems: Locks the Steering → Locks the	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Steering Components placed in Engine compartment: - → - Example of Engine No: HJEZXYZ891 → HJEZXYZ891		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
AIS-137 (Part 3), applicable amendments Real-driving emissions / route applicability	Base fuel type used for vehicle Type approval including blend level (Ethanol, Bio-diesel etc): Diesel (Maximum 7% bio-diesel blend) → Diesel (Maximum 7% bio-diesel blend) Name and address of engine manufacturing plant (In case of imported vehicles, above details shall be supplied for importer also): Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX → Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX Reference mass (for vehicles with GVW less than or equal to 3.5 ton) (kg.): Refer Annexure T7-B → Refer Annexure T7-B Make: APOLLO → APOLLO Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade description(s): CEAT WINMILE X3-R →	CEA / report needs review	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	CEAT WINMILE X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 Tyre rolling Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer - TC7719XX000 003XXXF → Refer - TC7719XX000 003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX Identification: TAC No. / BIS License No. / E-		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L- 756XXXX → CM/L-	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E- Marking: NA → NA Others (For articulated / combination vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00 R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other (for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 325 mm		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	→ 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26 kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Type: Fully synchronized 5 - forward gears with a Constant mesh reverse gear → Fully synchronized 5 - forward	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	gears with a Constant mesh reverse gear Model name (if any): XXX-XXX → XXX-XXX Model name (if any): YYYY-YY → YYYY-YY Gear shifting control system (Sketch showing gear shifting arrangement): → No. of gears: Forward 5 + 1 Rev → Forward 5 + 1 Rev Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Type of steering gear (Re-circulating ball / Worm & Roller / Rack & Pinion / Others): Re-circulating ball type → Re-circulating ball type Make: 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd → 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2. YYYY – 18.2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	:1 Fuel Gauge: LED Bars → LED Bars Engine coolant temperature: Red Color – Thermometer Symbol → Red Color – Thermometer Symbol Engine low oil pressure: Red Color – Oil Tank Symbol → Red Color – Oil Tank Symbol Installation drawing showing rear	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size, make, unladen pressure) etc., distance between tyre & RUPD RH & LH, frame rear width, P1, P2 points.: - → - Make: M/s. XXXXYYYY → M/s. XXXXYYYY Material (Metallic / Plastic etc.): Metallic → Metallic Nominal thickness mm: 1.0mm → 1.0mm Capacity, Litre: 40 Lts, 30 Lts → 40 Lts, 30 Lts Detailed drawing of the fuel tank assembly with material specifications: FHSXXXXY FHSXXXXY → FHSXXXXY FHSXXXXY Detailed drawing indicating the position / location of the fuel tank(s) in the vehicle.: A5LXXX00 (50 LTS) A5LXXX00 (40 LTS) → A5LXXX00 (50 LTS) A5LXXX00 (40 LTS) Identification No. / Part No.: FHS00544 FfffXXXcdHXXS00744 → FHS00544 FfffXXXcdHXXS00744 Detailed		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	engineering drawing as specified in AIS – 013 or photographs showing the mounting details with dimensions.: NA → NA Cross sectional Drawing of Gear shift lever (Knob): FXA0XXYY (Non AC) /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FXA0XXYY (AC) → FXA0XXYY (Non AC) / FXA0XXYY (AC) In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → -- Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side, distance between front extremity & FUPD, width of FUPD, ID marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Drawing / Images of position & arrangement of Engine: -- → -- Drawing/ Images of Under bonnet (Engine house) Components.: -- → -- 1st · Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st · Overall Ratio /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69 2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC: 12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56 4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 → 20.44 Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No Wheel type		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(single or twin) and tyre type (e.g. structure, category of use, size): -- → -- Gearbox type (e.g. manual, automated manual, semi-automatic, automatic): -- → -- Management of the engine or any other source(s) of motive power e.g.Torque	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Management, control of fuel supply etc.: -- → -- Make: Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. → Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. Model name / identification: ALXYZABC0 → ALXYZABC0 Type: Compression Ignition → Compression Ignition Bore x stroke (mm): 80 X 98 → 80 X 98 No. of cylinders: 3 → 3 Displacement (cm³): 1468 cc → 1468 cc Compression ratio: 16.5±1.0 : 1 → 16.5±1.0 : 1 Max. Engine output (KW @ rpm): 55.1 Kw @ 3300 rpm → 55.1 Kw @ 3300 rpm Max. Torque (Nm @ rpm): 170Nm @ 1600 to 2400 rpm → 170Nm @ 1600 to 2400 rpm Air cleaner type: Dry type → Dry type Type of Fuel: Diesel → Diesel Make and model: RSB		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Transmission (T210R) → RSB Transmission (T210R) Make and model: AVTEC (A5M210) → AVTEC (A5M210) Type: Manual gearbox with fully synchro mesh & Rev with sliding mesh → Manual	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	gearbox with fully synchro mesh & Rev with sliding mesh No. of gears: 5 Forward and 1 Reverse → 5 Forward and 1 Reverse 1st: 4.62 → 4.62 1st: 4.60 → 4.60 1st: 21.53 → 21.53 1st: 21.44 → 21.44 2nd: 2.68 → 2.68 2nd: 2.69 → 2.69 2nd: 12.67 → 12.67 2nd: 12.71 → 12.71 3rd: 1.51 → 1.51 3rd: 1.49 → 1.49 3rd: 7.34 → 7.34 3rd: 7.25 → 7.25 4th: 1.0 → 1.0 4th: 1.0 → 1.0 4th: 4.56 → 4.56 4th: 4.56 → 4.56 5th: 0.70 → 0.70 5th: 0.71 → 0.71 5th: 3.64 → 3.64 5th: 3.67 → 3.67 Reverse: 4.39 → 4.39 Reverse: 20.36 → 20.36 Reverse: 4.38 → 4.38 Reverse: 20.34 → 20.34 Drive Axle (Front / Rear / All): Rear → Rear Front axle ratio: NA → NA Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Wheels and tyres: Apollo → Apollo Wheels and tyres:		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT 12 PR Tyre size designation including ply rating: 215/75 R15 LT → 215/75 R15 LT Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2 → 3.26 kg/cm2 Material: Metallic → Metallic Capacity (l): 50 lts , 40 lts → 50 lts , 40 lts Dimensions: 2490WB → 2490WB Dimensions: 2640WB → 2640WB Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Overall width (mm): 1842 → 1842 Overall Length (mm): CC → CC Overall Length (mm): 4750 → 4750 Overall Length (mm): 4988 → 4988 Overall Length (mm): FSD / HSD → FSD / HSD Overall Length (mm): 4925 → 4925 Overall Length (mm): 5225 → 5225 Overall height (mm): 1961 → 1961	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Overall height (mm): 1948 → 1948 Maximum gradeability in 1st gear: GVW 2890 kg → GVW 2890 kg Maximum gradeability in 1st gear: GVW 3390 kg → GVW 3390 kg		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Maximum gradeability in 1st gear: 10 Deg → 10 Deg Maximum gradeability in 1st gear: 8.5 Deg → 8.5 Deg Maximum GVW kg (for rigid vehicles): Refer Annexure T7 – A Details for Recommended load body - Refer Annexure T7 - B → Refer Annexure T7 – A Details for Recommended load body - Refer Annexure T7 - B Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Description of energy converters and power plant. (In the case of a vehicle that can run either on petrol, diesel, etc., or also in combination with another fuel, items shall be repeated.1): Diesel → Diesel Engine Manufacturer:: Ashok Leyland Limited → Ashok Leyland Limited Manufacturer's engine model code (as marked on the engine, or other means of identification): Model Code: ALXYZABC0 Enclosure – 4A → Model Code: ALXYZABC0 Enclosure – 4A Location of engine model code and engine serial number on block /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	crankcase (drawing to be attached): Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A → Engine Model Code Location - On Engine Head Cover Engine Serial Number		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Location - On Engine Block Refer Enclosure – 4A Working principle: positive ignition / compression- ignition, four-stroke / two-stroke / rotary cycle (DI/IDI) (NA/TC/TCIC/Any other): Compression Ignition, four-stroke, DI, TCIC → Compression Ignition, four-stroke, DI, TCIC Number, arrangement of cylinders:: 3, Inline → 3, Inline Bore----------(mm): 80 → 80 Stroke: ----------(mm): 98 → 98 Firing order:: 1-2-3 → 1-2-3 Engine capacity:----------(cm3): 1468 → 1468 Volumetric compression ratio (specify tolerance): 17.5±1.0 : 1 → 17.5±1.0 : 1 Drawings of combustion chamber and piston crown and, in the case of positive ignition engine, piston rings:: Refer Enclosure – 4B → Refer Enclosure – 4B Vehicle with Defeat Device Yes / No: No	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	→ No If yes – Details of the Defeat device: NA → NA Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane equivalent) in PPM content by volume in		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	the exhaust gas with the engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum net power:..... kW at min-1: 55.1 KW @ 3100 ± 50 rpm → 55.1 KW @ 3100 ± 50 rpm Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm Maximum net torque: .....Nm at: -------min-1 ...(manufacturer's declared value): 170 Nm @ 1600 – 2400 rpm → 170 Nm @ 1600 – 2400 rpm Maximum rated speed (Specify tolerance): 3100 rpm ± 50 rpm → 3100 rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Type of fuel used : Mono Fuel / Bi-Fuel /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Flex Fuel / Dual Fuel: Mono Fuel → Mono Fuel Fuel type: Diesel / Gasoline / LPG / CNG / Biomethane / Bio-H methane / HCNG / Biogas / LNG / Ethanol ((E85) / 2 (E100)) /ED (95) Flex fuel Methanol M15 / M100 / MD95 / Biodiesel up to 100% /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Hydrogen.: Diesel → Diesel Maximum amount of biofuel acceptable in fuel (manufacturer's declared value): per cent by volume: 7% → 7% By fuel injection (compression-ignition only): (Yes/No): Yes → Yes System description:: Unit Pump Common Rail Direct Injection system → Unit Pump Common Rail Direct Injection system Working principle: direct-injection/pre-chamber/swirl Chamber: Direct Injection → Direct Injection Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Plunger Type & FHL0XYZ0 → Plunger Type & FHL0XYZ0 Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Injection advance curve (Specify the	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	tolerances):: ECU Controlled → ECU Controlled Governor: ECU Controlled → ECU Controlled Cut-off point under load:…(min-1): 3300 →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	3300 Cut-off point without load (Maximum no-load speed)…(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Multi-Hole Injectors (7 holes) FKXXYYZZ → Multi-Hole Injectors (7 holes) FKXXYYZZ Opening pressure (kPa) (specify the tolerances) or characteristic diagram: 15000 to 23000 kPa → 15000 to 23000 kPa Length and internal diameter…(mm): Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm → Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm Common rail make & type: M/s. XXXXYYY, Hot Forged Rail Type → M/s. XXXXYYY, Hot Forged Rail Type Cold start system: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA Description:: NA → NA Auxiliary starting aid: NA → NA Make(s):: NA → NA Type(s):: NA → NA System description:: NA → NA Electronic controlled injection: (Yes/No): Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Rail, Direct Injection with Pressure Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA → NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s. XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and Metering type	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	of air temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type By fuel injection (positive ignition only): (Yes/No): No → No Working principle: intake manifold (single/multi- point)/direct injection/other (specify): NA → NA Make(s) of Injection Pump :: NA → NA Type(s) & Identification Number of Injection Pump:: NA → NA System description (in the case of systems other than continuous injection give equivalent details):: NA → NA Make and type of the control unit:: NA → NA Make and type of the fuel regulator:: NA → NA Make and type of the air-flow sensor:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make and type of the micro-switch:: NA → NA Make and type of the throttle housing:: NA → NA Make and type of the water temperature sensor:: NA → NA Make and type of the air temperature sensor:: NA → NA Injectors: Opening pressure (Specify the		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	tolerances):,..... (kPa) or characteristic diagram:: NA → NA Make(s):: NA → NA Type(s):: NA → NA Identification Number: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA Cut-off point under load (min-1): NA → NA Injection timing:: NA → NA Cold start system:: NA → NA Operating principle(s):: NA → NA Operating limits/settings(Specify the tolerances): NA → NA Pressure, (Specify the tolerances).....(kPa) or characteristic diagram:: 320 ± 50 kPa → 320 ± 50 kPa Rated voltage:...V, positive/negative ground: 12V, Negative Ground → 12V,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Negative Ground Generator/ Alternator: Alternator → Alternator Type:: DC – Alternator with Voltage regulator → DC – Alternator with Voltage regulator Nominal output: …VA: 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps → 1205 VA (For Non		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps Ignition: NA → NA Make(s):: NA → NA Type(s):: NA → NA Working principle:: NA → NA Ignition advance curve or map(Specify the tolerances):: NA → NA Static ignition timing (Specify the tolerances):……..degrees before TDC: NA → NA Spark plugs: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Gap setting----------(mm): NA → NA Nominal resistance (kilo ohm) (if resistive type): NA → NA Ignition coil(s): NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Ignition condenser: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make: NA → NA Type: NA → NA Identification number: NA → NA EMI suppressor cap / Device / Electronic unit: NA → NA Make: NA → NA Type (Resistive/Capacitive): NA → NA Identification number: NA → NA Nominal resistance (kilo ohm): NA → NA Terminology and Drawing of interference Suppression equipment: NA → NA H.T. Cable: NA → NA Make and Place: NA → NA Type (Resistive/Non-resistive): NA → NA Length (mm) (if resistive type):		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	NA → NA Outside dia. (mm) (if resistive type): NA → NA Nominal resistance kilo ohm, (if resistive type): NA → NA Radiator drawing(s): Refer enclosure 4C2 & 4C4 → Refer enclosure 4C2 & 4C4 Radiator drawing(s): Refer enclosurev 4C1 & 4C3 → Refer enclosurev 4C1 & 4C3 Make: Mahle Behr → Mahle Behr Make: Tata Toyo → Tata Toyo Type (s): Cross flow → Cross flow Relief valve pressure setting: 0.9 kg/cm2 → 0.9 kg/cm2 Fan characteristics (Fan power, kW) Enclose the fan power curve	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part No: 9 → 9 Material of blades (metal / plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ± 150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(AC) Radiator core open area (cm^2): 1906 cm2 → 1906 cm2 Cooling system: liquid/air: Liquid → Liquid Nominal setting of the engine temperature control o mechanism (C)::: 85 deg C → 85 deg C Liquid: Yes → Yes Nature of liquid:: Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L → Option 1: Water + 70 %	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L Circulating pump(s): (Yes/No): Yes → Yes Characteristics of Circulating pump: Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm → Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm Make(s):: XXXXX Fasteners Ltd → XXXXX Fasteners Ltd Type(s):: Impeller Type → Impeller Type Drive ratio(s):: 1: 1.12 → 1: 1.12 Description of the fan and its drive mechanism:: Electric Fan , Motor Driven → Electric Fan , Motor Driven Blower: (Yes/No): No → No Characteristics of blower:		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	NA → NA Make(s):: NA → NA Type(s):: NA → NA Drive ratio(s):: NA → NA Pressure charger: (Yes/No) Turbocharger / Supercharger: Yes, Turbo charger → Yes, Turbo charger Make(s):: M/s. XYZ, India → M/s. XYZ, India Type(s) & Identification Number:: Waste Gate with Vacuum Actuator, X3M1XXXX → Waste Gate with Vacuum Actuator,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	X3M1XXXX Description of the system (maximum charge pressure:.....(kPa), waste-gate, if applicable): Waste gate turbocharger with vacuum actuator Maximum Charge pressure - 145 kPa → Waste gate turbocharger with vacuum actuator Maximum Charge pressure - 145 kPa Inter-cooler: (Yes/No): Yes → Yes Type: air-air / air-water: Air – Air → Air – Air Make :: Mahle Behr → Mahle Behr Make :: Tata Toyo → Tata Toyo Identification Number:: B1V42XXX → B1V42XXX Identification Number:: B1V42YYY → B1V42YYY Air pressure drop across the intercooler: 15 kPa → 15 kPa Intake depression at rated engine speed and at 100 per cent load (Specify location of measurement) Minimum allowable:... (kPa)		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Maximum allowable:... (kPa): Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Description and drawings of inlet pipes and their accessories (plenum chamber,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	heating device, additional air intakes, etc.):: Intake manifold with Electronic throttle valve Refer Enclosure 4D → Intake manifold with Electronic throttle valve Refer Enclosure 4D Intake manifold description including dimensional drawings and/or photographs):: Aluminium Intake manifold with pressure sensor Refer Enclosure 4D → Aluminium Intake manifold with pressure sensor Refer Enclosure 4D Air filter, Dimensional drawings: Refer Enclosure 4E → Refer Enclosure 4E Make(s):: M/s. Fleet guard → M/s. Fleet guard Type(s) & Identification Number:: Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 → Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 Intake silencer, Dimensional drawing: NA → NA Make(s)::		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	NA → NA Type(s) & Identification Number:: NA → NA Description and Dimensional drawing of the exhaust manifold:: Cast iron manifold with adaptation for Turbocharger and EGR inlet	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer Enclosure 4F → Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F Description and drawing with dimensions of the exhaust system:(Description with general arrangement of exhaust system along with its routing indicating the lengths of exhaust pipe, tail pipe and: Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly → Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	provided across DPF and O2 sensors provided before and after LNT-DPF assembly Effective volume of exhaust-system (specify the tolerance & range) in liters	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(from exhaust manifold / TC outlet to tail pipe end), Enclose the exhaust system dimensional drawing and indicate the volume of each parts clearly: 13.05 ± 1 litre Refer Enclosure 4G → 13.05 ± 1 litre Refer Enclosure 4G Description and drawing of the exhaust Muffler/Silencer:: NA → NA Type(s) & Identification Number:: NA → NA Make: NA → NA Number:.....: NA → NA Internal diameter of exhaust pipe (mm)(min): diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G → diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G Minimum distance between exhaust pipe(s) and the fuel line: >150 mm → >150 mm Auxiliary Noise shields for compliance to IS 3028 and / OR AIS-020 (If Provided): NA → NA Material: NA → NA Layout of noise shield / Photographs / Diagram Showing arrangements indicating fitment on vehicle.: NA → NA Maximum allowable exhaust back		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	pressure at rated engine speed and at 100 per cent load (location of measurement): ... (kPa): 30.0 kpa (150	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Minimum cross-sectional areas of inlet and outlet ports:: Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm → Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm Inlet (mm): 8.00 → 8.00 Exhaust (mm): 8.00 → 8.00 Opening: 17° bTDC → 17° bTDC Closing: 51° aBDC → 51° aBDC Opening: 66° bBDC → 66° bBDC Closing: 25° aTDC → 25° aTDC Transfer: NA → NA Opening: NA → NA Closing: NA → NA Reference and/or setting ranges (Specify the tolerances):: On Cold Engine → On Cold Engine Inlet: 0.3 to 0.4 mm → 0.3 to 0.4 mm Exhaust: 0.3 to 0.4 mm → 0.3 to 0.4 mm Distribution by ports: NA → NA Volume of crank-case cavity with piston at TDC: NA → NA Reed valve fitted (Yes / No): No → No Description of inlet ports, scavenging and exhaust ports with corresponding timing.: NA → NA Device for recycling crankcase gases (description and		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic

(continued)	drawings):: Closed Loop Crankcase ventilation with external oil separator	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer Enclosure 4J → Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx Catalyst (if any, and if not covered by another heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF / DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	means of identification): XXX / DDMMYY / X	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No Element – 1 No. → Catalytic Converter – 1No Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type for LNT (periodic / continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:……cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap Bulk Density (GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle and Characteristics:: Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY Type and Identification number:: Type - Passive NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite → Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA Make:: NA → NA Type and Idetification number :: NA → NA NOx Sensor (Before /After of SCR):: No → No Make:: NA → NA Type and Identification number:: NA → NA Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems, description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent engine test bench cycles, between two cycles where regenerative phases occur under the conditions equivalent to Type I test (Distance "D" in t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration (Distance “d” in AIS 137): 2 cycles → 2 cycles Description of method employed to determine the number of cycles between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter (DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA International standard (where appropriate):: NA → NA Frequency of reagent refill: continuous/maintenance (where appropriate):: NA → NA Normal operating temperature range (°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	YYYYYYY Air injection: (Yes/No): No → No Type (pulse air, air pump, etc.) & Identification Number:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other): Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY → FEW0XXYY Characteristics (flow rate, etc.):: Flow rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes EGR Cooler Make and Identification Number: M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:……..g: NA → NA Canister working capacity:……..g: NA → NA Canister bed volume:……..litre: NA → NA Canister Purge and vent strategy description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve : (Yes/No): NA → NA Make and Identification Number: NA → NA Operating pressure (kPa): NA → NA Liquid fuel hoses material :: NA → NA Schematic drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or drawing (Periodic / Continuous): NA → NA Particulate trap: (Yes/No): Yes → Yes	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dimensions and shape of the particulate trap (capacity):: Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic Regeneration Type	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh: 90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type Material - Silicon Carbide → Structure - Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI Electronic Control Unit (ECU): Yes → Yes Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA Misfire detection:: NA → NA Oxygen sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	→ Refer Enclosure 4K Particulate traps monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method):: Refer Enclosure 4K → Refer Enclosure 4K List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K → Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K A description of the type and number of the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3 EUDC A description of the type of the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing all sensed components with the strategy for fault detection and MI activation	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(fixed number of driving cycles or statistical method), including a list of relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output		
(continued)	codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive explanation for the data given in service $05 Test ID $21 to FF and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles – Diagnostics on Controller Area Network (CAN) – Part 4: Requirements for emissions- related systems", a comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137): Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA → NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA Systems to ensure the correct operation of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent deactivation of the driver inducement, for use by the rescue services [Note: cross- reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling /disable after parking: NA → NA Lowest concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	operation of NOx control measures: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Details of any devices designed to influence fuel economy (if not covered by other items):: NA → NA LPG fuelling system: (Yes/No): No → No Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to LPG or back:: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.): NA → NA NG fuelling system: (Yes/No): No → No Approval number: NA → NA Electronic engine management control unit for NG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation:: NA → NA Description of the safeguarding of the		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	catalyst at switch- over from petrol to NG or back: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.):: NA → NA Hydrogen fuelling system: (Yes/No): No → No Electronic engine management control unit for hydrogen fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to hydrogen or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA H2NG fuelling system: (Yes/No): No → No Percentage of hydrogen in the fuel (maximum specified by the manufacturer): NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to H2NG or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA Electric motor(Traction): No → No Type (winding, excitation):: NA → NA Maximum hourly output: kW(manufacturer’s declared value): NA → NA Maximum net power:kW(manufacturer’s declared value): NA → NA Maximum 30 minutes power: kW (manufacturer’s declared value): NA → NA Operating voltage: V: NA → NA Make and Identification Number:: → Battery (Traction): NA → NA Number of cells:: NA → NA Mass: ……(kg): NA → NA Capacity: Ah (Amp-hours): NA → NA Position:: NA → NA Engines or motor combinations: NA → NA Hybrid Electric Vehicle: (Yes/No): No → No Category of Hybrid Electric vehicle Off Vehicle Charging/Not Off Vehicle Charging9: NA → NA Operating mode switch: with/without: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Selectable modes: NA → NA Pure electric: (Yes/No): NA → NA Pure fuel consuming: (Yes/No): NA → NA Hybrid modes: (Yes/No) (if yes, short description): NA → NA Description of the energy storage device: (traction battery, capacitor, flywheel/generator...): NA → NA Make(s):: NA → NA Type(s) :: NA → NA Identification number:: NA → NA Kind of electrochemical couple:: NA → NA Energy: ..... (for battery: voltage and capacity Ah in 2 h, for capacitor: J): NA → NA Charger: on board/external/without: NA → NA Electric machines (describe each type of electric machine separately): NA → NA Make:: NA → NA Type:: NA → NA Primary use: traction motor/generator: NA → NA When used as traction motor: mono motor / multi motors (number):: NA → NA Maximum power: kW: NA → NA Working principle:: NA → NA Direct current/alternating current/number of phases:: NA → NA Separate excitation/series/compound: NA → NA Synchronous/asynchronous: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification number:: NA → NA Power controller: NA → NA Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Vehicle electric range..... km (according to AIS 137):: NA → NA Manufacturer’s recommendation for preconditioning:: NA → NA CO2 (g/km) (Applicable for category M1 with GVW<3.5 T): NA → NA Declared : (rounded to 3 decimal places): NA → NA Fuel consumption (l/100 km) for Petrol, LPG or Diesel and (kg/100km) for CNG and (kWh/100 km) for Electric Driven Vehicles(Applicable for category M1 with GVW<3.5 T) (rounded to 3 decimal places)*: NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA *calculated on the declared CO2: NA → NA CO2 Reducing technologies Available (Yes/No): No → No Regenerative braking: NA → NA Start-Stop System: NA → NA Tyre pressure monitoring system: NA → NA 6 or more Speed Transmission: NA → NA Any other technology (manufacture to	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	enclose detail): NA → NA Electric		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	energy consumption for electric vehicles Wh/km: NA → NA Electric energy consumption for pure electric vehicles Wh/km: NA → NA Electric energy consumption for externally chargeable hybrid electric vehicles: NA → NA Electric energy consumption (condition A, combined) Wh/km: NA → NA Electric energy consumption (condition B, combined) Wh/km: NA → NA Electric energy consumption (weighted combined) ….Wh/km: NA → NA Temperatures permitted by the manufacturer: NA → NA Cooling system: Yes → Yes Liquid cooling: Yes → Yes Maximum temperature at outlet: …….(°C): 100 °C → 100 °C Air cooling: NA → NA Reference point:: NA → NA Maximum temperature at reference point: …….(°C): NA → NA Maximum outlet temperature of the inlet intercooler: .. (°C): 45 °C → 45 °C Maximum exhaust temperature at the point in the exhaust pipe(s) adjacent to the outer flange(s) of the exhaust manifold: ...(°C): 740 °C → 740 °C Minimum: .. (°C): - 8 → - 8 Maximum: .. (°C): 42 → 42	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Minimum: ...(°C): - 5 → - 5 Maximum: ...(°C): 120 →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	120 Lubrication system: Yes → Yes Description of the system: Forced Lubrication system → Forced Lubrication system Position of the lubricant reservoir:: Sump beneath crankcase → Sump beneath crankcase Feed system (by pump/injection into intake/mixing with fuel, etc.): Pump → Pump Lubricating pump: Yes → Yes Make(s):: M/s. ZXWA → M/s. ZXWA Type(s):: XXYY Type → XXYY Type Mixture with fuel: No → No Percentage:: → Oil cooler: (Yes/No): Yes → Yes Drawing(s):..., or: Enclosure 4L → Enclosure 4L Make(s):: M/s. XXYY or Any other equivalent → M/s. XXYY or Any other equivalent Type(s):: DONUT → DONUT Moment of inertia of engine flywheel:: 0.1212 kg/m2 → 0.1212 kg/m2 Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2 Schematic Diagram of Engine cooling system: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Description and drawings/photographs of the shapes and constituent materials of the part of the body forming the engine compartment and the part of the passenger compartment nearest to it:: -- → --		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Drawings or photographs of the position of the metal components housed in the engine compartment (e.g. heating appliances, spare wheel, air filter, steering mechanism, etc.):: -- → -- Name of the manufacturer of the system (Steering column shaft / Gear shift control / Transmission) on which the Protective Device functions: XXYY Steering System → XXYY Steering System Type/Part no./Identification number(ID)/Drawing No.of the device on which the Protective Device functions: FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly → FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable only in case of steering column locking systems: Locks the Steering → Locks the	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Steering Components placed in Engine compartment: - → - Example of Engine No: HJEZXYZ891 → HJEZXYZ891		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
CMVR 120; IS 3028; current edition review Exterior / pass-by noise	Name and address of engine manufacturing plant (In case of imported vehicles, above details shall be supplied for importer also): Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX → Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX Vehicle kerb weight kg: Refer Annexure T7 - B Details for Recommended load body Refer Annexure T7 – C → Refer Annexure T7 - B Details for Recommended load body Refer Annexure T7 – C Other axle: NA → NA Gross combination weight kg (applicable for articulated / Tractor Trailer combination vehicles): NA → NA Front axle: NA → NA Rear axle: NA → NA Trailer axle / other axles: NA → NA If applicable, maximum mass of trailer whic5h may be coupled: NA → NA Unbraked trailer: NA → NA Reference mass (for vehicles with GVW	CEA / report needs review	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	less than or equal to 3.5 ton) (kg.): Refer Annexure T7-B → Refer Annexure T7-B Maximum (Including driver) for completely built vehicles: D+2 → D+2 Maximum (Including driver) for completely built vehicles: → Make: APOLLO → APOLLO Make: CEAT → CEAT Make:		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade description(s): CEAT WINMILE X3-R → CEAT WINMILE X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 Tyre rolling Resistance: 7.84 N/kN → 7.84 N/kN	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA /		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer - TC7719XX000 003XXXF → Refer - TC7719XX000 003XXXF	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L-756XXXX → CM/L-756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E-		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Marking: NA → NA Others (For articulated / combination vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00 R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other (for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for M2, M3, N and T category of vehicles.):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	325 mm → 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26 kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Type: Fully synchronized 5 - forward gears with a Constant mesh reverse gear → Fully synchronized 5 - forward gears with a Constant mesh reverse gear Model name (if any): XXX-XXX → XXX-XXX Model name (if any): YYYY-YY → YYYY-YY Gear shifting control system (Sketch showing gear shifting arrangement): → No. of gears: Forward 5 + 1 Rev →		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Forward 5 + 1 Rev Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Rear axle ratio: 4.56 → 5.13 Type of steering gear (Re-circulating ball / Worm & Roller / Rack & Pinion /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Others): Re-circulating ball type → Re-circulating ball type Make: 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd → 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 Engine coolant temperature: Red Color – Thermometer Symbol → Red Color – Thermometer Symbol Engine low oil pressure: Red Color – Oil Tank Symbol → Red Color – Oil Tank Symbol Weight of seats: Driver seat : 11.5 Kg & Co-driver seat : 11Kg → Driver seat : 11.5 Kg & Co-driver seat : 11Kg Seat Identification No. / Part No/Drawing Number.: ID Number/ Part Number/D rawing number Weight Description Make (Seat (kg) Manufactu rer/ Vehicle Manufactu rer): -- → -- Installation drawing showing rear extremity of vehicle, chassis rear		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	overhang, chassis cross section details. tyre details (size, make, unladen pressure) etc., distance between tyre & RUPD RH & LH, frame rear width, P1, P2 points.: - → - Detailed engineering drawing as specified in AIS – 013 or photographs showing the mounting details with	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	dimensions.: NA → NA Cross sectional Drawing of Gear shift lever (Knob): FXA0XXYY (Non AC) / FXA0XXYY (AC) → FXA0XXYY (Non AC) / FXA0XXYY (AC) In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → -- Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side, distance between front extremity & FUPD, width of FUPD, ID marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Drawing / Images of position & arrangement of Engine: -- → -- Drawing/ Images of Under bonnet		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Engine house) Components.: -- → -- 1st · Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st · Overall Ratio / AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69 2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC: 12.71 → 12.71	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56 4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC:		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	20.44 → 20.44 Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No Wheel type (single or twin) and tyre type (e.g. structure, category of use, size): -- → -- Gearbox type (e.g. manual, automated manual, semi-automatic, automatic): --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	→ -- Management of the engine or any other source(s) of motive power e.g.Torque Management, control of fuel supply etc.: -- → -- Make: Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. → Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. Model name / identification: ALXYZABC0 → ALXYZABC0 Type: Compression Ignition → Compression Ignition Bore x stroke (mm): 80 X 98 → 80 X 98 No. of cylinders: 3 → 3 Displacement (cm^3): 1468 cc → 1468 cc Compression ratio: 16.5±1.0 : 1 → 16.5±1.0 : 1 Max. Engine output (KW @ rpm): 55.1 Kw @ 3300 rpm → 55.1 Kw @ 3300		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	rpm Max. Torque (Nm @ rpm): 170Nm @ 1600 to 2400 rpm → 170Nm @ 1600 to 2400 rpm Air cleaner type: Dry type → Dry type Type of Fuel: Diesel → Diesel Make and model: RSB Transmission (T210R) → RSB Transmission (T210R) Make and model: AVTEC (A5M210) →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	AVTEC (A5M210) Type: Manual gearbox with fully synchro mesh & Rev with sliding mesh → Manual gearbox with fully synchro mesh & Rev with sliding mesh No. of gears: 5 Forward and 1 Reverse → 5 Forward and 1 Reverse 1st: 4.62 → 4.62 1st: 4.60 → 4.60 1st: 21.53 → 21.53 1st: 21.44 → 21.44 2nd: 2.68 → 2.68 2nd: 2.69 → 2.69 2nd: 12.67 → 12.67 2nd: 12.71 → 12.71 3rd: 1.51 → 1.51 3rd: 1.49 → 1.49 3rd: 7.34 → 7.34 3rd: 7.25 → 7.25 4th: 1.0 → 1.0 4th: 1.0 → 1.0 4th: 4.56 → 4.56 4th: 4.56 → 4.56 5th: 0.70 → 0.70 5th: 0.71 → 0.71 5th: 3.64 → 3.64 5th: 3.67 → 3.67 Reverse: 4.39 → 4.39 Reverse: 20.36 → 20.36 Reverse: 4.38 → 4.38 Reverse: 20.34 → 20.34 Drive Axle (Front / Rear / All): Rear → Rear Front axle ratio: NA → NA Steering Gear Ratio: 1. XXXX		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT 12 PR Tyre size designation including ply rating: 215/75 R15 LT → 215/75 R15 LT Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2 → 3.26 kg/cm2 Maximum gradeability in 1st gear: GVW 2890 kg → GVW 2890 kg Maximum gradeability in 1st gear: GVW 3390 kg → GVW 3390 kg Maximum gradeability in 1st gear: 10 Deg → 10 Deg Maximum gradeability in 1st gear: 8.5 Deg → 8.5 Deg Weights: Base → Base Weights: Variant 1 → Variant 1 Weights: Variant 2 → Variant 2 Weights: Variant 3 → Variant 3 Weights: Variant 4 → Variant 4 Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Engine Manufacturer:: Ashok Leyland Limited → Ashok Leyland Limited Manufacturer's engine model code (as marked on the engine, or other means of identification): Model Code: ALXYZABC0 Enclosure – 4A → Model Code: ALXYZABC0 Enclosure – 4A Location of engine model code and engine serial number on block / crankcase (drawing to be attached): Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A → Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A Working principle: positive ignition / compression- ignition, four-stroke / two-stroke / rotary cycle (DI/IDI) (NA/TC/TCIC/Any other): Compression Ignition, four-stroke, DI, TCIC → Compression Ignition,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	four-stroke, DI, TCIC Number, arrangement of cylinders:: 3, Inline → 3, Inline Bore----------(mm): 80 → 80 Stroke: ----------(mm): 98 → 98 Firing order:: 1-2-3 → 1-2-3 Engine capacity:----------(cm3): 1468 → 1468 Volumetric compression ratio (specify tolerance): 17.5±1.0 : 1 → 17.5±1.0 : 1 Drawings of combustion chamber and piston crown and, in the case of positive ignition engine, piston rings:: Refer Enclosure – 4B → Refer Enclosure – 4B Vehicle with Defeat Device Yes / No: No → No If yes – Details of the Defeat device: NA → NA Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane equivalent) in PPM content by volume in the exhaust gas with the engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum net power:..... kW at	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	min-1: 55.1 KW @ 3100 ± 50 rpm → 55.1 KW @ 3100 ± 50 rpm Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm Maximum net torque: .....Nm at: -------min-1 ...(manufacturer's declared value): 170 Nm @ 1600 – 2400 rpm → 170 Nm @ 1600 – 2400 rpm Maximum rated speed (Specify tolerance): 3100 rpm ± 50 rpm → 3100 rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Type of fuel used : Mono Fuel / Bi-Fuel / Flex Fuel / Dual Fuel: Mono Fuel → Mono Fuel Fuel type: Diesel / Gasoline / LPG / CNG / Biomethane / Bio-H methane / HCNG / Biogas / LNG / Ethanol ((E85) / 2 (E100)) /ED (95) Flex fuel Methanol M15 / M100 / MD95 / Biodiesel up to 100% / Hydrogen.: Diesel → Diesel Maximum amount of biofuel acceptable in fuel (manufacturer's declared value): per cent by volume: 7% → 7% By fuel injection (compression-ignition only): (Yes/No): Yes → Yes System description:: Unit Pump Common Rail Direct	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Injection system → Unit Pump Common Rail Direct Injection system Working principle: direct-injection/pre-chamber/swirl Chamber: Direct Injection → Direct Injection Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Plunger Type & FHL0XYZ0 → Plunger Type & FHL0XYZ0 Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Injection advance curve (Specify the tolerances):: ECU Controlled → ECU Controlled Governor: ECU Controlled → ECU Controlled Cut-off point under load:…(min-1): 3300 → 3300 Cut-off point without load (Maximum no-load speed)…(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Multi-Hole Injectors (7 holes) FKXXYYZZ → Multi-Hole Injectors (7 holes) FKXXYYZZ	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Opening pressure (kPa) (specify the tolerances) or characteristic diagram: 15000 to 23000 kPa → 15000 to 23000 kPa Length and internal diameter...(mm): Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm → Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm Common rail make & type: M/s. XXXXYYY, Hot Forged Rail Type → M/s. XXXXYYY, Hot Forged Rail Type Cold start system: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description:: NA → NA Auxiliary starting aid: NA → NA Make(s):: NA → NA Type(s):: NA → NA System description:: NA → NA Electronic controlled injection: (Yes/No): Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s. XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and Metering type Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type By fuel injection	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(positive ignition only): (Yes/No): No → No Working principle: intake manifold (single/multi- point)/direct injection/other (specify): NA → NA Make(s) of Injection Pump :: NA → NA Type(s) & Identification Number of Injection Pump:: NA → NA System description (in the case of systems other than continuous injection give equivalent details):: NA → NA Make and type of the control unit:: NA → NA Make and type of the fuel regulator:: NA → NA Make and type of the air-flow sensor:: NA → NA Make and type of the micro-switch:: NA → NA Make and type of the throttle housing:: NA → NA Make and type of the water temperature sensor:: NA → NA Make and type of the air temperature sensor:: NA → NA Injectors: Opening pressure (Specify the tolerances):,..... (kPa) or characteristic diagram:: NA → NA Make(s):: NA → NA Type(s):: NA → NA Identification Number: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA Cut-off point under load (min-1): NA → NA Injection timing:: NA → NA Cold start system:: NA → NA Operating principle(s):: NA → NA Operating limits/settings(Specify the tolerances): NA → NA Pressure, (Specify the tolerances).....(kPa) or characteristic diagram:: 320 ± 50 kPa → 320 ± 50 kPa Rated voltage:...V, positive/negative ground: 12V, Negative Ground → 12V, Negative Ground Generator/ Alternator: Alternator → Alternator Type:: DC – Alternator with Voltage regulator → DC – Alternator with Voltage regulator Nominal output: ...VA: 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps → 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps Ignition: NA → NA Make(s):: NA → NA Type(s):: NA → NA Working principle:: NA → NA Ignition advance curve or map(Specify the tolerances):: NA → NA Static ignition timing (Specify the	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	tolerances):.....degrees before TDC: NA → NA Spark plugs: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Gap setting----------(mm): NA → NA Nominal resistance (kilo ohm) (if resistive type): NA → NA Ignition coil(s): NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Ignition condenser: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA EMI suppressor cap / Device / Electronic unit: NA → NA Make: NA → NA Type (Resistive/Capacitive): NA → NA Identification number: NA → NA Nominal resistance (kilo ohm): NA → NA Terminology and Drawing of interference Suppression equipment: NA → NA H.T. Cable: NA → NA Make and Place: NA → NA Type (Resistive/Non-resistive): NA → NA Length (mm) (if resistive type): NA → NA Outside dia. (mm) (if resistive type): NA → NA Nominal resistance kilo ohm, (if resistive type): NA → NA Radiator drawing(s): Refer enclosure 4C2 & 4C4 → Refer enclosure 4C2 & 4C4 Radiator drawing(s): Refer enclosurev	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	4C1 & 4C3 →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Refer enclosurev 4C1 & 4C3 Make: Mahle Behr → Mahle Behr Make: Tata Toyo → Tata Toyo Type (s): Cross flow → Cross flow Relief valve pressure setting: 0.9 kg/cm2 → 0.9 kg/cm2 Fan characteristics (Fan power, kW) Enclose the fan power curve corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part No: 9 → 9 Material of blades (metal / plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ± 150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150 (AC) Radiator core open area (cm²): 1906 cm2 → 1906 cm2 Cooling system: liquid/air: Liquid → Liquid Nominal setting of the engine temperature control o mechanism (C)::::	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	85 deg C → 85 deg C Liquid: Yes → Yes Nature of liquid:: Option 1:		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L → Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L Circulating pump(s): (Yes/No): Yes → Yes Characteristics of Circulating pump: Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm → Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm Make(s):: XXXXX Fasteners Ltd → XXXXX Fasteners Ltd Type(s):: Impeller Type → Impeller Type Drive ratio(s):: 1: 1.12 → 1: 1.12 Description of the fan and its drive mechanism:: Electric Fan , Motor Driven → Electric Fan , Motor Driven Blower: (Yes/No): No → No Characteristics of blower: NA → NA Make(s):: NA → NA Type(s):: NA → NA Drive ratio(s):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Pressure charger: (Yes/No) Turbocharger / Supercharger: Yes, Turbo charger → Yes, Turbo charger Make(s):: M/s. XYZ, India → M/s. XYZ, India Type(s) & Identification		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number:: Waste Gate with Vacuum Actuator, X3M1XXXX → Waste Gate with Vacuum Actuator, X3M1XXXX Description of the system (maximum charge pressure:.....(kPa), waste-gate, if applicable): Waste gate turbocharger with vacuum actuator Maximum Charge pressure - 145 kPa → Waste gate turbocharger with vacuum actuator Maximum Charge pressure - 145 kPa Inter-cooler: (Yes/No): Yes → Yes Type: air-air / air-water: Air – Air → Air – Air Make :: Mahle Behr → Mahle Behr Make :: Tata Toyo → Tata Toyo Identification Number:: B1V42XXX → B1V42XXX Identification Number:: B1V42YYY → B1V42YYY Air pressure drop across the intercooler: 15 kPa → 15 kPa Intake depression at rated engine speed and at 100 per cent load (Specify location of measurement) Minimum allowable:... (kPa) Maximum allowable:...	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(kPa): Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Description and drawings of inlet pipes and their accessories (plenum		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	chamber, heating device, additional air intakes, etc.):: Intake manifold with Electronic throttle valve Refer Enclosure 4D → Intake manifold with Electronic throttle valve Refer Enclosure 4D Intake manifold description including dimensional drawings and/or photographs):: Aluminium Intake manifold with pressure sensor Refer Enclosure 4D → Aluminium Intake manifold with pressure sensor Refer Enclosure 4D Air filter, Dimensional drawings: Refer Enclosure 4E → Refer Enclosure 4E Make(s):: M/s. Fleet guard → M/s. Fleet guard Type(s) & Identification Number:: Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 → Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 Intake silencer, Dimensional drawing: NA → NA Make(s):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Type(s) & Identification Number:: NA → NA Description and Dimensional drawing of the exhaust manifold:: Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F → Cast iron manifold with adaptation for		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Turbocharger and EGR inlet Refer Enclosure 4F Description and drawing with dimensions of the exhaust system:(Description with general arrangement of exhaust system along with its routing indicating the lengths of exhaust pipe, tail pipe and: Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly → Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly Effective volume of exhaust-system (specify the tolerance & range) in liters (from exhaust manifold / TC outlet to tail pipe end), Enclose the exhaust system		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	dimensional drawing and indicate the volume of each parts clearly: 13.05 ± 1 litre Refer Enclosure 4G → 13.05 ± 1 litre Refer Enclosure 4G Description and drawing of the exhaust Muffler/Silencer:: NA → NA Type(s) & Identification Number:: NA → NA Make: NA → NA Number:……..: NA → NA Internal diameter of exhaust pipe (mm)(min): diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G → diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G Minimum distance between exhaust pipe(s) and the fuel line: >150 mm → >150 mm Auxiliary Noise shields for compliance to IS 3028 and / OR AIS-020 (If Provided): NA → NA Material: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Layout of noise shield / Photographs / Diagram Showing arrangements indicating fitment on vehicle.: NA → NA Maximum allowable exhaust back pressure at rated engine speed and at 100 per cent load (location of measurement): ... (kPa): 30.0 kpa (150 mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Minimum cross-sectional areas of inlet and outlet ports:: Inlet – 31.5		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm → Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm Inlet (mm): 8.00 → 8.00 Exhaust (mm): 8.00 → 8.00 Opening: 17° bTDC → 17° bTDC Closing: 51° aBDC → 51° aBDC Opening: 66° bBDC → 66° bBDC Closing: 25° aTDC → 25° aTDC Transfer: NA → NA Opening: NA → NA Closing: NA → NA Reference and/or setting ranges (Specify the tolerances):: On Cold Engine → On Cold Engine Inlet: 0.3 to 0.4 mm → 0.3 to 0.4 mm Exhaust: 0.3 to 0.4 mm → 0.3 to 0.4 mm Distribution by ports: NA → NA Volume of crank-case cavity with piston at TDC: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Reed valve fitted (Yes / No): No → No Description of inlet ports, scavenging and exhaust ports with corresponding timing.: NA → NA Device for recycling crankcase gases (description and drawings):: Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J → Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Catalyst (if any, and if not covered by another heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF / DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Element – 1 No. → Catalytic Converter – 1No Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type for LNT (periodic /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:……cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap Bulk Density		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle and Characteristics:: Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY Type and Identification number:: Type - Passive		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite → Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA Make:: NA → NA Type and Idetification number :: NA → NA NOx Sensor (Before /After of SCR):: No → No Make:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Type and Identification number:: NA → NA Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems, description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	engine test bench cycles, between two cycles where regenerative phases occur under the conditions equivalent to Type I test (Distance "D" in t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration (Distance “d” in AIS 137):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2 cycles → 2 cycles Description of method employed to determine the number of cycles between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter (DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA International standard (where appropriate):: NA → NA Frequency of reagent refill: continuous/maintenance (where appropriate):: NA → NA Normal operating temperature range	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY → YYYYYYY Air injection: (Yes/No): No → No Type (pulse air, air pump, etc.) & Identification Number:: NA → NA Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other):		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY → FEW0XXYY Characteristics (flow rate, etc.):: Flow rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	EGR Cooler Make and Identification Number: M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:.....g: NA → NA Canister working capacity:.....g: NA → NA Canister bed volume:.....litre: NA → NA Canister Purge		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	and vent strategy description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve :	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(Yes/No): NA → NA Make and Identification Number: NA → NA Operating pressure (kPa): NA → NA Liquid fuel hoses material :: NA → NA Schematic drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or drawing (Periodic / Continuous): NA → NA Particulate trap: (Yes/No): Yes → Yes Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY Unique Identification (Serial No. or Month and Year of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	DDMMYY / X (Serial No. / Date / Shift) Dimensions and shape of the particulate trap (capacity):: Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure 4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh: 90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type Material - Silicon Carbide → Structure -		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Electronic Control Unit (ECU): Yes → Yes Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number: P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA Misfire detection:: NA → NA Oxygen sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K → Refer Enclosure 4K Particulate traps monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method):: Refer Enclosure 4K → Refer Enclosure 4K List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	A description of the type and number of the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3 EUDC A description of the type of the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing all sensed components with the strategy for fault detection and MI activation (fixed number of driving cycles or statistical method), including a list of relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	explanation for the data given in service $05 Test ID $21 to FF and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles –	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Diagnostics on Controller Area Network (CAN) – Part 4: Requirements for emissions- related systems", a comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137): Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	→ NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA Systems to ensure the correct operation of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent deactivation of the driver inducement, for use by the	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	rescue services [Note: cross- reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling /disable after parking: NA → NA Lowest concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct operation of NOx control measures: NA → NA Details of any devices designed to influence fuel economy (if not covered by other items):: NA → NA LPG fuelling system: (Yes/No): No → No Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to LPG or back:: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NG fuelling system: (Yes/No): No → No Approval number: NA → NA Electronic engine management control unit for NG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation:: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to NG or back: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.):: NA → NA Hydrogen fuelling system: (Yes/No): No → No Electronic engine management control unit for hydrogen fuelling: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to hydrogen or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA H2NG fuelling system: (Yes/No): No → No Percentage of hydrogen in the fuel (maximum specified by the manufacturer): NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to H2NG or back:: NA → NA System lay-out (electrical connections, vacuum		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	connections compensation hoses, etc.):: NA → NA Engines or motor combinations: NA → NA Hybrid Electric Vehicle: (Yes/No): No → No	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Category of Hybrid Electric vehicle Off Vehicle Charging/Not Off Vehicle Charging9: NA → NA Operating mode switch: with/without: NA → NA Selectable modes: NA → NA Pure electric: (Yes/No): NA → NA Pure fuel consuming: (Yes/No): NA → NA Hybrid modes: (Yes/No) (if yes, short description): NA → NA Description of the energy storage device: (traction battery, capacitor, flywheel/generator...): NA → NA Make(s):: NA → NA Type(s) :: NA → NA Identification number:: NA → NA Kind of electrochemical couple:: NA → NA Energy: ..... (for battery: voltage and capacity Ah in 2 h, for capacitor: J): NA → NA Charger: on board/external/without: NA → NA Electric machines (describe each type of electric machine separately): NA → NA Make:: NA → NA Type:: NA → NA Primary use: traction motor/generator: NA → NA When used as traction motor: mono motor / multi motors (number):: NA		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	→ NA Maximum power: kW: NA → NA Working principle:: NA → NA Direct current/alternating current/number of phases:: NA → NA Separate excitation/series/compound: NA → NA Synchronous/asynchronous: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA Identification number:: NA → NA Power controller: NA → NA Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Vehicle electric range..... km (according to AIS 137):: NA → NA Manufacturer’s recommendation for preconditioning:: NA → NA Tyre pressure monitoring system: NA → NA Electric energy consumption (weighted combined) ….Wh/km: NA → NA Maximum exhaust temperature at the point in the exhaust pipe(s) adjacent to the outer flange(s) of the exhaust manifold: …(°C): 740 °C → 740 °C Feed system (by pump/injection into intake/mixing with fuel, etc.): Pump → Pump Moment of inertia of engine flywheel:: 0.1212 kg/m2 → 0.1212 kg/m2 Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	exhaust		
(continued)	outlet location, indicated in a Schematic dimensional drawing.): Refer Enclosure 4G → Refer Enclosure 4G Schematic Diagram of Engine cooling system: - → - Description and drawings/photographs of the shapes and constituent materials of the part of the body forming the engine compartment and the part of the passenger compartment nearest to it:: -- → -- Drawings or photographs of the position of the metal components housed in the engine compartment (e.g. heating appliances, spare wheel, air filter, steering mechanism, etc.):: -- → -- Name of the manufacturer of the system (Steering column shaft / Gear shift control / Transmission) on which the Protective Device functions: XXYY Steering System → XXYY Steering System Type/Part no./Identification number(ID)/Drawing No.of the device on which the Protective Device functions: FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly → FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable only in case of steering column locking systems: Locks the Steering → Locks the Steering Components placed in Engine compartment: - → - Example of Engine No: HJEZXYZ891 → HJEZXYZ891	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
CMVR 96; applicable IS 11852 / IS 15986 route Secondary / residual braking	Name and address of engine manufacturing plant (In case of imported vehicles, above details shall be supplied for importer also): Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX → Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Axle spacing in case of multi axle vehicles.: NA → NA Other axle: NA → NA Gross combination weight kg (applicable for articulated / Tractor Trailer combination vehicles): NA → NA Front axle: NA → NA Rear axle: NA → NA Trailer axle / other axles: NA → NA If applicable, maximum mass of trailer	CEA / report needs review	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	whic5h may be coupled: NA → NA Unbraked trailer: NA → NA Reference mass (for vehicles with GVW less than or equal to 3.5 ton) (kg.): Refer Annexure T7-B → Refer Annexure T7-B Make: APOLLO → APOLLO Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade description(s): CEAT WINMILE X3-R → CEAT WINMILE		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 Tyre rolling Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	7.86 N/kN Tyre rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing:		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Refer - TC7719XX000 003XXXF → Refer - TC7719XX000 003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022-	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022- 2023/300002XXX X/TA/XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L- 756XXXX → CM/L- 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E- Marking: NA → NA		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Others (For articulated / combination vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00 R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 325 mm → 325 mm Dynamic rolling radius, mm, as per IS	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26 kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Final Drive (4x2 or 4x4 or 6x2 or 6x4 or others) Front / Rear / Both: (4 x 2) Rear → (4 x 2) Rear Rear axle make: Single Reduction, Hypoid, Semi-floating Banjo type → Single Reduction, Hypoid, Semi-floating Banjo type Rear axle ratio: 4.56 → 5.13 Differential type: Single Reduction, Hypoid, Semi-floating Banjo type → Single Reduction, Hypoid, Semi-floating Banjo type Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2.		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	YYYY – 18.2 :1 Engine coolant temperature: Red Color – Thermometer Symbol → Red Color – Thermometer Symbol Engine low oil pressure: Red Color – Oil Tank Symbol → Red Color – Oil Tank Symbol Brake failure: Red Color – Parking and Oil symbol together → Red Color – Parking and Oil symbol together Brake failure: → Installation drawing showing rear extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size, make, unladen pressure) etc., distance between tyre & RUPD RH & LH, frame rear width, P1, P2 points.: - → - Parking brake: Hand operated [LHS] → Hand operated [LHS] Service Brake: Foot operated → Foot operated Service Brake: → Type of coupling device for brake: NA → NA Detailed engineering drawing as specified in AIS – 013 or photographs showing the mounting details with dimensions.: NA → NA Distance between the contour points of the orthogonal projections on to plane “P” of the accelerator pedal and service brake pedal bearing surfaces, “E” in	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	mm.: -- → -- Distance between the projection of the service brake pedal on to the reference plane “P”, to the right, “H” in mm.: -- → -- Distance between the projection of the service brake pedal on to the0 reference plane “P”, to the left, “J” in mm.: -- → -- In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → -- Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side, distance between front extremity & FUPD, width of FUPD, ID marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Drawing / Images of position & arrangement of Engine: -- → -- Drawing/ Images of Under bonnet (Engine house) Components.: -- → -- 1st · Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	· Overall Ratio / AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC: 12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56 4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 → 20.44 Make: M/s Brakes India Private Limited → M/s Brakes India Private Limited Type (Mechanical/hydraulic/air/air assisted/vacuum assisted/others):		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Vacuum assisted hydraulic → Vacuum assisted hydraulic Control system & braking wheel: Hydraulic acting on all wheels →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Hydraulic acting on all wheels Schematic layout indicating method of split of brake system, location of valves, reservoirs, ABS components etc. (Attach drawing and indicate the drawing number): AJ1XXX00 → AJ1XXX00 Anti-Lock braking system Provided (Yes / No / Optional): NA → NA If yes, details of ABS: - → - Make: - → - Category of ABS: - → - Nos. of directly controlled wheel(s): - → - Brief description of failure warning tell–tale: - → - Wheel Speed Sensors: - → - No. of sensors: - → - Make of sensors: - → - Type of sensors: - → - Modulator: - → - Nos. of Modulators: - → - Make of Modulators: - → - Identification No. / Part No. of Modulator: - → - Brief description and features: - → - Controller: - → - Nos. of Controller: - → - Make of Controller: - → - Identification No. / Part No. of Controller: - → - Brief description and features: - → - Height of Center of Gravity		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(mm): - → - Un laden condition: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Laden condition: - → - Slack adjuster (only applicable for M2, M3, N and T category of vehicles.): - → - Front (Automatic / Manual): - → - Make of slack adjuster: - → - Model and Type: - → - Rear (Automatic / Manual): - → - Make of slack adjuster: - → - Model and Type: - → - Front wheel: 132x47.3x10.35 → 132x47.3x10.35 Rear wheel: LEADING - 314x51.8x5.6 TRAILING – 246x51.8x5.6 → LEADING - 314x51.8x5.6 TRAILING – 246x51.8x5.6 Others: NA → NA Type of liner wear indicator (window/ acoustic/ optical/ any other): Window → Window Front: 250 sq.cm → 250 sq.cm Rear: 570 sq.cm → 570 sq.cm Others: NA → NA Front wheel / axle: Option 1: SYY 6XXXYA → Option 1: SYY 6XXXYA Rear wheel / axle: APACHE 4059M → APACHE 4059M Others: NA → NA Whether asbestos or asbestos-free: Asbestos Free → Asbestos Free Front axle (Disc / Drum): DISC → DISC Effective Diameter (mm): 212 → 212 Rear axle (Disc / drum): DRUM → DRUM Effective Diameter (mm): 282 → 282 Other axle (		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Disc / Drum): NA → NA Effective diameter (mm): NA → NA Material (if the braking surface is not ferrous): NA → NA Front wheel: -- → -- Rear wheel: -- → -- Others: -- → -- Make: M/s XXXX YYYY → M/s XXXX YYYY Type (only applicable for M2, M3, N and T category of vehicles.): N Category → N Category Inner diameter of the master cylinder (mm): 22.81 → 22.81 Operating stroke (mm): 41 → 41 Type of supply tank (only applicable for M2, M3, N and T category of vehicles.): → Brake Chamber: Rear → Rear Brake Chamber: Parking → Parking Make and type: NA → NA Make and type: NA → NA Size mm: NA → NA Size mm: NA → NA Inner diameter mm: NA → NA Inner diameter mm: NA → NA Stroke mm: NA → NA Stroke mm: NA → NA Make and Identification No.: MAKE: XXX GIRLING TAC No: AB2XX1. CoP No: SYL/29/2020-2021/300000YY60/ADDM/XXXX → MAKE: XXX GIRLING TAC No: AB2XX1. CoP No: SYL/29/2020-2021/300000YY60/ADDM/X	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	XXX Type of expansion (HL / HR): HL → HL Free Length of hoses: FOR NON ABS. FRONT : L1 – 470, L2 -417.7 REAR :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	L1 – 405, L2 – 360.9 → FOR NON ABS. FRONT : L1 – 470, L2 -417.7 REAR : L1 – 405, L2 – 360.9 Thickness of lining (mm): 0.6 → 0.6 Nominal bore dia. (mm): 3.1 → 3.1 Type (Visual display/ audible/others): Visual display → Visual display Operation pressure (kg/cm2 / bar / kPa): NA → NA Type of safety device: NA → NA Make: M/s XXXXXYY → M/s XXXXXYY Type (mechanical/spring brake): MECHANICAL → MECHANICAL Acting on Transmission/wheel: WHEEL → WHEEL Control System & Braking wheel: Hand operated & Rear wheel → Hand operated & Rear wheel Lining / pad: Lining → Lining Name of producer: M/s XXXXYYY → M/s XXXXYYY Dimension (mm) (D x W x T): LEADING - 305x48.8x5.6 TRAILING – 256x48.8x5.6 → LEADING - 305x48.8x5.6 TRAILING – 256x48.8x5.6 Area (cm 2): 279 sq.cm → 279 sq.cm Material: XYZ 420 → XYZ 420	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Diameter of brake drum/disc (mm): 272 → 272 Type: UNFAILED PART OF THE SERVICE BRAKE → UNFAILED PART OF THE SERVICE BRAKE Make: Ms. GULF Oil → Ms. GULF Oil Trade name: DOT 4 → DOT 4 Specification/ grade as		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	per Indian standard: DOT 4 → DOT 4 Brake fluid level indicator (Yes/No): Yes → Yes Type (Manual/Auto) Front and Rear: Manual → Manual Lever length: → Pneumatic Brake Actuator: NA → NA Type (as per IS): NA → NA Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No Brake Assist System (Fitted : Yes/No): No → No Category of Brake Assist System A / B: -- → -- For category A systems, define the force threshold at which the ratio between pedal force and brake pressure increases: -- → -- For category B systems, define the brake pedal speed which must be achieved in	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	order to activate the Brake Assist System (e.g. pedal stroke speed (mm/s) during a given time interval): -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Motor Vehicle stability function information document: -- → -- System: -- → -- System variants: -- → -- System options: -- → -- Control function (directional/roll-over/both) including an explanation of the basic function and/or philosophy of the control: -- → -- System configurations (where appropriate): -- → -- System identification including software level identifier: -- → -- List of motor vehicles by description and configuration that are covered by the information document: -- → -- Schematic diagrams of the respective configurations installed on the motor vehicles defined in item 2.1. above with consideration given to the following:: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Lift axles: -- → -- Steering axles: -- → -- Anti-lock braking configurations: -- → -- Wheelbase, track, centre of gravity height: -- → -- Wheel type (single or twin) and tyre type (e.g. structure, category of use, size): -- → --		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Management of the engine or any other source(s) of motive power e.g.Torque Management, control of fuel supply etc.: -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Pneumatic circuits System schematics including anti-lock braking configurations associated with the trailer types defined in paragraph 6.2.1. of this annex.: -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies that	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Vehicle is/ is not equipped to		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	tow a trailer with electric braking system (Yes/No): No → No Make: Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. → Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. Model name / identification: ALXYZABC0 → ALXYZABC0 Type: Compression Ignition → Compression Ignition Bore x stroke (mm): 80 X 98 → 80 X 98 No. of cylinders: 3 → 3 Displacement (cm³): 1468 cc → 1468 cc Compression ratio: 16.5±1.0 : 1 → 16.5±1.0 : 1 Max. Engine output (KW @ rpm): 55.1 Kw @ 3300 rpm → 55.1 Kw @ 3300 rpm Max. Torque (Nm @ rpm): 170Nm @ 1600 to 2400 rpm → 170Nm @ 1600 to 2400 rpm Air cleaner type: Dry type → Dry type Type of Fuel: Diesel → Diesel Front axle ratio: NA → NA Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2. YYYY Pvt.Ltd Service brake Type (Mechanical/hydraulic/air/air assisted/vacuum assisted/others): Vacuum assisted hydraulic → Vacuum assisted		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	hydraulic Auto Slack Adjuster Fitted (Yes / No / Optional): Yes (Automatic wear adjustment) → Yes (Automatic wear adjustment) Anti-Lock braking system Provided(Yes/No/Optional): Optional → Optional Front (Disc / Drum): Disc → Disc Rear (Disc / Drum): Drum → Drum Total braking area (cm²): Front axle-254 sq.cm Rear axle-580 sq.cm → Front axle-254 sq.cm Rear axle-580 sq.cm Parking brake type: Mechanical → Mechanical Secondary brake type: Un failed part of the service brake → Un failed part of the service brake Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	12 PR Tyre size designation including ply rating: 215/75 R15 LT → 215/75 R15 LT Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	→ 3.26 kg/cm2 Wiper motor: XYZ Ltd. → XYZ Ltd. Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Maximum GVW kg (for rigid vehicles): Refer Annexure T7 – A Details for Recommended load body - Refer Annexure T7 - B → Refer Annexure T7 – A Details for Recommended load body - Refer Annexure T7 - B Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Engine Manufacturer::		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Ashok Leyland Limited → Ashok Leyland Limited Manufacturer's engine model code (as marked on the engine, or other means of identification): Model Code: ALXYZABC0 Enclosure – 4A → Model Code: ALXYZABC0 Enclosure – 4A Location of engine model code and engine serial number on block / crankcase (drawing to be attached): Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A → Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer Enclosure – 4A Working principle: positive ignition / compression- ignition, four-stroke / two-stroke / rotary cycle (DI/IDI) (NA/TC/TCIC/Any other): Compression Ignition, four-stroke, DI, TCIC → Compression Ignition, four-stroke, DI, TCIC Number, arrangement of cylinders:: 3, Inline → 3, Inline Bore----------(mm): 80 → 80 Stroke: ----------(mm): 98 → 98 Firing order:: 1-2-3 → 1-2-3 Engine		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	capacity:----------(cm3): 1468 → 1468 Volumetric compression ratio (specify tolerance): 17.5±1.0 : 1 → 17.5±1.0 : 1 Drawings of combustion chamber and piston crown and, in the case of positive ignition engine, piston rings:: Refer Enclosure – 4B → Refer Enclosure – 4B Vehicle with Defeat Device Yes / No: No → No If yes – Details of the Defeat device: NA → NA Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	equivalent) in PPM content by volume in the exhaust gas with the engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum net power:..... kW at min-1: 55.1 KW @ 3100 ± 50 rpm → 55.1 KW @ 3100 ± 50 rpm Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm Maximum net torque: .....Nm at: -------min-1 ...(manufacturer's		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	declared value): 170 Nm @ 1600 – 2400 rpm → 170 Nm @ 1600 – 2400 rpm Maximum rated speed (Specify tolerance): 3100 rpm ± 50 rpm → 3100 rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Type of fuel used : Mono Fuel / Bi-Fuel / Flex Fuel / Dual Fuel: Mono Fuel → Mono Fuel Fuel type: Diesel / Gasoline / LPG / CNG / Biomethane / Bio-H methane / HCNG / Biogas / LNG / Ethanol ((E85) / 2 (E100)) /ED (95) Flex fuel Methanol M15 / M100 / MD95 / Biodiesel up to 100% /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Hydrogen.: Diesel → Diesel Maximum amount of biofuel acceptable in fuel (manufacturer's declared value): per cent by volume: 7% → 7% By fuel injection (compression-ignition only): (Yes/No): Yes → Yes System description:: Unit Pump Common Rail Direct Injection system → Unit Pump Common Rail Direct Injection system Working principle: direct-injection/pre-chamber/swirl Chamber: Direct Injection → Direct Injection Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) &		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: Plunger Type & FHL0XYZ0 → Plunger Type & FHL0XYZ0 Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Injection advance curve (Specify the tolerances):: ECU Controlled → ECU Controlled Governor: ECU Controlled → ECU Controlled Cut-off point under load:...(min-1): 3300 → 3300 Cut-off point without load (Maximum	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	no-load speed)…(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Multi-Hole Injectors (7 holes) FKXXYYZZ → Multi-Hole Injectors (7 holes) FKXXYYZZ Opening pressure (kPa) (specify the tolerances) or characteristic diagram: 15000 to 23000 kPa → 15000 to 23000 kPa Length and internal diameter…(mm): Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) -		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	193.5± 3 Internal Diameter – 3mm → Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm Common rail make & type: M/s. XXXXYYY, Hot Forged Rail Type → M/s. XXXXYYY, Hot Forged Rail Type Cold start system: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description:: NA → NA Auxiliary starting aid: NA → NA Make(s):: NA → NA Type(s):: NA → NA System description:: NA → NA Electronic controlled injection: (Yes/No):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA → NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s. XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and Metering type Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make and type of air temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type By fuel injection (positive ignition only): (Yes/No): No → No Working principle: intake manifold (single/multi- point)/direct injection/other (specify): NA → NA Make(s) of Injection Pump :: NA → NA Type(s) & Identification		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number of Injection Pump:: NA → NA System description (in the case of systems other than continuous injection give equivalent details):: NA → NA Make and type of the control unit:: NA → NA Make and type of the fuel regulator:: NA → NA Make and type of the air-flow sensor:: NA → NA Make and type of the micro-switch:: NA → NA Make and type of the throttle housing:: NA → NA Make and type of the water temperature sensor:: NA → NA Make and type of the air temperature	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	sensor:: NA → NA Injectors: Opening pressure (Specify the tolerances):,..... (kPa) or characteristic diagram:: NA → NA Make(s):: NA → NA Type(s):: NA → NA Identification Number: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA Cut-off point under load (min-1): NA → NA Injection timing:: NA → NA Cold start system:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Operating principle(s):: NA → NA Operating limits/settings(Specify the tolerances): NA → NA Pressure, (Specify the tolerances).....(kPa) or characteristic diagram:: 320 ± 50 kPa → 320 ± 50 kPa Rated voltage:...V, positive/negative ground: 12V, Negative Ground → 12V, Negative Ground Generator/ Alternator: Alternator → Alternator Type:: DC – Alternator with Voltage regulator → DC – Alternator with Voltage regulator Nominal output: ...VA: 1205 VA (For Non AC Variants) 850 Amps	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	1745 VA (For AC Variants) 120 Amps → 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps Ignition: NA → NA Make(s):: NA → NA Type(s):: NA → NA Working principle:: NA → NA Ignition advance curve or map(Specify the tolerances):: NA → NA Static ignition timing (Specify the tolerances):.....degrees before TDC: NA → NA Spark plugs: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Gap setting----------(mm): NA → NA Nominal resistance (kilo ohm) (if resistive type): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Ignition coil(s): NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Ignition condenser: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA EMI suppressor cap / Device / Electronic unit: NA → NA Make: NA → NA Type (Resistive/Capacitive): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Identification number: NA → NA Nominal resistance (kilo ohm): NA → NA Terminology and Drawing of interference Suppression equipment: NA → NA H.T. Cable: NA → NA Make and Place: NA → NA Type (Resistive/Non-resistive): NA → NA Length (mm) (if resistive type): NA → NA Outside dia. (mm) (if resistive type): NA → NA Nominal resistance kilo ohm, (if resistive type): NA → NA Radiator drawing(s): Refer enclosure 4C2 & 4C4 → Refer enclosure 4C2 & 4C4 Radiator drawing(s): Refer enclosurev 4C1 & 4C3 → Refer enclosurev 4C1 & 4C3 Make: Mahle Behr → Mahle Behr Make: Tata Toyo → Tata Toyo Type (s): Cross flow → Cross flow Relief valve pressure setting: 0.9 kg/cm2 → 0.9 kg/cm2 Fan characteristics (Fan power, kW) Enclose the fan		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	power curve corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part No: 9 → 9	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Material of blades (metal / plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ± 150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150 (AC) Radiator core open area (cm²): 1906 cm2 → 1906 cm2 Cooling system: liquid/air: Liquid → Liquid Nominal setting of the engine temperature control o mechanism (C)::: 85 deg C → 85 deg C Liquid: Yes → Yes Nature of liquid:: Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L → Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	coolant capacity: approx..8 L Circulating pump(s): (Yes/No): Yes → Yes Characteristics of Circulating pump:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm → Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm Make(s):: XXXXX Fasteners Ltd → XXXXX Fasteners Ltd Type(s):: Impeller Type → Impeller Type Drive ratio(s):: 1: 1.12 → 1: 1.12 Description of the fan and its drive mechanism:: Electric Fan , Motor Driven → Electric Fan , Motor Driven Blower: (Yes/No): No → No Characteristics of blower: NA → NA Make(s):: NA → NA Type(s):: NA → NA Drive ratio(s):: NA → NA Pressure charger: (Yes/No) Turbocharger / Supercharger: Yes, Turbo charger → Yes, Turbo charger Make(s):: M/s. XYZ, India → M/s. XYZ, India Type(s) & Identification Number:: Waste Gate with Vacuum Actuator, X3M1XXXX → Waste Gate with Vacuum Actuator, X3M1XXXX Description of the system (maximum charge pressure:.....(kPa), waste-gate, if applicable): Waste gate turbocharger with vacuum		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	actuator Maximum Charge pressure - 145 kPa → Waste gate turbocharger with vacuum actuator	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	actuator Maximum Charge pressure - 145 kPa Inter-cooler: (Yes/No): Yes → Yes Type: air-air / air-water: Air – Air → Air – Air Make :: Mahle Behr → Mahle Behr Make :: Tata Toyo → Tata Toyo Identification Number:: B1V42XXX → B1V42XXX Identification Number:: B1V42YYY → B1V42YYY Air pressure drop across the intercooler: 15 kPa → 15 kPa Intake depression at rated engine speed and at 100 per cent load (Specify location of measurement) Minimum allowable:... (kPa) Maximum allowable:... (kPa): Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Description and drawings of inlet pipes and their accessories (plenum chamber, heating device, additional air intakes, etc.):: Intake manifold with Electronic throttle valve Refer Enclosure 4D → Intake manifold with Electronic throttle valve Refer Enclosure 4D Intake manifold description		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	including dimensional drawings and/or photographs):: Aluminium Intake manifold with pressure sensor Refer Enclosure 4D → Aluminium Intake manifold with pressure sensor Refer Enclosure 4D Air filter, Dimensional drawings: Refer Enclosure 4E → Refer Enclosure 4E Make(s):: M/s. Fleet guard → M/s. Fleet guard Type(s) & Identification Number:: Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 → Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 Intake silencer, Dimensional drawing: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description and Dimensional drawing of the exhaust manifold:: Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F → Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F Description and drawing with dimensions of the exhaust system:(Description with general arrangement of exhaust system along with its routing indicating the lengths of exhaust	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	pipe, tail pipe and: Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly → Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly Effective volume of exhaust-system (specify the tolerance & range) in liters (from exhaust manifold / TC outlet to tail pipe end), Enclose the exhaust system dimensional drawing and indicate the volume of each parts clearly: 13.05 ± 1 litre Refer Enclosure 4G → 13.05 ± 1 litre Refer Enclosure 4G Description and drawing of the exhaust Muffler/Silencer:: NA → NA Type(s) & Identification	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number:: NA → NA Make: NA → NA Number:.....: NA → NA Internal diameter of exhaust pipe (mm)(min): diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G → diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G Minimum distance between exhaust pipe(s) and the fuel line: >150 mm → >150 mm Auxiliary Noise shields for compliance to IS 3028 and / OR AIS-020 (If Provided): NA → NA Material: NA → NA Layout of noise shield / Photographs / Diagram Showing arrangements indicating fitment on vehicle.: NA → NA Maximum allowable exhaust back pressure at rated engine speed and at 100 per cent load (location of measurement): ... (kPa): 30.0 kpa (150 mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Minimum cross-sectional areas of inlet and outlet ports:: Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm → Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm Inlet (mm): 8.00 → 8.00 Exhaust (mm): 8.00 → 8.00 Opening: 17° bTDC → 17° bTDC Closing: 51° aBDC → 51° aBDC Opening: 66° bBDC → 66°	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	bBDC Closing: 25° aTDC → 25° aTDC Transfer: NA → NA Opening: NA → NA Closing: NA → NA Reference and/or setting ranges (Specify the tolerances):: On Cold Engine → On Cold Engine Inlet: 0.3 to 0.4 mm → 0.3 to 0.4 mm Exhaust: 0.3 to 0.4 mm → 0.3 to 0.4 mm Distribution by ports: NA → NA Volume of crank-case cavity with piston at TDC: NA → NA Reed valve fitted (Yes / No): No → No Description of inlet ports, scavenging and exhaust ports with corresponding timing.: NA → NA Device for recycling crankcase gases (description and drawings):: Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J → Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx Catalyst (if any, and if not covered by another heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	integrated and Passive Selective catalytic reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF / DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No Element – 1 No. → Catalytic Converter – 1No Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type for LNT (periodic / continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:……cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap Bulk Density (GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	and Characteristics:: Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY Type and Identification number:: Type - Passive NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Honey Comb type Material: Cordierite → Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make:: NA → NA Type and Idetification number :: NA → NA NOx Sensor		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Before /After of SCR):: No → No Make:: NA → NA Type and Identification number:: NA → NA Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems, description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent engine test bench cycles, between two cycles where regenerative	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	phases occur under the conditions equivalent to Type I test (Distance "D" in t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Distance “d” in AIS 137): 2 cycles → 2 cycles Description of method employed to determine the number of cycles between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter (DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	International standard (where appropriate):: NA → NA Frequency of reagent refill: continuous/maintenance (where		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	appropriate):: NA → NA Normal operating temperature range (°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY → YYYYYYY Air injection: (Yes/No): No → No Type (pulse air, air pump, etc.) & Identification Number:: NA → NA Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other): Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FEW0XXYY Characteristics (flow rate, etc.):: Flow rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes EGR		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Cooler Make and Identification Number: M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:.....g: NA → NA Canister working capacity:.....g: NA → NA Canister bed	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	volume:.....litre: NA → NA Canister Purge and vent strategy description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve : (Yes/No): NA → NA Make and Identification Number: NA → NA Operating pressure (kPa): NA → NA Liquid		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	fuel hoses material :: NA → NA Schematic drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or drawing (Periodic / Continuous): NA → NA Particulate trap: (Yes/No): Yes → Yes Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dimensions and shape of the particulate trap (capacity):: Dimension :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure 4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type Material - Silicon Carbide → Structure - Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI Electronic Control Unit (ECU): Yes → Yes Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number: P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Misfire detection:: NA → NA Oxygen sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K → Refer Enclosure 4K Particulate traps monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method)::	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer Enclosure 4K → Refer Enclosure 4K List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K → Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K A description of the type and number of the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	EUDC A description of the type of the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing all sensed components with the strategy for fault detection and MI activation (fixed number of driving cycles or statistical method), including a list of relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive explanation for the data	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	given in service $05 Test ID $21 to FF and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles –		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Diagnostics on Controller Area Network (CAN) – Part 4: Requirements for emissions- related systems", a comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137): Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA → NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Systems to ensure the correct operation of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	deactivation of the driver inducement, for use by the rescue services [Note: cross- reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling /disable after parking: NA → NA Lowest concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct operation of NOx control measures: NA → NA Details of any devices designed to influence fuel economy (if not covered by other items):: NA → NA LPG fuelling system: (Yes/No): No → No Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to LPG or back:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	System layout (electrical connections, vacuum connections, compensation hoses, etc.): NA → NA NG fuelling system: (Yes/No): No → No Approval number: NA → NA Electronic engine management control unit for NG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation:: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to NG or back: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.):: NA → NA Hydrogen fuelling system: (Yes/No): No → No Electronic engine management control unit for hydrogen fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Description of the safeguarding of the catalyst at switch- over from petrol to hydrogen or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA H2NG fuelling system: (Yes/No): No → No Percentage of hydrogen in the fuel (maximum specified by the manufacturer): NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to H2NG or back:: NA → NA System lay-out (electrical connections,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	vacuum connections compensation hoses, etc.):: NA → NA Electric motor(Traction): No → No Type (winding, excitation):: NA → NA Maximum hourly output: kW(manufacturer’s declared value): NA → NA Maximum net power:kW(manufacturer’s declared		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	value): NA → NA Maximum 30 minutes power: kW (manufacturer’s declared value): NA → NA Operating voltage: V: NA → NA Make and Identification Number:: → Battery (Traction): NA → NA Number of cells:: NA → NA Mass: ……(kg): NA → NA Capacity: Ah (Amp-hours): NA → NA Position:: NA → NA Engines or motor combinations: NA → NA Hybrid Electric Vehicle: (Yes/No): No → No Category of Hybrid Electric vehicle Off Vehicle Charging/Not Off Vehicle Charging9: NA → NA Operating mode switch: with/without: NA → NA Selectable modes: NA → NA Pure electric: (Yes/No): NA → NA Pure fuel consuming: (Yes/No): NA → NA Hybrid modes: (Yes/No) (if yes, short description): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Description of the energy storage device: (traction battery, capacitor, flywheel/generator...): NA → NA Make(s):: NA → NA Type(s) :: NA → NA Identification number:: NA → NA Kind of electrochemical couple:: NA → NA Energy: ..... (for battery: voltage and capacity Ah in 2 h, for capacitor: J): NA → NA Charger: on		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	board/external/without: NA → NA Electric machines (describe each type of electric machine separately): NA → NA Make:: NA → NA Type:: NA → NA Primary use: traction motor/generator: NA → NA When used as traction motor: mono motor / multi motors (number):: NA → NA Maximum power: kW: NA → NA Working principle:: NA → NA Direct current/alternating current/number of phases:: NA → NA Separate excitation/series/compound: NA → NA Synchronous/asynchronous: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA Identification number:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Power controller: NA → NA Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Vehicle electric range..... km (according to AIS 137):: NA → NA Manufacturer’s recommendation for preconditioning:: NA → NA CO2 (g/km) (Applicable for category M1 with GVW<3.5 T): NA → NA Declared : (rounded to 3 decimal places): NA → NA Fuel consumption (l/100 km) for Petrol, LPG or Diesel and (kg/100km) for CNG and (kWh/100 km) for Electric Driven		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Vehicles(Applicable for category M1 with GVW<3.5 T) (rounded to 3 decimal places)*: NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA *calculated on the declared CO2: NA → NA CO2 Reducing technologies Available (Yes/No): No → No Regenerative braking: NA → NA Start-Stop System: NA → NA Tyre pressure monitoring system: NA → NA 6 or more Speed Transmission: NA → NA Any other technology (manufacture to	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	enclose detail): NA → NA Electric energy consumption for electric vehicles Wh/km: NA → NA Electric energy consumption for pure electric vehicles Wh/km: NA → NA Electric energy consumption for externally chargeable hybrid electric vehicles: NA → NA Electric energy consumption (condition A, combined) Wh/km: NA → NA Electric energy consumption (condition B, combined) Wh/km: NA → NA Electric energy consumption (weighted combined) ….Wh/km: NA → NA Moment of inertia of engine flywheel:: 0.1212 kg/m2 → 0.1212 kg/m2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2 Make: XXYY System → XXYY System Type: 12V → 12V Identification mark: X5XYZ100 → X5XYZ100 Rated voltage: 12V → 12V Number of sweep Frequencies: 2 → 2 Highest sweep frequency (Cycles / min): 62±8 rpm → 62±8 rpm Lowest sweep frequency (Cycles / min): 40±5 rpm → 40±5 rpm Washer tank motor / Washer pump: ELECTRIC MOTOR WITH PUMP → ELECTRIC MOTOR WITH PUMP Make: XYZ → XYZ	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Model: PEAD64XYZS00 → PEAD64XYZS00 Schematic Diagram of Engine cooling system: - → - Maximum vehicle speed for continuous operation (km/h) (Only for AC horns): 80 Kmph → 80 Kmph A list, specifying the parts which constitute the headlamp cleaner and drawings thereof, (e.g. pumps, nozzles, valves, motors and wipers);: -- → -- Description and drawings/photographs of the shapes and constituent materials of the part of the body forming the engine compartment and the part of the passenger compartment nearest to it:: -- → -- Drawings or photographs of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	the position of the metal components housed in the engine compartment (e.g. heating appliances, spare wheel, air filter, steering mechanism, etc.):: -- → -- List of all Electrical components, which include Broadband EMI sources (like HAVC Motor, Wiper Motor, Window motor and Horn etc.) The list shall include Device name, Make/s and Part No. /Type ID/Drawing No: -- → -- System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	only in case of steering column locking systems: Locks the Steering → Locks the Steering Components placed in Engine compartment: - → - Frequency bands (Hz) · Max Output Power (W): Antenna Position at Vehicle, specific conditions for installation and/or use → Antenna Position at Vehicle, specific conditions for installation and/or use Example of Engine No: HJEZXYZ891 → HJEZXYZ891		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
CMVR 96; applicable IS 11852 / IS 15986 route Brake fade, recovery and laden/unladen performance	Name and address of engine manufacturing plant (In case of imported vehicles, above details shall be supplied for importer also): Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX → Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Axle spacing in case of multi axle vehicles.: NA → NA Other axle: NA → NA Gross combination weight kg (applicable for articulated / Tractor Trailer combination vehicles): NA → NA Front axle: NA → NA	CEA / report needs review	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Rear axle: NA → NA Trailer axle / other axles: NA → NA If applicable, maximum mass of trailer whic5h may be coupled: NA → NA Unbraked trailer: NA → NA Reference mass (for vehicles with GVW less than or equal to 3.5 ton) (kg.): Refer Annexure T7-B → Refer Annexure T7-B Make: APOLLO → APOLLO Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade description(s): CEAT WINMILE X3-R → CEAT WINMILE		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	142 Tyre rolling Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing:		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Refer - TC7719XX000 003XXXF → Refer - TC7719XX000	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L-756XXXX → CM/L-756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Spare wheel: 1 → 1 Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E- Marking: NA → NA Others (For articulated / combination vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00 R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	M2, M3, N and T category of vehicles.): 325 mm → 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26 kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Final Drive (4x2 or 4x4 or 6x2 or 6x4 or others) Front / Rear / Both: (4 x 2) Rear → (4 x 2) Rear Rear axle make: Single Reduction, Hypoid, Semi-floating Banjo type → Single Reduction, Hypoid, Semi-floating Banjo type Rear axle ratio: 4.56 → 5.13 Differential type: Single Reduction, Hypoid, Semi-floating Banjo type → Single Reduction, Hypoid, Semi-floating Banjo type Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2.	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	YYYY – 18.2 :1 Engine coolant temperature: Red Color – Thermometer Symbol → Red Color – Thermometer Symbol Engine low oil pressure: Red Color – Oil Tank Symbol → Red Color – Oil Tank Symbol Brake failure: Red Color – Parking and Oil symbol together → Red Color – Parking and Oil symbol together Brake failure: → Installation drawing showing rear extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size, make, unladen pressure) etc., distance between tyre & RUPD RH & LH, frame rear width, P1, P2 points.: - → - Parking brake: Hand operated [LHS] → Hand operated [LHS] Service Brake: Foot operated → Foot operated Service Brake: → Type of coupling device for brake: NA → NA Detailed engineering drawing as specified in AIS – 013 or photographs showing the mounting details with dimensions.: NA → NA Distance between the contour points of the orthogonal projections on to plane “P” of the accelerator pedal and service brake pedal bearing surfaces, “E” in	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	mm.: -- → -- Distance between the projection of the service brake pedal on to the reference plane “P”, to the right, “H” in mm.: -- → -- Distance between the projection of the service brake pedal on to the0 reference plane “P”, to the left, “J” in mm.: -- → -- In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → -- Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side, distance between front extremity & FUPD, width of FUPD, ID marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Drawing / Images of position & arrangement of Engine: -- → -- Drawing/ Images of Under bonnet (Engine house) Components.: -- → -- 1st · Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	· Overall Ratio / AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC: 12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56 4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 → 20.44 Make: M/s Brakes India Private Limited → M/s Brakes India Private Limited Type (Mechanical/hydraulic/air/air assisted/vacuum assisted/others):		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Vacuum assisted hydraulic → Vacuum assisted hydraulic Control system & braking wheel: Hydraulic acting on all wheels →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Hydraulic acting on all wheels Schematic layout indicating method of split of brake system, location of valves, reservoirs, ABS components etc. (Attach drawing and indicate the drawing number): AJ1XXX00 → AJ1XXX00 Anti-Lock braking system Provided (Yes / No / Optional): NA → NA If yes, details of ABS: - → - Make: - → - Category of ABS: - → - Nos. of directly controlled wheel(s): - → - Brief description of failure warning tell–tale: - → - Wheel Speed Sensors: - → - No. of sensors: - → - Make of sensors: - → - Type of sensors: - → - Modulator: - → - Nos. of Modulators: - → - Make of Modulators: - → - Identification No. / Part No. of Modulator: - → - Brief description and features: - → - Controller: - → - Nos. of Controller: - → - Make of Controller: - → - Identification No. / Part No. of Controller: - → - Brief description and features: - → - Height of Center of Gravity		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(mm): - → - Un laden condition: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Laden condition: - → - Slack adjuster (only applicable for M2, M3, N and T category of vehicles.): - → - Front (Automatic / Manual): - → - Make of slack adjuster: - → - Model and Type: - → - Rear (Automatic / Manual): - → - Make of slack adjuster: - → - Model and Type: - → - Front wheel: 132x47.3x10.35 → 132x47.3x10.35 Rear wheel: LEADING - 314x51.8x5.6 TRAILING – 246x51.8x5.6 → LEADING - 314x51.8x5.6 TRAILING – 246x51.8x5.6 Others: NA → NA Type of liner wear indicator (window/ acoustic/ optical/ any other): Window → Window Front: 250 sq.cm → 250 sq.cm Rear: 570 sq.cm → 570 sq.cm Others: NA → NA Front wheel / axle: Option 1: SYY 6XXXYA → Option 1: SYY 6XXXYA Rear wheel / axle: APACHE 4059M → APACHE 4059M Others: NA → NA Whether asbestos or asbestos-free: Asbestos Free → Asbestos Free Front axle (Disc / Drum): DISC → DISC Effective Diameter (mm): 212 → 212 Rear axle (Disc / drum): DRUM → DRUM Effective Diameter (mm): 282 → 282 Other axle (		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Disc / Drum): NA → NA Effective diameter (mm): NA → NA Material (if the braking surface is not ferrous): NA → NA Front wheel: -- → -- Rear wheel: -- → -- Others: -- → -- Make: M/s XXXX YYYY → M/s XXXX YYYY Type (only applicable for M2, M3, N and T category of vehicles.): N Category → N Category Inner diameter of the master cylinder (mm): 22.81 → 22.81 Operating stroke (mm): 41 → 41 Type of supply tank (only applicable for M2, M3, N and T category of vehicles.): → Brake Chamber: Rear → Rear Brake Chamber: Parking → Parking Make and type: NA → NA Make and type: NA → NA Size mm: NA → NA Size mm: NA → NA Inner diameter mm: NA → NA Inner diameter mm: NA → NA Stroke mm: NA → NA Stroke mm: NA → NA Make and Identification No.: MAKE: XXX GIRLING TAC No: AB2XX1. CoP No: SYL/29/2020-2021/300000YY60/ADDM/XXXX → MAKE: XXX GIRLING TAC No: AB2XX1. CoP No: SYL/29/2020-2021/300000YY60/ADDM/X	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	XXX Type of expansion (HL / HR): HL → HL Free Length of hoses: FOR NON ABS. FRONT : L1 – 470, L2 -417.7 REAR :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	L1 – 405, L2 – 360.9 → FOR NON ABS. FRONT : L1 – 470, L2 -417.7 REAR : L1 – 405, L2 – 360.9 Thickness of lining (mm): 0.6 → 0.6 Nominal bore dia. (mm): 3.1 → 3.1 Type (Visual display/ audible/others): Visual display → Visual display Operation pressure (kg/cm2 / bar / kPa): NA → NA Type of safety device: NA → NA Make: M/s XXXXXYY → M/s XXXXXYY Type (mechanical/spring brake): MECHANICAL → MECHANICAL Acting on Transmission/wheel: WHEEL → WHEEL Control System & Braking wheel: Hand operated & Rear wheel → Hand operated & Rear wheel Lining / pad: Lining → Lining Name of producer: M/s XXXXYYY → M/s XXXXYYY Dimension (mm) (D x W x T): LEADING - 305x48.8x5.6 TRAILING – 256x48.8x5.6 → LEADING - 305x48.8x5.6 TRAILING – 256x48.8x5.6 Area (cm 2): 279 sq.cm → 279 sq.cm Material: XYZ 420 → XYZ 420	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Diameter of brake drum/disc (mm): 272 → 272 Type: UNFAILED PART OF THE SERVICE BRAKE → UNFAILED PART OF THE SERVICE BRAKE Make: Ms. GULF Oil → Ms. GULF Oil Trade name: DOT 4 → DOT 4 Specification/ grade as		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	per Indian standard: DOT 4 → DOT 4 Brake fluid level indicator (Yes/No): Yes → Yes Type (Manual/Auto) Front and Rear: Manual → Manual Lever length: → Pneumatic Brake Actuator: NA → NA Type (as per IS): NA → NA Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No Brake Assist System (Fitted : Yes/No): No → No Category of Brake Assist System A / B: -- → -- For category A systems, define the force threshold at which the ratio between pedal force and brake pressure increases: -- → -- For category B systems, define the brake pedal speed which must be achieved in	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	order to activate the Brake Assist System (e.g. pedal stroke speed (mm/s) during a given time interval): -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Motor Vehicle stability function information document: -- → -- System: -- → -- System variants: -- → -- System options: -- → -- Control function (directional/roll-over/both) including an explanation of the basic function and/or philosophy of the control: -- → -- System configurations (where appropriate): -- → -- System identification including software level identifier: -- → -- List of motor vehicles by description and configuration that are covered by the information document: -- → -- Schematic diagrams of the respective configurations installed on the motor vehicles defined in item 2.1. above with consideration given to the following:: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Lift axles: -- → -- Steering axles: -- → -- Anti-lock braking configurations: -- → -- Wheelbase, track, centre of gravity height: -- → -- Wheel type (single or twin) and tyre type (e.g. structure, category of use, size): -- → --		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Management of the engine or any other source(s) of motive power e.g.Torque Management, control of fuel supply etc.: -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Pneumatic circuits System schematics including anti-lock braking configurations associated with the trailer types defined in paragraph 6.2.1. of this annex.: -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies that	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Vehicle is/ is not equipped to		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	tow a trailer with electric braking system (Yes/No): No → No Make: Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. → Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. Model name / identification: ALXYZABC0 → ALXYZABC0 Type: Compression Ignition → Compression Ignition Bore x stroke (mm): 80 X 98 → 80 X 98 No. of cylinders: 3 → 3 Displacement (cm³): 1468 cc → 1468 cc Compression ratio: 16.5±1.0 : 1 → 16.5±1.0 : 1 Max. Engine output (KW @ rpm): 55.1 Kw @ 3300 rpm → 55.1 Kw @ 3300 rpm Max. Torque (Nm @ rpm): 170Nm @ 1600 to 2400 rpm → 170Nm @ 1600 to 2400 rpm Air cleaner type: Dry type → Dry type Type of Fuel: Diesel → Diesel Front axle ratio: NA → NA Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2. YYYY Pvt.Ltd Service brake Type (Mechanical/hydraulic/air/air assisted/vacuum assisted/others): Vacuum assisted hydraulic → Vacuum assisted		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	hydraulic Auto Slack Adjuster Fitted (Yes / No / Optional): Yes (Automatic wear adjustment) → Yes (Automatic wear adjustment) Anti-Lock braking system Provided(Yes/No/Optional): Optional → Optional Front (Disc / Drum): Disc → Disc Rear (Disc / Drum): Drum → Drum Total braking area (cm²): Front axle-254 sq.cm Rear axle-580 sq.cm → Front axle-254 sq.cm Rear axle-580 sq.cm Parking brake type: Mechanical → Mechanical Secondary brake type: Un failed part of the service brake → Un failed part of the service brake Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	12 PR Tyre size designation including ply rating: 215/75 R15 LT → 215/75 R15 LT Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	→ 3.26 kg/cm2 Wiper motor: XYZ Ltd. → XYZ Ltd. Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Maximum GVW kg (for rigid vehicles): Refer Annexure T7 – A Details for Recommended load body - Refer Annexure T7 - B → Refer Annexure T7 – A Details for Recommended load body - Refer Annexure T7 - B Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Engine Manufacturer::		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Ashok Leyland Limited → Ashok Leyland Limited Manufacturer's engine model code (as marked on the engine, or other means of identification): Model Code: ALXYZABC0 Enclosure – 4A → Model Code: ALXYZABC0 Enclosure – 4A Location of engine model code and engine serial number on block / crankcase (drawing to be attached): Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A → Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer Enclosure – 4A Working principle: positive ignition / compression- ignition, four-stroke / two-stroke / rotary cycle (DI/IDI) (NA/TC/TCIC/Any other): Compression Ignition, four-stroke, DI, TCIC → Compression Ignition, four-stroke, DI, TCIC Number, arrangement of cylinders:: 3, Inline → 3, Inline Bore----------(mm): 80 → 80 Stroke: ----------(mm): 98 → 98 Firing order:: 1-2-3 → 1-2-3 Engine		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	capacity:----------(cm3): 1468 → 1468 Volumetric compression ratio (specify tolerance): 17.5±1.0 : 1 → 17.5±1.0 : 1 Drawings of combustion chamber and piston crown and, in the case of positive ignition engine, piston rings:: Refer Enclosure – 4B → Refer Enclosure – 4B Vehicle with Defeat Device Yes / No: No → No If yes – Details of the Defeat device: NA → NA Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	equivalent) in PPM content by volume in the exhaust gas with the engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum net power:..... kW at min-1: 55.1 KW @ 3100 ± 50 rpm → 55.1 KW @ 3100 ± 50 rpm Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm Maximum net torque: .....Nm at: -------min-1 ...(manufacturer's		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	declared value): 170 Nm @ 1600 – 2400 rpm → 170 Nm @ 1600 – 2400 rpm Maximum rated speed (Specify tolerance): 3100 rpm ± 50 rpm → 3100 rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Type of fuel used : Mono Fuel / Bi-Fuel / Flex Fuel / Dual Fuel: Mono Fuel → Mono Fuel Fuel type: Diesel / Gasoline / LPG / CNG / Biomethane / Bio-H methane / HCNG / Biogas / LNG / Ethanol ((E85) / 2 (E100)) /ED (95) Flex fuel Methanol M15 / M100 / MD95 / Biodiesel up to 100% /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Hydrogen.: Diesel → Diesel Maximum amount of biofuel acceptable in fuel (manufacturer's declared value): per cent by volume: 7% → 7% By fuel injection (compression-ignition only): (Yes/No): Yes → Yes System description:: Unit Pump Common Rail Direct Injection system → Unit Pump Common Rail Direct Injection system Working principle: direct-injection/pre-chamber/swirl Chamber: Direct Injection → Direct Injection Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) &		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: Plunger Type & FHL0XYZ0 → Plunger Type & FHL0XYZ0 Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Injection advance curve (Specify the tolerances):: ECU Controlled → ECU Controlled Governor: ECU Controlled → ECU Controlled Cut-off point under load:…(min-1): 3300 → 3300 Cut-off point without load (Maximum	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	no-load speed)…(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Multi-Hole Injectors (7 holes) FKXXYYZZ → Multi-Hole Injectors (7 holes) FKXXYYZZ Opening pressure (kPa) (specify the tolerances) or characteristic diagram: 15000 to 23000 kPa → 15000 to 23000 kPa Length and internal diameter…(mm): Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) -		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	193.5± 3 Internal Diameter – 3mm → Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm Common rail make & type: M/s. XXXXYYY, Hot Forged Rail Type → M/s. XXXXYYY, Hot Forged Rail Type Cold start system: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description:: NA → NA Auxiliary starting aid: NA → NA Make(s):: NA → NA Type(s):: NA → NA System description:: NA → NA Electronic controlled injection: (Yes/No):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA → NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s. XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and Metering type Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make and type of air temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type By fuel injection (positive ignition only): (Yes/No): No → No Working principle: intake manifold (single/multi- point)/direct injection/other (specify): NA → NA Make(s) of Injection Pump :: NA → NA Type(s) & Identification		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number of Injection Pump:: NA → NA System description (in the case of systems other than continuous injection give equivalent details):: NA → NA Make and type of the control unit:: NA → NA Make and type of the fuel regulator:: NA → NA Make and type of the air-flow sensor:: NA → NA Make and type of the micro-switch:: NA → NA Make and type of the throttle housing:: NA → NA Make and type of the water temperature sensor:: NA → NA Make and type of the air temperature	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	sensor:: NA → NA Injectors: Opening pressure (Specify the tolerances):,..... (kPa) or characteristic diagram:: NA → NA Make(s):: NA → NA Type(s):: NA → NA Identification Number: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA Cut-off point under load (min-1): NA → NA Injection timing:: NA → NA Cold start system:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Operating principle(s):: NA → NA Operating limits/settings(Specify the tolerances): NA → NA Pressure, (Specify the tolerances).....(kPa) or characteristic diagram:: 320 ± 50 kPa → 320 ± 50 kPa Rated voltage:...V, positive/negative ground: 12V, Negative Ground → 12V, Negative Ground Generator/ Alternator: Alternator → Alternator Type:: DC – Alternator with Voltage regulator → DC – Alternator with Voltage regulator Nominal output: ...VA: 1205 VA (For Non AC Variants) 850 Amps	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	1745 VA (For AC Variants) 120 Amps → 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps Ignition: NA → NA Make(s):: NA → NA Type(s):: NA → NA Working principle:: NA → NA Ignition advance curve or map(Specify the tolerances):: NA → NA Static ignition timing (Specify the tolerances):.....degrees before TDC: NA → NA Spark plugs: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Gap setting----------(mm): NA → NA Nominal resistance (kilo ohm) (if resistive type): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Ignition coil(s): NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Ignition condenser: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA EMI suppressor cap / Device / Electronic unit: NA → NA Make: NA → NA Type (Resistive/Capacitive): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Identification number: NA → NA Nominal resistance (kilo ohm): NA → NA Terminology and Drawing of interference Suppression equipment: NA → NA H.T. Cable: NA → NA Make and Place: NA → NA Type (Resistive/Non-resistive): NA → NA Length (mm) (if resistive type): NA → NA Outside dia. (mm) (if resistive type): NA → NA Nominal resistance kilo ohm, (if resistive type): NA → NA Radiator drawing(s): Refer enclosure 4C2 & 4C4 → Refer enclosure 4C2 & 4C4 Radiator drawing(s): Refer enclosurev 4C1 & 4C3 → Refer enclosurev 4C1 & 4C3 Make: Mahle Behr → Mahle Behr Make: Tata Toyo → Tata Toyo Type (s): Cross flow → Cross flow Relief valve pressure setting: 0.9 kg/cm2 → 0.9 kg/cm2 Fan characteristics (Fan power, kW) Enclose the fan		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	power curve corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part No: 9 → 9	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Material of blades (metal / plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ± 150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150 (AC) Radiator core open area (cm²): 1906 cm2 → 1906 cm2 Cooling system: liquid/air: Liquid → Liquid Nominal setting of the engine temperature control o mechanism (C)::: 85 deg C → 85 deg C Liquid: Yes → Yes Nature of liquid:: Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L → Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	coolant capacity: approx..8 L Circulating pump(s): (Yes/No): Yes → Yes Characteristics of Circulating pump:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm → Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm Make(s):: XXXXX Fasteners Ltd → XXXXX Fasteners Ltd Type(s):: Impeller Type → Impeller Type Drive ratio(s):: 1: 1.12 → 1: 1.12 Description of the fan and its drive mechanism:: Electric Fan , Motor Driven → Electric Fan , Motor Driven Blower: (Yes/No): No → No Characteristics of blower: NA → NA Make(s):: NA → NA Type(s):: NA → NA Drive ratio(s):: NA → NA Pressure charger: (Yes/No) Turbocharger / Supercharger: Yes, Turbo charger → Yes, Turbo charger Make(s):: M/s. XYZ, India → M/s. XYZ, India Type(s) & Identification Number:: Waste Gate with Vacuum Actuator, X3M1XXXX → Waste Gate with Vacuum Actuator, X3M1XXXX Description of the system (maximum charge pressure:.....(kPa), waste-gate, if applicable): Waste gate turbocharger with vacuum		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	actuator Maximum Charge pressure - 145 kPa → Waste gate turbocharger with vacuum actuator	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	actuator Maximum Charge pressure - 145 kPa Inter-cooler: (Yes/No): Yes → Yes Type: air-air / air-water: Air – Air → Air – Air Make :: Mahle Behr → Mahle Behr Make :: Tata Toyo → Tata Toyo Identification Number:: B1V42XXX → B1V42XXX Identification Number:: B1V42YYY → B1V42YYY Air pressure drop across the intercooler: 15 kPa → 15 kPa Intake depression at rated engine speed and at 100 per cent load (Specify location of measurement) Minimum allowable:... (kPa) Maximum allowable:... (kPa): Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Description and drawings of inlet pipes and their accessories (plenum chamber, heating device, additional air intakes, etc.):: Intake manifold with Electronic throttle valve Refer Enclosure 4D → Intake manifold with Electronic throttle valve Refer Enclosure 4D Intake manifold description		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	including dimensional drawings and/or photographs):: Aluminium Intake manifold with pressure sensor Refer Enclosure 4D → Aluminium Intake manifold with pressure sensor Refer Enclosure 4D Air filter, Dimensional drawings: Refer Enclosure 4E → Refer Enclosure 4E Make(s):: M/s. Fleet guard → M/s. Fleet guard Type(s) & Identification Number:: Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 → Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 Intake silencer, Dimensional drawing: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description and Dimensional drawing of the exhaust manifold:: Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F → Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F Description and drawing with dimensions of the exhaust system:(Description with general arrangement of exhaust system along with its routing indicating the lengths of exhaust	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	pipe, tail pipe and: Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly → Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly Effective volume of exhaust-system (specify the tolerance & range) in liters (from exhaust manifold / TC outlet to tail pipe end), Enclose the exhaust system dimensional drawing and indicate the volume of each parts clearly: 13.05 ± 1 litre Refer Enclosure 4G → 13.05 ± 1 litre Refer Enclosure 4G Description and drawing of the exhaust Muffler/Silencer:: NA → NA Type(s) & Identification	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number:: NA → NA Make: NA → NA Number:.....: NA → NA Internal diameter of exhaust pipe (mm)(min): diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G → diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G Minimum distance between exhaust pipe(s) and the fuel line: >150 mm → >150 mm Auxiliary Noise shields for compliance to IS 3028 and / OR AIS-020 (If Provided): NA → NA Material: NA → NA Layout of noise shield / Photographs / Diagram Showing arrangements indicating fitment on vehicle.: NA → NA Maximum allowable exhaust back pressure at rated engine speed and at 100 per cent load (location of measurement): ... (kPa): 30.0 kpa (150 mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Minimum cross-sectional areas of inlet and outlet ports:: Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm → Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm Inlet (mm): 8.00 → 8.00 Exhaust (mm): 8.00 → 8.00 Opening: 17° bTDC → 17° bTDC Closing: 51° aBDC → 51° aBDC Opening: 66° bBDC → 66°	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	bBDC Closing: 25° aTDC → 25° aTDC Transfer: NA → NA Opening: NA → NA Closing: NA → NA Reference and/or setting ranges (Specify the tolerances):: On Cold Engine → On Cold Engine Inlet: 0.3 to 0.4 mm → 0.3 to 0.4 mm Exhaust: 0.3 to 0.4 mm → 0.3 to 0.4 mm Distribution by ports: NA → NA Volume of crank-case cavity with piston at TDC: NA → NA Reed valve fitted (Yes / No): No → No Description of inlet ports, scavenging and exhaust ports with corresponding timing.: NA → NA Device for recycling crankcase gases (description and drawings):: Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J → Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx Catalyst (if any, and if not covered by another heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	integrated and Passive Selective catalytic reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF / DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No Element – 1 No. → Catalytic Converter – 1No Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type for LNT (periodic / continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:……cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap Bulk Density (GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	and Characteristics:: Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY Type and Identification number:: Type - Passive NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Honey Comb type Material: Cordierite → Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make:: NA → NA Type and Idetification number :: NA → NA NOx Sensor		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Before /After of SCR):: No → No Make:: NA → NA Type and Identification number:: NA → NA Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems, description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent engine test bench cycles, between two cycles where regenerative	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	phases occur under the conditions equivalent to Type I test (Distance "D" in t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Distance “d” in AIS 137): 2 cycles → 2 cycles Description of method employed to determine the number of cycles between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter (DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	International standard (where appropriate):: NA → NA Frequency of reagent refill: continuous/maintenance (where		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	appropriate):: NA → NA Normal operating temperature range (°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY → YYYYYYY Air injection: (Yes/No): No → No Type (pulse air, air pump, etc.) & Identification Number:: NA → NA Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other): Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FEW0XXYY Characteristics (flow rate, etc.):: Flow rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes EGR		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Cooler Make and Identification Number: M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:.....g: NA → NA Canister working capacity:.....g: NA → NA Canister bed	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	volume:.....litre: NA → NA Canister Purge and vent strategy description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve : (Yes/No): NA → NA Make and Identification Number: NA → NA Operating pressure (kPa): NA → NA Liquid		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	fuel hoses material :: NA → NA Schematic drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or drawing (Periodic / Continuous): NA → NA Particulate trap: (Yes/No): Yes → Yes Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dimensions and shape of the particulate trap (capacity):: Dimension :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure 4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type Material - Silicon Carbide → Structure - Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI Electronic Control Unit (ECU): Yes → Yes Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number: P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Misfire detection:: NA → NA Oxygen sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K → Refer Enclosure 4K Particulate traps monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method)::	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer Enclosure 4K → Refer Enclosure 4K List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K → Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K A description of the type and number of the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	EUDC A description of the type of the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing all sensed components with the strategy for fault detection and MI activation (fixed number of driving cycles or statistical method), including a list of relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive explanation for the data	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	given in service $05 Test ID $21 to FF and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles –		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Diagnostics on Controller Area Network (CAN) – Part 4: Requirements for emissions- related systems", a comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137): Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA → NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Systems to ensure the correct operation of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	deactivation of the driver inducement, for use by the rescue services [Note: cross- reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling /disable after parking: NA → NA Lowest concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct operation of NOx control measures: NA → NA Details of any devices designed to influence fuel economy (if not covered by other items):: NA → NA LPG fuelling system: (Yes/No): No → No Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to LPG or back:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	System layout (electrical connections, vacuum connections, compensation hoses, etc.): NA → NA NG fuelling system: (Yes/No): No → No Approval number: NA → NA Electronic engine management control unit for NG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation:: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to NG or back: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.):: NA → NA Hydrogen fuelling system: (Yes/No): No → No Electronic engine management control unit for hydrogen fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Description of the safeguarding of the catalyst at switch- over from petrol to hydrogen or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA H2NG fuelling system: (Yes/No): No → No Percentage of hydrogen in the fuel (maximum specified by the manufacturer): NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to H2NG or back:: NA → NA System lay-out (electrical connections,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	vacuum connections compensation hoses, etc.):: NA → NA Electric motor(Traction): No → No Type (winding, excitation):: NA → NA Maximum hourly output: kW(manufacturer’s declared value): NA → NA Maximum net power:kW(manufacturer’s declared		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	value): NA → NA Maximum 30 minutes power: kW (manufacturer’s declared value): NA → NA Operating voltage: V: NA → NA Make and Identification Number:: → Battery (Traction): NA → NA Number of cells:: NA → NA Mass: ……(kg): NA → NA Capacity: Ah (Amp-hours): NA → NA Position:: NA → NA Engines or motor combinations: NA → NA Hybrid Electric Vehicle: (Yes/No): No → No Category of Hybrid Electric vehicle Off Vehicle Charging/Not Off Vehicle Charging9: NA → NA Operating mode switch: with/without: NA → NA Selectable modes: NA → NA Pure electric: (Yes/No): NA → NA Pure fuel consuming: (Yes/No): NA → NA Hybrid modes: (Yes/No) (if yes, short description): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Description of the energy storage device: (traction battery, capacitor, flywheel/generator...): NA → NA Make(s):: NA → NA Type(s) :: NA → NA Identification number:: NA → NA Kind of electrochemical couple:: NA → NA Energy: ..... (for battery: voltage and capacity Ah in 2 h, for capacitor: J): NA → NA Charger: on		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	board/external/without: NA → NA Electric machines (describe each type of electric machine separately): NA → NA Make:: NA → NA Type:: NA → NA Primary use: traction motor/generator: NA → NA When used as traction motor: mono motor / multi motors (number):: NA → NA Maximum power: kW: NA → NA Working principle:: NA → NA Direct current/alternating current/number of phases:: NA → NA Separate excitation/series/compound: NA → NA Synchronous/asynchronous: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA Identification number:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Power controller: NA → NA Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Vehicle electric range..... km (according to AIS 137):: NA → NA Manufacturer’s recommendation for preconditioning:: NA → NA CO2 (g/km) (Applicable for category M1 with GVW<3.5 T): NA → NA Declared : (rounded to 3 decimal places): NA → NA Fuel consumption (l/100 km) for Petrol, LPG or Diesel and (kg/100km) for CNG and (kWh/100 km) for Electric Driven		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Vehicles(Applicable for category M1 with GVW<3.5 T) (rounded to 3 decimal places)*: NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA *calculated on the declared CO2: NA → NA CO2 Reducing technologies Available (Yes/No): No → No Regenerative braking: NA → NA Start-Stop System: NA → NA Tyre pressure monitoring system: NA → NA 6 or more Speed Transmission: NA → NA Any other technology (manufacture to	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	enclose detail): NA → NA Electric energy consumption for electric vehicles Wh/km: NA → NA Electric energy consumption for pure electric vehicles Wh/km: NA → NA Electric energy consumption for externally chargeable hybrid electric vehicles: NA → NA Electric energy consumption (condition A, combined) Wh/km: NA → NA Electric energy consumption (condition B, combined) Wh/km: NA → NA Electric energy consumption (weighted combined) ….Wh/km: NA → NA Moment of inertia of engine flywheel:: 0.1212 kg/m2 → 0.1212 kg/m2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2 Make: XXYY System → XXYY System Type: 12V → 12V Identification mark: X5XYZ100 → X5XYZ100 Rated voltage: 12V → 12V Number of sweep Frequencies: 2 → 2 Highest sweep frequency (Cycles / min): 62±8 rpm → 62±8 rpm Lowest sweep frequency (Cycles / min): 40±5 rpm → 40±5 rpm Washer tank motor / Washer pump: ELECTRIC MOTOR WITH PUMP → ELECTRIC MOTOR WITH PUMP Make: XYZ → XYZ	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Model: PEAD64XYZS00 → PEAD64XYZS00 Schematic Diagram of Engine cooling system: - → - Maximum vehicle speed for continuous operation (km/h) (Only for AC horns): 80 Kmph → 80 Kmph A list, specifying the parts which constitute the headlamp cleaner and drawings thereof, (e.g. pumps, nozzles, valves, motors and wipers);: -- → -- Description and drawings/photographs of the shapes and constituent materials of the part of the body forming the engine compartment and the part of the passenger compartment nearest to it:: -- → -- Drawings or photographs of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	the position of the metal components housed in the engine compartment (e.g. heating appliances, spare wheel, air filter, steering mechanism, etc.):: -- → -- List of all Electrical components, which include Broadband EMI sources (like HAVC Motor, Wiper Motor, Window motor and Horn etc.) The list shall include Device name, Make/s and Part No. /Type ID/Drawing No: -- → -- System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	only in case of steering column locking systems: Locks the Steering → Locks the Steering Components placed in Engine compartment: - → - Frequency bands (Hz) · Max Output Power (W): Antenna Position at Vehicle, specific conditions for installation and/or use → Antenna Position at Vehicle, specific conditions for installation and/or use Example of Engine No: HJEZXYZ891 → HJEZXYZ891		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
CMVR 96; applicable braking standard Brake circuit, warning, control and installation checks	Name and address of engine manufacturing plant (In case of imported vehicles, above details shall be supplied for importer also): Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX → Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Axle spacing in case of multi axle vehicles.: NA → NA Other axle: NA → NA Gross combination weight kg (applicable for articulated / Tractor Trailer combination vehicles): NA → NA Front axle: NA → NA	CEA / report needs review	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Rear axle: NA → NA Trailer axle / other axles: NA → NA If applicable, maximum mass of trailer whic5h may be coupled: NA → NA Unbraked trailer: NA → NA Reference mass (for vehicles with GVW less than or equal to 3.5 ton) (kg.): Refer Annexure T7-B → Refer Annexure T7-B Make: APOLLO → APOLLO Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade description(s): CEAT WINMILE X3-R → CEAT WINMILE		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	142 Tyre rolling Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing:		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Refer - TC7719XX000 003XXXF → Refer - TC7719XX000	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L-756XXXX → CM/L-756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Spare wheel: 1 → 1 Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E- Marking: NA → NA Others (For articulated / combination vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00 R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	M2, M3, N and T category of vehicles.): 325 mm → 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26 kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Final Drive (4x2 or 4x4 or 6x2 or 6x4 or others) Front / Rear / Both: (4 x 2) Rear → (4 x 2) Rear Rear axle make: Single Reduction, Hypoid, Semi-floating Banjo type → Single Reduction, Hypoid, Semi-floating Banjo type Rear axle ratio: 4.56 → 5.13 Differential type: Single Reduction, Hypoid, Semi-floating Banjo type → Single Reduction, Hypoid, Semi-floating Banjo type Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2.	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	YYYY – 18.2 :1 Engine coolant temperature: Red Color – Thermometer Symbol → Red Color – Thermometer Symbol Engine low oil pressure: Red Color – Oil Tank Symbol → Red Color – Oil Tank Symbol Brake failure: Red Color – Parking and Oil symbol together → Red Color – Parking and Oil symbol together Brake failure: → Installation drawing showing rear extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size, make, unladen pressure) etc., distance between tyre & RUPD RH & LH, frame rear width, P1, P2 points.: - → - Parking brake: Hand operated [LHS] → Hand operated [LHS] Service Brake: Foot operated → Foot operated Service Brake: → Type of coupling device for brake: NA → NA Detailed engineering drawing as specified in AIS – 013 or photographs showing the mounting details with dimensions.: NA → NA Distance between the contour points of the orthogonal projections on to plane “P” of the accelerator pedal and service brake pedal bearing surfaces, “E” in	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	mm.: -- → -- Distance between the projection of the service brake pedal on to the reference plane “P”, to the right, “H” in mm.: -- → -- Distance between the projection of the service brake pedal on to the0 reference plane “P”, to the left, “J” in mm.: -- → -- In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → -- Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side, distance between front extremity & FUPD, width of FUPD, ID marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Drawing / Images of position & arrangement of Engine: -- → -- Drawing/ Images of Under bonnet (Engine house) Components.: -- → -- 1st · Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	· Overall Ratio / AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC: 12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56 4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 → 20.44 Make: M/s Brakes India Private Limited → M/s Brakes India Private Limited Type (Mechanical/hydraulic/air/air assisted/vacuum assisted/others):		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Vacuum assisted hydraulic → Vacuum assisted hydraulic Control system & braking wheel: Hydraulic acting on all wheels →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Hydraulic acting on all wheels Schematic layout indicating method of split of brake system, location of valves, reservoirs, ABS components etc. (Attach drawing and indicate the drawing number): AJ1XXX00 → AJ1XXX00 Anti-Lock braking system Provided (Yes / No / Optional): NA → NA If yes, details of ABS: - → - Make: - → - Category of ABS: - → - Nos. of directly controlled wheel(s): - → - Brief description of failure warning tell–tale: - → - Wheel Speed Sensors: - → - No. of sensors: - → - Make of sensors: - → - Type of sensors: - → - Modulator: - → - Nos. of Modulators: - → - Make of Modulators: - → - Identification No. / Part No. of Modulator: - → - Brief description and features: - → - Controller: - → - Nos. of Controller: - → - Make of Controller: - → - Identification No. / Part No. of Controller: - → - Brief description and features: - → - Height of Center of Gravity		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(mm): - → - Un laden condition: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Laden condition: - → - Slack adjuster (only applicable for M2, M3, N and T category of vehicles.): - → - Front (Automatic / Manual): - → - Make of slack adjuster: - → - Model and Type: - → - Rear (Automatic / Manual): - → - Make of slack adjuster: - → - Model and Type: - → - Front wheel: 132x47.3x10.35 → 132x47.3x10.35 Rear wheel: LEADING - 314x51.8x5.6 TRAILING – 246x51.8x5.6 → LEADING - 314x51.8x5.6 TRAILING – 246x51.8x5.6 Others: NA → NA Type of liner wear indicator (window/ acoustic/ optical/ any other): Window → Window Front: 250 sq.cm → 250 sq.cm Rear: 570 sq.cm → 570 sq.cm Others: NA → NA Front wheel / axle: Option 1: SYY 6XXXYA → Option 1: SYY 6XXXYA Rear wheel / axle: APACHE 4059M → APACHE 4059M Others: NA → NA Whether asbestos or asbestos-free: Asbestos Free → Asbestos Free Front axle (Disc / Drum): DISC → DISC Effective Diameter (mm): 212 → 212 Rear axle (Disc / drum): DRUM → DRUM Effective Diameter (mm): 282 → 282 Other axle (		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Disc / Drum): NA → NA Effective diameter (mm): NA → NA Material (if the braking surface is not ferrous): NA → NA Front wheel: -- → -- Rear wheel: -- → -- Others: -- → -- Make: M/s XXXX YYYY → M/s XXXX YYYY Type (only applicable for M2, M3, N and T category of vehicles.): N Category → N Category Inner diameter of the master cylinder (mm): 22.81 → 22.81 Operating stroke (mm): 41 → 41 Type of supply tank (only applicable for M2, M3, N and T category of vehicles.): → Brake Chamber: Rear → Rear Brake Chamber: Parking → Parking Make and type: NA → NA Make and type: NA → NA Size mm: NA → NA Size mm: NA → NA Inner diameter mm: NA → NA Inner diameter mm: NA → NA Stroke mm: NA → NA Stroke mm: NA → NA Make and Identification No.: MAKE: XXX GIRLING TAC No: AB2XX1. CoP No: SYL/29/2020-2021/300000YY60/ADDM/XXXX → MAKE: XXX GIRLING TAC No: AB2XX1. CoP No: SYL/29/2020-2021/300000YY60/ADDM/X	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	XXX Type of expansion (HL / HR): HL → HL Free Length of hoses: FOR NON ABS. FRONT : L1 – 470, L2 -417.7 REAR :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	L1 – 405, L2 – 360.9 → FOR NON ABS. FRONT : L1 – 470, L2 -417.7 REAR : L1 – 405, L2 – 360.9 Thickness of lining (mm): 0.6 → 0.6 Nominal bore dia. (mm): 3.1 → 3.1 Type (Visual display/ audible/others): Visual display → Visual display Operation pressure (kg/cm2 / bar / kPa): NA → NA Type of safety device: NA → NA Make: M/s XXXXXYY → M/s XXXXXYY Type (mechanical/spring brake): MECHANICAL → MECHANICAL Acting on Transmission/wheel: WHEEL → WHEEL Control System & Braking wheel: Hand operated & Rear wheel → Hand operated & Rear wheel Lining / pad: Lining → Lining Name of producer: M/s XXXXYYY → M/s XXXXYYY Dimension (mm) (D x W x T): LEADING - 305x48.8x5.6 TRAILING – 256x48.8x5.6 → LEADING - 305x48.8x5.6 TRAILING – 256x48.8x5.6 Area (cm 2): 279 sq.cm → 279 sq.cm Material: XYZ 420 → XYZ 420	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Diameter of brake drum/disc (mm): 272 → 272 Type: UNFAILED PART OF THE SERVICE BRAKE → UNFAILED PART OF THE SERVICE BRAKE Make: Ms. GULF Oil → Ms. GULF Oil Trade name: DOT 4 → DOT 4 Specification/ grade as		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	per Indian standard: DOT 4 → DOT 4 Brake fluid level indicator (Yes/No): Yes → Yes Type (Manual/Auto) Front and Rear: Manual → Manual Lever length: → Pneumatic Brake Actuator: NA → NA Type (as per IS): NA → NA Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No Brake Assist System (Fitted : Yes/No): No → No Category of Brake Assist System A / B: -- → -- For category A systems, define the force threshold at which the ratio between pedal force and brake pressure increases: -- → -- For category B systems, define the brake pedal speed which must be achieved in	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	order to activate the Brake Assist System (e.g. pedal stroke speed (mm/s) during a given time interval): -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Motor Vehicle stability function information document: -- → -- System: -- → -- System variants: -- → -- System options: -- → -- Control function (directional/roll-over/both) including an explanation of the basic function and/or philosophy of the control: -- → -- System configurations (where appropriate): -- → -- System identification including software level identifier: -- → -- List of motor vehicles by description and configuration that are covered by the information document: -- → -- Schematic diagrams of the respective configurations installed on the motor vehicles defined in item 2.1. above with consideration given to the following:: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Lift axles: -- → -- Steering axles: -- → -- Anti-lock braking configurations: -- → -- Wheelbase, track, centre of gravity height: -- → -- Wheel type (single or twin) and tyre type (e.g. structure, category of use, size): -- → --		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Management of the engine or any other source(s) of motive power e.g.Torque Management, control of fuel supply etc.: -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Pneumatic circuits System schematics including anti-lock braking configurations associated with the trailer types defined in paragraph 6.2.1. of this annex.: -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies that	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Vehicle is/ is not equipped to		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	tow a trailer with electric braking system (Yes/No): No → No Make: Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. → Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. Model name / identification: ALXYZABC0 → ALXYZABC0 Type: Compression Ignition → Compression Ignition Bore x stroke (mm): 80 X 98 → 80 X 98 No. of cylinders: 3 → 3 Displacement (cm³): 1468 cc → 1468 cc Compression ratio: 16.5±1.0 : 1 → 16.5±1.0 : 1 Max. Engine output (KW @ rpm): 55.1 Kw @ 3300 rpm → 55.1 Kw @ 3300 rpm Max. Torque (Nm @ rpm): 170Nm @ 1600 to 2400 rpm → 170Nm @ 1600 to 2400 rpm Air cleaner type: Dry type → Dry type Type of Fuel: Diesel → Diesel Front axle ratio: NA → NA Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2. YYYY Pvt.Ltd Service brake Type (Mechanical/hydraulic/air/air assisted/vacuum assisted/others): Vacuum assisted hydraulic → Vacuum assisted		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	hydraulic Auto Slack Adjuster Fitted (Yes / No / Optional): Yes (Automatic wear adjustment) → Yes (Automatic wear adjustment) Anti-Lock braking system Provided(Yes/No/Optional): Optional → Optional Front (Disc / Drum): Disc → Disc Rear (Disc / Drum): Drum → Drum Total braking area (cm²): Front axle-254 sq.cm Rear axle-580 sq.cm → Front axle-254 sq.cm Rear axle-580 sq.cm Parking brake type: Mechanical → Mechanical Secondary brake type: Un failed part of the service brake → Un failed part of the service brake Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	12 PR Tyre size designation including ply rating: 215/75 R15 LT → 215/75 R15 LT Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	→ 3.26 kg/cm2 Wiper motor: XYZ Ltd. → XYZ Ltd. Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Maximum GVW kg (for rigid vehicles): Refer Annexure T7 – A Details for Recommended load body - Refer Annexure T7 - B → Refer Annexure T7 – A Details for Recommended load body - Refer Annexure T7 - B Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Engine Manufacturer::		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Ashok Leyland Limited → Ashok Leyland Limited Manufacturer's engine model code (as marked on the engine, or other means of identification): Model Code: ALXYZABC0 Enclosure – 4A → Model Code: ALXYZABC0 Enclosure – 4A Location of engine model code and engine serial number on block / crankcase (drawing to be attached): Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A → Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer Enclosure – 4A Working principle: positive ignition / compression- ignition, four-stroke / two-stroke / rotary cycle (DI/IDI) (NA/TC/TCIC/Any other): Compression Ignition, four-stroke, DI, TCIC → Compression Ignition, four-stroke, DI, TCIC Number, arrangement of cylinders:: 3, Inline → 3, Inline Bore----------(mm): 80 → 80 Stroke: ----------(mm): 98 → 98 Firing order:: 1-2-3 → 1-2-3 Engine		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	capacity:----------(cm3): 1468 → 1468 Volumetric compression ratio (specify tolerance): 17.5±1.0 : 1 → 17.5±1.0 : 1 Drawings of combustion chamber and piston crown and, in the case of positive ignition engine, piston rings:: Refer Enclosure – 4B → Refer Enclosure – 4B Vehicle with Defeat Device Yes / No: No → No If yes – Details of the Defeat device: NA → NA Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	equivalent) in PPM content by volume in the exhaust gas with the engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum net power:..... kW at min-1: 55.1 KW @ 3100 ± 50 rpm → 55.1 KW @ 3100 ± 50 rpm Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm Maximum net torque: .....Nm at: -------min-1 ...(manufacturer's		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	declared value): 170 Nm @ 1600 – 2400 rpm → 170 Nm @ 1600 – 2400 rpm Maximum rated speed (Specify tolerance): 3100 rpm ± 50 rpm → 3100 rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Type of fuel used : Mono Fuel / Bi-Fuel / Flex Fuel / Dual Fuel: Mono Fuel → Mono Fuel Fuel type: Diesel / Gasoline / LPG / CNG / Biomethane / Bio-H methane / HCNG / Biogas / LNG / Ethanol ((E85) / 2 (E100)) /ED (95) Flex fuel Methanol M15 / M100 / MD95 / Biodiesel up to 100% /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Hydrogen.: Diesel → Diesel Maximum amount of biofuel acceptable in fuel (manufacturer's declared value): per cent by volume: 7% → 7% By fuel injection (compression-ignition only): (Yes/No): Yes → Yes System description:: Unit Pump Common Rail Direct Injection system → Unit Pump Common Rail Direct Injection system Working principle: direct-injection/pre-chamber/swirl Chamber: Direct Injection → Direct Injection Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) &		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: Plunger Type & FHL0XYZ0 → Plunger Type & FHL0XYZ0 Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Injection advance curve (Specify the tolerances):: ECU Controlled → ECU Controlled Governor: ECU Controlled → ECU Controlled Cut-off point under load:...(min-1): 3300 → 3300 Cut-off point without load (Maximum	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	no-load speed)…(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Multi-Hole Injectors (7 holes) FKXXYYZZ → Multi-Hole Injectors (7 holes) FKXXYYZZ Opening pressure (kPa) (specify the tolerances) or characteristic diagram: 15000 to 23000 kPa → 15000 to 23000 kPa Length and internal diameter…(mm): Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) -		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	193.5± 3 Internal Diameter – 3mm → Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm Common rail make & type: M/s. XXXXYYY, Hot Forged Rail Type → M/s. XXXXYYY, Hot Forged Rail Type Cold start system: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description:: NA → NA Auxiliary starting aid: NA → NA Make(s):: NA → NA Type(s):: NA → NA System description:: NA → NA Electronic controlled injection: (Yes/No):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA → NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s. XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and Metering type Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make and type of air temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type By fuel injection (positive ignition only): (Yes/No): No → No Working principle: intake manifold (single/multi- point)/direct injection/other (specify): NA → NA Make(s) of Injection Pump :: NA → NA Type(s) & Identification		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number of Injection Pump:: NA → NA System description (in the case of systems other than continuous injection give equivalent details):: NA → NA Make and type of the control unit:: NA → NA Make and type of the fuel regulator:: NA → NA Make and type of the air-flow sensor:: NA → NA Make and type of the micro-switch:: NA → NA Make and type of the throttle housing:: NA → NA Make and type of the water temperature sensor:: NA → NA Make and type of the air temperature	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	sensor:: NA → NA Injectors: Opening pressure (Specify the tolerances):,..... (kPa) or characteristic diagram:: NA → NA Make(s):: NA → NA Type(s):: NA → NA Identification Number: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA Cut-off point under load (min-1): NA → NA Injection timing:: NA → NA Cold start system:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Operating principle(s):: NA → NA Operating limits/settings(Specify the tolerances): NA → NA Pressure, (Specify the tolerances).....(kPa) or characteristic diagram:: 320 ± 50 kPa → 320 ± 50 kPa Rated voltage:...V, positive/negative ground: 12V, Negative Ground → 12V, Negative Ground Generator/ Alternator: Alternator → Alternator Type:: DC – Alternator with Voltage regulator → DC – Alternator with Voltage regulator Nominal output: ...VA: 1205 VA (For Non AC Variants) 850 Amps	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	1745 VA (For AC Variants) 120 Amps → 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps Ignition: NA → NA Make(s):: NA → NA Type(s):: NA → NA Working principle:: NA → NA Ignition advance curve or map(Specify the tolerances):: NA → NA Static ignition timing (Specify the tolerances):.....degrees before TDC: NA → NA Spark plugs: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Gap setting----------(mm): NA → NA Nominal resistance (kilo ohm) (if resistive type): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Ignition coil(s): NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Ignition condenser: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA EMI suppressor cap / Device / Electronic unit: NA → NA Make: NA → NA Type (Resistive/Capacitive): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Identification number: NA → NA Nominal resistance (kilo ohm): NA → NA Terminology and Drawing of interference Suppression equipment: NA → NA H.T. Cable: NA → NA Make and Place: NA → NA Type (Resistive/Non-resistive): NA → NA Length (mm) (if resistive type): NA → NA Outside dia. (mm) (if resistive type): NA → NA Nominal resistance kilo ohm, (if resistive type): NA → NA Radiator drawing(s): Refer enclosure 4C2 & 4C4 → Refer enclosure 4C2 & 4C4 Radiator drawing(s): Refer enclosurev 4C1 & 4C3 → Refer enclosurev 4C1 & 4C3 Make: Mahle Behr → Mahle Behr Make: Tata Toyo → Tata Toyo Type (s): Cross flow → Cross flow Relief valve pressure setting: 0.9 kg/cm2 → 0.9 kg/cm2 Fan characteristics (Fan power, kW) Enclose the fan		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	power curve corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part No: 9 → 9	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Material of blades (metal / plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ± 150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150 (AC) Radiator core open area (cm²): 1906 cm2 → 1906 cm2 Cooling system: liquid/air: Liquid → Liquid Nominal setting of the engine temperature control o mechanism (C)::: 85 deg C → 85 deg C Liquid: Yes → Yes Nature of liquid:: Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L → Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	coolant capacity: approx..8 L Circulating pump(s): (Yes/No): Yes → Yes Characteristics of Circulating pump:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm → Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm Make(s):: XXXXX Fasteners Ltd → XXXXX Fasteners Ltd Type(s):: Impeller Type → Impeller Type Drive ratio(s):: 1: 1.12 → 1: 1.12 Description of the fan and its drive mechanism:: Electric Fan , Motor Driven → Electric Fan , Motor Driven Blower: (Yes/No): No → No Characteristics of blower: NA → NA Make(s):: NA → NA Type(s):: NA → NA Drive ratio(s):: NA → NA Pressure charger: (Yes/No) Turbocharger / Supercharger: Yes, Turbo charger → Yes, Turbo charger Make(s):: M/s. XYZ, India → M/s. XYZ, India Type(s) & Identification Number:: Waste Gate with Vacuum Actuator, X3M1XXXX → Waste Gate with Vacuum Actuator, X3M1XXXX Description of the system (maximum charge pressure:.....(kPa), waste-gate, if applicable): Waste gate turbocharger with vacuum		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	actuator Maximum Charge pressure - 145 kPa → Waste gate turbocharger with vacuum actuator	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	actuator Maximum Charge pressure - 145 kPa Inter-cooler: (Yes/No): Yes → Yes Type: air-air / air-water: Air – Air → Air – Air Make :: Mahle Behr → Mahle Behr Make :: Tata Toyo → Tata Toyo Identification Number:: B1V42XXX → B1V42XXX Identification Number:: B1V42YYY → B1V42YYY Air pressure drop across the intercooler: 15 kPa → 15 kPa Intake depression at rated engine speed and at 100 per cent load (Specify location of measurement) Minimum allowable:... (kPa) Maximum allowable:... (kPa): Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Description and drawings of inlet pipes and their accessories (plenum chamber, heating device, additional air intakes, etc.):: Intake manifold with Electronic throttle valve Refer Enclosure 4D → Intake manifold with Electronic throttle valve Refer Enclosure 4D Intake manifold description		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	including dimensional drawings and/or photographs):: Aluminium Intake manifold with pressure sensor Refer Enclosure 4D → Aluminium Intake manifold with pressure sensor Refer Enclosure 4D Air filter, Dimensional drawings: Refer Enclosure 4E → Refer Enclosure 4E Make(s):: M/s. Fleet guard → M/s. Fleet guard Type(s) & Identification Number:: Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 → Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 Intake silencer, Dimensional drawing: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description and Dimensional drawing of the exhaust manifold:: Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F → Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F Description and drawing with dimensions of the exhaust system:(Description with general arrangement of exhaust system along with its routing indicating the lengths of exhaust	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	pipe, tail pipe and: Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly → Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly Effective volume of exhaust-system (specify the tolerance & range) in liters (from exhaust manifold / TC outlet to tail pipe end), Enclose the exhaust system dimensional drawing and indicate the volume of each parts clearly: 13.05 ± 1 litre Refer Enclosure 4G → 13.05 ± 1 litre Refer Enclosure 4G Description and drawing of the exhaust Muffler/Silencer:: NA → NA Type(s) & Identification	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number:: NA → NA Make: NA → NA Number:.....: NA → NA Internal diameter of exhaust pipe (mm)(min): diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G → diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G Minimum distance between exhaust pipe(s) and the fuel line: >150 mm → >150 mm Auxiliary Noise shields for compliance to IS 3028 and / OR AIS-020 (If Provided): NA → NA Material: NA → NA Layout of noise shield / Photographs / Diagram Showing arrangements indicating fitment on vehicle.: NA → NA Maximum allowable exhaust back pressure at rated engine speed and at 100 per cent load (location of measurement): ... (kPa): 30.0 kpa (150 mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Minimum cross-sectional areas of inlet and outlet ports:: Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm → Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm Inlet (mm): 8.00 → 8.00 Exhaust (mm): 8.00 → 8.00 Opening: 17° bTDC → 17° bTDC Closing: 51° aBDC → 51° aBDC Opening: 66° bBDC → 66°	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	bBDC Closing: 25° aTDC → 25° aTDC Transfer: NA → NA Opening: NA → NA Closing: NA → NA Reference and/or setting ranges (Specify the tolerances):: On Cold Engine → On Cold Engine Inlet: 0.3 to 0.4 mm → 0.3 to 0.4 mm Exhaust: 0.3 to 0.4 mm → 0.3 to 0.4 mm Distribution by ports: NA → NA Volume of crank-case cavity with piston at TDC: NA → NA Reed valve fitted (Yes / No): No → No Description of inlet ports, scavenging and exhaust ports with corresponding timing.: NA → NA Device for recycling crankcase gases (description and drawings):: Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J → Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx Catalyst (if any, and if not covered by another heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	integrated and Passive Selective catalytic reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF / DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No Element – 1 No. → Catalytic Converter – 1No Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type for LNT (periodic / continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:……cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap Bulk Density (GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	and Characteristics:: Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY Type and Identification number:: Type - Passive NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Honey Comb type Material: Cordierite → Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make:: NA → NA Type and Idetification number :: NA → NA NOx Sensor		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Before /After of SCR):: No → No Make:: NA → NA Type and Identification number:: NA → NA Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems, description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent engine test bench cycles, between two cycles where regenerative	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	phases occur under the conditions equivalent to Type I test (Distance "D" in t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Distance “d” in AIS 137): 2 cycles → 2 cycles Description of method employed to determine the number of cycles between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter (DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	International standard (where appropriate):: NA → NA Frequency of reagent refill: continuous/maintenance (where		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	appropriate):: NA → NA Normal operating temperature range (°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY → YYYYYYY Air injection: (Yes/No): No → No Type (pulse air, air pump, etc.) & Identification Number:: NA → NA Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other): Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FEW0XXYY Characteristics (flow rate, etc.):: Flow rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes EGR		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Cooler Make and Identification Number: M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:.....g: NA → NA Canister working capacity:.....g: NA → NA Canister bed	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	volume:.....litre: NA → NA Canister Purge and vent strategy description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve : (Yes/No): NA → NA Make and Identification Number: NA → NA Operating pressure (kPa): NA → NA Liquid		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	fuel hoses material :: NA → NA Schematic drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or drawing (Periodic / Continuous): NA → NA Particulate trap: (Yes/No): Yes → Yes Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dimensions and shape of the particulate trap (capacity):: Dimension :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure 4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type Material - Silicon Carbide → Structure - Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI Electronic Control Unit (ECU): Yes → Yes Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number: P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Misfire detection:: NA → NA Oxygen sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K → Refer Enclosure 4K Particulate traps monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method)::	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer Enclosure 4K → Refer Enclosure 4K List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K → Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K A description of the type and number of the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	EUDC A description of the type of the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing all sensed components with the strategy for fault detection and MI activation (fixed number of driving cycles or statistical method), including a list of relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive explanation for the data	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	given in service $05 Test ID $21 to FF and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles –		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Diagnostics on Controller Area Network (CAN) – Part 4: Requirements for emissions- related systems", a comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137): Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA → NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Systems to ensure the correct operation of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	deactivation of the driver inducement, for use by the rescue services [Note: cross- reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling /disable after parking: NA → NA Lowest concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct operation of NOx control measures: NA → NA Details of any devices designed to influence fuel economy (if not covered by other items):: NA → NA LPG fuelling system: (Yes/No): No → No Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to LPG or back:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	System layout (electrical connections, vacuum connections, compensation hoses, etc.): NA → NA NG fuelling system: (Yes/No): No → No Approval number: NA → NA Electronic engine management control unit for NG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation:: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to NG or back: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.):: NA → NA Hydrogen fuelling system: (Yes/No): No → No Electronic engine management control unit for hydrogen fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Description of the safeguarding of the catalyst at switch- over from petrol to hydrogen or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA H2NG fuelling system: (Yes/No): No → No Percentage of hydrogen in the fuel (maximum specified by the manufacturer): NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to H2NG or back:: NA → NA System lay-out (electrical connections,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	vacuum connections compensation hoses, etc.):: NA → NA Electric motor(Traction): No → No Type (winding, excitation):: NA → NA Maximum hourly output: kW(manufacturer’s declared value): NA → NA Maximum net power:kW(manufacturer’s declared		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	value): NA → NA Maximum 30 minutes power: kW (manufacturer’s declared value): NA → NA Operating voltage: V: NA → NA Make and Identification Number:: → Battery (Traction): NA → NA Number of cells:: NA → NA Mass: ……(kg): NA → NA Capacity: Ah (Amp-hours): NA → NA Position:: NA → NA Engines or motor combinations: NA → NA Hybrid Electric Vehicle: (Yes/No): No → No Category of Hybrid Electric vehicle Off Vehicle Charging/Not Off Vehicle Charging9: NA → NA Operating mode switch: with/without: NA → NA Selectable modes: NA → NA Pure electric: (Yes/No): NA → NA Pure fuel consuming: (Yes/No): NA → NA Hybrid modes: (Yes/No) (if yes, short description): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Description of the energy storage device: (traction battery, capacitor, flywheel/generator...): NA → NA Make(s):: NA → NA Type(s) :: NA → NA Identification number:: NA → NA Kind of electrochemical couple:: NA → NA Energy: ..... (for battery: voltage and capacity Ah in 2 h, for capacitor: J): NA → NA Charger: on		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	board/external/without: NA → NA Electric machines (describe each type of electric machine separately): NA → NA Make:: NA → NA Type:: NA → NA Primary use: traction motor/generator: NA → NA When used as traction motor: mono motor / multi motors (number):: NA → NA Maximum power: kW: NA → NA Working principle:: NA → NA Direct current/alternating current/number of phases:: NA → NA Separate excitation/series/compound: NA → NA Synchronous/asynchronous: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA Identification number:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Power controller: NA → NA Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Vehicle electric range..... km (according to AIS 137):: NA → NA Manufacturer’s recommendation for preconditioning:: NA → NA CO2 (g/km) (Applicable for category M1 with GVW<3.5 T): NA → NA Declared : (rounded to 3 decimal places): NA → NA Fuel consumption (l/100 km) for Petrol, LPG or Diesel and (kg/100km) for CNG and (kWh/100 km) for Electric Driven		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Vehicles(Applicable for category M1 with GVW<3.5 T) (rounded to 3 decimal places)*: NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA *calculated on the declared CO2: NA → NA CO2 Reducing technologies Available (Yes/No): No → No Regenerative braking: NA → NA Start-Stop System: NA → NA Tyre pressure monitoring system: NA → NA 6 or more Speed Transmission: NA → NA Any other technology (manufacture to	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	enclose detail): NA → NA Electric energy consumption for electric vehicles Wh/km: NA → NA Electric energy consumption for pure electric vehicles Wh/km: NA → NA Electric energy consumption for externally chargeable hybrid electric vehicles: NA → NA Electric energy consumption (condition A, combined) Wh/km: NA → NA Electric energy consumption (condition B, combined) Wh/km: NA → NA Electric energy consumption (weighted combined) ….Wh/km: NA → NA Moment of inertia of engine flywheel:: 0.1212 kg/m2 → 0.1212 kg/m2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2 Make: XXYY System → XXYY System Type: 12V → 12V Identification mark: X5XYZ100 → X5XYZ100 Rated voltage: 12V → 12V Number of sweep Frequencies: 2 → 2 Highest sweep frequency (Cycles / min): 62±8 rpm → 62±8 rpm Lowest sweep frequency (Cycles / min): 40±5 rpm → 40±5 rpm Washer tank motor / Washer pump: ELECTRIC MOTOR WITH PUMP → ELECTRIC MOTOR WITH PUMP Make: XYZ → XYZ	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Model: PEAD64XYZS00 → PEAD64XYZS00 Schematic Diagram of Engine cooling system: - → - Maximum vehicle speed for continuous operation (km/h) (Only for AC horns): 80 Kmph → 80 Kmph A list, specifying the parts which constitute the headlamp cleaner and drawings thereof, (e.g. pumps, nozzles, valves, motors and wipers);: -- → -- Description and drawings/photographs of the shapes and constituent materials of the part of the body forming the engine compartment and the part of the passenger compartment nearest to it:: -- → -- Drawings or photographs of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	the position of the metal components housed in the engine compartment (e.g. heating appliances, spare wheel, air filter, steering mechanism, etc.):: -- → -- List of all Electrical components, which include Broadband EMI sources (like HAVC Motor, Wiper Motor, Window motor and Horn etc.) The list shall include Device name, Make/s and Part No. /Type ID/Drawing No: -- → -- System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	only in case of steering column locking systems: Locks the Steering → Locks the Steering Components placed in Engine compartment: - → - Frequency bands (Hz) · Max Output Power (W): Antenna Position at Vehicle, specific conditions for installation and/or use → Antenna Position at Vehicle, specific conditions for installation and/or use Example of Engine No: HJEZXYZ891 → HJEZXYZ891		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
CMVR 96; IS 11852:2019; GSR 834(E), 11 Nov 2025 2027 braking standard transition	Name and address of engine manufacturing plant (In case of imported vehicles, above details shall be supplied for importer also): Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX → Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Axle spacing in case of multi axle vehicles.: NA → NA Other axle: NA → NA Gross combination weight kg (applicable for articulated / Tractor Trailer combination vehicles): NA → NA Front axle: NA → NA	CEA / report needs review	Confirm manufacture date and whether this is a new or existing model. Final notified route and edition still need confirmation.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Rear axle: NA → NA Trailer axle / other axles: NA → NA If applicable, maximum mass of trailer whic5h may be coupled: NA → NA Unbraked trailer: NA → NA Reference mass (for vehicles with GVW less than or equal to 3.5 ton) (kg.): Refer Annexure T7-B → Refer Annexure T7-B Make: APOLLO → APOLLO Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade description(s): CEAT WINMILE X3-R → CEAT WINMILE		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	142 Tyre rolling Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing:		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Refer - TC7719XX000 003XXXF → Refer - TC7719XX000	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L-756XXXX → CM/L-756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Spare wheel: 1 → 1 Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E- Marking: NA → NA Others (For articulated / combination vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00 R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	M2, M3, N and T category of vehicles.): 325 mm → 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26 kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Final Drive (4x2 or 4x4 or 6x2 or 6x4 or others) Front / Rear / Both: (4 x 2) Rear → (4 x 2) Rear Rear axle make: Single Reduction, Hypoid, Semi-floating Banjo type → Single Reduction, Hypoid, Semi-floating Banjo type Rear axle ratio: 4.56 → 5.13 Differential type: Single Reduction, Hypoid, Semi-floating Banjo type → Single Reduction, Hypoid, Semi-floating Banjo type Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2.	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	YYYY – 18.2 :1 Engine coolant temperature: Red Color – Thermometer Symbol → Red Color – Thermometer Symbol Engine low oil pressure: Red Color – Oil Tank Symbol → Red Color – Oil Tank Symbol Brake failure: Red Color – Parking and Oil symbol together → Red Color – Parking and Oil symbol together Brake failure: → Installation drawing showing rear extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size, make, unladen pressure) etc., distance between tyre & RUPD RH & LH, frame rear width, P1, P2 points.: - → - Parking brake: Hand operated [LHS] → Hand operated [LHS] Service Brake: Foot operated → Foot operated Service Brake: → Type of coupling device for brake: NA → NA Detailed engineering drawing as specified in AIS – 013 or photographs showing the mounting details with dimensions.: NA → NA Distance between the contour points of the orthogonal projections on to plane “P” of the accelerator pedal and service brake pedal bearing surfaces, “E” in	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	mm.: -- → -- Distance between the projection of the service brake pedal on to the reference plane “P”, to the right, “H” in mm.: -- → -- Distance between the projection of the service brake pedal on to the0 reference plane “P”, to the left, “J” in mm.: -- → -- In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → -- Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side, distance between front extremity & FUPD, width of FUPD, ID marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Drawing / Images of position & arrangement of Engine: -- → -- Drawing/ Images of Under bonnet (Engine house) Components.: -- → -- 1st · Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	· Overall Ratio / AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC: 12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56 4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 → 20.44 Make: M/s Brakes India Private Limited → M/s Brakes India Private Limited Type (Mechanical/hydraulic/air/air assisted/vacuum assisted/others):		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Vacuum assisted hydraulic → Vacuum assisted hydraulic Control system & braking wheel: Hydraulic acting on all wheels →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Hydraulic acting on all wheels Schematic layout indicating method of split of brake system, location of valves, reservoirs, ABS components etc. (Attach drawing and indicate the drawing number): AJ1XXX00 → AJ1XXX00 Anti-Lock braking system Provided (Yes / No / Optional): NA → NA If yes, details of ABS: - → - Make: - → - Category of ABS: - → - Nos. of directly controlled wheel(s): - → - Brief description of failure warning tell–tale: - → - Wheel Speed Sensors: - → - No. of sensors: - → - Make of sensors: - → - Type of sensors: - → - Modulator: - → - Nos. of Modulators: - → - Make of Modulators: - → - Identification No. / Part No. of Modulator: - → - Brief description and features: - → - Controller: - → - Nos. of Controller: - → - Make of Controller: - → - Identification No. / Part No. of Controller: - → - Brief description and features: - → - Height of Center of Gravity		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(mm): - → - Un laden condition: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Laden condition: - → - Slack adjuster (only applicable for M2, M3, N and T category of vehicles.): - → - Front (Automatic / Manual): - → - Make of slack adjuster: - → - Model and Type: - → - Rear (Automatic / Manual): - → - Make of slack adjuster: - → - Model and Type: - → - Front wheel: 132x47.3x10.35 → 132x47.3x10.35 Rear wheel: LEADING - 314x51.8x5.6 TRAILING – 246x51.8x5.6 → LEADING - 314x51.8x5.6 TRAILING – 246x51.8x5.6 Others: NA → NA Type of liner wear indicator (window/ acoustic/ optical/ any other): Window → Window Front: 250 sq.cm → 250 sq.cm Rear: 570 sq.cm → 570 sq.cm Others: NA → NA Front wheel / axle: Option 1: SYY 6XXXYA → Option 1: SYY 6XXXYA Rear wheel / axle: APACHE 4059M → APACHE 4059M Others: NA → NA Whether asbestos or asbestos-free: Asbestos Free → Asbestos Free Front axle (Disc / Drum): DISC → DISC Effective Diameter (mm): 212 → 212 Rear axle (Disc / drum): DRUM → DRUM Effective Diameter (mm): 282 → 282 Other axle (		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Disc / Drum): NA → NA Effective diameter (mm): NA → NA Material (if the braking surface is not ferrous): NA → NA Front wheel: -- → -- Rear wheel: -- → -- Others: -- → -- Make: M/s XXXX YYYY → M/s XXXX YYYY Type (only applicable for M2, M3, N and T category of vehicles.): N Category → N Category Inner diameter of the master cylinder (mm): 22.81 → 22.81 Operating stroke (mm): 41 → 41 Type of supply tank (only applicable for M2, M3, N and T category of vehicles.): → Brake Chamber: Rear → Rear Brake Chamber: Parking → Parking Make and type: NA → NA Make and type: NA → NA Size mm: NA → NA Size mm: NA → NA Inner diameter mm: NA → NA Inner diameter mm: NA → NA Stroke mm: NA → NA Stroke mm: NA → NA Make and Identification No.: MAKE: XXX GIRLING TAC No: AB2XX1. CoP No: SYL/29/2020-2021/300000YY60/ADDM/XXXX → MAKE: XXX GIRLING TAC No: AB2XX1. CoP No: SYL/29/2020-2021/300000YY60/ADDM/X	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	XXX Type of expansion (HL / HR): HL → HL Free Length of hoses: FOR NON ABS. FRONT : L1 – 470, L2 -417.7 REAR :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	L1 – 405, L2 – 360.9 → FOR NON ABS. FRONT : L1 – 470, L2 -417.7 REAR : L1 – 405, L2 – 360.9 Thickness of lining (mm): 0.6 → 0.6 Nominal bore dia. (mm): 3.1 → 3.1 Type (Visual display/ audible/others): Visual display → Visual display Operation pressure (kg/cm2 / bar / kPa): NA → NA Type of safety device: NA → NA Make: M/s XXXXXYY → M/s XXXXXYY Type (mechanical/spring brake): MECHANICAL → MECHANICAL Acting on Transmission/wheel: WHEEL → WHEEL Control System & Braking wheel: Hand operated & Rear wheel → Hand operated & Rear wheel Lining / pad: Lining → Lining Name of producer: M/s XXXXYYY → M/s XXXXYYY Dimension (mm) (D x W x T): LEADING - 305x48.8x5.6 TRAILING – 256x48.8x5.6 → LEADING - 305x48.8x5.6 TRAILING – 256x48.8x5.6 Area (cm 2): 279 sq.cm → 279 sq.cm Material: XYZ 420 → XYZ 420	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Diameter of brake drum/disc (mm): 272 → 272 Type: UNFAILED PART OF THE SERVICE BRAKE → UNFAILED PART OF THE SERVICE BRAKE Make: Ms. GULF Oil → Ms. GULF Oil Trade name: DOT 4 → DOT 4 Specification/ grade as		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	per Indian standard: DOT 4 → DOT 4 Brake fluid level indicator (Yes/No): Yes → Yes Type (Manual/Auto) Front and Rear: Manual → Manual Lever length: → Pneumatic Brake Actuator: NA → NA Type (as per IS): NA → NA Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No Brake Assist System (Fitted : Yes/No): No → No Category of Brake Assist System A / B: -- → -- For category A systems, define the force threshold at which the ratio between pedal force and brake pressure increases: -- → -- For category B systems, define the brake pedal speed which must be achieved in	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	order to activate the Brake Assist System (e.g. pedal stroke speed (mm/s) during a given time interval): -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Motor Vehicle stability function information document: -- → -- System: -- → -- System variants: -- → -- System options: -- → -- Control function (directional/roll-over/both) including an explanation of the basic function and/or philosophy of the control: -- → -- System configurations (where appropriate): -- → -- System identification including software level identifier: -- → -- List of motor vehicles by description and configuration that are covered by the information document: -- → -- Schematic diagrams of the respective configurations installed on the motor vehicles defined in item 2.1. above with consideration given to the following:: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Lift axles: -- → -- Steering axles: -- → -- Anti-lock braking configurations: -- → -- Wheelbase, track, centre of gravity height: -- → -- Wheel type (single or twin) and tyre type (e.g. structure, category of use, size): -- → --		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Management of the engine or any other source(s) of motive power e.g.Torque Management, control of fuel supply etc.: -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Pneumatic circuits System schematics including anti-lock braking configurations associated with the trailer types defined in paragraph 6.2.1. of this annex.: -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies that	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Vehicle is/ is not equipped to		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	tow a trailer with electric braking system (Yes/No): No → No Make: Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. → Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. Model name / identification: ALXYZABC0 → ALXYZABC0 Type: Compression Ignition → Compression Ignition Bore x stroke (mm): 80 X 98 → 80 X 98 No. of cylinders: 3 → 3 Displacement (cm³): 1468 cc → 1468 cc Compression ratio: 16.5±1.0 : 1 → 16.5±1.0 : 1 Max. Engine output (KW @ rpm): 55.1 Kw @ 3300 rpm → 55.1 Kw @ 3300 rpm Max. Torque (Nm @ rpm): 170Nm @ 1600 to 2400 rpm → 170Nm @ 1600 to 2400 rpm Air cleaner type: Dry type → Dry type Type of Fuel: Diesel → Diesel Front axle ratio: NA → NA Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2. YYYY Pvt.Ltd Service brake Type (Mechanical/hydraulic/air/air assisted/vacuum assisted/others): Vacuum assisted hydraulic → Vacuum assisted		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	hydraulic Auto Slack Adjuster Fitted (Yes / No / Optional): Yes (Automatic wear adjustment) → Yes (Automatic wear adjustment) Anti-Lock braking system Provided(Yes/No/Optional): Optional → Optional Front (Disc / Drum): Disc → Disc Rear (Disc / Drum): Drum → Drum Total braking area (cm²): Front axle-254 sq.cm Rear axle-580 sq.cm → Front axle-254 sq.cm Rear axle-580 sq.cm Parking brake type: Mechanical → Mechanical Secondary brake type: Un failed part of the service brake → Un failed part of the service brake Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	12 PR Tyre size designation including ply rating: 215/75 R15 LT → 215/75 R15 LT Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	→ 3.26 kg/cm2 Wiper motor: XYZ Ltd. → XYZ Ltd. Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Maximum GVW kg (for rigid vehicles): Refer Annexure T7 – A Details for Recommended load body - Refer Annexure T7 - B → Refer Annexure T7 – A Details for Recommended load body - Refer Annexure T7 - B Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Engine Manufacturer::		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Ashok Leyland Limited → Ashok Leyland Limited Manufacturer's engine model code (as marked on the engine, or other means of identification): Model Code: ALXYZABC0 Enclosure – 4A → Model Code: ALXYZABC0 Enclosure – 4A Location of engine model code and engine serial number on block / crankcase (drawing to be attached): Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A → Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer Enclosure – 4A Working principle: positive ignition / compression- ignition, four-stroke / two-stroke / rotary cycle (DI/IDI) (NA/TC/TCIC/Any other): Compression Ignition, four-stroke, DI, TCIC → Compression Ignition, four-stroke, DI, TCIC Number, arrangement of cylinders:: 3, Inline → 3, Inline Bore----------(mm): 80 → 80 Stroke: ----------(mm): 98 → 98 Firing order:: 1-2-3 → 1-2-3 Engine		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	capacity:----------(cm3): 1468 → 1468 Volumetric compression ratio (specify tolerance): 17.5±1.0 : 1 → 17.5±1.0 : 1 Drawings of combustion chamber and piston crown and, in the case of positive ignition engine, piston rings:: Refer Enclosure – 4B → Refer Enclosure – 4B Vehicle with Defeat Device Yes / No: No → No If yes – Details of the Defeat device: NA → NA Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	equivalent) in PPM content by volume in the exhaust gas with the engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum net power:..... kW at min-1: 55.1 KW @ 3100 ± 50 rpm → 55.1 KW @ 3100 ± 50 rpm Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm Maximum net torque: .....Nm at: -------min-1 ...(manufacturer's		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	declared value): 170 Nm @ 1600 – 2400 rpm → 170 Nm @ 1600 – 2400 rpm Maximum rated speed (Specify tolerance): 3100 rpm ± 50 rpm → 3100 rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Type of fuel used : Mono Fuel / Bi-Fuel / Flex Fuel / Dual Fuel: Mono Fuel → Mono Fuel Fuel type: Diesel / Gasoline / LPG / CNG / Biomethane / Bio-H methane / HCNG / Biogas / LNG / Ethanol ((E85) / 2 (E100)) /ED (95) Flex fuel Methanol M15 / M100 / MD95 / Biodiesel up to 100% /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Hydrogen.: Diesel → Diesel Maximum amount of biofuel acceptable in fuel (manufacturer's declared value): per cent by volume: 7% → 7% By fuel injection (compression-ignition only): (Yes/No): Yes → Yes System description:: Unit Pump Common Rail Direct Injection system → Unit Pump Common Rail Direct Injection system Working principle: direct-injection/pre-chamber/swirl Chamber: Direct Injection → Direct Injection Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) &		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: Plunger Type & FHL0XYZ0 → Plunger Type & FHL0XYZ0 Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Injection advance curve (Specify the tolerances):: ECU Controlled → ECU Controlled Governor: ECU Controlled → ECU Controlled Cut-off point under load:...(min-1): 3300 → 3300 Cut-off point without load (Maximum	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	no-load speed)…(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Multi-Hole Injectors (7 holes) FKXXYYZZ → Multi-Hole Injectors (7 holes) FKXXYYZZ Opening pressure (kPa) (specify the tolerances) or characteristic diagram: 15000 to 23000 kPa → 15000 to 23000 kPa Length and internal diameter…(mm): Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) -		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	193.5± 3 Internal Diameter – 3mm → Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm Common rail make & type: M/s. XXXXYYY, Hot Forged Rail Type → M/s. XXXXYYY, Hot Forged Rail Type Cold start system: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description:: NA → NA Auxiliary starting aid: NA → NA Make(s):: NA → NA Type(s):: NA → NA System description:: NA → NA Electronic controlled injection: (Yes/No):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA → NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s. XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and Metering type Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make and type of air temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type By fuel injection (positive ignition only): (Yes/No): No → No Working principle: intake manifold (single/multi- point)/direct injection/other (specify): NA → NA Make(s) of Injection Pump :: NA → NA Type(s) & Identification		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number of Injection Pump:: NA → NA System description (in the case of systems other than continuous injection give equivalent details):: NA → NA Make and type of the control unit:: NA → NA Make and type of the fuel regulator:: NA → NA Make and type of the air-flow sensor:: NA → NA Make and type of the micro-switch:: NA → NA Make and type of the throttle housing:: NA → NA Make and type of the water temperature sensor:: NA → NA Make and type of the air temperature	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	sensor:: NA → NA Injectors: Opening pressure (Specify the tolerances):,..... (kPa) or characteristic diagram:: NA → NA Make(s):: NA → NA Type(s):: NA → NA Identification Number: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA Cut-off point under load (min-1): NA → NA Injection timing:: NA → NA Cold start system:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Operating principle(s):: NA → NA Operating limits/settings(Specify the tolerances): NA → NA Pressure, (Specify the tolerances).....(kPa) or characteristic diagram:: 320 ± 50 kPa → 320 ± 50 kPa Rated voltage:...V, positive/negative ground: 12V, Negative Ground → 12V, Negative Ground Generator/ Alternator: Alternator → Alternator Type:: DC – Alternator with Voltage regulator → DC – Alternator with Voltage regulator Nominal output: ...VA: 1205 VA (For Non AC Variants) 850 Amps	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	1745 VA (For AC Variants) 120 Amps → 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps Ignition: NA → NA Make(s):: NA → NA Type(s):: NA → NA Working principle:: NA → NA Ignition advance curve or map(Specify the tolerances):: NA → NA Static ignition timing (Specify the tolerances):.....degrees before TDC: NA → NA Spark plugs: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Gap setting----------(mm): NA → NA Nominal resistance (kilo ohm) (if resistive type): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Ignition coil(s): NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Ignition condenser: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA EMI suppressor cap / Device / Electronic unit: NA → NA Make: NA → NA Type (Resistive/Capacitive): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Identification number: NA → NA Nominal resistance (kilo ohm): NA → NA Terminology and Drawing of interference Suppression equipment: NA → NA H.T. Cable: NA → NA Make and Place: NA → NA Type (Resistive/Non-resistive): NA → NA Length (mm) (if resistive type): NA → NA Outside dia. (mm) (if resistive type): NA → NA Nominal resistance kilo ohm, (if resistive type): NA → NA Radiator drawing(s): Refer enclosure 4C2 & 4C4 → Refer enclosure 4C2 & 4C4 Radiator drawing(s): Refer enclosurev 4C1 & 4C3 → Refer enclosurev 4C1 & 4C3 Make: Mahle Behr → Mahle Behr Make: Tata Toyo → Tata Toyo Type (s): Cross flow → Cross flow Relief valve pressure setting: 0.9 kg/cm2 → 0.9 kg/cm2 Fan characteristics (Fan power, kW) Enclose the fan		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	power curve corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part No: 9 → 9	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Material of blades (metal / plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ± 150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150 (AC) Radiator core open area (cm²): 1906 cm2 → 1906 cm2 Cooling system: liquid/air: Liquid → Liquid Nominal setting of the engine temperature control o mechanism (C)::: 85 deg C → 85 deg C Liquid: Yes → Yes Nature of liquid:: Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L → Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	coolant capacity: approx..8 L Circulating pump(s): (Yes/No): Yes → Yes Characteristics of Circulating pump:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm → Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm Make(s):: XXXXX Fasteners Ltd → XXXXX Fasteners Ltd Type(s):: Impeller Type → Impeller Type Drive ratio(s):: 1: 1.12 → 1: 1.12 Description of the fan and its drive mechanism:: Electric Fan , Motor Driven → Electric Fan , Motor Driven Blower: (Yes/No): No → No Characteristics of blower: NA → NA Make(s):: NA → NA Type(s):: NA → NA Drive ratio(s):: NA → NA Pressure charger: (Yes/No) Turbocharger / Supercharger: Yes, Turbo charger → Yes, Turbo charger Make(s):: M/s. XYZ, India → M/s. XYZ, India Type(s) & Identification Number:: Waste Gate with Vacuum Actuator, X3M1XXXX → Waste Gate with Vacuum Actuator, X3M1XXXX Description of the system (maximum charge pressure:.....(kPa), waste-gate, if applicable): Waste gate turbocharger with vacuum		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic

(continued)	actuator Maximum Charge pressure - 145 kPa → Waste gate turbocharger with vacuum actuator	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	actuator Maximum Charge pressure - 145 kPa Inter-cooler: (Yes/No): Yes → Yes Type: air-air / air-water: Air – Air → Air – Air Make :: Mahle Behr → Mahle Behr Make :: Tata Toyo → Tata Toyo Identification Number:: B1V42XXX → B1V42XXX Identification Number:: B1V42YYY → B1V42YYY Air pressure drop across the intercooler: 15 kPa → 15 kPa Intake depression at rated engine speed and at 100 per cent load (Specify location of measurement) Minimum allowable:... (kPa) Maximum allowable:... (kPa): Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Description and drawings of inlet pipes and their accessories (plenum chamber, heating device, additional air intakes, etc.):: Intake manifold with Electronic throttle valve Refer Enclosure 4D → Intake manifold with Electronic throttle valve Refer Enclosure 4D Intake manifold description		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	including dimensional drawings and/or photographs):: Aluminium Intake manifold with pressure sensor Refer Enclosure 4D → Aluminium Intake manifold with pressure sensor Refer Enclosure 4D Air filter, Dimensional drawings: Refer Enclosure 4E → Refer Enclosure 4E Make(s):: M/s. Fleet guard → M/s. Fleet guard Type(s) & Identification Number:: Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 → Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 Intake silencer, Dimensional drawing: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description and Dimensional drawing of the exhaust manifold:: Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F → Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F Description and drawing with dimensions of the exhaust system:(Description with general arrangement of exhaust system along with its routing indicating the lengths of exhaust	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	pipe, tail pipe and: Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly → Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly Effective volume of exhaust-system (specify the tolerance & range) in liters (from exhaust manifold / TC outlet to tail pipe end), Enclose the exhaust system dimensional drawing and indicate the volume of each parts clearly: 13.05 ± 1 litre Refer Enclosure 4G → 13.05 ± 1 litre Refer Enclosure 4G Description and drawing of the exhaust Muffler/Silencer:: NA → NA Type(s) & Identification	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number:: NA → NA Make: NA → NA Number:.....: NA → NA Internal diameter of exhaust pipe (mm)(min): diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G → diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G Minimum distance between exhaust pipe(s) and the fuel line: >150 mm → >150 mm Auxiliary Noise shields for compliance to IS 3028 and / OR AIS-020 (If Provided): NA → NA Material: NA → NA Layout of noise shield / Photographs / Diagram Showing arrangements indicating fitment on vehicle.: NA → NA Maximum allowable exhaust back pressure at rated engine speed and at 100 per cent load (location of measurement): ... (kPa): 30.0 kpa (150 mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Minimum cross-sectional areas of inlet and outlet ports:: Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm → Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm Inlet (mm): 8.00 → 8.00 Exhaust (mm): 8.00 → 8.00 Opening: 17° bTDC → 17° bTDC Closing: 51° aBDC → 51° aBDC Opening: 66° bBDC → 66°	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	bBDC Closing: 25° aTDC → 25° aTDC Transfer: NA → NA Opening: NA → NA Closing: NA → NA Reference and/or setting ranges (Specify the tolerances):: On Cold Engine → On Cold Engine Inlet: 0.3 to 0.4 mm → 0.3 to 0.4 mm Exhaust: 0.3 to 0.4 mm → 0.3 to 0.4 mm Distribution by ports: NA → NA Volume of crank-case cavity with piston at TDC: NA → NA Reed valve fitted (Yes / No): No → No Description of inlet ports, scavenging and exhaust ports with corresponding timing.: NA → NA Device for recycling crankcase gases (description and drawings):: Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J → Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx Catalyst (if any, and if not covered by another heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	integrated and Passive Selective catalytic reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF / DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No Element – 1 No. → Catalytic Converter – 1No Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type for LNT (periodic / continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:……cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap Bulk Density (GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	and Characteristics:: Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY Type and Identification number:: Type - Passive NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Honey Comb type Material: Cordierite → Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make:: NA → NA Type and Idetification number :: NA → NA NOx Sensor		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Before /After of SCR):: No → No Make:: NA → NA Type and Identification number:: NA → NA Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems, description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent engine test bench cycles, between two cycles where regenerative	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	phases occur under the conditions equivalent to Type I test (Distance "D" in t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Distance “d” in AIS 137): 2 cycles → 2 cycles Description of method employed to determine the number of cycles between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter (DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	International standard (where appropriate):: NA → NA Frequency of reagent refill: continuous/maintenance (where		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	appropriate):: NA → NA Normal operating temperature range (°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY → YYYYYYY Air injection: (Yes/No): No → No Type (pulse air, air pump, etc.) & Identification Number:: NA → NA Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other): Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FEW0XXYY Characteristics (flow rate, etc.):: Flow rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes EGR		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Cooler Make and Identification Number: M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:.....g: NA → NA Canister working capacity:.....g: NA → NA Canister bed	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	volume:.....litre: NA → NA Canister Purge and vent strategy description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve : (Yes/No): NA → NA Make and Identification Number: NA → NA Operating pressure (kPa): NA → NA Liquid		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	fuel hoses material :: NA → NA Schematic drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or drawing (Periodic / Continuous): NA → NA Particulate trap: (Yes/No): Yes → Yes Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dimensions and shape of the particulate trap (capacity):: Dimension :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure 4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type Material - Silicon Carbide → Structure - Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI Electronic Control Unit (ECU): Yes → Yes Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number: P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Misfire detection:: NA → NA Oxygen sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K → Refer Enclosure 4K Particulate traps monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method)::	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer Enclosure 4K → Refer Enclosure 4K List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K → Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K A description of the type and number of the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	EUDC A description of the type of the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing all sensed components with the strategy for fault detection and MI activation (fixed number of driving cycles or statistical method), including a list of relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive explanation for the data	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	given in service $05 Test ID $21 to FF and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles –		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Diagnostics on Controller Area Network (CAN) – Part 4: Requirements for emissions- related systems", a comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137): Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA → NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Systems to ensure the correct operation of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	deactivation of the driver inducement, for use by the rescue services [Note: cross- reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling /disable after parking: NA → NA Lowest concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct operation of NOx control measures: NA → NA Details of any devices designed to influence fuel economy (if not covered by other items):: NA → NA LPG fuelling system: (Yes/No): No → No Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to LPG or back:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	System layout (electrical connections, vacuum connections, compensation hoses, etc.): NA → NA NG fuelling system: (Yes/No): No → No Approval number: NA → NA Electronic engine management control unit for NG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation:: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to NG or back: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.):: NA → NA Hydrogen fuelling system: (Yes/No): No → No Electronic engine management control unit for hydrogen fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Description of the safeguarding of the catalyst at switch- over from petrol to hydrogen or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA H2NG fuelling system: (Yes/No): No → No Percentage of hydrogen in the fuel (maximum specified by the manufacturer): NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to H2NG or back:: NA → NA System lay-out (electrical connections,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	vacuum connections compensation hoses, etc.):: NA → NA Electric motor(Traction): No → No Type (winding, excitation):: NA → NA Maximum hourly output: kW(manufacturer’s declared value): NA → NA Maximum net power:kW(manufacturer’s declared		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	value): NA → NA Maximum 30 minutes power: kW (manufacturer’s declared value): NA → NA Operating voltage: V: NA → NA Make and Identification Number:: → Battery (Traction): NA → NA Number of cells:: NA → NA Mass: ……(kg): NA → NA Capacity: Ah (Amp-hours): NA → NA Position:: NA → NA Engines or motor combinations: NA → NA Hybrid Electric Vehicle: (Yes/No): No → No Category of Hybrid Electric vehicle Off Vehicle Charging/Not Off Vehicle Charging9: NA → NA Operating mode switch: with/without: NA → NA Selectable modes: NA → NA Pure electric: (Yes/No): NA → NA Pure fuel consuming: (Yes/No): NA → NA Hybrid modes: (Yes/No) (if yes, short description): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Description of the energy storage device: (traction battery, capacitor, flywheel/generator...): NA → NA Make(s):: NA → NA Type(s) :: NA → NA Identification number:: NA → NA Kind of electrochemical couple:: NA → NA Energy: ..... (for battery: voltage and capacity Ah in 2 h, for capacitor: J): NA → NA Charger: on		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	board/external/without: NA → NA Electric machines (describe each type of electric machine separately): NA → NA Make:: NA → NA Type:: NA → NA Primary use: traction motor/generator: NA → NA When used as traction motor: mono motor / multi motors (number):: NA → NA Maximum power: kW: NA → NA Working principle:: NA → NA Direct current/alternating current/number of phases:: NA → NA Separate excitation/series/compound: NA → NA Synchronous/asynchronous: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA Identification number:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Power controller: NA → NA Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Vehicle electric range..... km (according to AIS 137):: NA → NA Manufacturer’s recommendation for preconditioning:: NA → NA CO2 (g/km) (Applicable for category M1 with GVW<3.5 T): NA → NA Declared : (rounded to 3 decimal places): NA → NA Fuel consumption (l/100 km) for Petrol, LPG or Diesel and (kg/100km) for CNG and (kWh/100 km) for Electric Driven		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Vehicles(Applicable for category M1 with GVW<3.5 T) (rounded to 3 decimal places)*: NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA *calculated on the declared CO2: NA → NA CO2 Reducing technologies Available (Yes/No): No → No Regenerative braking: NA → NA Start-Stop System: NA → NA Tyre pressure monitoring system: NA → NA 6 or more Speed Transmission: NA → NA Any other technology (manufacture to	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	enclose detail): NA → NA Electric energy consumption for electric vehicles Wh/km: NA → NA Electric energy consumption for pure electric vehicles Wh/km: NA → NA Electric energy consumption for externally chargeable hybrid electric vehicles: NA → NA Electric energy consumption (condition A, combined) Wh/km: NA → NA Electric energy consumption (condition B, combined) Wh/km: NA → NA Electric energy consumption (weighted combined) ….Wh/km: NA → NA Moment of inertia of engine flywheel:: 0.1212 kg/m2 → 0.1212 kg/m2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2 Make: XXYY System → XXYY System Type: 12V → 12V Identification mark: X5XYZ100 → X5XYZ100 Rated voltage: 12V → 12V Number of sweep Frequencies: 2 → 2 Highest sweep frequency (Cycles / min): 62±8 rpm → 62±8 rpm Lowest sweep frequency (Cycles / min): 40±5 rpm → 40±5 rpm Washer tank motor / Washer pump: ELECTRIC MOTOR WITH PUMP → ELECTRIC MOTOR WITH PUMP Make: XYZ → XYZ	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Model: PEAD64XYZS00 → PEAD64XYZS00 Schematic Diagram of Engine cooling system: - → - Maximum vehicle speed for continuous operation (km/h) (Only for AC horns): 80 Kmph → 80 Kmph A list, specifying the parts which constitute the headlamp cleaner and drawings thereof, (e.g. pumps, nozzles, valves, motors and wipers);: -- → -- Description and drawings/photographs of the shapes and constituent materials of the part of the body forming the engine compartment and the part of the passenger compartment nearest to it:: -- → -- Drawings or photographs of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	the position of the metal components housed in the engine compartment (e.g. heating appliances, spare wheel, air filter, steering mechanism, etc.):: -- → -- List of all Electrical components, which include Broadband EMI sources (like HAVC Motor, Wiper Motor, Window motor and Horn etc.) The list shall include Device name, Make/s and Part No. /Type ID/Drawing No: -- → -- System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	only in case of steering column locking systems: Locks the Steering → Locks the Steering Components placed in Engine compartment: - → - Frequency bands (Hz) · Max Output Power (W): Antenna Position at Vehicle, specific conditions for installation and/or use → Antenna Position at Vehicle, specific conditions for installation and/or use Example of Engine No: HJEZXYZ891 → HJEZXYZ891		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
CMVR 118; AIS-018 Speed limiting function / device and tamper protection	Control (Forward / semi-forward / normal / others): Forward → Forward Vehicle Max Speed in unladen condition (km/h): Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Vehicle Max Speed in laden condition (km/h): Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Make: APOLLO → APOLLO Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade	CEA / report needs review	Confirm Transport vehicle. Unknown fitment / scope is retained for review.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	description(s): CEAT WINMILE X3-R → CEAT WINMILE X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 Tyre rolling		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	/ XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer - TC7719XX000 003XXXF → Refer - TC7719XX000 003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. /		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	BIS License No. / E- Marking: CM/L-756XXXX → CM/L-	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E- Marking: NA → NA Others (For articulated / combination vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other (for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 325 mm → 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26 kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Type: Fully synchronized 5 - forward gears with a Constant mesh reverse gear → Fully synchronized 5 - forward	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	gears with a Constant mesh reverse gear Model name (if any): XXX-XXX → XXX-XXX Model name (if any): YYYY-YY → YYYY-YY Gear shifting control system (Sketch showing gear shifting arrangement): → No. of gears: Forward 5 + 1 Rev → Forward 5 + 1 Rev Control system: NA → NA Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Detailed drawing of the mounting arrangement of Steering control assembly showing vertical / tilt / actual angle.: FZY0XX00 → FZY0XX00 Type of steering gear (Re-circulating ball / Worm & Roller / Rack & Pinion / Others): Re-circulating ball type → Re-circulating ball type Make: 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd → 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2. YYYY – 18.2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	:1 Head lamp – upper / lower control: 0 – 3 Switch provided → 0 – 3 Switch provided Ignition cut-off: Not Provided → Not	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Provided Turn signal: Green Color - LH and RH Direction Symbol → Green Color - LH and RH Direction Symbol Fuel Gauge: LED Bars → LED Bars Engine coolant temperature: Red Color – Thermometer Symbol → Red Color – Thermometer Symbol Engine low oil pressure: Red Color – Oil Tank Symbol → Red Color – Oil Tank Symbol High beam indicator: Blue Color – High Beam Symbol → Blue Color – High Beam Symbol Electrical charge indicator (Battery charge): Red Color – Battery Symbol → Red Color – Battery Symbol Brake failure: Red Color – Parking and Oil symbol together → Red Color – Parking and Oil symbol together Brake failure: → Front fog light, if provided: Not applicable. → Not applicable. Front fog light, if provided: → Rear fog light, if provided: Not applicable. → Not applicable. Horn: On steering Wheel → On steering Wheel ABS: Red Color – ABS function alert → Red Color –		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	ABS function alert Seat Belt Reminder: Not Applicable → Not Applicable	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Others (such as, Airbag, HVAC, Content gauge, LPG / CNG changeover switch etc.,): Not Applicable → Not Applicable Speedometer:: 7.00 R15 LT → 7.00 R15 LT Speedometer:: 215/75 R15 LT → 215/75 R15 LT Type: Electronic → Electronic Make, and Identification No. / Part No./ Drawing No.: Pricol Limited → Pricol Limited Make, and Identification No. / Part No./ Drawing No.: Minda stoneridge → Minda stoneridge Make, and Identification No. / Part No./ Drawing No.: Pricol Limited → Pricol Limited Make, and Identification No. / Part No./ Drawing No.: Minda stoneri dge → Minda stoneri dge Make, and Identification No. / Part No./ Drawing No.: Pricol Limited → Pricol Limited Make, and Identification No. / Part No./ Drawing No.: Minda stoneridge → Minda stoneridge Make, and Identification No. / Part No./ Drawing No.: FL102XXX → FL102XXX		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make, and Identification No. / Part No./ Drawing No.: FL103XXX → FL103XXX Make, and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification No. / Part No./ Drawing No.: FL104XXX → FL104XXX Make, and Identification No. / Part No./ Drawing No.: FL103 XXX → FL103 XXX Make, and Identification No. / Part No./ Drawing No.: FL102XXX → FL102XXX Make, and Identification No. / Part No./ Drawing No.: FL103XXX → FL103XXX Range: 0 - 130 km/h → 0 - 130 km/h Major marking: NA (Digital Indicator) → NA (Digital Indicator) Minor marking: NA (Digital Indicator) → NA (Digital Indicator) Speedometer ratio, Pulses / km RAR: 2440 → 2440 Speedometer ratio, Pulses / km RAR: 2424 → 2424 Speedometer ratio, Pulses / km RAR: 2590 → 2590 Ratio of Speedo drive: 0.60 → 0.60 Ratio of Speedo drive: 0.60 → 0.60 Ratio of Speedo drive: 0.64 → 0.64 Odometer: (only in case of CNG buses): NA → NA Make: -- → -- Type and Identification No. / Part No.: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Installation drawing showing rear extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size, make, unladen pressure) etc., distance between tyre & RUPD RH &		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	LH, frame rear width, P1, P2 points.: - → - Controls - Specify method of operation, hand operated – left / right, foot operated – left / right Ref. SS12.1 or corresponding Indian Standard).: - → - Ignition: Hand operated [Rotary Switch RHS] → Hand operated [Rotary Switch RHS] Horn: Hand operated [Horn pad on steering wheel] → Hand operated [Horn pad on steering wheel] Lamps (Head lamp, Tail lamp, Parking lamp and Number plate lamp): Hand operated [Rotary switch – RHS] → Hand operated [Rotary switch – RHS] Turn signal: Hand operated [Swivel arm switch – RHS] → Hand operated [Swivel arm switch – RHS] Transmission shift lever: Hand operated [LHS] → Hand operated [LHS] Wind shield wiper: Hand operated [Swivel arm switch – LHS] → Hand operated [Swivel arm switch – LHS] High beam/low beam: Hand operated [Swivel arm switch – RHS] → Hand operated [Swivel arm switch – RHS]	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Parking brake: Hand operated [LHS] → Hand operated [LHS] Master switch for electrical: NA → NA Hazard warning signal: Hand		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	operated [Push button Switch -LHS] → Hand operated [Push button Switch -LHS] Service Brake: Foot operated → Foot operated Service Brake: → Accelerator Pedal (Floor hinged/hanging type): Foot operated – [Floor hanging Type] → Foot operated – [Floor hanging Type] Others: NA → NA Instrument Panel (Dash Board): → Drawing showing the mounting details, overall size and all control switches with dimensions: BWR0XX01 → BWR0XX01 Instrument Panel Variants with photographs (With / without Airbag, Music system, AC): NA → NA Material used for instrument Panel: PPTYY & PCXXS → PPTYY & PCXXS Instrument Panel mounting (With hardware details): BWRXXYY1 → BWRXXYY1 Cross sectional Drawing of Gear shift lever (Knob): FXA0XXYY (Non AC) / FXA0XXYY (AC) → FXA0XXYY (Non AC) / FXA0XXYY (AC) Arrangement of front controls (For M1	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	category as per AIS – 035): NA → NA Distance between the contour points of the orthogonal projections on to plane “P” of the accelerator pedal and service brake pedal bearing		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	surfaces, “E” in mm.: -- → -- Distance between the projection of the service brake pedal on to the reference plane “P”, to the right, “H” in mm.: -- → -- Distance between the projection of the service brake pedal on to the0 reference plane “P”, to the left, “J” in mm.: -- → -- Drawing showing the parts and arrangement of the foot controls along with dimensions “E”, “H” and “J”: -- → -- In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → -- Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side, distance between front extremity & FUPD, width of FUPD, ID marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Type of Speed Governor (SLD / SLF): SLF → SLF Speed Limiting Device (SLD) [For	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	vehicles with max. speed above 80 kmph in		
(continued)	unladen condition as per AIS-018]: NA → NA Make: M/s. XXXYYY → M/s. XXXYYY Model: DXM 8.1AP → DXM 8.1AP Identification: TAC No. / BIS License No.: AL Development ID : F6NXYZW1 AL Production ID : F6NXYZW0 → AL Development ID : F6NXYZW1 AL Production ID : F6NXYZW0 SLD approved for speed/s (Annexure to be provided including information for every Vehicle Model / Variants indicating various SLD approved speeds contributed due to declared transmission options): -- → -- Information for Speed alert: NA → NA Details of audible alert (Alert above 80 km/h): -- → -- Details of audible alert Continuous /intermittent.(Alert above 120 km/h): -- → -- if intermittent, Interval between Alerts: -- → -- 1st · Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st · Overall Ratio / AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69 2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56 4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 → 20.44 Control system & braking wheel: Hydraulic acting on all wheels → Hydraulic acting on all wheels Nos. of directly controlled wheel(s): - → - Brief description of failure warning tell–tale: - → - Wheel Speed Sensors: - → - No. of sensors: - → - Make of sensors: - → - Type of sensors: - → - Controller: - → - Nos. of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Controller: - → - Make of Controller: - → - Identification No. / Part No. of Controller: - → - Brief description and features: - → - Control System & Braking wheel: Hand operated & Rear wheel → Hand operated & Rear wheel Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No For category B systems, define the brake pedal speed which must be achieved in order to activate the Brake Assist System (e.g. pedal stroke speed (mm/s) during a given time interval): -- → -- Adequate documentation according to annex G (Complex electronics vehicle control systems) provided in respect of the following system(s): ..... Yes / No / Not applicable: -- → -- List of complex functions covered:: -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Manufacturer’s	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- A list of input and output variables, including their working range, is provided.: -- → -- Documentation includes an inventory of components, outlines the function of each unit, its interconnection/interaction with other systems and defines signal flow priorities.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software version as appropriate.: -- → -- Safety concept statement verifies that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- System architecture, design methods and tools identified.: -- → -- Information regarding design provisions in case of failure is provided including any error messages, warning signals, partial performance conditions, back-up modes and/or removal of high- level functions.: -- → -- Additional material and	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	analysis (FMEA, FTA) of fault conditions made available for inspection and maintained by the manufacturer.: -- → -- Document reference and date of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	inspection:: -- → -- Vehicle complies with all other performance requirements of IS 15986 : 2015 and meets manufacturer’s specifications under non-fault conditions.: -- → -- Under fault conditions, vehicle response corresponds to that described in the manufacturer’s documents / failure analysis and safety concept is verified.: -- → -- Electronic Stability Control system as per AIS:133 (Fitted : Yes/No): NO → NO If yes, details of ESC: -- → -- ESC System Technical Documentation. To ensure a vehicle is equipped with an ESC system that meets the definition of "ESC System" in paragraph 2.0, the vehicle manufacturer shall make available to the test agency, upon request, the documentation specified below.: -- → -- System diagram identifying all ESC system hardware.: -- → -- A brief written explanation sufficient to describe the ESC system basic		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	operational characteristics.: -- → -- Logic diagram: -- → -- Understeer information.: -- → -- Static Stability Factor: -- → -- Make and Country of manufacturer(if imported): -- → -- Software Id / version: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Hardware Id: -- → -- Brief description of failure warning tell–tale: -- → -- Control function (Directional / Roll / Directional and Roll): -- → -- Steering Angle Sensor: -- → -- Make and Country of manufacturer(if imported): -- → -- Identification No. / Part No.: -- → -- Brief description and features: -- → -- Yaw Rate Sensor: -- → -- Make and Country of manufacturer(if imported): -- → -- Identification No. / Part No.: -- → -- Brief description and features: -- → -- Control function (directional/roll-over/both) including an explanation of the basic function and/or philosophy of the control: -- → -- Additional information (if applicable) to the application of the directional control and roll-over control functions, for example:: -- → -- Wheelbase, track, centre of gravity		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	height: -- → -- Wheel type (single or twin) and tyre type (e.g. structure, category of use, size): -- → -- Gearbox type (e.g. manual, automated manual, semi-automatic, automatic): -- → -- Drive train options (e.g. retarder): -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Differential type/differential lock(s) (e.g. standard or self-locking, automatic or driver selected): -- → -- Management of the engine or any other source(s) of motive power e.g.Torque Management, control of fuel supply etc.: -- → -- Sensor(s) External to the Controller: -- → -- Function: -- → -- Limitations on the location of the sensors: -- → -- Identification (e.g. part numbers): -- → -- Controller(s): -- → -- General description and function: -- → -- Functionality of internal sensors (if applicable): -- → -- Hardware identification (e.g. part numbers): -- → -- Limitations on the location of the controller(s): -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	the system directly controls output variables.: -- → -- Control function (directional/roll-over/both) including an explanation of the basic function and/or philosophy of the control: -- → -- Additional information (if applicable) to	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	the application of the directional control and/or the roll-over control function(s): -- → -- Sensors external to the controller (a) Function; (b) Limitations on the location of the sensors; (c) Identification, e.g. part numbers.: -- → -- Controller(s) (a) General description and function; (b) Identification e.g. part numbers; (c) Limitations on the location of the controller(s); (d) Additional features.: -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Make and model: RSB Transmission (T210R) → RSB Transmission (T210R) Make and model: AVTEC (A5M210) → AVTEC (A5M210) Type: Manual gearbox with fully synchro		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	mesh & Rev with sliding mesh → Manual gearbox with fully synchro mesh & Rev with sliding mesh No. of gears: 5 Forward and 1 Reverse → 5 Forward and 1 Reverse 1st: 4.62 → 4.62 1st: 4.60 → 4.60 1st: 21.53 → 21.53 1st: 21.44 → 21.44 2nd: 2.68 → 2.68 2nd: 2.69 → 2.69	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2nd: 12.67 → 12.67 2nd: 12.71 → 12.71 3rd: 1.51 → 1.51 3rd: 1.49 → 1.49 3rd: 7.34 → 7.34 3rd: 7.25 → 7.25 4th: 1.0 → 1.0 4th: 1.0 → 1.0 4th: 4.56 → 4.56 4th: 4.56 → 4.56 5th: 0.70 → 0.70 5th: 0.71 → 0.71 5th: 3.64 → 3.64 5th: 3.67 → 3.67 Reverse: 4.39 → 4.39 Reverse: 20.36 → 20.36 Reverse: 4.38 → 4.38 Reverse: 20.34 → 20.34 Drive Axle (Front / Rear / All): Rear → Rear Front axle ratio: NA → NA Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT 12 PR Tyre size designation including ply		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	rating: 215/75 R15 LT → 215/75 R15 LT Front: Q → Q Front: L → L Front: S → S Front: S → S Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2 → 3.26 kg/cm2 Maximum gradeability in 1st gear: GVW 2890 kg → GVW 2890 kg Maximum gradeability in 1st gear: GVW 3390 kg → GVW 3390 kg Maximum gradeability in 1st gear: 10 Deg → 10 Deg Maximum gradeability in 1st gear: 8.5 Deg → 8.5 Deg Vehicle Max Speed in unladen condition (km/h): Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Vehicle Max Speed in laden condition (km/h): Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane equivalent) in PPM content by volume in the exhaust gas with the	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm Maximum rated speed (Specify tolerance): 3100 rpm ± 50 rpm → 3100 rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Cut-off point without load (Maximum no-load speed)…(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Electronic controlled injection: (Yes/No): Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common Rail, Direct Injection with Pressure	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA → NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s. XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and Metering type Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air temperature sensor:: M/s.	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Continental and Thermistor Type → M/s. Continental and Thermistor Type	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type Make and type of the control unit:: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA Fan characteristics (Fan power, kW) Enclose the fan power curve corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part No: 9 → 9 Material of blades (metal / plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ±		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150 (AC)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Nominal setting of the engine temperature control o mechanism (C)::: 85 deg C → 85 deg C Intake depression at rated engine speed and at 100 per cent load (Specify location of measurement) Minimum allowable:... (kPa) Maximum allowable:... (kPa): Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Maximum allowable exhaust back pressure at rated engine speed and at 100 per cent load (location of measurement): ... (kPa): 30.0 kpa (150 mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx Catalyst (if any, and if not covered by another heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No Element – 1 No. → Catalytic Converter – 1No Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type for LNT (periodic / continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:.....cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap Bulk Density (GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle and Characteristics:: Passive selective	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY Type and Identification number:: Type - Passive NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA Make:: NA → NA Type and Idetification number :: NA →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA NOx Sensor (Before /After of SCR):: No → No Make:: NA →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	NA Type and Identification number:: NA → NA Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems, description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent engine test bench cycles, between two cycles where regenerative phases occur under the conditions equivalent to Type I test (Distance "D" in	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration (Distance “d” in AIS 137): 2 cycles → 2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	cycles Description of method employed to determine the number of cycles between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter (DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA International standard (where appropriate):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Frequency of reagent refill: continuous/maintenance (where appropriate):: NA → NA Normal operating		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	temperature range (°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY → YYYYYYY Air injection: (Yes/No): No → No Type (pulse air, air pump, etc.) & Identification Number:: NA → NA Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other): Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY → FEW0XXYY Characteristics (flow rate, etc.):: Flow	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes EGR Cooler Make and Identification Number:		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:.....g: NA → NA Canister working capacity:.....g: NA → NA Canister bed volume:.....litre: NA → NA Canister Purge and vent strategy	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve : (Yes/No): NA → NA Make and Identification Number: NA → NA Operating pressure (kPa): NA → NA Liquid fuel hoses material :: NA → NA Schematic		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or drawing (Periodic / Continuous): NA → NA Particulate trap: (Yes/No): Yes → Yes Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY Unique Identification (Serial No. or	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dimensions and shape of the particulate trap (capacity):: Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH)		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure 4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh: 90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Material - Silicon Carbide → Structure - Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI Electronic Control Unit (ECU): Yes → Yes Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID : XSSSZ 5.1XX, AL Development ID :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number: P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA Misfire detection:: NA → NA Oxygen		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K → Refer Enclosure 4K Particulate traps monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method):: Refer Enclosure 4K → Refer Enclosure 4K	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K → Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K A description of the type and number of the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3 EUDC A description of the type of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing all sensed components with the strategy for fault detection and MI activation (fixed number of driving cycles or statistical method), including a list of relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive explanation for the data given in service $05 Test ID $21 to FF	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles – Diagnostics on Controller Area Network		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(CAN) – Part 4: Requirements for emissions- related systems", a comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137): Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA → NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA Systems to ensure the correct operation	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent deactivation of the driver inducement, for		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	use by the rescue services [Note: cross-reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling /disable after parking: NA → NA Lowest concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct operation of NOx control measures: NA → NA Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Electronic engine management control unit for NG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Electronic		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	engine management control unit for hydrogen fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA Identification number:: NA → NA Power controller: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Tyre pressure monitoring system: NA → NA 6 or more Speed Transmission: NA → NA Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2 Transmission Control Unit (ECU) (Yes/No):: No → No Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Verification Number (CVN): NA → NA Maximum vehicle speed for continuous operation (km/h) (Only for AC horns): 80 Kmph → 80 Kmph Drawing showing type and controls of dipped beam leveling device with table showing positions of switch for various vehicle loading condition: VFTXYZ507 → VFTXYZ507 Tell-Tale for leveling switch: Yes → Yes Description and sketch showing the detail positions of Tell-Tale and seating layout (For M category vehicles): NA → NA Name of the manufacturer of the system (Steering column shaft / Gear shift	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	control / Transmission) on which the Protective Device functions: XXYY Steering System → XXYY Steering System Type/Part no./Identification number(ID)/Drawing No.of the device on which the Protective Device functions: FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly → FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable only in case of steering column locking systems: Locks the Steering → Locks the Steering VLT to Control Centre: -- → -- No. of speeds offered in blower (s): 4 → 4 No. of speeds offered in blower (s): 4 → 4 No. of speeds offered in blower (s): 4 → 4 Whether air vents have been provided for direction control, If yes, their locations: Yes (4) → Yes (4) Whether air vents have been provided for direction control, If yes, their locations: Yes (4) LH/CLH/CRH/RH → Yes (4) LH/CLH/CRH/RH Whether air vents have been provided	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	for direction control, If yes, their locations: Yes (4) → Yes (4) Whether air vents have been provided for direction control, If yes, their locations: LH/CLH/CRH/RH → LH/CLH/CRH/RH Whether air vents have been provided for direction control, If yes, their locations: LH/CLH/CRH/RH → LH/CLH/CRH/RH		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
AIS-071; applicable category scope Identification of controls, tell-tales and indicators	Control (Forward / semi-forward / normal / others): Forward → Forward Vehicle Max Speed in unladen condition (km/h): Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Vehicle Max Speed in laden condition (km/h): Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Make: APOLLO → APOLLO Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade description(s): CEAT WINMILE X3-R → CEAT WINMILE X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu	CEA / report needs review	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Maxx LT Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 Tyre rolling		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer - TC7719XX000 003XXXF → Refer - TC7719XX000 003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	BIS License No. / E- Marking: CM/L-756XXXX → CM/L-756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L – 670007XXXX	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E- Marking: NA → NA Others (For articulated / combination vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS:		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other (for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 325 mm → 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Type: Fully synchronized 5 - forward gears with a Constant mesh reverse gear → Fully synchronized 5 - forward gears with a Constant mesh reverse gear Model name (if any): XXX-XXX → XXX-XXX	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Model name (if any): YYYY-YY → YYYY-YY Gear shifting control system (Sketch showing gear shifting arrangement): → No. of gears: Forward 5 + 1 Rev → Forward 5 + 1 Rev Control system: NA → NA Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Detailed drawing of the mounting arrangement of Steering control assembly showing vertical / tilt / actual angle.: FZY0XX00 → FZY0XX00 Type of steering gear (Re-circulating ball / Worm & Roller / Rack & Pinion / Others): Re-circulating ball type → Re-circulating ball type Make: 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd → 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2. YYYY – 18.2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	:1 Head lamp – upper / lower control: 0 – 3 Switch provided → 0 – 3 Switch provided Ignition cut-off: Not Provided → Not Provided Turn signal: Green Color - LH and RH Direction Symbol → Green Color - LH and	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	RH Direction Symbol Fuel Gauge: LED Bars → LED Bars Engine coolant temperature: Red Color – Thermometer Symbol → Red Color – Thermometer Symbol Engine low oil pressure: Red Color – Oil Tank Symbol → Red Color – Oil Tank Symbol High beam indicator: Blue Color – High Beam Symbol → Blue Color – High Beam Symbol Electrical charge indicator (Battery charge): Red Color – Battery Symbol → Red Color – Battery Symbol Brake failure: Red Color – Parking and Oil symbol together → Red Color – Parking and Oil symbol together Brake failure: → Front fog light, if provided: Not applicable. → Not applicable. Front fog light, if provided: → Rear fog light, if provided: Not applicable. → Not applicable. Horn: On steering Wheel → On steering Wheel ABS: Red Color – ABS function alert → Red Color –		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	ABS function alert Seat Belt Reminder: Not Applicable → Not Applicable Others (such as, Airbag, HVAC, Content gauge, LPG / CNG changeover switch etc.,): Not Applicable → Not Applicable	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Speedometer:: 7.00 R15 LT → 7.00 R15 LT Speedometer:: 215/75 R15 LT → 215/75 R15 LT Type: Electronic → Electronic Make, and Identification No. / Part No./ Drawing No.: Pricol Limited → Pricol Limited Make, and Identification No. / Part No./ Drawing No.: Minda stoneridge → Minda stoneridge Make, and Identification No. / Part No./ Drawing No.: Pricol Limited → Pricol Limited Make, and Identification No. / Part No./ Drawing No.: Minda stoneri dge → Minda stoneri dge Make, and Identification No. / Part No./ Drawing No.: Pricol Limited → Pricol Limited Make, and Identification No. / Part No./ Drawing No.: Minda stoneridge → Minda stoneridge Make, and Identification No. / Part No./ Drawing No.: FL102XXX → FL102XXX Make, and Identification No. / Part No./ Drawing No.: FL103XXX → FL103XXX Make, and		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification No. / Part No./ Drawing No.: FL104XXX → FL104XXX Make, and Identification No. / Part No./ Drawing No.: FL103 XXX → FL103 XXX Make, and Identification No. / Part No./ Drawing No.: FL102XXX → FL102XXX Make, and Identification No. / Part No./ Drawing No.: FL103XXX → FL103XXX Range: 0 - 130 km/h → 0 - 130 km/h Major marking: NA (Digital Indicator) → NA (Digital Indicator) Minor marking: NA (Digital Indicator) → NA (Digital Indicator) Speedometer ratio, Pulses / km RAR: 2440 → 2440 Speedometer ratio, Pulses / km RAR: 2424 → 2424 Speedometer ratio, Pulses / km RAR: 2590 → 2590 Ratio of Speedo drive: 0.60 → 0.60 Ratio of Speedo drive: 0.60 → 0.60 Ratio of Speedo drive: 0.64 → 0.64 Odometer: (only in case of CNG buses): NA → NA Make: -- → -- Type and Identification No. / Part No.: -- → -- Installation drawing showing rear extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size, make, unladen	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	pressure) etc., distance between tyre & RUPD RH &		
(continued)	LH, frame rear width, P1, P2 points.: - → - Controls - Specify method of operation, hand operated – left / right, foot operated – left / right Ref. SS12.1 or corresponding Indian Standard).: - → - Ignition: Hand operated [Rotary Switch RHS] → Hand operated [Rotary Switch RHS] Horn: Hand operated [Horn pad on steering wheel] → Hand operated [Horn pad on steering wheel] Lamps (Head lamp, Tail lamp, Parking lamp and Number plate lamp): Hand operated [Rotary switch – RHS] → Hand operated [Rotary switch – RHS] Turn signal: Hand operated [Swivel arm switch – RHS] → Hand operated [Swivel arm switch – RHS] Transmission shift lever: Hand operated [LHS] → Hand operated [LHS] Wind shield wiper: Hand operated [Swivel arm switch – LHS] → Hand operated [Swivel arm switch – LHS] High beam/low beam: Hand operated [Swivel arm switch – RHS] → Hand operated [Swivel arm switch – RHS] Parking brake: Hand operated [LHS] → Hand operated [LHS] Master switch for electrical: NA → NA Hazard warning signal: Hand	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	operated [Push button Switch -LHS] → Hand operated [Push button Switch -LHS] Service Brake: Foot operated → Foot operated Service Brake: → Accelerator Pedal (Floor hinged/hanging type): Foot operated – [Floor hanging Type] → Foot operated – [Floor hanging Type] Others: NA → NA Instrument Panel (Dash Board): → Drawing showing the mounting details, overall size and all control switches with dimensions: BWR0XX01 → BWR0XX01 Instrument Panel Variants with photographs (With / without Airbag, Music system, AC): NA → NA Material used for instrument Panel: PPTYY & PCXXS → PPTYY & PCXXS Instrument Panel mounting (With hardware details): BWRXXYY1 → BWRXXYY1 Cross sectional Drawing of Gear shift lever (Knob): FXA0XXYY (Non AC) / FXA0XXYY (AC) → FXA0XXYY (Non AC) / FXA0XXYY (AC) Arrangement of front controls (For M1 category as per AIS – 035): NA → NA Distance between the contour points of the orthogonal projections on to plane “P” of the accelerator pedal and service brake pedal bearing	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	surfaces, “E” in mm.: -- → -- Distance between the projection of the service brake pedal on to the reference plane “P”, to the right, “H” in mm.: -- → -- Distance between the projection of the service brake pedal on to the0 reference plane “P”, to the left, “J” in mm.: -- → -- Drawing showing the parts and arrangement of the foot controls along with dimensions “E”, “H” and “J”: -- → -- In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → -- Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side, distance between front extremity & FUPD, width of FUPD, ID marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Type of Speed Governor (SLD / SLF): SLF → SLF Speed Limiting Device (SLD) [For vehicles with max. speed above 80 kmph in	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	unladen condition as per AIS-018]: NA → NA Make: M/s. XXXYYY → M/s. XXXYYY	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Model: DXM 8.1AP → DXM 8.1AP Identification: TAC No. / BIS License No.: AL Development ID : F6NXYZW1 AL Production ID : F6NXYZW0 → AL Development ID : F6NXYZW1 AL Production ID : F6NXYZW0 SLD approved for speed/s (Annexure to be provided including information for every Vehicle Model / Variants indicating various SLD approved speeds contributed due to declared transmission options): -- → -- Information for Speed alert: NA → NA Details of audible alert (Alert above 80 km/h): -- → -- Details of audible alert Continuous /intermittent.(Alert above 120 km/h): -- → -- if intermittent, Interval between Alerts: -- → -- 1st · Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st · Overall Ratio / AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69 2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC:		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56 4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 → 20.44 Control system & braking wheel: Hydraulic acting on all wheels → Hydraulic acting on all wheels Nos. of directly controlled wheel(s): - → - Brief description of failure warning tell–tale: - → - Wheel Speed Sensors: - → - No. of sensors: - → - Make of sensors: - → - Type of sensors: - → - Controller: - → - Nos. of		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Controller: - → - Make of Controller: - → - Identification No. / Part No. of Controller: - → - Brief description and features: - → - Control System & Braking wheel: Hand operated & Rear wheel → Hand operated & Rear wheel Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No For category B systems, define the brake pedal speed which must be achieved in order to activate the Brake Assist System (e.g. pedal stroke speed (mm/s) during a given time interval): -- → -- Adequate documentation according to annex G (Complex electronics vehicle control systems) provided in respect of the following system(s): ..... Yes / No / Not applicable: -- → -- List of complex functions covered:: -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Manufacturer’s	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- A list of input and output variables, including their working range, is provided.: -- → -- Documentation includes an inventory of components, outlines the function of each unit, its interconnection/interaction with other systems and defines signal flow priorities.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software version as appropriate.: -- → -- Safety concept statement verifies that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- System architecture, design methods and tools identified.: -- → -- Information regarding design provisions in case of failure is provided including any error messages, warning signals, partial performance conditions, back-up modes and/or removal of high- level functions.: -- → -- Additional material and	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	analysis (FMEA, FTA) of fault conditions made available for inspection and maintained by the manufacturer.: -- → -- Document reference and date of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	inspection:: -- → -- Vehicle complies with all other performance requirements of IS 15986 : 2015 and meets manufacturer’s specifications under non-fault conditions.: -- → -- Under fault conditions, vehicle response corresponds to that described in the manufacturer’s documents / failure analysis and safety concept is verified.: -- → -- Electronic Stability Control system as per AIS:133 (Fitted : Yes/No): NO → NO If yes, details of ESC: -- → -- ESC System Technical Documentation. To ensure a vehicle is equipped with an ESC system that meets the definition of "ESC System" in paragraph 2.0, the vehicle manufacturer shall make available to the test agency, upon request, the documentation specified below.: -- → -- System diagram identifying all ESC system hardware.: -- → -- A brief written explanation sufficient to describe the ESC system basic		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	operational characteristics.: -- → -- Logic diagram: -- → -- Understeer information.: -- → -- Static Stability Factor: -- → -- Make and Country of manufacturer(if imported): -- → -- Software Id / version: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Hardware Id: -- → -- Brief description of failure warning tell–tale: -- → -- Control function (Directional / Roll / Directional and Roll): -- → -- Steering Angle Sensor: -- → -- Make and Country of manufacturer(if imported): -- → -- Identification No. / Part No.: -- → -- Brief description and features: -- → -- Yaw Rate Sensor: -- → -- Make and Country of manufacturer(if imported): -- → -- Identification No. / Part No.: -- → -- Brief description and features: -- → -- Control function (directional/roll-over/both) including an explanation of the basic function and/or philosophy of the control: -- → -- Additional information (if applicable) to the application of the directional control and roll-over control functions, for example:: -- → -- Wheelbase, track, centre of gravity		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	height: -- → -- Wheel type (single or twin) and tyre type (e.g. structure, category of use, size): -- → -- Gearbox type (e.g. manual, automated manual, semi-automatic, automatic): -- → -- Drive train options (e.g. retarder): -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Differential type/differential lock(s) (e.g. standard or self-locking, automatic or driver selected): -- → -- Management of the engine or any other source(s) of motive power e.g.Torque Management, control of fuel supply etc.: -- → -- Sensor(s) External to the Controller: -- → -- Function: -- → -- Limitations on the location of the sensors: -- → -- Identification (e.g. part numbers): -- → -- Controller(s): -- → -- General description and function: -- → -- Functionality of internal sensors (if applicable): -- → -- Hardware identification (e.g. part numbers): -- → -- Limitations on the location of the controller(s): -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	the system directly controls output variables.: -- → -- Control function (directional/roll-over/both) including an explanation of the basic function and/or philosophy of the control: -- → -- Additional information (if applicable) to	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	the application of the directional control and/or the roll-over control function(s): -- → -- Sensors external to the controller (a) Function; (b) Limitations on the location of the sensors; (c) Identification, e.g. part numbers.: -- → -- Controller(s) (a) General description and function; (b) Identification e.g. part numbers; (c) Limitations on the location of the controller(s); (d) Additional features.: -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Make and model: RSB Transmission (T210R) → RSB Transmission (T210R) Make and model: AVTEC (A5M210) → AVTEC (A5M210) Type: Manual gearbox with fully synchro		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	mesh & Rev with sliding mesh → Manual gearbox with fully synchro mesh & Rev with sliding mesh No. of gears: 5 Forward and 1 Reverse → 5 Forward and 1 Reverse 1st: 4.62 → 4.62 1st: 4.60 → 4.60 1st: 21.53 → 21.53 1st: 21.44 → 21.44 2nd: 2.68 → 2.68 2nd: 2.69 → 2.69	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2nd: 12.67 → 12.67 2nd: 12.71 → 12.71 3rd: 1.51 → 1.51 3rd: 1.49 → 1.49 3rd: 7.34 → 7.34 3rd: 7.25 → 7.25 4th: 1.0 → 1.0 4th: 1.0 → 1.0 4th: 4.56 → 4.56 4th: 4.56 → 4.56 5th: 0.70 → 0.70 5th: 0.71 → 0.71 5th: 3.64 → 3.64 5th: 3.67 → 3.67 Reverse: 4.39 → 4.39 Reverse: 20.36 → 20.36 Reverse: 4.38 → 4.38 Reverse: 20.34 → 20.34 Drive Axle (Front / Rear / All): Rear → Rear Front axle ratio: NA → NA Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT 12 PR Tyre size designation including ply		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	rating: 215/75 R15 LT → 215/75 R15 LT Front: Q → Q Front: L → L Front: S → S Front: S → S Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2 → 3.26 kg/cm2 Maximum gradeability in 1st gear: GVW 2890 kg → GVW 2890 kg Maximum gradeability in 1st gear: GVW 3390 kg → GVW 3390 kg Maximum gradeability in 1st gear: 10 Deg → 10 Deg Maximum gradeability in 1st gear: 8.5 Deg → 8.5 Deg Vehicle Max Speed in unladen condition (km/h): Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Vehicle Max Speed in laden condition (km/h): Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane equivalent) in PPM content by volume in the exhaust gas with the	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm Maximum rated speed (Specify tolerance): 3100 rpm ± 50 rpm → 3100 rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Cut-off point without load (Maximum no-load speed)…(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Electronic controlled injection: (Yes/No): Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common Rail, Direct Injection with Pressure	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA → NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s. XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and Metering type Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air temperature sensor:: M/s.	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Continental and Thermistor Type → M/s. Continental and Thermistor Type	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type Make and type of the control unit:: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA Fan characteristics (Fan power, kW) Enclose the fan power curve corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part No: 9 → 9 Material of blades (metal / plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ±		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150 (AC)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Nominal setting of the engine temperature control o mechanism (C)::: 85 deg C → 85 deg C Intake depression at rated engine speed and at 100 per cent load (Specify location of measurement) Minimum allowable:... (kPa) Maximum allowable:... (kPa): Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Maximum allowable exhaust back pressure at rated engine speed and at 100 per cent load (location of measurement): ... (kPa): 30.0 kpa (150 mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx Catalyst (if any, and if not covered by another heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No Element – 1 No. → Catalytic Converter – 1No Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type for LNT (periodic / continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:.....cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap Bulk Density (GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle and Characteristics:: Passive selective	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY Type and Identification number:: Type - Passive NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA Make:: NA → NA Type and Idetification number :: NA →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA NOx Sensor (Before /After of SCR):: No → No Make:: NA →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	NA Type and Identification number:: NA → NA Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems, description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent engine test bench cycles, between two cycles where regenerative phases occur under the conditions equivalent to Type I test (Distance "D" in	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration (Distance “d” in AIS 137): 2 cycles → 2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	cycles Description of method employed to determine the number of cycles between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter (DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA International standard (where appropriate):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Frequency of reagent refill: continuous/maintenance (where appropriate):: NA → NA Normal operating		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	temperature range (°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY → YYYYYYY Air injection: (Yes/No): No → No Type (pulse air, air pump, etc.) & Identification Number:: NA → NA Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other): Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY → FEW0XXYY Characteristics (flow rate, etc.):: Flow	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes EGR Cooler Make and Identification Number:		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:.....g: NA → NA Canister working capacity:.....g: NA → NA Canister bed volume:.....litre: NA → NA Canister Purge and vent strategy	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve : (Yes/No): NA → NA Make and Identification Number: NA → NA Operating pressure (kPa): NA → NA Liquid fuel hoses material :: NA → NA Schematic		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or drawing (Periodic / Continuous): NA → NA Particulate trap: (Yes/No): Yes → Yes Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY Unique Identification (Serial No. or	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dimensions and shape of the particulate trap (capacity):: Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH)		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure 4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh: 90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Material - Silicon Carbide → Structure - Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI Electronic Control Unit (ECU): Yes → Yes Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID : XSSSZ 5.1XX, AL Development ID :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number: P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA Misfire detection:: NA → NA Oxygen		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K → Refer Enclosure 4K Particulate traps monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method):: Refer Enclosure 4K → Refer Enclosure 4K	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K → Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K A description of the type and number of the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3 EUDC A description of the type of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing all sensed components with the strategy for fault detection and MI activation (fixed number of driving cycles or statistical method), including a list of relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive explanation for the data given in service $05 Test ID $21 to FF	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles – Diagnostics on Controller Area Network		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(CAN) – Part 4: Requirements for emissions- related systems", a comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137): Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA → NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA Systems to ensure the correct operation	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent deactivation of the driver inducement, for		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	use by the rescue services [Note: cross-reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling /disable after parking: NA → NA Lowest concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct operation of NOx control measures: NA → NA Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Electronic engine management control unit for NG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Electronic		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	engine management control unit for hydrogen fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA Identification number:: NA → NA Power controller: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Tyre pressure monitoring system: NA → NA 6 or more Speed Transmission: NA → NA Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2 Transmission Control Unit (ECU) (Yes/No):: No → No Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Verification Number (CVN): NA → NA Maximum vehicle speed for continuous operation (km/h) (Only for AC horns): 80 Kmph → 80 Kmph Drawing showing type and controls of dipped beam leveling device with table showing positions of switch for various vehicle loading condition: VFTXYZ507 → VFTXYZ507 Tell-Tale for leveling switch: Yes → Yes Description and sketch showing the detail positions of Tell-Tale and seating layout (For M category vehicles): NA → NA Name of the manufacturer of the system (Steering column shaft / Gear shift	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	control / Transmission) on which the Protective Device functions: XXYY Steering System → XXYY Steering System Type/Part no./Identification number(ID)/Drawing No.of the device on which the Protective Device functions: FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly → FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable only in case of steering column locking systems: Locks the Steering → Locks the Steering VLT to Control Centre: -- → -- No. of speeds offered in blower (s): 4 → 4 No. of speeds offered in blower (s): 4 → 4 No. of speeds offered in blower (s): 4 → 4 Whether air vents have been provided for direction control, If yes, their locations: Yes (4) → Yes (4) Whether air vents have been provided for direction control, If yes, their locations: Yes (4) LH/CLH/CRH/RH → Yes (4) LH/CLH/CRH/RH Whether air vents have been provided	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	for direction control, If yes, their locations: Yes (4) → Yes (4) Whether air vents have been provided for direction control, If yes, their locations: LH/CLH/CRH/RH → LH/CLH/CRH/RH Whether air vents have been provided for direction control, If yes, their locations: LH/CLH/CRH/RH → LH/CLH/CRH/RH		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
AIS-137 Part 3; CMVR 115 Emission family, road-load and worst-case selection	Base fuel type used for vehicle Type approval including blend level (Ethanol, Bio-diesel etc): Diesel (Maximum 7% bio-diesel blend) → Diesel (Maximum 7% bio-diesel blend) Name and address of engine manufacturing plant (In case of imported vehicles, above details shall be supplied for importer also): Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX → Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX Max. Recommended Gradeability (Stand-start) – in degrees: 10 degs → 10 degs Max. Recommended Gradeability (Stand-start) – in degrees: 8.5 degs → 8.5 degs Vehicle kerb weight kg: Refer Annexure T7 - B	CEA / report needs review	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Details for Recommended load body Refer Annexure T7 – C → Refer Annexure T7 - B Details for Recommended load body Refer Annexure T7 – C Other axle: NA → NA Gross combination weight kg (applicable for articulated / Tractor Trailer combination vehicles): NA → NA Front axle: NA → NA Rear axle: NA → NA Trailer axle / other axles: NA → NA If applicable, maximum mass of trailer whic5h may be coupled: NA →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	NA Unbraked trailer: NA → NA Reference mass (for vehicles with GVW less than or equal to 3.5 ton) (kg.): Refer Annexure T7-B → Refer Annexure T7-B Maximum (Including driver) for completely built vehicles: D+2 → D+2 Maximum (Including driver) for completely built vehicles: → Make: APOLLO → APOLLO Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	description(s): CEAT WINMILE X3-R → CEAT WINMILE X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 Tyre rolling Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer - TC7719XX000 003XXXF → Refer - TC7719XX000 003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022- 2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022- 2023/300002XXX X/TA/XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L- 756XXXX → CM/L- 756XXXX Identification: TAC No. /		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E- Marking: NA → NA Others (For articulated / combination vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00 R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube /		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other (for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 325 mm → 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26 kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Type: Fully		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	synchronized 5 - forward gears with a Constant mesh reverse gear → Fully synchronized 5 - forward gears with a	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Constant mesh reverse gear Model name (if any): XXX-XXX → XXX-XXX Model name (if any): YYYY-YY → YYYY-YY Gear shifting control system (Sketch showing gear shifting arrangement): → No. of gears: Forward 5 + 1 Rev → Forward 5 + 1 Rev Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Rear axle ratio: 4.56 → 5.13 Type of steering gear (Re-circulating ball / Worm & Roller / Rack & Pinion / Others): Re-circulating ball type → Re-circulating ball type Make: 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd → 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 Type and description (Leaf / Coil / Air / Semi- pneumatic/ Torsion bar): Leaf → Leaf Front: Parabolic spring- Rigid axle suspension → Parabolic spring- Rigid axle suspension Rear: Semi elliptic spring-		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Rigid axle suspension → Semi elliptic spring- Rigid axle suspension	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Fuel Gauge: LED Bars → LED Bars Engine coolant temperature: Red Color – Thermometer Symbol → Red Color – Thermometer Symbol Engine low oil pressure: Red Color – Oil Tank Symbol → Red Color – Oil Tank Symbol Weight of seats: Driver seat : 11.5 Kg & Co-driver seat : 11Kg → Driver seat : 11.5 Kg & Co-driver seat : 11Kg Description of seats: -- → -- Make: -- → -- Number of seats fitted or capable of being fitted with head restraints, adjustable or not adjustable.: -- → -- Description of the adjustment, displacement and locking systems of the seat or of its parts and a description of occupant protection system against displacement of luggage.: -- → -- Description of seat anchorage: -- → -- Longitudinal / Vertical (If applicable) position of the seats during the tests.: -- → -- Drawings, diagrams and plans of the seats, their anchorages on the vehicle, the floor layout, the adjustment and displacement system of the seats and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	their parts, and their locking devices and of additional occupant protection system against displacement of luggage.: -- → -- In the case of seats fitted with head	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	restraints, the head restraint shall be shown on all drawings, diagrams and photographs.: -- → -- Seat Identification No. / Part No/Drawing Number.: ID Number/ Part Number/D rawing number Weight Description Make (Seat (kg) Manufactu rer/ Vehicle Manufactu rer): -- → -- Description: Make → Make Description: ID Number/ Part Number/D rawing number (Seat Manufactu rer/ Vehicle Manufactu rer) → ID Number/ Part Number/D rawing number (Seat Manufactu rer/ Vehicle Manufactu rer) Description: Weight (kg) → Weight (kg) Brief description and drawings of the		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	seat type, its attachment fittings and its adjustment, displacement and locking systems including the minimum distance between fitting points.: D+1 → D+1 Installation drawing showing rear extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	make, unladen pressure) etc., distance between tyre & RUPD RH & LH, frame rear width, P1, P2 points.: - → - Make: M/s. XXXXYYYY → M/s. XXXXYYYY Material (Metallic / Plastic etc.): Metallic → Metallic Nominal thickness mm: 1.0mm → 1.0mm Capacity, Litre: 40 Lts, 30 Lts → 40 Lts, 30 Lts Detailed drawing of the fuel tank assembly with material specifications: FHSXXXXY FHSXXXXY → FHSXXXXY FHSXXXXY Detailed drawing indicating the position / location of the fuel tank(s) in the vehicle.: A5LXXX00 (50 LTS) A5LXXX00 (40 LTS) → A5LXXX00 (50 LTS) A5LXXX00 (40 LTS) Identification No. / Part No.: FHS00544 FfffXXXcdHXXS00744 → FHS00544 FfffXXXcdHXXS00744 Detailed engineering drawing as specified in AIS – 013 or photographs	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	showing the mounting details with dimensions.: NA → NA Cross sectional Drawing of Gear shift lever (Knob): FXA0XXYY (Non AC) / FXA0XXYY (AC) → FXA0XXYY (Non AC) / FXA0XXYY (AC) Drawings and/or photographs of the locations of the statutory plates and inscriptions and of the vehicle		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	identification number: X4W0XYZ4 / VFT0XYZ2 → X4W0XYZ4 / VFT0XYZ2 Drawings and / or photographs of the official part of the plates and inscriptions (completed example with dimensions): X4WXYZ44 → X4WXYZ44 Location of Vehicle Descriptor Section (VDS) in Vehicle Identification Number (VIN): 4thto 9thCharacter of 17 Characters VIN → 4thto 9thCharacter of 17 Characters VIN Brief description of the Provision for emergency exit provided in case of sealed windows: -- → -- Dimensions of the instructions describing each motion necessary to unlatch and open the exit are provided from the release mechanism.: -- → -- In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side, distance between front extremity & FUPD, width of FUPD, ID		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Brief description of the vehicle type as regards its structure, dimensions, lines and constituent materials:: -- → -- Description of the protective system installed in the vehicle:: -- → -- Description of the protective system installed in the vehicle:: → Description of interior arrangements or fittings that might affect the tests:: -- → -- A detailed description, including photographs and/or drawings, of the vehicle type with respect to the structure, the dimensions, the design and the constituent materials, the side walls of the passenger compartment (exterior and interior), including specific details of the protection system, where applicable:: -- → -- Drawing / Images of position & arrangement of Engine: -- → -- Drawing/ Images of Under bonnet (Engine house) Components.: -- → -- Construction: Integrated / Attached with	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Mounts: 4 Mounts, Fixed CABIN → 4 Mounts, Fixed CABIN 1st ·		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st · Overall Ratio / AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69 2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC: 12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56 4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	20.44 Description · Make: ID		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number/ Part Number/Drawing number (Seat Manufacturer/ Vehicle Manufacturer) → ID Number/ Part Number/Drawing number (Seat Manufacturer/ Vehicle Manufacturer) Description · Make: Weight (kg) → Weight (kg) B 51.0 B49.0 · Pedestrian protection as per AIS-100: The distance, measured longitudinally on a horizontal plane, between the transverse centerline of the front axle & the R point of the driver's seat is less than 1100 mm (Yes / No).: NA → NA Brief description of failure warning tell–tale: - → - Brief description and features: - → - Brief description and features: - → - Description: -- → -- Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No Manufacturer's documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	-- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	version as appropriate.: -- → -- Under fault conditions, vehicle response corresponds to that described in the manufacturer’s documents / failure analysis and safety concept is verified.: -- → -- A brief written explanation sufficient to describe the ESC system basic operational characteristics.: -- → -- Logic diagram: -- → -- Understeer information.: -- → -- Static Stability Factor: -- → -- Make and Country of manufacturer(if imported): -- → -- Software Id / version: -- → -- Brief description of failure warning tell–tale: -- → -- Brief description and features: -- → -- Brief description and features: -- → -- System identification including software level identifier: -- → -- List of motor vehicles by description and configuration that are covered by the information document: -- → -- Wheel type (single or twin) and tyre type (e.g. structure, category of use, size): -- → -- Gearbox type (e.g. manual, automated manual, semi-automatic, automatic): --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	→ -- Management of the engine or any		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	other source(s) of motive power e.g.Torque Management, control of fuel supply etc.: -- → -- Component description:: -- → -- Sensor(s) External to the Controller: -- → -- Function: -- → -- Limitations on the location of the sensors: -- → -- Identification (e.g. part numbers): -- → -- Controller(s): -- → -- General description and function: -- → -- Functionality of internal sensors (if applicable): -- → -- Hardware identification (e.g. part numbers): -- → -- Software identification: -- → -- Limitations on the location of the controller(s): -- → -- Additional features: -- → -- Modulators: -- → -- General description and function: -- → -- Hardware identification (e.g. part numbers): -- → -- Software identification (if applicable): -- → -- Limitations: -- → -- Electrical Equipment: -- → -- Circuit diagrams: -- → -- Powering methods: -- → -- Pneumatic circuits: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Safety aspects of the electronic system in accordance with Annex S of IS 11852: -- → -- List of complex functions covered::		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	-- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Manufacturer’s documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- A list of input and output variables, including their working range, is provided.: -- → -- Documentation includes an inventory of components, outlines the function of each unit, its interconnection/interaction with other systems and defines signal flow priorities.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software version as appropriate.: -- → -- Safety concept statement verifies that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- System architecture, design methods	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	and tools identified.: -- → -- Information regarding		
(continued)	design provisions in case of failure is provided including any error messages, warning signals, partial performance conditions, back-up modes and/or removal of high- level functions.: -- → -- Additional material and analysis (FMEA, FTA) of fault conditions made available for inspection and maintained by the manufacturer.: -- → -- Document reference and date of inspection:: -- → -- Vehicle complies with all other performance requirements of IS 15986:2015 and meets manufacturer’s specifications under non-fault conditions.: -- → -- Under fault conditions, vehicle response corresponds to that described in the manufacturer’s documents / failure analysis and safety concept is verified.: -- → -- Electro-magnetic compatibility: -- → -- Documentation demonstrating compliance with AIS-004(Part 3).: -- → -- Component description: -- → -- Controller(s) (a) General description and function; (b) Identification e.g. part numbers; (c) Limitations on the location of the controller(s); (d) Additional	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	features.: -- → -- Modulators (a) General description and function; (b) Identification; (c) Limitations.: -- → -- Manufacturer’s documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software version as appropriate.: -- → -- (see annex 4 of this report for test data): -- → -- (see annex 4 of this report for test data): -- → -- Type / Description: Fully built vehicle, D+2, Non AC, 3.39 Ton GVW → Fully built vehicle, D+2, Non AC, 3.39 Ton GVW Make: Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. → Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. Model name / identification: ALXYZABC0 → ALXYZABC0 Type: Compression Ignition → Compression Ignition Bore x stroke (mm): 80 X 98 → 80 X 98 No. of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	cylinders: 3 → 3 Displacement (cm³): 1468 cc → 1468 cc Compression ratio: 16.5±1.0 : 1 → 16.5±1.0 : 1 Max. Engine output (KW @ rpm): 55.1 Kw @ 3300 rpm → 55.1 Kw @ 3300 rpm Max. Torque (Nm @ rpm): 170Nm @ 1600 to 2400 rpm → 170Nm @ 1600 to 2400 rpm Air cleaner type: Dry type → Dry type Type of Fuel: Diesel → Diesel Make and model: RSB Transmission (T210R) → RSB Transmission (T210R) Make and model: AVTEC (A5M210) → AVTEC (A5M210) Type: Manual gearbox with fully synchro mesh & Rev with sliding mesh → Manual gearbox with fully synchro mesh & Rev with sliding mesh No. of gears: 5 Forward and 1 Reverse → 5 Forward and 1 Reverse 1st: 4.62 → 4.62 1st: 4.60 → 4.60 1st: 21.53 → 21.53 1st: 21.44 → 21.44 2nd: 2.68 → 2.68 2nd: 2.69 → 2.69 2nd: 12.67 → 12.67 2nd: 12.71 → 12.71 3rd: 1.51 → 1.51 3rd: 1.49 → 1.49 3rd: 7.34 → 7.34 3rd: 7.25 → 7.25 4th: 1.0 → 1.0 4th: 1.0 → 1.0 4th: 4.56 → 4.56	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	4th: 4.56 → 4.56 5th: 0.70 → 0.70 5th: 0.71 → 0.71 5th: 3.64 → 3.64 5th: 3.67 → 3.67 Reverse: 4.39 → 4.39		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Reverse: 20.36 → 20.36 Reverse: 4.38 → 4.38 Reverse: 20.34 → 20.34 Drive Axle (Front / Rear / All): Rear → Rear Front axle ratio: NA → NA Type / Description: Re-circulating ball type → Re-circulating ball type Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT 12 PR Tyre size designation including ply rating: 215/75 R15 LT → 215/75 R15 LT Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2 → 3.26 kg/cm2 Wiping system (Brief description): Twin blade, Tandem Pattern → Twin blade, Tandem Pattern Material: Metallic → Metallic Capacity (l): 50 lts , 40 lts → 50 lts , 40 lts Dimensions: 2490WB → 2490WB		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Dimensions: 2640WB → 2640WB Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Overall width (mm): 1842 → 1842 Overall Length (mm): CC → CC Overall Length (mm): 4750 → 4750 Overall Length (mm): 4988 → 4988 Overall Length (mm): FSD / HSD → FSD / HSD Overall Length (mm): 4925 → 4925 Overall Length (mm): 5225 → 5225 Overall height (mm): 1961 → 1961 Overall height (mm): 1948 → 1948 Maximum gradeability in 1st gear: GVW 2890 kg → GVW 2890 kg Maximum gradeability in 1st gear: GVW 3390 kg → GVW 3390 kg Maximum gradeability in 1st gear: 10 Deg → 10 Deg Maximum gradeability in 1st gear: 8.5 Deg → 8.5 Deg Weights: Base → Base Weights: Variant 1 → Variant 1	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Weights: Variant 2 → Variant 2 Weights: Variant 3 → Variant 3 Weights: Variant 4 → Variant 4 Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	spare wheel, tools, etc.) (kg): - → - Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Description of energy converters and power plant. (In the case of a vehicle that can run either on petrol, diesel, etc., or also in combination with another fuel, items shall be repeated.1): Diesel →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Diesel Engine Manufacturer:: Ashok Leyland Limited → Ashok Leyland Limited Manufacturer's engine model code (as marked on the engine, or other means of identification): Model Code: ALXYZABC0 Enclosure – 4A → Model Code: ALXYZABC0 Enclosure – 4A Location of engine model code and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	engine serial number on block / crankcase (drawing to be attached): Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A → Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A Working principle: positive ignition / compression- ignition, four-stroke / two-stroke / rotary cycle (DI/IDI) (NA/TC/TCIC/Any other): Compression Ignition, four-stroke, DI, TCIC → Compression Ignition, four-stroke, DI, TCIC Number, arrangement of cylinders:: 3, Inline → 3, Inline Bore----------(mm): 80 → 80 Stroke: ----------(mm): 98 →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	98 Firing order:: 1-2-3 → 1-2-3 Engine capacity:----------(cm3): 1468 → 1468 Volumetric compression ratio (specify tolerance): 17.5±1.0 : 1 → 17.5±1.0 : 1 Drawings of combustion chamber and piston crown and, in the case of positive ignition engine, piston rings:: Refer Enclosure – 4B → Refer		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure – 4B Vehicle with Defeat Device Yes / No: No → No If yes – Details of the Defeat device: NA → NA Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane equivalent) in PPM content by volume in the exhaust gas with the engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum net power:..... kW at min-1: 55.1 KW @ 3100 ± 50 rpm → 55.1 KW @ 3100 ± 50 rpm Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Maximum net torque: .....Nm at: -------min-1 ...(manufacturer's declared value): 170 Nm @ 1600 – 2400 rpm → 170 Nm @ 1600 – 2400 rpm Maximum rated speed (Specify tolerance): 3100 rpm ± 50 rpm → 3100 rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Type of fuel used : Mono		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Fuel / Bi-Fuel / Flex Fuel / Dual Fuel: Mono Fuel → Mono Fuel Fuel type: Diesel / Gasoline / LPG / CNG / Biomethane / Bio-H methane / HCNG / Biogas / LNG / Ethanol ((E85) / 2 (E100)) /ED (95) Flex fuel Methanol M15 / M100 / MD95 / Biodiesel up to 100% / Hydrogen.: Diesel → Diesel Maximum amount of biofuel acceptable in fuel (manufacturer's declared value): per cent by volume: 7% → 7% By fuel injection (compression-ignition only): (Yes/No): Yes → Yes System description:: Unit Pump Common Rail Direct Injection system → Unit Pump Common Rail Direct Injection system Working principle: direct-injection/pre-chamber/swirl Chamber: Direct Injection → Direct Injection	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Plunger Type & FHL0XYZ0 → Plunger Type & FHL0XYZ0 Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Injection advance curve (Specify the tolerances):: ECU Controlled → ECU Controlled Governor: ECU Controlled → ECU Controlled Cut-off point under load:…(min-1): 3300 → 3300 Cut-off point without load (Maximum no-load speed)…(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Multi-Hole Injectors (7 holes) FKXXYYZZ → Multi-Hole Injectors (7 holes) FKXXYYZZ Opening pressure (kPa) (specify the tolerances) or characteristic diagram: 15000 to 23000 kPa → 15000 to 23000	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	kPa Length and internal diameter...(mm): Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm → Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm Common rail make & type: M/s. XXXXYYY, Hot Forged Rail Type → M/s. XXXXYYY, Hot Forged Rail Type Cold start system: NA → NA Make(s):: NA → NA Type(s) &		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: NA → NA Description:: NA → NA Auxiliary starting aid: NA → NA Make(s):: NA → NA Type(s):: NA → NA System description:: NA → NA Electronic controlled injection: (Yes/No): Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA → NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s.	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Metering type Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type By fuel injection (positive ignition only): (Yes/No): No → No Working principle: intake manifold (single/multi- point)/direct	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	injection/other (specify): NA → NA Make(s) of Injection Pump :: NA → NA Type(s) & Identification Number of Injection Pump:: NA → NA System description (in the case of systems other than continuous injection give equivalent details):: NA → NA Make and type of the control unit:: NA → NA Make and type of the fuel regulator:: NA → NA Make and type of the air-flow sensor:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Make and type of the micro-switch:: NA → NA Make and type of the throttle housing:: NA → NA Make and type of the water temperature sensor:: NA → NA Make and type of the air temperature sensor:: NA → NA Injectors: Opening pressure (Specify the tolerances):,..... (kPa) or characteristic diagram:: NA → NA Make(s):: NA → NA Type(s):: NA → NA Identification Number: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Cut-off point under load (min-1): NA → NA Injection timing:: NA → NA Cold start system:: NA → NA Operating principle(s):: NA → NA Operating limits/settings(Specify the tolerances): NA → NA Pressure, (Specify the tolerances).....(kPa) or characteristic diagram:: 320 ± 50 kPa → 320 ± 50 kPa Rated voltage:...V, positive/negative ground: 12V, Negative Ground → 12V, Negative Ground		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Generator/ Alternator: Alternator → Alternator Type:: DC – Alternator with Voltage regulator → DC – Alternator with Voltage regulator Nominal output: ...VA: 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps → 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps Ignition: NA → NA Make(s):: NA → NA Type(s):: NA → NA Working principle:: NA → NA Ignition advance curve or map(Specify the tolerances):: NA → NA Static ignition timing (Specify the tolerances):.....degrees before TDC: NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	→ NA Spark plugs: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Gap setting----------(mm): NA → NA Nominal resistance (kilo ohm) (if resistive type): NA → NA Ignition coil(s): NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Ignition condenser: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA EMI suppressor cap / Device / Electronic unit: NA → NA Make: NA → NA Type (Resistive/Capacitive): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification number: NA → NA Nominal resistance (kilo ohm): NA → NA Terminology and Drawing of interference Suppression equipment: NA → NA H.T. Cable: NA → NA Make and Place: NA → NA Type (Resistive/Non-resistive): NA → NA Length (mm) (if resistive type): NA → NA Outside dia. (mm) (if resistive type): NA → NA Nominal resistance kilo ohm, (if resistive type): NA → NA Radiator drawing(s): Refer enclosure	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	4C2 & 4C4 → Refer enclosure 4C2 & 4C4 Radiator drawing(s): Refer enclosurev 4C1 & 4C3 → Refer enclosurev 4C1 & 4C3 Make: Mahle Behr → Mahle Behr Make: Tata Toyo → Tata Toyo Type (s): Cross flow → Cross flow Relief valve pressure setting: 0.9 kg/cm2 → 0.9 kg/cm2 Fan characteristics (Fan power, kW) Enclose the fan power curve corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part No: 9 → 9 Material of blades (metal /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ± 150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150 (AC) Radiator core open area (cm^2): 1906 cm2 → 1906 cm2 Cooling system: liquid/air: Liquid → Liquid	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Nominal setting of the engine temperature control o mechanism (C)::: 85 deg C → 85 deg C Liquid: Yes → Yes Nature of liquid:: Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L → Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L Circulating pump(s): (Yes/No): Yes → Yes Characteristics of Circulating pump: Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm → Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm Make(s):: XXXXX		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Fasteners Ltd → XXXXX Fasteners Ltd Type(s):: Impeller Type → Impeller Type Drive ratio(s):: 1: 1.12 → 1: 1.12 Description of the fan and its drive mechanism:: Electric Fan , Motor Driven → Electric Fan , Motor Driven Blower: (Yes/No): No → No Characteristics of blower: NA → NA Make(s):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Type(s):: NA → NA Drive ratio(s):: NA → NA Pressure charger: (Yes/No) Turbocharger / Supercharger: Yes, Turbo charger → Yes, Turbo charger Make(s):: M/s. XYZ, India → M/s. XYZ, India Type(s) & Identification Number:: Waste Gate with Vacuum Actuator, X3M1XXXX → Waste Gate with Vacuum Actuator, X3M1XXXX Description of the system (maximum charge pressure:.....(kPa), waste-gate, if applicable): Waste gate turbocharger with vacuum actuator Maximum Charge pressure - 145 kPa → Waste gate turbocharger with vacuum actuator Maximum Charge pressure - 145 kPa Inter-cooler: (Yes/No): Yes → Yes Type: air-air / air-water: Air – Air → Air – Air Make :: Mahle Behr → Mahle Behr Make :: Tata Toyo → Tata Toyo		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: B1V42XXX → B1V42XXX Identification Number:: B1V42YYY → B1V42YYY Air pressure drop across the intercooler: 15 kPa → 15 kPa Intake depression at rated engine speed and at 100 per cent load (Specify	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	location of measurement) Minimum allowable:... (kPa) Maximum allowable:... (kPa): Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Description and drawings of inlet pipes and their accessories (plenum chamber, heating device, additional air intakes, etc.):: Intake manifold with Electronic throttle valve Refer Enclosure 4D → Intake manifold with Electronic throttle valve Refer Enclosure 4D Intake manifold description including dimensional drawings and/or photographs):: Aluminium Intake manifold with pressure sensor Refer Enclosure 4D → Aluminium Intake manifold with pressure sensor Refer Enclosure 4D Air filter, Dimensional drawings: Refer Enclosure 4E → Refer Enclosure 4E		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Make(s):: M/s. Fleet guard → M/s. Fleet guard Type(s) & Identification Number:: Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 → Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 Intake silencer, Dimensional drawing: NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	→ NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description and Dimensional drawing of the exhaust manifold:: Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F → Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F Description and drawing with dimensions of the exhaust system:(Description with general arrangement of exhaust system along with its routing indicating the lengths of exhaust pipe, tail pipe and: Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	O2 sensors provided before and after LNT-DPF assembly → Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly Effective volume of exhaust-system (specify the tolerance & range) in liters (from exhaust manifold / TC outlet to tail pipe end), Enclose the exhaust system dimensional drawing and indicate the volume of each parts clearly: 13.05 ± 1 litre Refer Enclosure 4G → 13.05 ± 1 litre Refer Enclosure 4G Description and drawing of the exhaust Muffler/Silencer:: NA → NA Type(s) & Identification Number:: NA → NA Make: NA → NA Number:.....: NA → NA Internal diameter of exhaust pipe (mm)(min): diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G → diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G Minimum distance between exhaust pipe(s) and the fuel		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	line: >150 mm → >150 mm Auxiliary Noise shields for compliance to IS 3028 and / OR AIS-020 (If Provided):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA → NA Material: NA → NA Layout of noise shield / Photographs / Diagram Showing arrangements indicating fitment on vehicle.: NA → NA Maximum allowable exhaust back pressure at rated engine speed and at 100 per cent load (location of measurement): ... (kPa): 30.0 kpa (150 mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Minimum cross-sectional areas of inlet and outlet ports:: Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm → Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm Inlet (mm): 8.00 → 8.00 Exhaust (mm): 8.00 → 8.00 Opening: 17° bTDC → 17° bTDC Closing: 51° aBDC → 51° aBDC Opening: 66° bBDC → 66° bBDC Closing: 25° aTDC → 25° aTDC Transfer: NA → NA Opening: NA → NA Closing: NA → NA Reference and/or setting ranges (Specify the tolerances):: On Cold Engine → On Cold Engine Inlet: 0.3 to 0.4 mm → 0.3 to 0.4 mm Exhaust: 0.3 to 0.4 mm → 0.3 to 0.4 mm		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Distribution by ports: NA → NA Volume of crank-case cavity with piston	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	at TDC: NA → NA Reed valve fitted (Yes / No): No → No Description of inlet ports, scavenging and exhaust ports with corresponding timing.: NA → NA Device for recycling crankcase gases (description and drawings):: Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J → Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx Catalyst (if any, and if not covered by another heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF / DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No Element – 1 No. → Catalytic Converter – 1No Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	for LNT (periodic / continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:……cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap Bulk Density (GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle and Characteristics:: Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	downstream SCR catalyst for the reduction of NOX during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY Type and Identification number:: Type - Passive NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite → Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA Make:: NA → NA Type and Idetification number :: NA → NA NOx Sensor (Before /After of SCR):: No → No Make:: NA → NA Type and Identification number:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent engine test bench cycles, between two cycles where regenerative phases occur under the conditions equivalent to Type I test (Distance "D" in t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration (Distance “d” in AIS 137): 2 cycles → 2 cycles Description of method employed to determine the number of cycles	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA International standard (where appropriate):: NA → NA Frequency of reagent refill: continuous/maintenance (where appropriate):: NA → NA Normal operating temperature range (°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY → YYYYYYY Air injection: (Yes/No): No → No Type (pulse air, air pump, etc.) & Identification Number:: NA → NA Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other): Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY → FEW0XXYY Characteristics (flow rate, etc.):: Flow rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes EGR Cooler Make and Identification Number: M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:.....g: NA → NA Canister working capacity:.....g: NA → NA Canister bed volume:.....litre: NA → NA Canister Purge and vent strategy description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve : (Yes/No): NA → NA Make and Identification Number: NA →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA Operating pressure (kPa): NA → NA Liquid fuel hoses material :: NA → NA Schematic drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	drawing (Periodic / Continuous): NA → NA Particulate trap: (Yes/No): Yes → Yes Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Dimensions and shape of the particulate trap (capacity):: Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Regeneration Type Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure 4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh: 90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type Material - Silicon Carbide → Structure - Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI Electronic Control Unit (ECU): Yes → Yes	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number: P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA Misfire detection:: NA → NA Oxygen sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K → Refer Enclosure 4K Particulate traps		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method):: Refer Enclosure 4K → Refer Enclosure 4K List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K → Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K A description of the type and number of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3 EUDC A description of the type of the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	all sensed components with the strategy for fault detection and MI activation (fixed number of driving cycles or statistical method), including a list of relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive explanation for the data given in service $05 Test ID $21 to FF and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles – Diagnostics on Controller Area Network (CAN) – Part 4: Requirements for emissions- related systems", a	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137): Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA → NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA Systems to ensure the correct operation of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent deactivation of the driver inducement, for use by the rescue services [Note: cross- reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	/disable after parking: NA → NA Lowest		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct operation of NOx control measures: NA → NA Details of any devices designed to influence fuel economy (if not covered by other items):: NA → NA LPG fuelling system: (Yes/No): No → No Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to LPG or back:: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.): NA → NA NG fuelling system: (Yes/No): No → No Approval number: NA → NA Electronic engine management control	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	unit for NG fuelling: NA → NA Make(s):: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation:: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to NG or back: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.):: NA → NA Hydrogen fuelling system: (Yes/No): No → No Electronic engine management control unit for hydrogen fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to hydrogen or back:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA H2NG fuelling system: (Yes/No): No → No		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Percentage of hydrogen in the fuel (maximum specified by the manufacturer): NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to H2NG or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA Electric motor(Traction): No → No Type (winding, excitation):: NA → NA Maximum hourly output: kW(manufacturer’s declared value): NA → NA Maximum net power:kW(manufacturer’s declared value): NA → NA Maximum 30 minutes power: kW	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(manufacturer’s declared value): NA → NA Operating voltage: V: NA → NA Make and Identification Number:: → Battery (Traction): NA → NA Number of cells:: NA → NA Mass: .....(kg): NA → NA Capacity: Ah		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Amp-hours): NA → NA Position:: NA → NA Engines or motor combinations: NA → NA Hybrid Electric Vehicle: (Yes/No): No → No Category of Hybrid Electric vehicle Off Vehicle Charging/Not Off Vehicle Charging9: NA → NA Operating mode switch: with/without: NA → NA Selectable modes: NA → NA Pure electric: (Yes/No): NA → NA Pure fuel consuming: (Yes/No): NA → NA Hybrid modes: (Yes/No) (if yes, short description): NA → NA Description of the energy storage device: (traction battery, capacitor, flywheel/generator...): NA → NA Make(s):: NA → NA Type(s) :: NA → NA Identification number:: NA → NA Kind of electrochemical couple:: NA → NA Energy: ..... (for battery: voltage and capacity Ah in 2 h, for capacitor: J): NA →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA Charger: on board/external/without: NA → NA Electric machines (describe each type of electric machine separately): NA → NA Make:: NA → NA Type:: NA → NA Primary use: traction motor/generator: NA → NA When used as traction motor: mono motor / multi motors (number):: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Maximum power: kW: NA → NA Working principle:: NA → NA Direct current/alternating current/number of phases:: NA → NA Separate excitation/series/compound: NA → NA Synchronous/asynchronous: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA Identification number:: NA → NA Power controller: NA → NA Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Vehicle electric range..... km (according to AIS 137):: NA → NA Manufacturer’s recommendation for preconditioning:: NA → NA CO2 (g/km) (Applicable for category M1 with GVW<3.5 T): NA → NA Declared : (rounded to 3 decimal	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	places): NA → NA Fuel consumption (l/100 km) for Petrol, LPG or Diesel and (kg/100km) for CNG and (kWh/100 km) for Electric Driven Vehicles(Applicable for category M1 with GVW<3.5 T) (rounded to 3 decimal places)*: NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA *calculated on the declared CO2: NA → NA CO2 Reducing		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	technologies Available (Yes/No): No → No Regenerative braking: NA → NA Start-Stop System: NA → NA Tyre pressure monitoring system: NA → NA 6 or more Speed Transmission: NA → NA Any other technology (manufacture to enclose detail): NA → NA Electric energy consumption for electric vehicles Wh/km: NA → NA Electric energy consumption for pure electric vehicles Wh/km: NA → NA Electric energy consumption for externally chargeable hybrid electric vehicles: NA → NA Electric energy consumption (condition A, combined) Wh/km: NA → NA Electric energy consumption (condition	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	B, combined) Wh/km: NA → NA Electric energy consumption (weighted combined) ….Wh/km: NA → NA Temperatures permitted by the manufacturer: NA → NA Cooling system: Yes → Yes Liquid cooling: Yes → Yes Maximum temperature at outlet: …….(°C): 100 °C → 100 °C Air cooling: NA → NA Reference point:: NA → NA Maximum temperature at reference point: …….(°C): NA → NA Maximum outlet temperature of the inlet intercooler: .. (°C): 45 °C → 45		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	°C Maximum exhaust temperature at the point in the exhaust pipe(s) adjacent to the outer flange(s) of the exhaust manifold: …(°C): 740 °C → 740 °C Minimum: .. (°C): - 8 → - 8 Maximum: .. (°C): 42 → 42 Minimum: …(°C): - 5 → - 5 Maximum: …(°C): 120 → 120 Lubrication system: Yes → Yes Description of the system: Forced Lubrication system → Forced Lubrication system Position of the lubricant reservoir:: Sump beneath crankcase → Sump beneath crankcase Feed system (by pump/injection into intake/mixing with fuel, etc.): Pump →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Pump Lubricating pump: Yes → Yes Make(s):: M/s. ZXWA → M/s. ZXWA Type(s):: XXYY Type → XXYY Type Mixture with fuel: No → No Percentage:: → Oil cooler: (Yes/No): Yes → Yes Drawing(s):..., or: Enclosure 4L → Enclosure 4L Make(s):: M/s. XXYY or Any other equivalent → M/s. XXYY or Any other equivalent Type(s):: DONUT → DONUT Moment of inertia of engine flywheel:: 0.1212 kg/m2 → 0.1212 kg/m2 Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Transmission Control Unit (ECU) (Yes/No):: No → No Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA OBD IUPR Family if any: XX+XYY → XX+XYY exhaust outlet location, indicated in a Schematic dimensional drawing.): Refer Enclosure 4G → Refer Enclosure 4G 1.2.11.2.1.1 4.4 · 3. · Calibration Verification Number:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Catalyst · P0420 · Oxygen sensor 1 and 2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Engine speed, engine load, A/F mode, catalyst temperature → Engine speed, engine load, A/F mode, catalyst temperature Catalyst · P0420 · Oxygen sensor 1 and 2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Two Type I cycles → Two Type I cycles Catalyst · P0420 · Oxygen sensor 1 and 2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Type I → Type I Type (Manual / power): Manual → Manual No. of wipers: Two → Two Make: XXYY System →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	XXYY System Type: 12V → 12V Identification mark: X5XYZ100 → X5XYZ100 Rated voltage: 12V → 12V Number of sweep Frequencies: 2 → 2 Highest sweep frequency (Cycles / min): 62±8 rpm → 62±8 rpm	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Lowest sweep frequency (Cycles / min): 40±5 rpm → 40±5 rpm Wiper arm: Provided → Provided Length: 500mm → 500mm Make: M/s XYZ → M/s XYZ Length: 450 mm → 450 mm Make: BCS (SYNDICATE WIPER SYSTEM) → BCS (SYNDICATE WIPER SYSTEM) Identification: X5XY0258 → X5XY0258 Washer tank: BWXY4501 → BWXY4501 Type: Plastic → Plastic Make: XYZ → XYZ Identification No. / Part No.: X5BXYZ168 → X5BXYZ168 Capacity l: 1.5Ltr → 1.5Ltr Material: H.D.P.E → H.D.P.E Washer tank motor / Washer pump: ELECTRIC MOTOR WITH PUMP → ELECTRIC MOTOR WITH PUMP Make: XYZ → XYZ Model: PEAD64XYZS00 → PEAD64XYZS00 Nozzle(s): XYZ → XYZ No. Of Nozzles: 2 → 2 Defroster: NA → NA Type: - → - Make: - → - Details of Demisting and / or Defrosting system: - → - Schematic Diagram of Engine cooling system: - → - Schematic Diagram of Heating system: - →		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	- Cross sectional view of complete air handling system including position of louvers (vents) on dash board: - → - Blower identification: - → - Blower Voltage and capacity (Watts): - → - If defrost system is fitted, specify defrost test temperature (-8 +/- 2 deg C / -18 +/- 3 deg C): → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	& related dimensions as per related standards.: → Description and sketch showing the detail positions of Tell-Tale and seating layout (For M category vehicles): NA → NA A brief technical description indicating the length of the cleaning period, the consumption of cleaning fluid during the cleaning period and the minimum capacity of the container provided;: -- → -- Description and drawings/photographs	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	of the shapes and constituent materials of the part of the body forming the engine compartment and the part of the passenger compartment nearest to it:: -- → -- Drawings or photographs of the position of the metal components housed in the engine compartment (e.g. heating appliances, spare wheel, air filter, steering mechanism, etc.):: -- → -- List of all subassemblies, which includes an electronic oscillator or switching frequency greater than 9 kHz (like ECU, steering, suspension, Body: -- → -- List of all Electrical components, which include Broadband EMI sources (like HAVC Motor,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Wiper Motor, Window motor and Horn etc.) The list shall include Device name, Make/s and Part No. /Type ID/Drawing No: -- → -- Name of the manufacturer of the system (Steering column shaft / Gear shift control / Transmission) on which the Protective Device functions: XXYY Steering System → XXYY Steering System Type/Part no./Identification number(ID)/Drawing No.of the device on which the Protective Device functions: FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable only in case of steering column locking systems: Locks the Steering → Locks the Steering SYSTEM SOFTWARE (If fitted): NA → NA Make: -- → -- Version: -- → -- Operating System Details with Version: -- → -- DESCRIPTION OF DEVICE (If fitted): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Description of the VAS: - → - A detailed description of the alarm system and of the vehicle parts related to alarm system installed with Block Diagram: → A list of main components comprising the alarm system: - → - Components placed in Engine compartment: - → - Components placed in Passenger compartment: - → - Availability of radio/wireless transmission Yes/No: - → - Details of radio/wireless transmission if any: - → - frequency: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Availability of WPC approval for wireless technology Yes/No: - → - The measures taken against false alarm: - → - Range of protection offered by the device. Details about the detection mechanism (opening of Door, Bonnet, luggage compartment, intrusion etc.): - → - Details & specification of audible alarm: - → - Details & specification of optical alarm: - → - Details & specification of radio alarm if fitted: - → - Method of setting/ unsetting the device: - → - Type of key used for unsetting (Mechanical Key / electronic device): - → - Number of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - Exit delay in Sec: - → - Entry Delay in sec: - → - Details of status display (0.5cd): - → - Details of self check , automatic failure indication: - → - Details & specification of audible alarm: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Details & specification of optical alarm: - → - Details & specification of radio alarm if fitted: - → - Method of setting/ unsetting the device: - → - Type of key used for unsetting (Mechanical Key / electronic device): - → - Number of interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - Exit delay in Sec: - → - Entry Delay in sec: - → - Details of status display (0.5cd): - → - Details of self check , automatic failure indication: - → - Details about the panic alarm , if provided: - → - Description of the Immobilizer: - → - A detailed description of the vehicle immobilizer and of the vehicle and the measures		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	taken against inadvertent activation with block diagram.: → The vehicle system(s) on which the immobilizer acts.: - → - Details of radio/wireless transmission if any: - → - Method of setting/ unsetting the device.: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Number of interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - A list of main components comprising the device: - → - Description: AL Part No. → AL Part No. Description · AL Part No.: Hella Part No → Hella Part No ECU: FXX01000 → FXX01000 ECU · FXX01000: 015.XYZ-01 → 015.XYZ-01 a. · Make: - → - b. · Type: - → - c. · Serial Number: - → - d. · Name and Address of manufacture: - → - a. · Make: - → - b. · Type: - → - c. · Serial Number: - → - d. · Name and Address of manufacture: - → - Example of Engine No: HJEZXYZ891 → HJEZXYZ891		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
AIS-137 Part 3; CMVR 115 Mass-emission conformity of production	Base fuel type used for vehicle Type approval including blend level (Ethanol, Bio-diesel etc): Diesel (Maximum 7% bio-diesel blend) → Diesel (Maximum 7% bio-diesel blend) Name and address of engine manufacturing plant (In case of imported vehicles, above details shall be supplied	CEA / report needs review	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	for importer also): Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX → Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX Max. Recommended Gradeability (Stand-start) – in degrees: 10 degs → 10 degs Max. Recommended Gradeability (Stand-start) – in degrees: 8.5 degs → 8.5 degs Vehicle kerb weight kg: Refer Annexure T7 - B Details for Recommended load body Refer Annexure T7 – C → Refer Annexure T7 - B Details for Recommended load body Refer Annexure T7 – C Other axle: NA → NA Gross combination weight kg (applicable for articulated / Tractor Trailer combination vehicles): NA → NA Front axle: NA → NA Rear axle: NA → NA Trailer axle / other axles: NA → NA If applicable, maximum mass of trailer whic5h may be coupled: NA →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	NA Unbraked trailer: NA → NA Reference mass (for vehicles with GVW less than or equal to 3.5 ton) (kg.): Refer	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Annexure T7-B → Refer Annexure T7-B Maximum (Including driver) for completely built vehicles: D+2 → D+2 Maximum (Including driver) for completely built vehicles: → Make: APOLLO → APOLLO Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade description(s): CEAT WINMILE X3-R → CEAT WINMILE X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 Tyre rolling Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer - TC7719XX000 003XXXF → Refer - TC7719XX000 003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022-2023/300002XXX	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	X/TA/XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L- 756XXXX → CM/L- 756XXXX Identification: TAC No. /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E- Marking: NA → NA Others (For articulated / combination	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00 R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other (for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 325 mm → 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	kg/cm2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26 kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Type: Fully		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	synchronized 5 - forward gears with a Constant mesh reverse gear → Fully synchronized 5 - forward gears with a Constant mesh reverse gear Model name (if any): XXX-XXX → XXX-XXX Model name (if any): YYYY-YY → YYYY-YY Gear shifting control system (Sketch showing gear shifting arrangement): → No. of gears: Forward 5 + 1 Rev → Forward 5 + 1 Rev Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Rear axle ratio: 4.56 → 5.13 Type of steering gear (Re-circulating ball / Worm & Roller / Rack & Pinion / Others): Re-circulating ball type → Re-circulating ball type Make: 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd → 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2. YYYY – 18.2 :1	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Type and description (Leaf / Coil / Air / Semi- pneumatic/ Torsion bar): Leaf → Leaf Front: Parabolic spring- Rigid axle suspension → Parabolic spring- Rigid axle suspension Rear: Semi elliptic spring-		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Rigid axle suspension → Semi elliptic spring- Rigid axle suspension Fuel Gauge: LED Bars → LED Bars Engine coolant temperature: Red Color – Thermometer Symbol → Red Color – Thermometer Symbol Engine low oil pressure: Red Color – Oil Tank Symbol → Red Color – Oil Tank Symbol Weight of seats: Driver seat : 11.5 Kg & Co-driver seat : 11Kg → Driver seat : 11.5 Kg & Co-driver seat : 11Kg Description of seats: -- → -- Make: -- → -- Number of seats fitted or capable of being fitted with head restraints, adjustable or not adjustable.: -- → -- Description of the adjustment, displacement and locking systems of the seat or of its parts and a description of occupant protection system against displacement of luggage.: -- → -- Description of seat anchorage: -- → -- Longitudinal / Vertical (If applicable) position of the seats during the tests.: --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	→ -- Drawings, diagrams and plans of the seats, their anchorages on the vehicle, the floor layout, the adjustment and displacement system of the seats and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	their parts, and their locking devices and of additional occupant protection system against displacement of luggage.: -- → -- In the case of seats fitted with head restraints, the head restraint shall be shown on all drawings, diagrams and photographs.: -- → -- Seat Identification No. / Part No/Drawing Number.: ID Number/ Part Number/D rawing number Weight Description Make (Seat (kg) Manufactu rer/ Vehicle Manufactu rer): -- → -- Description: Make → Make Description: ID Number/ Part Number/D rawing number (Seat Manufactu rer/ Vehicle Manufactu rer) → ID Number/ Part Number/D	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	rawing number (Seat Manufactu rer/ Vehicle Manufactu rer) Description: Weight (kg) → Weight (kg) Brief description and drawings of the seat type, its attachment fittings and its adjustment, displacement and locking systems including the minimum distance between fitting points.: D+1 → D+1 Installation drawing showing rear extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	make, unladen pressure) etc., distance between tyre & RUPD RH & LH, frame rear width, P1, P2 points.: - → - Make: M/s. XXXXYYYY → M/s. XXXXYYYY Material (Metallic / Plastic etc.): Metallic → Metallic Nominal thickness mm: 1.0mm → 1.0mm Capacity, Litre: 40 Lts, 30 Lts → 40 Lts, 30 Lts Detailed drawing of the fuel tank assembly with material specifications: FHSXXXXY	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FHSXXXXY → FHSXXXXY FHSXXXXY Detailed drawing indicating the position / location of the fuel tank(s) in the vehicle.: A5LXXX00 (50 LTS) A5LXXX00 (40 LTS) → A5LXXX00 (50 LTS) A5LXXX00 (40 LTS) Identification No. / Part No.: FHS00544 FfffXXXcdHXXS00744 → FHS00544 FfffXXXcdHXXS00744 Detailed engineering drawing as specified in AIS – 013 or photographs showing the mounting details with dimensions.: NA → NA Cross sectional Drawing of Gear shift lever (Knob): FXA0XXYY (Non AC) / FXA0XXYY (AC) → FXA0XXYY (Non AC) / FXA0XXYY (AC) Drawings and/or photographs of the locations of the statutory plates and inscriptions and of the vehicle		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	identification number: X4W0XYZ4 / VFT0XYZ2 → X4W0XYZ4 / VFT0XYZ2 Drawings and / or photographs of the official part of the plates and inscriptions (completed example with dimensions): X4WXYZ44 → X4WXYZ44 Location of Vehicle Descriptor Section (VDS) in Vehicle Identification Number (VIN): 4thto 9thCharacter of 17 Characters VIN → 4thto 9thCharacter of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	17 Characters VIN Brief description of the Provision for emergency exit provided in case of sealed windows: -- → -- Dimensions of the instructions describing each motion necessary to unlatch and open the exit are provided from the release mechanism.: -- → -- In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → -- Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side, distance between front extremity & FUPD, width of FUPD, ID		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Brief description of the vehicle type as regards its structure, dimensions, lines and constituent materials:: -- → -- Description of the protective system installed in the vehicle:: -- → -- Description of the protective system installed in the vehicle:: → Description of interior arrangements or fittings that might affect the tests:: -- → -- A detailed description, including	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	photographs and/or drawings, of the vehicle type with respect to the structure, the dimensions, the design and the constituent materials, the side walls of the passenger compartment (exterior and interior), including specific details of the protection system, where applicable:: -- → -- Drawing / Images of position & arrangement of Engine: -- → -- Drawing/ Images of Under bonnet (Engine house) Components.: -- → -- Construction: Integrated / Attached with Mounts: 4 Mounts, Fixed CABIN → 4 Mounts, Fixed CABIN 1st ·		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st · Overall Ratio / AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69 2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC: 12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 → 20.44 Description · Make: ID		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number/ Part Number/Drawing number (Seat Manufacturer/ Vehicle Manufacturer) → ID Number/ Part Number/Drawing number (Seat Manufacturer/ Vehicle Manufacturer) Description · Make: Weight (kg) → Weight (kg) B 51.0 B49.0 · Pedestrian protection as per AIS-100: The distance, measured longitudinally on a horizontal plane, between the transverse centerline of the front axle & the R point of the driver’s seat is less than 1100 mm (Yes / No).: NA → NA Brief description of failure warning	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	tell–tale: - → - Brief description and features: - → - Brief description and features: - → - Description: -- → -- Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No Manufacturer’s documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	version as appropriate.: -- → -- Under fault conditions, vehicle response corresponds to that described in the manufacturer’s documents / failure analysis and safety concept is verified.: -- → -- A brief written explanation sufficient to describe the ESC system basic operational characteristics.: -- → -- Logic diagram: -- → -- Understeer information.: -- → -- Static Stability Factor: -- → -- Make and Country of manufacturer(if imported): -- → -- Software Id / version: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Brief description of failure warning tell–tale: -- → -- Brief description and features: -- → -- Brief description and features: -- → -- System identification including software level identifier: -- → -- List of motor vehicles by description and configuration that are covered by the information document: -- → -- Wheel type (single or twin) and tyre type (e.g. structure, category of use, size): -- → -- Gearbox type (e.g. manual, automated manual, semi-automatic, automatic): -- → -- Management of the engine or any		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	other source(s) of motive power e.g.Torque Management, control of fuel supply etc.: -- → -- Component description:: -- → -- Sensor(s) External to the Controller: -- → -- Function: -- → -- Limitations on the location of the sensors: -- → -- Identification (e.g. part numbers): -- → -- Controller(s): -- → -- General description and function: -- → -- Functionality of internal sensors (if applicable): -- → -- Hardware identification (e.g. part numbers): -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Software identification: -- → -- Limitations on the location of the controller(s): -- → -- Additional features: -- → -- Modulators: -- → -- General description and function: -- → -- Hardware identification (e.g. part numbers): -- → -- Software identification (if applicable): -- → -- Limitations: -- → -- Electrical Equipment: -- → -- Circuit diagrams: -- → -- Powering methods: -- → -- Pneumatic circuits: -- → -- Safety aspects of the electronic system in accordance with Annex S of IS 11852: -- → -- List of complex functions covered::		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	-- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Manufacturer’s documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- A list of input and output variables, including their working range, is	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	provided.: -- → -- Documentation includes an inventory of components, outlines the function of each unit, its interconnection/interaction with other systems and defines signal flow priorities.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software version as appropriate.: -- → -- Safety concept statement verifies that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- System architecture, design methods and tools identified.: -- → -- Information regarding		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	design provisions in case of failure is provided including any error messages, warning signals, partial performance conditions, back-up modes and/or removal of high- level functions.: -- → -- Additional material and analysis (FMEA, FTA) of fault conditions made available for inspection and maintained by the manufacturer.: -- → -- Document reference and date of inspection:: -- → -- Vehicle complies with all other performance requirements of IS 15986:2015 and meets manufacturer’s specifications under non-fault conditions.: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Under fault conditions, vehicle response corresponds to that described in the manufacturer’s documents / failure analysis and safety concept is verified.: -- → -- Electro-magnetic compatibility: -- → -- Documentation demonstrating compliance with AIS-004(Part 3).: -- → -- Component description: -- → -- Controller(s) (a) General description and function; (b) Identification e.g. part numbers; (c) Limitations on the location of the controller(s); (d) Additional		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	features.: -- → -- Modulators (a) General description and function; (b) Identification; (c) Limitations.: -- → -- Manufacturer’s documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software version as appropriate.: -- → -- (see annex 4 of this report for test data): -- → -- (see annex 4 of this report for test data): -- → -- Type / Description: Fully built vehicle, D+2, Non AC, 3.39 Ton GVW → Fully	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	built vehicle, D+2, Non AC, 3.39 Ton GVW Make: Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. → Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. Model name / identification: ALXYZABC0 → ALXYZABC0 Type: Compression Ignition → Compression Ignition Bore x stroke (mm): 80 X 98 → 80 X 98 No. of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	cylinders: 3 → 3 Displacement (cm³): 1468 cc → 1468 cc Compression ratio: 16.5±1.0 : 1 → 16.5±1.0 : 1 Max. Engine output (KW @ rpm): 55.1 Kw @ 3300 rpm → 55.1 Kw @ 3300 rpm Max. Torque (Nm @ rpm): 170Nm @ 1600 to 2400 rpm → 170Nm @ 1600 to 2400 rpm Air cleaner type: Dry type → Dry type Type of Fuel: Diesel → Diesel Make and model: RSB Transmission (T210R) → RSB Transmission (T210R) Make and model: AVTEC (A5M210) → AVTEC (A5M210) Type: Manual gearbox with fully synchro mesh & Rev with sliding mesh → Manual	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	gearbox with fully synchro mesh & Rev with sliding mesh No. of gears: 5 Forward and 1 Reverse → 5 Forward and 1 Reverse 1st: 4.62 → 4.62 1st: 4.60 → 4.60 1st: 21.53 → 21.53 1st: 21.44 → 21.44 2nd: 2.68 → 2.68 2nd: 2.69 → 2.69 2nd: 12.67 → 12.67 2nd: 12.71 → 12.71 3rd: 1.51 → 1.51 3rd: 1.49 → 1.49 3rd: 7.34 → 7.34 3rd: 7.25 → 7.25 4th: 1.0 → 1.0 4th: 1.0 → 1.0 4th: 4.56 → 4.56 4th: 4.56 → 4.56 5th: 0.70 → 0.70 5th: 0.71 → 0.71 5th: 3.64 → 3.64 5th: 3.67 → 3.67 Reverse: 4.39 → 4.39		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Reverse: 20.36 → 20.36 Reverse: 4.38 → 4.38 Reverse: 20.34 → 20.34 Drive Axle (Front / Rear / All): Rear → Rear Front axle ratio: NA → NA Type / Description: Re-circulating ball type → Re-circulating ball type	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT 12 PR Tyre size designation including ply rating: 215/75 R15 LT → 215/75 R15 LT Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2 → 3.26 kg/cm2 Wiping system (Brief description): Twin blade, Tandem Pattern → Twin blade, Tandem Pattern Material: Metallic → Metallic Capacity (l): 50 lts , 40 lts → 50 lts , 40 lts Dimensions: 2490WB → 2490WB		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Dimensions: 2640WB → 2640WB Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Overall width (mm): 1842 → 1842 Overall Length (mm): CC → CC	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Overall Length (mm): 4750 → 4750 Overall Length (mm): 4988 → 4988 Overall Length (mm): FSD / HSD → FSD / HSD Overall Length (mm): 4925 → 4925 Overall Length (mm): 5225 → 5225 Overall height (mm): 1961 → 1961 Overall height (mm): 1948 → 1948 Maximum gradeability in 1st gear: GVW 2890 kg → GVW 2890 kg Maximum gradeability in 1st gear: GVW 3390 kg → GVW 3390 kg Maximum gradeability in 1st gear: 10 Deg → 10 Deg Maximum gradeability in 1st gear: 8.5 Deg → 8.5 Deg Weights: Base → Base Weights: Variant 1 → Variant 1 Weights: Variant 2 → Variant 2 Weights: Variant 3 → Variant 3 Weights: Variant 4 → Variant 4 Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	spare wheel, tools, etc.) (kg): - → - Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Description of energy converters and power plant. (In the case of a vehicle that can run either on petrol, diesel, etc., or also in combination with another fuel, items shall be repeated.1): Diesel → Diesel Engine Manufacturer:: Ashok Leyland Limited → Ashok Leyland Limited Manufacturer's engine model code (as marked on the engine, or other means of identification): Model Code: ALXYZABC0 Enclosure – 4A → Model Code: ALXYZABC0 Enclosure – 4A Location of engine model code and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	engine serial number on block / crankcase (drawing to be attached): Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Engine Block Refer Enclosure – 4A → Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A Working principle: positive ignition / compression- ignition, four-stroke / two-stroke / rotary cycle (DI/IDI) (NA/TC/TCIC/Any other): Compression Ignition, four-stroke, DI, TCIC → Compression Ignition, four-stroke, DI, TCIC Number, arrangement of cylinders:: 3, Inline → 3, Inline Bore----------(mm): 80 → 80 Stroke: ----------(mm): 98 → 98 Firing order:: 1-2-3 → 1-2-3 Engine capacity:----------(cm3): 1468 → 1468 Volumetric compression ratio (specify tolerance): 17.5±1.0 : 1 → 17.5±1.0 : 1 Drawings of combustion chamber and piston crown and, in the case of positive ignition engine, piston rings:: Refer Enclosure – 4B → Refer		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure – 4B Vehicle with Defeat Device Yes / No: No → No If yes – Details of the Defeat device: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane equivalent) in PPM content by volume in the exhaust gas with the engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum net power:..... kW at min-1: 55.1 KW @ 3100 ± 50 rpm → 55.1 KW @ 3100 ± 50 rpm Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm Maximum net torque: .....Nm at: -------min-1 ...(manufacturer's declared value): 170 Nm @ 1600 – 2400 rpm → 170 Nm @ 1600 – 2400 rpm Maximum rated speed (Specify tolerance): 3100 rpm ± 50 rpm → 3100 rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Type of fuel used : Mono		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Fuel / Bi-Fuel / Flex Fuel / Dual Fuel: Mono Fuel → Mono Fuel	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Fuel type: Diesel / Gasoline / LPG / CNG / Biomethane / Bio-H methane / HCNG / Biogas / LNG / Ethanol ((E85) / 2 (E100)) /ED (95) Flex fuel Methanol M15 / M100 / MD95 / Biodiesel up to 100% / Hydrogen.: Diesel → Diesel Maximum amount of biofuel acceptable in fuel (manufacturer's declared value): per cent by volume: 7% → 7% By fuel injection (compression-ignition only): (Yes/No): Yes → Yes System description:: Unit Pump Common Rail Direct Injection system → Unit Pump Common Rail Direct Injection system Working principle: direct-injection/pre-chamber/swirl Chamber: Direct Injection → Direct Injection Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Plunger Type & FHL0XYZ0 → Plunger Type & FHL0XYZ0 Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Injection advance curve (Specify the	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	tolerances):: ECU Controlled → ECU Controlled Governor: ECU Controlled → ECU Controlled Cut-off point under load:…(min-1): 3300 → 3300 Cut-off point without load (Maximum no-load speed)…(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Multi-Hole Injectors (7 holes) FKXXYYZZ → Multi-Hole Injectors (7 holes) FKXXYYZZ Opening pressure (kPa) (specify the tolerances) or characteristic diagram: 15000 to 23000 kPa → 15000 to 23000 kPa Length and internal diameter…(mm): Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm → Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm Common rail make & type: M/s. XXXXYYY, Hot Forged Rail Type → M/s. XXXXYYY, Hot Forged Rail Type Cold start system: NA → NA Make(s):: NA → NA Type(s) &		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: NA → NA Description:: NA → NA Auxiliary starting aid: NA → NA Make(s):: NA → NA Type(s):: NA → NA System description:: NA → NA Electronic controlled injection: (Yes/No): Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA → NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s. XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Metering type Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type By fuel injection (positive ignition only): (Yes/No): No → No Working principle: intake manifold (single/multi- point)/direct injection/other (specify): NA → NA Make(s) of Injection Pump :: NA → NA Type(s) & Identification Number of Injection Pump:: NA → NA System description (in the case of systems other than continuous injection give equivalent details):: NA → NA Make and type of the control unit:: NA → NA Make and type of the fuel regulator:: NA → NA Make and type of the air-flow sensor:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Make and type of the micro-switch:: NA → NA Make and type of the throttle housing:: NA → NA Make and type of the water temperature sensor:: NA → NA Make and type of the air temperature sensor:: NA → NA Injectors: Opening pressure (Specify the tolerances):,..... (kPa) or characteristic diagram:: NA → NA Make(s):: NA → NA Type(s):: NA → NA Identification Number: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA Cut-off point under load (min-1): NA → NA Injection timing:: NA → NA Cold start system:: NA → NA Operating principle(s):: NA → NA Operating limits/settings(Specify the tolerances): NA → NA Pressure, (Specify the tolerances).....(kPa) or characteristic diagram:: 320 ± 50 kPa → 320 ± 50 kPa Rated voltage:...V, positive/negative ground: 12V, Negative Ground → 12V, Negative Ground	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Generator/ Alternator: Alternator → Alternator Type:: DC – Alternator with Voltage regulator → DC – Alternator with Voltage regulator Nominal output: ...VA: 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps → 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps Ignition: NA → NA Make(s):: NA → NA Type(s):: NA → NA Working principle:: NA → NA Ignition advance curve or map(Specify the tolerances):: NA → NA Static ignition timing (Specify the tolerances):.....degrees before TDC: NA → NA Spark plugs: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Gap setting----------(mm): NA → NA Nominal resistance (kilo ohm) (if resistive type): NA → NA Ignition coil(s): NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Ignition condenser: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA EMI suppressor cap / Device / Electronic	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	unit: NA → NA Make: NA → NA Type (Resistive/Capacitive): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification number: NA → NA Nominal resistance (kilo ohm): NA → NA Terminology and Drawing of interference Suppression equipment: NA → NA H.T. Cable: NA → NA Make and Place: NA → NA Type (Resistive/Non-resistive): NA → NA Length (mm) (if resistive type): NA → NA Outside dia. (mm) (if resistive type): NA → NA Nominal resistance kilo ohm, (if resistive type): NA → NA Radiator drawing(s): Refer enclosure 4C2 & 4C4 → Refer enclosure 4C2 & 4C4 Radiator drawing(s): Refer enclosurev 4C1 & 4C3 → Refer enclosurev 4C1 & 4C3 Make: Mahle Behr → Mahle Behr Make: Tata Toyo → Tata Toyo Type (s): Cross flow → Cross flow Relief valve pressure setting: 0.9 kg/cm2 → 0.9 kg/cm2 Fan characteristics (Fan power, kW) Enclose the fan power curve corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	No: 9 → 9 Material of blades (metal /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ± 150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150 (AC) Radiator core open area (cm²): 1906 cm2 → 1906 cm2 Cooling system: liquid/air: Liquid → Liquid Nominal setting of the engine temperature control o mechanism (C)::: 85 deg C → 85 deg C Liquid: Yes → Yes Nature of liquid:: Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L → Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L Circulating pump(s): (Yes/No): Yes → Yes Characteristics of Circulating pump:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm → Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm Make(s):: XXXXX		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Fasteners Ltd → XXXXX Fasteners Ltd Type(s):: Impeller Type → Impeller Type Drive ratio(s):: 1: 1.12 → 1: 1.12 Description of the fan and its drive mechanism:: Electric Fan , Motor Driven → Electric Fan , Motor Driven Blower: (Yes/No): No → No Characteristics of blower: NA → NA Make(s):: NA → NA Type(s):: NA → NA Drive ratio(s):: NA → NA Pressure charger: (Yes/No) Turbocharger / Supercharger: Yes, Turbo charger → Yes, Turbo charger Make(s):: M/s. XYZ, India → M/s. XYZ, India Type(s) & Identification Number:: Waste Gate with Vacuum Actuator, X3M1XXXX → Waste Gate with Vacuum Actuator, X3M1XXXX Description of the system (maximum charge pressure:.....(kPa), waste-gate, if applicable): Waste gate turbocharger with vacuum actuator Maximum Charge pressure - 145 kPa → Waste gate turbocharger with vacuum actuator	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Maximum Charge pressure - 145 kPa Inter-cooler: (Yes/No): Yes → Yes Type: air-air / air-water: Air – Air → Air – Air Make :: Mahle Behr → Mahle Behr Make :: Tata Toyo → Tata Toyo		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: B1V42XXX → B1V42XXX Identification Number:: B1V42YYY → B1V42YYY Air pressure drop across the intercooler: 15 kPa → 15 kPa Intake depression at rated engine speed and at 100 per cent load (Specify location of measurement) Minimum allowable:... (kPa) Maximum allowable:... (kPa): Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Description and drawings of inlet pipes and their accessories (plenum chamber, heating device, additional air intakes, etc.):: Intake manifold with Electronic throttle valve Refer Enclosure 4D → Intake manifold with Electronic throttle valve Refer Enclosure 4D Intake manifold description including dimensional drawings and/or	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	photographs):: Aluminium Intake manifold with pressure sensor Refer Enclosure 4D → Aluminium Intake manifold with pressure sensor Refer Enclosure 4D Air filter, Dimensional drawings: Refer Enclosure 4E → Refer Enclosure 4E		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Make(s):: M/s. Fleet guard → M/s. Fleet guard Type(s) & Identification Number:: Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 → Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 Intake silencer, Dimensional drawing: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description and Dimensional drawing of the exhaust manifold:: Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F → Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F Description and drawing with dimensions of the exhaust system:(Description with general arrangement of exhaust system along with its routing	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	indicating the lengths of exhaust pipe, tail pipe and: Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	O2 sensors provided before and after LNT-DPF assembly → Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly Effective volume of exhaust-system (specify the tolerance & range) in liters (from exhaust manifold / TC outlet to tail pipe end), Enclose the exhaust system dimensional drawing and indicate the volume of each parts clearly: 13.05 ± 1 litre Refer Enclosure 4G → 13.05 ± 1 litre Refer Enclosure 4G Description and drawing of the exhaust Muffler/Silencer:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Type(s) & Identification Number:: NA → NA Make: NA → NA Number:……: NA → NA Internal diameter of exhaust pipe (mm)(min): diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G → diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G Minimum distance between exhaust pipe(s) and the fuel		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	line: >150 mm → >150 mm Auxiliary Noise shields for compliance to IS 3028 and / OR AIS-020 (If Provided): NA → NA Material: NA → NA Layout of noise shield / Photographs / Diagram Showing arrangements indicating fitment on vehicle.: NA → NA Maximum allowable exhaust back pressure at rated engine speed and at 100 per cent load (location of measurement): … (kPa): 30.0 kpa (150 mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Minimum cross-sectional areas of inlet and outlet ports:: Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm → Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Inlet (mm): 8.00 → 8.00 Exhaust (mm): 8.00 → 8.00 Opening: 17° bTDC → 17° bTDC Closing: 51° aBDC → 51° aBDC Opening: 66° bBDC → 66° bBDC Closing: 25° aTDC → 25° aTDC Transfer: NA → NA Opening: NA → NA Closing: NA → NA Reference and/or setting ranges (Specify the tolerances):: On Cold Engine → On Cold Engine Inlet: 0.3 to 0.4 mm → 0.3 to 0.4 mm Exhaust: 0.3 to 0.4 mm → 0.3 to 0.4 mm		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Distribution by ports: NA → NA Volume of crank-case cavity with piston at TDC: NA → NA Reed valve fitted (Yes / No): No → No Description of inlet ports, scavenging and exhaust ports with corresponding timing.: NA → NA Device for recycling crankcase gases (description and drawings):: Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J → Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx Catalyst (if any, and if not covered by another	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF / DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No Element – 1 No. → Catalytic Converter – 1No	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	for LNT (periodic / continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:.....cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Bulk Density (GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle and Characteristics:: Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	downstream SCR catalyst for the reduction of NOX during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Type and Identification number:: Type - Passive NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite → Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA Make:: NA → NA Type and Idetification number :: NA → NA NOx Sensor (Before /After of SCR):: No → No Make:: NA → NA Type and Identification number:: NA → NA Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent engine test bench cycles, between two cycles where regenerative phases occur under the conditions equivalent to Type I test (Distance "D" in t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration (Distance “d” in AIS 137): 2 cycles → 2 cycles Description of method employed to determine the number of cycles between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA International standard (where appropriate):: NA → NA Frequency of reagent refill: continuous/maintenance (where appropriate):: NA → NA Normal operating temperature range (°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY → YYYYYYY Air injection: (Yes/No): No → No	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Type (pulse air, air pump, etc.) & Identification Number:: NA → NA Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other): Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY → FEW0XXYY Characteristics (flow rate, etc.):: Flow rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes EGR Cooler Make and Identification Number: M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:.....g: NA → NA Canister working capacity:.....g: NA → NA Canister bed volume:.....litre: NA → NA Canister Purge and vent strategy description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve : (Yes/No): NA → NA Make and Identification Number: NA → NA Operating pressure (kPa): NA → NA Liquid fuel hoses material :: NA → NA Schematic drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	drawing (Periodic / Continuous): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Particulate trap: (Yes/No): Yes → Yes Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dimensions and shape of the particulate trap (capacity):: Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Regeneration Type Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure 4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh: 90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type Material - Silicon Carbide → Structure - Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI Electronic Control Unit (ECU): Yes → Yes Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number: P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA Misfire detection:: NA → NA Oxygen sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K → Refer Enclosure 4K Particulate traps	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method):: Refer Enclosure 4K → Refer Enclosure 4K List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K → Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K A description of the type and number of the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3 EUDC A description of the type of the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	all sensed components with the strategy for fault detection and MI activation (fixed number of driving cycles or statistical method), including a list of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive explanation for the data given in service $05 Test ID $21 to FF and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles – Diagnostics on Controller Area Network (CAN) – Part 4: Requirements for emissions- related systems", a comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA → NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA Systems to ensure the correct operation of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent deactivation of the driver inducement, for use by the rescue services [Note: cross- reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling /disable after parking: NA → NA Lowest		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct operation of NOx control measures: NA → NA Details of any devices designed to influence fuel economy (if not covered by other items):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	LPG fuelling system: (Yes/No): No → No Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to LPG or back:: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.): NA → NA NG fuelling system: (Yes/No): No → No Approval number: NA → NA Electronic engine management control unit for NG fuelling: NA → NA Make(s):: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation:: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NG or back: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.):: NA → NA Hydrogen fuelling system: (Yes/No): No → No Electronic engine management control unit for hydrogen fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to hydrogen or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA H2NG fuelling system: (Yes/No): No → No		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Percentage of hydrogen in the fuel (maximum specified by the manufacturer): NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to H2NG or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA Electric motor(Traction): No → No Type (winding, excitation):: NA → NA Maximum hourly output: kW(manufacturer’s declared value): NA → NA Maximum net power:kW(manufacturer’s declared value): NA → NA Maximum 30 minutes power: kW (manufacturer’s declared value): NA → NA Operating voltage: V: NA → NA Make and Identification Number:: → Battery (Traction): NA → NA Number of cells:: NA → NA Mass: ……(kg): NA → NA Capacity: Ah		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Amp-hours): NA → NA Position:: NA → NA Engines or motor combinations: NA → NA Hybrid Electric Vehicle: (Yes/No): No → No	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Category of Hybrid Electric vehicle Off Vehicle Charging/Not Off Vehicle Charging9: NA → NA Operating mode switch: with/without: NA → NA Selectable modes: NA → NA Pure electric: (Yes/No): NA → NA Pure fuel consuming: (Yes/No): NA → NA Hybrid modes: (Yes/No) (if yes, short description): NA → NA Description of the energy storage device: (traction battery, capacitor, flywheel/generator...): NA → NA Make(s):: NA → NA Type(s) :: NA → NA Identification number:: NA → NA Kind of electrochemical couple:: NA → NA Energy: ..... (for battery: voltage and capacity Ah in 2 h, for capacitor: J): NA → NA Charger: on board/external/without: NA → NA Electric machines (describe each type of electric machine separately): NA → NA Make:: NA → NA Type:: NA → NA Primary use: traction motor/generator: NA → NA When used as traction motor: mono motor / multi motors (number):: NA → NA		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Maximum power: kW: NA → NA Working principle:: NA → NA Direct current/alternating current/number of phases:: NA → NA Separate excitation/series/compound: NA → NA Synchronous/asynchronous: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA Identification number:: NA → NA Power controller: NA → NA Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Vehicle electric range..... km (according to AIS 137):: NA → NA Manufacturer’s recommendation for preconditioning:: NA → NA CO2 (g/km) (Applicable for category M1 with GVW<3.5 T): NA → NA Declared : (rounded to 3 decimal places): NA → NA Fuel consumption (l/100 km) for Petrol, LPG or Diesel and (kg/100km) for CNG and (kWh/100 km) for Electric Driven Vehicles(Applicable for category M1 with GVW<3.5 T) (rounded to 3 decimal places)*: NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA *calculated on the declared CO2: NA → NA CO2 Reducing	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	technologies Available (Yes/No): No → No Regenerative braking: NA → NA Start-Stop System: NA → NA Tyre pressure monitoring system: NA → NA 6 or more Speed Transmission: NA → NA Any other technology (manufacture to enclose detail): NA → NA Electric energy consumption for electric vehicles Wh/km: NA → NA Electric energy consumption for pure electric vehicles Wh/km: NA → NA Electric energy consumption for externally chargeable hybrid electric vehicles: NA → NA Electric energy consumption (condition A, combined) Wh/km: NA → NA Electric energy consumption (condition B, combined) Wh/km: NA → NA Electric energy consumption (weighted combined) ….Wh/km: NA → NA Temperatures permitted by the manufacturer: NA → NA Cooling system: Yes → Yes Liquid cooling: Yes → Yes Maximum temperature at outlet: …….(°C): 100 °C → 100 °C Air cooling: NA → NA Reference point:: NA → NA Maximum temperature at reference point: …….(°C): NA → NA Maximum outlet temperature of the inlet intercooler: .. (°C): 45 °C → 45	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	°C Maximum exhaust temperature at the point in the exhaust pipe(s) adjacent to the outer flange(s) of the exhaust manifold: ...(°C): 740 °C → 740 °C Minimum: .. (°C): - 8 → - 8 Maximum: .. (°C): 42 → 42 Minimum: ...(°C): - 5 → - 5 Maximum: ...(°C): 120 → 120 Lubrication system: Yes → Yes Description of the system: Forced Lubrication system → Forced Lubrication system Position of the lubricant reservoir:: Sump beneath crankcase → Sump beneath crankcase Feed system (by pump/injection into intake/mixing with fuel, etc.): Pump → Pump Lubricating pump: Yes → Yes Make(s):: M/s. ZXWA → M/s. ZXWA Type(s):: XXYY Type → XXYY Type Mixture with fuel: No → No Percentage:: → Oil cooler: (Yes/No): Yes → Yes Drawing(s):..., or: Enclosure 4L → Enclosure 4L Make(s):: M/s. XXYY or Any other equivalent → M/s. XXYY or Any other equivalent Type(s):: DONUT → DONUT Moment of inertia of engine flywheel:: 0.1212 kg/m2 → 0.1212 kg/m2 Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Transmission Control Unit (ECU) (Yes/No):: No → No Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA OBD IUPR Family if any: XX+XYY → XX+XYY exhaust outlet location, indicated in a Schematic dimensional drawing.): Refer Enclosure 4G → Refer Enclosure 4G 1.2.11.2.1.1 4.4 · 3. · Calibration Verification Number:: NA → NA Catalyst · P0420 · Oxygen sensor 1 and 2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Engine speed, engine load, A/F mode, catalyst temperature → Engine speed, engine load, A/F mode, catalyst temperature Catalyst · P0420 · Oxygen sensor 1 and 2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Two Type I cycles → Two Type I cycles Catalyst · P0420 · Oxygen sensor 1 and	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Type I → Type I Type (Manual / power): Manual → Manual No. of wipers: Two → Two Make: XXYY System →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	XXYY System Type: 12V → 12V Identification mark: X5XYZ100 → X5XYZ100 Rated voltage: 12V → 12V Number of sweep Frequencies: 2 → 2 Highest sweep frequency (Cycles / min): 62±8 rpm → 62±8 rpm Lowest sweep frequency (Cycles / min): 40±5 rpm → 40±5 rpm Wiper arm: Provided → Provided Length: 500mm → 500mm Make: M/s XYZ → M/s XYZ Length: 450 mm → 450 mm Make: BCS (SYNDICATE WIPER SYSTEM) → BCS (SYNDICATE WIPER SYSTEM) Identification: X5XY0258 → X5XY0258 Washer tank: BWXY4501 → BWXY4501 Type: Plastic → Plastic Make: XYZ → XYZ Identification No. / Part No.: X5BXYZ168 → X5BXYZ168 Capacity l: 1.5Ltr → 1.5Ltr Material: H.D.P.E → H.D.P.E Washer tank motor / Washer pump: ELECTRIC MOTOR WITH PUMP → ELECTRIC MOTOR WITH PUMP	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make: XYZ → XYZ Model: PEAD64XYZS00 → PEAD64XYZS00 Nozzle(s): XYZ → XYZ No. Of Nozzles: 2 → 2 Defroster: NA → NA Type: - → - Make: - → - Details of Demisting and / or Defrosting system: - → - Schematic Diagram of Engine cooling system: - → - Schematic Diagram of Heating system: - →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	- Cross sectional view of complete air handling system including position of louvers (vents) on dash board: - → - Blower identification: - → - Blower Voltage and capacity (Watts): - → - If defrost system is fitted, specify defrost test temperature (-8 +/- 2 deg C / -18 +/- 3 deg C): → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar		
(continued)	& related dimensions as per related standards.: → Description and sketch showing the detail positions of Tell-Tale and seating layout (For M category vehicles): NA → NA A brief technical description indicating the length of the cleaning period, the consumption of cleaning fluid during the cleaning period and the minimum capacity of the container provided;: -- → -- Description and drawings/photographs of the shapes and constituent materials of the part of the body forming the engine compartment and the part of the passenger compartment nearest to it:: -- → -- Drawings or photographs of the position of the metal components housed in the engine compartment (e.g. heating appliances, spare wheel, air filter, steering mechanism, etc.):: -- → -- List of all subassemblies, which includes an electronic oscillator or switching frequency greater than 9 kHz (like ECU, steering, suspension, Body: -- → -- List of all Electrical components, which include Broadband EMI sources (like HAVC Motor,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Wiper Motor, Window motor and Horn etc.) The list shall include Device name, Make/s and Part No. /Type ID/Drawing No: -- → -- Name of the manufacturer of the system (Steering column shaft / Gear shift control / Transmission) on which the Protective Device functions: XXYY Steering System → XXYY Steering System Type/Part no./Identification number(ID)/Drawing No.of the device on which the Protective Device functions: FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly → FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable only in case of steering column locking systems: Locks the Steering → Locks the Steering SYSTEM SOFTWARE (If fitted): NA → NA Make: -- → -- Version: -- → -- Operating System Details with Version: -- → -- DESCRIPTION OF DEVICE (If fitted): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Description of the VAS: - → - A detailed description of the alarm system and of the vehicle parts related to alarm system installed with Block Diagram: → A list of main components comprising the alarm system: - → - Components placed in Engine compartment: - → - Components placed in Passenger compartment: - → - Availability of radio/wireless transmission Yes/No: - → - Details of radio/wireless transmission if any: - → - frequency: - → - Availability of WPC approval for wireless technology Yes/No: - → - The measures taken against false alarm: - → - Range of protection offered by the device. Details about the detection mechanism (opening of Door, Bonnet, luggage compartment, intrusion etc.): - → - Details & specification of audible alarm: - → - Details & specification of optical alarm: - → - Details & specification of radio alarm if fitted: - → - Method of setting/ unsetting the device: - → - Type of key used for unsetting (Mechanical Key / electronic device): - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Number of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - Exit delay in Sec: - → - Entry Delay in sec: - → - Details of status display (0.5cd): - → - Details of self check , automatic failure indication: - → - Details & specification of audible alarm: - → - Details & specification of optical alarm: - → - Details & specification of radio alarm if fitted: - → - Method of setting/ unsetting the device: - → - Type of key used for unsetting (Mechanical Key / electronic device): - → - Number of interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - Exit delay in Sec: - → - Entry Delay in sec: - → - Details of status display (0.5cd): - → - Details of self check , automatic failure indication: - → - Details about the panic alarm , if provided: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Description of the Immobilizer: - → - A detailed description of the vehicle immobilizer and of the vehicle and the measures		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	taken against inadvertent activation with block diagram.: → The vehicle system(s) on which the immobilizer acts.: - → - Details of radio/wireless transmission if any: - → - Method of setting/ unsetting the device.: - → - Number of interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - A list of main components comprising the device: - → - Description: AL Part No. → AL Part No. Description · AL Part No.: Hella Part No → Hella Part No ECU: FXX01000 → FXX01000 ECU · FXX01000: 015.XYZ-01 → 015.XYZ-01 a. · Make: - → - b. · Type: - → - c. · Serial Number: - → - d. · Name and Address of manufacture: - → - a. · Make: - → - b. · Type: - → - c. · Serial Number: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	d. · Name and Address of manufacture: - → - Example of Engine No: HJEZXYZ891 → HJEZXYZ891		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
CMVR 115; AIS-137 applicable in-service provisions In-service conformity / PUC applicability	Base fuel type used for vehicle Type approval including blend level (Ethanol, Bio-diesel etc): Diesel (Maximum 7% bio-diesel blend) → Diesel (Maximum 7% bio-diesel blend) Name and address of engine manufacturing plant (In case of imported vehicles, above details shall be supplied for importer also): Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX → Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX Max. Recommended Gradeability (Stand-start) – in degrees: 10 degs → 10 degs Max. Recommended Gradeability (Stand-start) – in degrees: 8.5 degs → 8.5 degs Vehicle kerb weight kg: Refer Annexure T7 - B Details for Recommended load body Refer Annexure T7 – C → Refer Annexure T7 - B Details for Recommended load body Refer Annexure T7 – C	CEA / report needs review	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Other axle: NA → NA Gross combination weight kg (applicable for articulated / Tractor Trailer combination vehicles): NA → NA Front axle: NA → NA Rear axle: NA → NA Trailer axle / other axles: NA → NA If applicable, maximum mass of trailer whic5h may be coupled: NA →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	NA Unbraked trailer: NA → NA Reference mass (for vehicles with GVW less than or equal to 3.5 ton) (kg.): Refer Annexure T7-B → Refer Annexure T7-B Maximum (Including driver) for completely built vehicles: D+2 → D+2 Maximum (Including driver) for completely built vehicles: → Make: APOLLO → APOLLO Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade description(s): CEAT WINMILE X3-R → CEAT WINMILE X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 Tyre rolling Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer - TC7719XX000 003XXXF → Refer - TC7719XX000 003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022- 2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022- 2023/300002XXX X/TA/XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L- 756XXXX → CM/L- 756XXXX Identification: TAC No. /		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E- Marking: NA → NA Others (For articulated / combination vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00 R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other (for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 325 mm → 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Type: Fully		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	synchronized 5 - forward gears with a Constant mesh reverse gear → Fully synchronized 5 - forward gears with a Constant mesh reverse gear Model name (if any): XXX-XXX → XXX-XXX Model name (if any): YYYY-YY → YYYY-YY	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Gear shifting control system (Sketch showing gear shifting arrangement): → No. of gears: Forward 5 + 1 Rev → Forward 5 + 1 Rev Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Rear axle ratio: 4.56 → 5.13 Type of steering gear (Re-circulating ball / Worm & Roller / Rack & Pinion / Others): Re-circulating ball type → Re-circulating ball type Make: 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd → 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 Type and description (Leaf / Coil / Air / Semi- pneumatic/ Torsion bar): Leaf → Leaf Front: Parabolic spring- Rigid axle suspension → Parabolic spring- Rigid axle suspension Rear: Semi elliptic spring-		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Rigid axle suspension → Semi elliptic spring- Rigid axle suspension Fuel Gauge: LED Bars → LED Bars Engine coolant temperature: Red Color – Thermometer Symbol → Red Color –	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Thermometer Symbol Engine low oil pressure: Red Color – Oil Tank Symbol → Red Color – Oil Tank Symbol Weight of seats: Driver seat : 11.5 Kg & Co-driver seat : 11Kg → Driver seat : 11.5 Kg & Co-driver seat : 11Kg Description of seats: -- → -- Make: -- → -- Number of seats fitted or capable of being fitted with head restraints, adjustable or not adjustable.: -- → -- Description of the adjustment, displacement and locking systems of the seat or of its parts and a description of occupant protection system against displacement of luggage.: -- → -- Description of seat anchorage: -- → -- Longitudinal / Vertical (If applicable) position of the seats during the tests.: -- → -- Drawings, diagrams and plans of the seats, their anchorages on the vehicle, the floor layout, the adjustment and displacement system of the seats and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	their parts, and their locking devices and of additional occupant protection system against displacement of luggage.: -- → -- In the case of seats fitted with head restraints, the head restraint shall be shown on all drawings, diagrams and photographs.: -- → -- Seat Identification No. / Part No/Drawing	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Number.: ID Number/ Part Number/D rawing number Weight Description Make (Seat (kg) Manufactu rer/ Vehicle Manufactu rer): -- → -- Description: Make → Make Description: ID Number/ Part Number/D rawing number (Seat Manufactu rer/ Vehicle Manufactu rer) → ID Number/ Part Number/D rawing number (Seat Manufactu rer/ Vehicle Manufactu rer) Description: Weight (kg) → Weight (kg) Brief description and drawings of the seat type, its attachment fittings and its adjustment, displacement and locking systems including the minimum distance between fitting points.: D+1 → D+1		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Installation drawing showing rear extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	make, unladen pressure) etc., distance between tyre & RUPD RH & LH, frame rear width, P1, P2 points.: - → - Make: M/s. XXXXYYYY → M/s. XXXXYYYY Material (Metallic / Plastic etc.): Metallic → Metallic Nominal thickness mm: 1.0mm → 1.0mm Capacity, Litre: 40 Lts, 30 Lts → 40 Lts, 30 Lts Detailed drawing of the fuel tank assembly with material specifications: FHSXXXXY FHSXXXXY → FHSXXXXY FHSXXXXY Detailed drawing indicating the position / location of the fuel tank(s) in the vehicle.: A5LXXX00 (50 LTS) A5LXXX00 (40 LTS) → A5LXXX00 (50 LTS) A5LXXX00 (40 LTS) Identification No. / Part No.: FHS00544 FfffXXXcdHXXS00744 → FHS00544 FfffXXXcdHXXS00744 Detailed engineering drawing as specified in AIS – 013 or photographs showing the mounting details with dimensions.: NA → NA Cross sectional Drawing of Gear shift	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	lever (Knob): FXA0XXYY (Non AC) / FXA0XXYY (AC) → FXA0XXYY (Non AC) / FXA0XXYY (AC) Drawings and/or photographs of the locations of the statutory plates and inscriptions and of the vehicle		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	identification number: X4W0XYZ4 / VFT0XYZ2 → X4W0XYZ4 / VFT0XYZ2 Drawings and / or photographs of the official part of the plates and inscriptions (completed example with dimensions): X4WXYZ44 → X4WXYZ44 Location of Vehicle Descriptor Section (VDS) in Vehicle Identification Number (VIN): 4thto 9thCharacter of 17 Characters VIN → 4thto 9thCharacter of 17 Characters VIN Brief description of the Provision for emergency exit provided in case of sealed windows: -- → -- Dimensions of the instructions describing each motion necessary to unlatch and open the exit are provided from the release mechanism.: -- → -- In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → -- Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	distance between front extremity & FUPD, width of FUPD, ID		
(continued)	marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Brief description of the vehicle type as regards its structure, dimensions, lines and constituent materials:: -- → -- Description of the protective system installed in the vehicle:: -- → -- Description of the protective system installed in the vehicle:: → Description of interior arrangements or fittings that might affect the tests:: -- → -- A detailed description, including photographs and/or drawings, of the vehicle type with respect to the structure, the dimensions, the design and the constituent materials, the side walls of the passenger compartment (exterior and interior), including specific details of the protection system, where applicable:: -- → -- Drawing / Images of position & arrangement of Engine: -- → -- Drawing/ Images of Under bonnet (Engine house) Components.: -- → -- Construction: Integrated / Attached with Mounts: 4 Mounts, Fixed CABIN → 4 Mounts, Fixed CABIN 1st ·	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st · Overall Ratio / AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69 2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC: 12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56 4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 → 20.44 Description · Make: ID	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number/ Part Number/Drawing number (Seat Manufacturer/ Vehicle Manufacturer) → ID Number/ Part Number/Drawing number (Seat Manufacturer/ Vehicle Manufacturer) Description · Make: Weight (kg) → Weight (kg) B 51.0 B49.0 · Pedestrian protection as per AIS-100: The distance, measured longitudinally on a horizontal plane, between the transverse centerline of the front axle & the R point of the driver’s seat is less than 1100 mm (Yes / No).: NA → NA Brief description of failure warning tell–tale: - → - Brief description and features: - → - Brief description and features: - → - Description: -- → -- Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No Manufacturer’s documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	version as appropriate.: -- → -- Under fault conditions, vehicle response corresponds to that described in the manufacturer’s documents / failure analysis and safety concept is verified.: -- → -- A brief written explanation sufficient to describe the ESC system basic operational characteristics.: -- → -- Logic diagram: -- → -- Understeer information.: -- → -- Static Stability Factor: -- → -- Make and Country of manufacturer(if imported): -- → -- Software Id / version: -- → -- Brief description of failure warning tell–tale: -- → -- Brief description and features: -- → -- Brief description and features: -- → -- System identification including software level identifier: -- → -- List of motor vehicles by description and configuration that are covered by the information document: -- → -- Wheel type (single or twin) and tyre type (e.g. structure, category of use, size): -- → -- Gearbox type (e.g. manual, automated manual, semi-automatic, automatic): -- → -- Management of the engine or any	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	other source(s) of motive power e.g.Torque Management, control of fuel	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	supply etc.: -- → -- Component description:: -- → -- Sensor(s) External to the Controller: -- → -- Function: -- → -- Limitations on the location of the sensors: -- → -- Identification (e.g. part numbers): -- → -- Controller(s): -- → -- General description and function: -- → -- Functionality of internal sensors (if applicable): -- → -- Hardware identification (e.g. part numbers): -- → -- Software identification: -- → -- Limitations on the location of the controller(s): -- → -- Additional features: -- → -- Modulators: -- → -- General description and function: -- → -- Hardware identification (e.g. part numbers): -- → -- Software identification (if applicable): -- → -- Limitations: -- → -- Electrical Equipment: -- → -- Circuit diagrams: -- → -- Powering methods: -- → -- Pneumatic circuits: -- → -- Safety aspects of the electronic system in accordance with Annex S of IS 11852: -- → -- List of complex functions covered::		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	-- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Manufacturer’s documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- A list of input and output variables, including their working range, is provided.: -- → -- Documentation includes an inventory of components, outlines the function of each unit, its interconnection/interaction with other systems and defines signal flow priorities.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software version as appropriate.: -- → -- Safety concept statement verifies that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- System architecture, design methods and tools identified.: -- → -- Information regarding	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	design provisions in case of failure is provided including any error messages, warning signals, partial performance	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	conditions, back-up modes and/or removal of high- level functions.: -- → -- Additional material and analysis (FMEA, FTA) of fault conditions made available for inspection and maintained by the manufacturer.: -- → -- Document reference and date of inspection:: -- → -- Vehicle complies with all other performance requirements of IS 15986:2015 and meets manufacturer’s specifications under non-fault conditions.: -- → -- Under fault conditions, vehicle response corresponds to that described in the manufacturer’s documents / failure analysis and safety concept is verified.: -- → -- Electro-magnetic compatibility: -- → -- Documentation demonstrating compliance with AIS-004(Part 3).: -- → -- Component description: -- → -- Controller(s) (a) General description and function; (b) Identification e.g. part numbers; (c) Limitations on the location of the controller(s); (d) Additional		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	features.: -- → -- Modulators (a) General description and function; (b) Identification; (c) Limitations.: -- → -- Manufacturer’s documents explain the functionality and safety concept of the system and describe how the	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	operational status may be checked.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software version as appropriate.: -- → -- (see annex 4 of this report for test data): -- → -- (see annex 4 of this report for test data): -- → -- Type / Description: Fully built vehicle, D+2, Non AC, 3.39 Ton GVW → Fully built vehicle, D+2, Non AC, 3.39 Ton GVW Make: Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. → Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. Model name / identification: ALXYZABC0 → ALXYZABC0 Type: Compression Ignition → Compression Ignition Bore x stroke (mm): 80 X 98 → 80 X 98 No. of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	cylinders: 3 → 3 Displacement (cm³): 1468 cc → 1468 cc Compression ratio: 16.5±1.0 : 1 → 16.5±1.0 : 1 Max. Engine output (KW @ rpm): 55.1	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Kw @ 3300 rpm → 55.1 Kw @ 3300 rpm Max. Torque (Nm @ rpm): 170Nm @ 1600 to 2400 rpm → 170Nm @ 1600 to 2400 rpm Air cleaner type: Dry type → Dry type Type of Fuel: Diesel → Diesel Make and model: RSB Transmission (T210R) → RSB Transmission (T210R) Make and model: AVTEC (A5M210) → AVTEC (A5M210) Type: Manual gearbox with fully synchro mesh & Rev with sliding mesh → Manual gearbox with fully synchro mesh & Rev with sliding mesh No. of gears: 5 Forward and 1 Reverse → 5 Forward and 1 Reverse 1st: 4.62 → 4.62 1st: 4.60 → 4.60 1st: 21.53 → 21.53 1st: 21.44 → 21.44 2nd: 2.68 → 2.68 2nd: 2.69 → 2.69 2nd: 12.67 → 12.67 2nd: 12.71 → 12.71 3rd: 1.51 → 1.51 3rd: 1.49 → 1.49 3rd: 7.34 → 7.34 3rd: 7.25 → 7.25 4th: 1.0 → 1.0 4th: 1.0 → 1.0 4th: 4.56 → 4.56 4th: 4.56 → 4.56 5th: 0.70 → 0.70 5th: 0.71 → 0.71 5th: 3.64 → 3.64 5th: 3.67 → 3.67		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Reverse: 4.39 → 4.39		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Reverse: 20.36 → 20.36 Reverse: 4.38 → 4.38 Reverse: 20.34 → 20.34 Drive Axle (Front / Rear / All): Rear → Rear Front axle ratio: NA → NA Type / Description: Re-circulating ball type → Re-circulating ball type Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT 12 PR Tyre size designation including ply rating: 215/75 R15 LT → 215/75 R15 LT Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2 → 3.26 kg/cm2 Wiping system (Brief description): Twin blade, Tandem Pattern → Twin blade, Tandem Pattern	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Material: Metallic → Metallic Capacity (l): 50 lts , 40 lts → 50 lts , 40 lts Dimensions: 2490WB → 2490WB		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Dimensions: 2640WB → 2640WB Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Overall width (mm): 1842 → 1842 Overall Length (mm): CC → CC Overall Length (mm): 4750 → 4750 Overall Length (mm): 4988 → 4988 Overall Length (mm): FSD / HSD → FSD / HSD Overall Length (mm): 4925 → 4925 Overall Length (mm): 5225 → 5225 Overall height (mm): 1961 → 1961 Overall height (mm): 1948 → 1948 Maximum gradeability in 1st gear: GVW 2890 kg → GVW 2890 kg Maximum gradeability in 1st gear: GVW 3390 kg → GVW 3390 kg Maximum gradeability in 1st gear: 10 Deg → 10 Deg Maximum gradeability in 1st gear: 8.5 Deg → 8.5 Deg Weights: Base → Base Weights: Variant 1 → Variant 1 Weights: Variant 2 → Variant 2 Weights: Variant 3 → Variant 3 Weights: Variant 4 → Variant 4 Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	spare wheel, tools, etc.) (kg): - → - Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Description of energy converters and power plant. (In the case of a vehicle that can run either on petrol, diesel, etc., or also in combination with another fuel, items shall be repeated.1): Diesel → Diesel Engine Manufacturer:: Ashok Leyland Limited → Ashok Leyland Limited Manufacturer's engine model code (as marked on the engine, or other means of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	identification): Model Code: ALXYZABC0 Enclosure – 4A → Model Code: ALXYZABC0 Enclosure – 4A Location of engine model code and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	engine serial number on block / crankcase (drawing to be attached): Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A → Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A Working principle: positive ignition / compression- ignition, four-stroke / two-stroke / rotary cycle (DI/IDI) (NA/TC/TCIC/Any other): Compression Ignition, four-stroke, DI, TCIC → Compression Ignition, four-stroke, DI, TCIC Number, arrangement of cylinders:: 3, Inline → 3, Inline Bore----------(mm): 80 → 80 Stroke: ----------(mm): 98 → 98 Firing order:: 1-2-3 → 1-2-3 Engine capacity:----------(cm3): 1468 → 1468 Volumetric compression ratio (specify	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	tolerance): 17.5±1.0 : 1 → 17.5±1.0 : 1 Drawings of combustion chamber and piston crown and, in the case of positive ignition engine, piston rings:: Refer Enclosure – 4B → Refer		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure – 4B Vehicle with Defeat Device Yes / No: No → No If yes – Details of the Defeat device: NA → NA Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane equivalent) in PPM content by volume in the exhaust gas with the engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum net power:..... kW at min-1: 55.1 KW @ 3100 ± 50 rpm → 55.1 KW @ 3100 ± 50 rpm Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm Maximum net torque: .....Nm at: -------min-1 ...(manufacturer's declared value): 170 Nm @ 1600 – 2400 rpm → 170 Nm @ 1600 – 2400 rpm Maximum rated speed (Specify	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	tolerance): 3100 rpm ± 50 rpm → 3100 rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Type of fuel used : Mono		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Fuel / Bi-Fuel / Flex Fuel / Dual Fuel: Mono Fuel → Mono Fuel Fuel type: Diesel / Gasoline / LPG / CNG / Biomethane / Bio-H methane / HCNG / Biogas / LNG / Ethanol ((E85) / 2 (E100)) /ED (95) Flex fuel Methanol M15 / M100 / MD95 / Biodiesel up to 100% / Hydrogen.: Diesel → Diesel Maximum amount of biofuel acceptable in fuel (manufacturer's declared value): per cent by volume: 7% → 7% By fuel injection (compression-ignition only): (Yes/No): Yes → Yes System description:: Unit Pump Common Rail Direct Injection system → Unit Pump Common Rail Direct Injection system Working principle: direct-injection/pre-chamber/swirl Chamber: Direct Injection → Direct Injection Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Plunger Type & FHL0XYZ0 → Plunger Type & FHL0XYZ0 Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Injection advance curve (Specify the tolerances):: ECU Controlled → ECU Controlled Governor: ECU Controlled → ECU Controlled Cut-off point under load:…(min-1): 3300 → 3300 Cut-off point without load (Maximum no-load speed)…(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Multi-Hole Injectors (7 holes) FKXXYYZZ → Multi-Hole Injectors (7 holes) FKXXYYZZ Opening pressure (kPa) (specify the tolerances) or characteristic diagram: 15000 to 23000 kPa → 15000 to 23000 kPa Length and internal diameter…(mm): Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm → Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Internal Diameter – 3mm Common rail make & type: M/s. XXXXYYY, Hot Forged Rail Type → M/s. XXXXYYY, Hot Forged Rail Type Cold start system: NA → NA Make(s):: NA → NA Type(s) &		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: NA → NA Description:: NA → NA Auxiliary starting aid: NA → NA Make(s):: NA → NA Type(s):: NA → NA System description:: NA → NA Electronic controlled injection: (Yes/No): Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA → NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s. XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Metering type Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type By fuel injection (positive ignition only): (Yes/No): No → No Working principle: intake manifold (single/multi- point)/direct injection/other (specify): NA → NA Make(s) of Injection Pump :: NA → NA Type(s) & Identification Number of Injection Pump:: NA → NA System description (in the case of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	systems other than continuous injection give equivalent details):: NA → NA Make and type of the control unit:: NA → NA Make and type of the fuel regulator:: NA → NA Make and type of the air-flow sensor:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Make and type of the micro-switch:: NA → NA Make and type of the throttle housing:: NA → NA Make and type of the water temperature sensor:: NA → NA Make and type of the air temperature sensor:: NA → NA Injectors: Opening pressure (Specify the tolerances):,..... (kPa) or characteristic diagram:: NA → NA Make(s):: NA → NA Type(s):: NA → NA Identification Number: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA Cut-off point under load (min-1): NA → NA Injection timing:: NA → NA Cold start system:: NA → NA Operating principle(s):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Operating limits/settings(Specify the tolerances): NA → NA Pressure, (Specify the tolerances).....(kPa) or characteristic diagram:: 320 ± 50 kPa → 320 ± 50 kPa Rated voltage:...V, positive/negative ground: 12V, Negative Ground → 12V, Negative Ground		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Generator/ Alternator: Alternator → Alternator Type:: DC – Alternator with Voltage regulator → DC – Alternator with Voltage regulator Nominal output: ...VA: 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps → 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps Ignition: NA → NA Make(s):: NA → NA Type(s):: NA → NA Working principle:: NA → NA Ignition advance curve or map(Specify the tolerances):: NA → NA Static ignition timing (Specify the tolerances):.....degrees before TDC: NA → NA Spark plugs: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Gap setting----------(mm): NA → NA Nominal resistance (kilo ohm) (if resistive type): NA → NA Ignition coil(s): NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Ignition condenser: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA EMI suppressor cap / Device / Electronic unit: NA → NA Make: NA → NA Type (Resistive/Capacitive): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification number: NA → NA Nominal resistance (kilo ohm): NA → NA Terminology and Drawing of interference Suppression equipment: NA → NA H.T. Cable: NA → NA Make and Place: NA → NA Type (Resistive/Non-resistive): NA → NA Length (mm) (if resistive type): NA → NA Outside dia. (mm) (if resistive type): NA → NA Nominal resistance kilo ohm, (if resistive type): NA → NA Radiator drawing(s): Refer enclosure 4C2 & 4C4 → Refer enclosure 4C2 & 4C4 Radiator drawing(s): Refer enclosurev 4C1 & 4C3 → Refer enclosurev 4C1 & 4C3	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make: Mahle Behr → Mahle Behr Make: Tata Toyo → Tata Toyo Type (s): Cross flow → Cross flow Relief valve pressure setting: 0.9 kg/cm2 → 0.9 kg/cm2 Fan characteristics (Fan power, kW) Enclose the fan power curve corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part No: 9 → 9 Material of blades (metal /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ± 150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150 (AC) Radiator core open area (cm²): 1906 cm2 → 1906 cm2 Cooling system: liquid/air: Liquid → Liquid Nominal setting of the engine temperature control o mechanism (C)::: 85 deg C → 85 deg C	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Liquid: Yes → Yes Nature of liquid:: Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L → Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L Circulating pump(s): (Yes/No): Yes → Yes Characteristics of Circulating pump: Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm → Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm Make(s):: XXXXX		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Fasteners Ltd → XXXXX Fasteners Ltd Type(s):: Impeller Type → Impeller Type Drive ratio(s):: 1: 1.12 → 1: 1.12 Description of the fan and its drive mechanism:: Electric Fan , Motor Driven → Electric Fan , Motor Driven Blower: (Yes/No): No → No Characteristics of blower: NA → NA Make(s):: NA → NA Type(s):: NA → NA Drive ratio(s):: NA → NA Pressure charger: (Yes/No) Turbocharger	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	/ Supercharger: Yes, Turbo charger → Yes, Turbo charger Make(s):: M/s. XYZ, India → M/s. XYZ, India Type(s) & Identification Number:: Waste Gate with Vacuum Actuator, X3M1XXXX → Waste Gate with Vacuum Actuator, X3M1XXXX Description of the system (maximum charge pressure:.....(kPa), waste-gate, if applicable): Waste gate turbocharger with vacuum actuator Maximum Charge pressure - 145 kPa → Waste gate turbocharger with vacuum actuator Maximum Charge pressure - 145 kPa Inter-cooler: (Yes/No): Yes → Yes Type: air-air / air-water: Air – Air → Air – Air Make :: Mahle Behr → Mahle Behr Make :: Tata Toyo → Tata Toyo		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: B1V42XXX → B1V42XXX Identification Number:: B1V42YYY → B1V42YYY Air pressure drop across the intercooler: 15 kPa → 15 kPa Intake depression at rated engine speed and at 100 per cent load (Specify location of measurement) Minimum allowable:... (kPa) Maximum allowable:... (kPa): Minimum : 0 kPa Maximum : 6.0 kPa	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Description and drawings of inlet pipes and their accessories (plenum chamber, heating device, additional air intakes, etc.):: Intake manifold with Electronic throttle valve Refer Enclosure 4D → Intake manifold with Electronic throttle valve Refer Enclosure 4D Intake manifold description including dimensional drawings and/or photographs):: Aluminium Intake manifold with pressure sensor Refer Enclosure 4D → Aluminium Intake manifold with pressure sensor Refer Enclosure 4D Air filter, Dimensional drawings: Refer Enclosure 4E → Refer Enclosure 4E		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Make(s):: M/s. Fleet guard → M/s. Fleet guard Type(s) & Identification Number:: Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 → Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 Intake silencer, Dimensional drawing: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA Description and Dimensional drawing of the exhaust manifold:: Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F → Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F Description and drawing with dimensions of the exhaust system:(Description with general arrangement of exhaust system along with its routing indicating the lengths of exhaust pipe, tail pipe and: Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	O2 sensors provided before and after LNT-DPF assembly → Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly Effective volume of exhaust-system (specify the tolerance & range) in liters (from exhaust manifold / TC outlet to tail pipe end), Enclose the exhaust system dimensional drawing and indicate the volume of each parts clearly: 13.05 ± 1 litre Refer Enclosure 4G → 13.05 ± 1 litre Refer Enclosure 4G Description and drawing of the exhaust Muffler/Silencer:: NA → NA Type(s) & Identification Number:: NA → NA Make: NA → NA Number:.....: NA → NA Internal diameter of exhaust pipe (mm)(min): diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G → diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G Minimum distance between exhaust pipe(s) and the fuel		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	line: >150 mm → >150 mm Auxiliary Noise shields for compliance to IS 3028 and / OR AIS-020 (If Provided): NA → NA Material: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Layout of noise shield / Photographs / Diagram Showing arrangements indicating fitment on vehicle.: NA → NA Maximum allowable exhaust back pressure at rated engine speed and at 100 per cent load (location of measurement): ... (kPa): 30.0 kpa (150 mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Minimum cross-sectional areas of inlet and outlet ports:: Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm → Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm Inlet (mm): 8.00 → 8.00 Exhaust (mm): 8.00 → 8.00 Opening: 17° bTDC → 17° bTDC Closing: 51° aBDC → 51° aBDC Opening: 66° bBDC → 66° bBDC Closing: 25° aTDC → 25° aTDC Transfer: NA → NA Opening: NA → NA Closing: NA → NA Reference and/or setting ranges (Specify the tolerances):: On Cold Engine → On Cold Engine Inlet: 0.3 to 0.4 mm → 0.3 to 0.4 mm Exhaust: 0.3 to 0.4 mm → 0.3 to 0.4 mm		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Distribution by ports: NA → NA Volume of crank-case cavity with piston at TDC: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Reed valve fitted (Yes / No): No → No Description of inlet ports, scavenging and exhaust ports with corresponding timing.: NA → NA Device for recycling crankcase gases (description and drawings):: Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J → Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx Catalyst (if any, and if not covered by another heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF / DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No Element – 1 No. → Catalytic Converter – 1No Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	for LNT (periodic / continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:……cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap Bulk Density (GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle and Characteristics:: Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	downstream SCR catalyst for the reduction of NOX during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY Type and Identification number:: Type - Passive NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite → Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA Make:: NA → NA Type and Idetification number :: NA → NA NOx Sensor (Before /After of SCR):: No → No Make:: NA → NA Type and Identification number:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent engine test bench cycles, between two cycles where regenerative phases occur under the conditions equivalent to Type I test (Distance "D" in t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration (Distance “d” in AIS 137): 2 cycles → 2 cycles Description of method employed to determine the number of cycles	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA International standard (where appropriate):: NA → NA Frequency of reagent refill: continuous/maintenance (where appropriate):: NA → NA Normal operating temperature range (°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY → YYYYYYY Air injection: (Yes/No): No → No Type (pulse air, air pump, etc.) & Identification Number:: NA → NA Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other): Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY → FEW0XXYY Characteristics (flow rate, etc.):: Flow rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes EGR Cooler Make and Identification Number: M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:.....g: NA → NA Canister working capacity:.....g: NA → NA Canister bed volume:.....litre: NA → NA Canister Purge and vent strategy description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve : (Yes/No): NA → NA Make and Identification Number: NA →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA Operating pressure (kPa): NA → NA Liquid fuel hoses material :: NA → NA Schematic drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	drawing (Periodic / Continuous): NA → NA Particulate trap: (Yes/No): Yes → Yes Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Dimensions and shape of the particulate trap (capacity):: Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Regeneration Type Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure 4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh: 90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type Material - Silicon Carbide → Structure - Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI Electronic Control Unit (ECU): Yes → Yes	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number: P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA Misfire detection:: NA → NA Oxygen sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K → Refer Enclosure 4K Particulate traps		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method):: Refer Enclosure 4K → Refer Enclosure 4K List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K → Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K A description of the type and number of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3 EUDC A description of the type of the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	all sensed components with the strategy for fault detection and MI activation (fixed number of driving cycles or statistical method), including a list of relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive explanation for the data given in service $05 Test ID $21 to FF and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles – Diagnostics on Controller Area Network (CAN) – Part 4: Requirements for emissions- related systems", a	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137): Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA → NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA Systems to ensure the correct operation of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent deactivation of the driver inducement, for use by the rescue services [Note: cross- reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	/disable after parking: NA → NA Lowest		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct operation of NOx control measures: NA → NA Details of any devices designed to influence fuel economy (if not covered by other items):: NA → NA LPG fuelling system: (Yes/No): No → No Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to LPG or back:: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.): NA → NA NG fuelling system: (Yes/No): No → No Approval number: NA → NA Electronic engine management control	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	unit for NG fuelling: NA → NA Make(s):: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation:: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to NG or back: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.):: NA → NA Hydrogen fuelling system: (Yes/No): No → No Electronic engine management control unit for hydrogen fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to hydrogen or back:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA H2NG fuelling system: (Yes/No): No → No		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Percentage of hydrogen in the fuel (maximum specified by the manufacturer): NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to H2NG or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA Electric motor(Traction): No → No Type (winding, excitation):: NA → NA Maximum hourly output: kW(manufacturer’s declared value): NA → NA Maximum net power:kW(manufacturer’s declared value): NA → NA Maximum 30 minutes power: kW	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(manufacturer’s declared value): NA → NA Operating voltage: V: NA → NA Make and Identification Number:: → Battery (Traction): NA → NA Number of cells:: NA → NA Mass: .....(kg): NA → NA Capacity: Ah		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Amp-hours): NA → NA Position:: NA → NA Engines or motor combinations: NA → NA Hybrid Electric Vehicle: (Yes/No): No → No Category of Hybrid Electric vehicle Off Vehicle Charging/Not Off Vehicle Charging9: NA → NA Operating mode switch: with/without: NA → NA Selectable modes: NA → NA Pure electric: (Yes/No): NA → NA Pure fuel consuming: (Yes/No): NA → NA Hybrid modes: (Yes/No) (if yes, short description): NA → NA Description of the energy storage device: (traction battery, capacitor, flywheel/generator...): NA → NA Make(s):: NA → NA Type(s) :: NA → NA Identification number:: NA → NA Kind of electrochemical couple:: NA → NA Energy: ..... (for battery: voltage and capacity Ah in 2 h, for capacitor: J): NA →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA Charger: on board/external/without: NA → NA Electric machines (describe each type of electric machine separately): NA → NA Make:: NA → NA Type:: NA → NA Primary use: traction motor/generator: NA → NA When used as traction motor: mono motor / multi motors (number):: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Maximum power: kW: NA → NA Working principle:: NA → NA Direct current/alternating current/number of phases:: NA → NA Separate excitation/series/compound: NA → NA Synchronous/asynchronous: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA Identification number:: NA → NA Power controller: NA → NA Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Vehicle electric range..... km (according to AIS 137):: NA → NA Manufacturer’s recommendation for preconditioning:: NA → NA CO2 (g/km) (Applicable for category M1 with GVW<3.5 T): NA → NA Declared : (rounded to 3 decimal	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	places): NA → NA Fuel consumption (l/100 km) for Petrol, LPG or Diesel and (kg/100km) for CNG and (kWh/100 km) for Electric Driven Vehicles(Applicable for category M1 with GVW<3.5 T) (rounded to 3 decimal places)*: NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA *calculated on the declared CO2: NA → NA CO2 Reducing		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	technologies Available (Yes/No): No → No Regenerative braking: NA → NA Start-Stop System: NA → NA Tyre pressure monitoring system: NA → NA 6 or more Speed Transmission: NA → NA Any other technology (manufacture to enclose detail): NA → NA Electric energy consumption for electric vehicles Wh/km: NA → NA Electric energy consumption for pure electric vehicles Wh/km: NA → NA Electric energy consumption for externally chargeable hybrid electric vehicles: NA → NA Electric energy consumption (condition A, combined) Wh/km: NA → NA Electric energy consumption (condition	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	B, combined) Wh/km: NA → NA Electric energy consumption (weighted combined) ….Wh/km: NA → NA Temperatures permitted by the manufacturer: NA → NA Cooling system: Yes → Yes Liquid cooling: Yes → Yes Maximum temperature at outlet: …….(°C): 100 °C → 100 °C Air cooling: NA → NA Reference point:: NA → NA Maximum temperature at reference point: …….(°C): NA → NA Maximum outlet temperature of the inlet intercooler: .. (°C): 45 °C → 45		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	°C Maximum exhaust temperature at the point in the exhaust pipe(s) adjacent to the outer flange(s) of the exhaust manifold: …(°C): 740 °C → 740 °C Minimum: .. (°C): - 8 → - 8 Maximum: .. (°C): 42 → 42 Minimum: …(°C): - 5 → - 5 Maximum: …(°C): 120 → 120 Lubrication system: Yes → Yes Description of the system: Forced Lubrication system → Forced Lubrication system Position of the lubricant reservoir:: Sump beneath crankcase → Sump beneath crankcase Feed system (by pump/injection into intake/mixing with fuel, etc.): Pump →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Pump Lubricating pump: Yes → Yes Make(s):: M/s. ZXWA → M/s. ZXWA Type(s):: XXYY Type → XXYY Type Mixture with fuel: No → No Percentage:: → Oil cooler: (Yes/No): Yes → Yes Drawing(s):..., or: Enclosure 4L → Enclosure 4L Make(s):: M/s. XXYY or Any other equivalent → M/s. XXYY or Any other equivalent Type(s):: DONUT → DONUT Moment of inertia of engine flywheel:: 0.1212 kg/m2 → 0.1212 kg/m2 Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Transmission Control Unit (ECU) (Yes/No):: No → No Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA OBD IUPR Family if any: XX+XYY → XX+XYY exhaust outlet location, indicated in a Schematic dimensional drawing.): Refer Enclosure 4G → Refer Enclosure 4G 1.2.11.2.1.1 4.4 · 3. · Calibration Verification Number:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Catalyst · P0420 · Oxygen sensor 1 and 2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Engine speed, engine load, A/F mode, catalyst temperature → Engine speed, engine load, A/F mode, catalyst temperature Catalyst · P0420 · Oxygen sensor 1 and 2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Two Type I cycles → Two Type I cycles Catalyst · P0420 · Oxygen sensor 1 and 2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Type I → Type I Type (Manual / power): Manual → Manual No. of wipers: Two → Two Make: XXYY System →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	XXYY System Type: 12V → 12V Identification mark: X5XYZ100 → X5XYZ100 Rated voltage: 12V → 12V Number of sweep Frequencies: 2 → 2 Highest sweep frequency (Cycles / min): 62±8 rpm → 62±8 rpm	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Lowest sweep frequency (Cycles / min): 40±5 rpm → 40±5 rpm Wiper arm: Provided → Provided Length: 500mm → 500mm Make: M/s XYZ → M/s XYZ Length: 450 mm → 450 mm Make: BCS (SYNDICATE WIPER SYSTEM) → BCS (SYNDICATE WIPER SYSTEM) Identification: X5XY0258 → X5XY0258 Washer tank: BWXY4501 → BWXY4501 Type: Plastic → Plastic Make: XYZ → XYZ Identification No. / Part No.: X5BXYZ168 → X5BXYZ168 Capacity l: 1.5Ltr → 1.5Ltr Material: H.D.P.E → H.D.P.E Washer tank motor / Washer pump: ELECTRIC MOTOR WITH PUMP → ELECTRIC MOTOR WITH PUMP Make: XYZ → XYZ Model: PEAD64XYZS00 → PEAD64XYZS00 Nozzle(s): XYZ → XYZ No. Of Nozzles: 2 → 2 Defroster: NA → NA Type: - → - Make: - → - Details of Demisting and / or Defrosting system: - → - Schematic Diagram of Engine cooling system: - → - Schematic Diagram of Heating system: - →		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	- Cross sectional view of complete air handling system including position of louvers (vents) on dash board: - → - Blower identification: - → - Blower Voltage and capacity (Watts): - → - If defrost system is fitted, specify defrost test temperature (-8 +/- 2 deg C / -18 +/- 3 deg C): → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	& related dimensions as per related standards.: → Description and sketch showing the detail positions of Tell-Tale and seating layout (For M category vehicles): NA → NA A brief technical description indicating the length of the cleaning period, the consumption of cleaning fluid during the cleaning period and the minimum capacity of the container provided;: -- → -- Description and drawings/photographs	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	of the shapes and constituent materials of the part of the body forming the engine compartment and the part of the passenger compartment nearest to it:: -- → -- Drawings or photographs of the position of the metal components housed in the engine compartment (e.g. heating appliances, spare wheel, air filter, steering mechanism, etc.):: -- → -- List of all subassemblies, which includes an electronic oscillator or switching frequency greater than 9 kHz (like ECU, steering, suspension, Body: -- → -- List of all Electrical components, which include Broadband EMI sources (like HAVC Motor,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Wiper Motor, Window motor and Horn etc.) The list shall include Device name, Make/s and Part No. /Type ID/Drawing No: -- → -- Name of the manufacturer of the system (Steering column shaft / Gear shift control / Transmission) on which the Protective Device functions: XXYY Steering System → XXYY Steering System Type/Part no./Identification number(ID)/Drawing No.of the device on which the Protective Device functions: FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable only in case of steering column locking systems: Locks the Steering → Locks the Steering SYSTEM SOFTWARE (If fitted): NA → NA Make: -- → -- Version: -- → -- Operating System Details with Version: -- → -- DESCRIPTION OF DEVICE (If fitted): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Description of the VAS: - → - A detailed description of the alarm system and of the vehicle parts related to alarm system installed with Block Diagram: → A list of main components comprising the alarm system: - → - Components placed in Engine compartment: - → - Components placed in Passenger compartment: - → - Availability of radio/wireless transmission Yes/No: - → - Details of radio/wireless transmission if any: - → - frequency: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Availability of WPC approval for wireless technology Yes/No: - → - The measures taken against false alarm: - → - Range of protection offered by the device. Details about the detection mechanism (opening of Door, Bonnet, luggage compartment, intrusion etc.): - → - Details & specification of audible alarm: - → - Details & specification of optical alarm: - → - Details & specification of radio alarm if fitted: - → - Method of setting/ unsetting the device: - → - Type of key used for unsetting (Mechanical Key / electronic device): - → - Number of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - Exit delay in Sec: - → - Entry Delay in sec: - → - Details of status display (0.5cd): - → - Details of self check , automatic failure indication: - → - Details & specification of audible alarm: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Details & specification of optical alarm: - → - Details & specification of radio alarm if fitted: - → - Method of setting/ unsetting the device: - → - Type of key used for unsetting (Mechanical Key / electronic device): - → - Number of interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - Exit delay in Sec: - → - Entry Delay in sec: - → - Details of status display (0.5cd): - → - Details of self check , automatic failure indication: - → - Details about the panic alarm , if provided: - → - Description of the Immobilizer: - → - A detailed description of the vehicle immobilizer and of the vehicle and the measures		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	taken against inadvertent activation with block diagram.: → The vehicle system(s) on which the immobilizer acts.: - → - Details of radio/wireless transmission if any: - → - Method of setting/ unsetting the device.: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Number of interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - A list of main components comprising the device: - → - Description: AL Part No. → AL Part No. Description · AL Part No.: Hella Part No → Hella Part No ECU: FXX01000 → FXX01000 ECU · FXX01000: 015.XYZ-01 → 015.XYZ-01 a. · Make: - → - b. · Type: - → - c. · Serial Number: - → - d. · Name and Address of manufacture: - → - a. · Make: - → - b. · Type: - → - c. · Serial Number: - → - d. · Name and Address of manufacture: - → - Example of Engine No: HJEZXYZ891 → HJEZXYZ891		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
CMVR 115; AIS-137 applicable diesel provisions Diesel free-acceleration smoke / opacity route	Base fuel type used for vehicle Type approval including blend level (Ethanol, Bio-diesel etc): Diesel (Maximum 7% bio-diesel blend) → Diesel (Maximum 7% bio-diesel blend) Name and address of engine manufacturing plant (In case of imported vehicles, above details shall be supplied	CEA / report needs review	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	for importer also): Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX → Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX Max. Recommended Gradeability (Stand-start) – in degrees: 10 degs → 10 degs Max. Recommended Gradeability (Stand-start) – in degrees: 8.5 degs → 8.5 degs Vehicle kerb weight kg: Refer Annexure T7 - B Details for Recommended load body Refer Annexure T7 – C → Refer Annexure T7 - B Details for Recommended load body Refer Annexure T7 – C Other axle: NA → NA Gross combination weight kg (applicable for articulated / Tractor Trailer combination vehicles): NA → NA Front axle: NA → NA Rear axle: NA → NA Trailer axle / other axles: NA → NA If applicable, maximum mass of trailer whic5h may be coupled: NA →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	NA Unbraked trailer: NA → NA Reference mass (for vehicles with GVW less than or equal to 3.5 ton) (kg.): Refer	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Annexure T7-B → Refer Annexure T7-B Maximum (Including driver) for completely built vehicles: D+2 → D+2 Maximum (Including driver) for completely built vehicles: → Make: APOLLO → APOLLO Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade description(s): CEAT WINMILE X3-R → CEAT WINMILE X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 Tyre rolling Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer - TC7719XX000 003XXXF → Refer - TC7719XX000 003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022-2023/300002XXX	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	X/TA/XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L- 756XXXX → CM/L- 756XXXX Identification: TAC No. /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E- Marking: NA → NA Others (For articulated / combination	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00 R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube /		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other (for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 325 mm → 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	kg/cm2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26 kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Type: Fully		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	synchronized 5 - forward gears with a Constant mesh reverse gear → Fully synchronized 5 - forward gears with a Constant mesh reverse gear Model name (if any): XXX-XXX → XXX-XXX Model name (if any): YYYY-YY → YYYY-YY Gear shifting control system (Sketch showing gear shifting arrangement): → No. of gears: Forward 5 + 1 Rev → Forward 5 + 1 Rev Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Rear axle ratio: 4.56 → 5.13 Type of steering gear (Re-circulating ball / Worm & Roller / Rack & Pinion / Others): Re-circulating ball type → Re-circulating ball type Make: 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd → 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2. YYYY – 18.2 :1	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Type and description (Leaf / Coil / Air / Semi- pneumatic/ Torsion bar): Leaf → Leaf Front: Parabolic spring- Rigid axle suspension → Parabolic spring- Rigid axle suspension Rear: Semi elliptic spring-		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Rigid axle suspension → Semi elliptic spring- Rigid axle suspension Fuel Gauge: LED Bars → LED Bars Engine coolant temperature: Red Color – Thermometer Symbol → Red Color – Thermometer Symbol Engine low oil pressure: Red Color – Oil Tank Symbol → Red Color – Oil Tank Symbol Weight of seats: Driver seat : 11.5 Kg & Co-driver seat : 11Kg → Driver seat : 11.5 Kg & Co-driver seat : 11Kg Description of seats: -- → -- Make: -- → -- Number of seats fitted or capable of being fitted with head restraints, adjustable or not adjustable.: -- → -- Description of the adjustment, displacement and locking systems of the seat or of its parts and a description of occupant protection system against displacement of luggage.: -- → -- Description of seat anchorage: -- → -- Longitudinal / Vertical (If applicable) position of the seats during the tests.: --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	→ -- Drawings, diagrams and plans of the seats, their anchorages on the vehicle, the floor layout, the adjustment and displacement system of the seats and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	their parts, and their locking devices and of additional occupant protection system against displacement of luggage.: -- → -- In the case of seats fitted with head restraints, the head restraint shall be shown on all drawings, diagrams and photographs.: -- → -- Seat Identification No. / Part No/Drawing Number.: ID Number/ Part Number/D rawing number Weight Description Make (Seat (kg) Manufactu rer/ Vehicle Manufactu rer): -- → -- Description: Make → Make Description: ID Number/ Part Number/D rawing number (Seat Manufactu rer/ Vehicle Manufactu rer) → ID Number/ Part Number/D	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	rawing number (Seat Manufactu rer/ Vehicle Manufactu rer) Description: Weight (kg) → Weight (kg) Brief description and drawings of the seat type, its attachment fittings and its adjustment, displacement and locking systems including the minimum distance between fitting points.: D+1 → D+1 Installation drawing showing rear extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	make, unladen pressure) etc., distance between tyre & RUPD RH & LH, frame rear width, P1, P2 points.: - → - Make: M/s. XXXXYYYY → M/s. XXXXYYYY Material (Metallic / Plastic etc.): Metallic → Metallic Nominal thickness mm: 1.0mm → 1.0mm Capacity, Litre: 40 Lts, 30 Lts → 40 Lts, 30 Lts Detailed drawing of the fuel tank assembly with material specifications: FHSXXXXY	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FHSXXXXY → FHSXXXXY FHSXXXXY Detailed drawing indicating the position / location of the fuel tank(s) in the vehicle.: A5LXXX00 (50 LTS) A5LXXX00 (40 LTS) → A5LXXX00 (50 LTS) A5LXXX00 (40 LTS) Identification No. / Part No.: FHS00544 FfffXXXcdHXXS00744 → FHS00544 FfffXXXcdHXXS00744 Detailed engineering drawing as specified in AIS – 013 or photographs showing the mounting details with dimensions.: NA → NA Cross sectional Drawing of Gear shift lever (Knob): FXA0XXYY (Non AC) / FXA0XXYY (AC) → FXA0XXYY (Non AC) / FXA0XXYY (AC) Drawings and/or photographs of the locations of the statutory plates and inscriptions and of the vehicle		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	identification number: X4W0XYZ4 / VFT0XYZ2 → X4W0XYZ4 / VFT0XYZ2 Drawings and / or photographs of the official part of the plates and inscriptions (completed example with dimensions): X4WXYZ44 → X4WXYZ44 Location of Vehicle Descriptor Section (VDS) in Vehicle Identification Number (VIN): 4thto 9thCharacter of 17 Characters VIN → 4thto 9thCharacter of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	17 Characters VIN Brief description of the Provision for emergency exit provided in case of sealed windows: -- → -- Dimensions of the instructions describing each motion necessary to unlatch and open the exit are provided from the release mechanism.: -- → -- In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → -- Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side, distance between front extremity & FUPD, width of FUPD, ID		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Brief description of the vehicle type as regards its structure, dimensions, lines and constituent materials:: -- → -- Description of the protective system installed in the vehicle:: -- → -- Description of the protective system installed in the vehicle:: → Description of interior arrangements or fittings that might affect the tests:: -- → -- A detailed description, including	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	photographs and/or drawings, of the vehicle type with respect to the structure, the dimensions, the design and the constituent materials, the side walls of the passenger compartment (exterior and interior), including specific details of the protection system, where applicable:: -- → -- Drawing / Images of position & arrangement of Engine: -- → -- Drawing/ Images of Under bonnet (Engine house) Components.: -- → -- Construction: Integrated / Attached with Mounts: 4 Mounts, Fixed CABIN → 4 Mounts, Fixed CABIN 1st ·		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st · Overall Ratio / AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69 2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC: 12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 → 20.44 Description · Make: ID		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number/ Part Number/Drawing number (Seat Manufacturer/ Vehicle Manufacturer) → ID Number/ Part Number/Drawing number (Seat Manufacturer/ Vehicle Manufacturer) Description · Make: Weight (kg) → Weight (kg) B 51.0 B49.0 · Pedestrian protection as per AIS-100: The distance, measured longitudinally on a horizontal plane, between the transverse centerline of the front axle & the R point of the driver’s seat is less than 1100 mm (Yes / No).: NA → NA Brief description of failure warning	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	tell–tale: - → - Brief description and features: - → - Brief description and features: - → - Description: -- → -- Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No Manufacturer’s documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	version as appropriate.: -- → -- Under fault conditions, vehicle response corresponds to that described in the manufacturer’s documents / failure analysis and safety concept is verified.: -- → -- A brief written explanation sufficient to describe the ESC system basic operational characteristics.: -- → -- Logic diagram: -- → -- Understeer information.: -- → -- Static Stability Factor: -- → -- Make and Country of manufacturer(if imported): -- → -- Software Id / version: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Brief description of failure warning tell–tale: -- → -- Brief description and features: -- → -- Brief description and features: -- → -- System identification including software level identifier: -- → -- List of motor vehicles by description and configuration that are covered by the information document: -- → -- Wheel type (single or twin) and tyre type (e.g. structure, category of use, size): -- → -- Gearbox type (e.g. manual, automated manual, semi-automatic, automatic): -- → -- Management of the engine or any		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	other source(s) of motive power e.g.Torque Management, control of fuel supply etc.: -- → -- Component description:: -- → -- Sensor(s) External to the Controller: -- → -- Function: -- → -- Limitations on the location of the sensors: -- → -- Identification (e.g. part numbers): -- → -- Controller(s): -- → -- General description and function: -- → -- Functionality of internal sensors (if applicable): -- → -- Hardware identification (e.g. part numbers): -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Software identification: -- → -- Limitations on the location of the controller(s): -- → -- Additional features: -- → -- Modulators: -- → -- General description and function: -- → -- Hardware identification (e.g. part numbers): -- → -- Software identification (if applicable): -- → -- Limitations: -- → -- Electrical Equipment: -- → -- Circuit diagrams: -- → -- Powering methods: -- → -- Pneumatic circuits: -- → -- Safety aspects of the electronic system in accordance with Annex S of IS 11852: -- → -- List of complex functions covered::		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	-- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Manufacturer’s documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- A list of input and output variables, including their working range, is	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	provided.: -- → -- Documentation includes an inventory of components, outlines the function of each unit, its interconnection/interaction with other systems and defines signal flow priorities.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software version as appropriate.: -- → -- Safety concept statement verifies that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- System architecture, design methods and tools identified.: -- → -- Information regarding		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	design provisions in case of failure is provided including any error messages, warning signals, partial performance conditions, back-up modes and/or removal of high- level functions.: -- → -- Additional material and analysis (FMEA, FTA) of fault conditions made available for inspection and maintained by the manufacturer.: -- → -- Document reference and date of inspection:: -- → -- Vehicle complies with all other performance requirements of IS 15986:2015 and meets manufacturer’s specifications under non-fault conditions.: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Under fault conditions, vehicle response corresponds to that described in the manufacturer’s documents / failure analysis and safety concept is verified.: -- → -- Electro-magnetic compatibility: -- → -- Documentation demonstrating compliance with AIS-004(Part 3).: -- → -- Component description: -- → -- Controller(s) (a) General description and function; (b) Identification e.g. part numbers; (c) Limitations on the location of the controller(s); (d) Additional		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	features.: -- → -- Modulators (a) General description and function; (b) Identification; (c) Limitations.: -- → -- Manufacturer’s documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software version as appropriate.: -- → -- (see annex 4 of this report for test data): -- → -- (see annex 4 of this report for test data): -- → -- Type / Description: Fully built vehicle, D+2, Non AC, 3.39 Ton GVW → Fully	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	built vehicle, D+2, Non AC, 3.39 Ton GVW Make: Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. → Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. Model name / identification: ALXYZABC0 → ALXYZABC0 Type: Compression Ignition → Compression Ignition Bore x stroke (mm): 80 X 98 → 80 X 98 No. of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	cylinders: 3 → 3 Displacement (cm³): 1468 cc → 1468 cc Compression ratio: 16.5±1.0 : 1 → 16.5±1.0 : 1 Max. Engine output (KW @ rpm): 55.1 Kw @ 3300 rpm → 55.1 Kw @ 3300 rpm Max. Torque (Nm @ rpm): 170Nm @ 1600 to 2400 rpm → 170Nm @ 1600 to 2400 rpm Air cleaner type: Dry type → Dry type Type of Fuel: Diesel → Diesel Make and model: RSB Transmission (T210R) → RSB Transmission (T210R) Make and model: AVTEC (A5M210) → AVTEC (A5M210) Type: Manual gearbox with fully synchro mesh & Rev with sliding mesh → Manual	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	gearbox with fully synchro mesh & Rev with sliding mesh No. of gears: 5 Forward and 1 Reverse → 5 Forward and 1 Reverse 1st: 4.62 → 4.62 1st: 4.60 → 4.60 1st: 21.53 → 21.53 1st: 21.44 → 21.44 2nd: 2.68 → 2.68 2nd: 2.69 → 2.69 2nd: 12.67 → 12.67 2nd: 12.71 → 12.71 3rd: 1.51 → 1.51 3rd: 1.49 → 1.49 3rd: 7.34 → 7.34 3rd: 7.25 → 7.25 4th: 1.0 → 1.0 4th: 1.0 → 1.0 4th: 4.56 → 4.56 4th: 4.56 → 4.56 5th: 0.70 → 0.70 5th: 0.71 → 0.71 5th: 3.64 → 3.64 5th: 3.67 → 3.67 Reverse: 4.39 → 4.39		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Reverse: 20.36 → 20.36 Reverse: 4.38 → 4.38 Reverse: 20.34 → 20.34 Drive Axle (Front / Rear / All): Rear → Rear Front axle ratio: NA → NA Type / Description: Re-circulating ball type → Re-circulating ball type	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT 12 PR Tyre size designation including ply rating: 215/75 R15 LT → 215/75 R15 LT Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2 → 3.26 kg/cm2 Wiping system (Brief description): Twin blade, Tandem Pattern → Twin blade, Tandem Pattern Material: Metallic → Metallic Capacity (l): 50 lts , 40 lts → 50 lts , 40 lts Dimensions: 2490WB → 2490WB		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Dimensions: 2640WB → 2640WB Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Overall width (mm): 1842 → 1842 Overall Length (mm): CC → CC	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Overall Length (mm): 4750 → 4750 Overall Length (mm): 4988 → 4988 Overall Length (mm): FSD / HSD → FSD / HSD Overall Length (mm): 4925 → 4925 Overall Length (mm): 5225 → 5225 Overall height (mm): 1961 → 1961 Overall height (mm): 1948 → 1948 Maximum gradeability in 1st gear: GVW 2890 kg → GVW 2890 kg Maximum gradeability in 1st gear: GVW 3390 kg → GVW 3390 kg Maximum gradeability in 1st gear: 10 Deg → 10 Deg Maximum gradeability in 1st gear: 8.5 Deg → 8.5 Deg Weights: Base → Base Weights: Variant 1 → Variant 1 Weights: Variant 2 → Variant 2 Weights: Variant 3 → Variant 3 Weights: Variant 4 → Variant 4 Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	spare wheel, tools, etc.) (kg): - → - Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Description of energy converters and power plant. (In the case of a vehicle that can run either on petrol, diesel, etc., or also in combination with another fuel, items shall be repeated.1): Diesel → Diesel Engine Manufacturer:: Ashok Leyland Limited → Ashok Leyland Limited Manufacturer's engine model code (as marked on the engine, or other means of identification): Model Code: ALXYZABC0 Enclosure – 4A → Model Code: ALXYZABC0 Enclosure – 4A Location of engine model code and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	engine serial number on block / crankcase (drawing to be attached): Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Engine Block Refer Enclosure – 4A → Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A Working principle: positive ignition / compression- ignition, four-stroke / two-stroke / rotary cycle (DI/IDI) (NA/TC/TCIC/Any other): Compression Ignition, four-stroke, DI, TCIC → Compression Ignition, four-stroke, DI, TCIC Number, arrangement of cylinders:: 3, Inline → 3, Inline Bore----------(mm): 80 → 80 Stroke: ----------(mm): 98 → 98 Firing order:: 1-2-3 → 1-2-3 Engine capacity:----------(cm3): 1468 → 1468 Volumetric compression ratio (specify tolerance): 17.5±1.0 : 1 → 17.5±1.0 : 1 Drawings of combustion chamber and piston crown and, in the case of positive ignition engine, piston rings:: Refer Enclosure – 4B → Refer		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure – 4B Vehicle with Defeat Device Yes / No: No → No If yes – Details of the Defeat device: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane equivalent) in PPM content by volume in the exhaust gas with the engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum net power:..... kW at min-1: 55.1 KW @ 3100 ± 50 rpm → 55.1 KW @ 3100 ± 50 rpm Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm Maximum net torque: .....Nm at: -------min-1 ...(manufacturer's declared value): 170 Nm @ 1600 – 2400 rpm → 170 Nm @ 1600 – 2400 rpm Maximum rated speed (Specify tolerance): 3100 rpm ± 50 rpm → 3100 rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Type of fuel used : Mono		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Fuel / Bi-Fuel / Flex Fuel / Dual Fuel: Mono Fuel → Mono Fuel	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Fuel type: Diesel / Gasoline / LPG / CNG / Biomethane / Bio-H methane / HCNG / Biogas / LNG / Ethanol ((E85) / 2 (E100)) /ED (95) Flex fuel Methanol M15 / M100 / MD95 / Biodiesel up to 100% / Hydrogen.: Diesel → Diesel Maximum amount of biofuel acceptable in fuel (manufacturer's declared value): per cent by volume: 7% → 7% By fuel injection (compression-ignition only): (Yes/No): Yes → Yes System description:: Unit Pump Common Rail Direct Injection system → Unit Pump Common Rail Direct Injection system Working principle: direct-injection/pre-chamber/swirl Chamber: Direct Injection → Direct Injection Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Plunger Type & FHL0XYZ0 → Plunger Type & FHL0XYZ0 Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Injection advance curve (Specify the	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	tolerances):: ECU Controlled → ECU Controlled Governor: ECU Controlled → ECU Controlled Cut-off point under load:…(min-1): 3300 → 3300 Cut-off point without load (Maximum no-load speed)…(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Multi-Hole Injectors (7 holes) FKXXYYZZ → Multi-Hole Injectors (7 holes) FKXXYYZZ Opening pressure (kPa) (specify the tolerances) or characteristic diagram: 15000 to 23000 kPa → 15000 to 23000 kPa Length and internal diameter…(mm): Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm → Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm Common rail make & type: M/s. XXXXYYY, Hot Forged Rail Type → M/s. XXXXYYY, Hot Forged Rail Type Cold start system: NA → NA Make(s):: NA → NA Type(s) &		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: NA → NA Description:: NA → NA Auxiliary starting aid: NA → NA Make(s):: NA → NA Type(s):: NA → NA System description:: NA → NA Electronic controlled injection: (Yes/No): Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA → NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s. XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Metering type Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type By fuel injection (positive ignition only): (Yes/No): No → No Working principle: intake manifold (single/multi- point)/direct injection/other (specify): NA → NA Make(s) of Injection Pump :: NA → NA Type(s) & Identification Number of Injection Pump:: NA → NA System description (in the case of systems other than continuous injection give equivalent details):: NA → NA Make and type of the control unit:: NA → NA Make and type of the fuel regulator:: NA → NA Make and type of the air-flow sensor:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Make and type of the micro-switch:: NA → NA Make and type of the throttle housing:: NA → NA Make and type of the water temperature sensor:: NA → NA Make and type of the air temperature sensor:: NA → NA Injectors: Opening pressure (Specify the tolerances):,..... (kPa) or characteristic diagram:: NA → NA Make(s):: NA → NA Type(s):: NA → NA Identification Number: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA Cut-off point under load (min-1): NA → NA Injection timing:: NA → NA Cold start system:: NA → NA Operating principle(s):: NA → NA Operating limits/settings(Specify the tolerances): NA → NA Pressure, (Specify the tolerances).....(kPa) or characteristic diagram:: 320 ± 50 kPa → 320 ± 50 kPa Rated voltage:...V, positive/negative ground: 12V, Negative Ground → 12V, Negative Ground	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Generator/ Alternator: Alternator → Alternator Type:: DC – Alternator with Voltage regulator → DC – Alternator with Voltage regulator Nominal output: ...VA: 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps → 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps Ignition: NA → NA Make(s):: NA → NA Type(s):: NA → NA Working principle:: NA → NA Ignition advance curve or map(Specify the tolerances):: NA → NA Static ignition timing (Specify the tolerances):.....degrees before TDC: NA → NA Spark plugs: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Gap setting----------(mm): NA → NA Nominal resistance (kilo ohm) (if resistive type): NA → NA Ignition coil(s): NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Ignition condenser: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA EMI suppressor cap / Device / Electronic	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	unit: NA → NA Make: NA → NA Type (Resistive/Capacitive): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification number: NA → NA Nominal resistance (kilo ohm): NA → NA Terminology and Drawing of interference Suppression equipment: NA → NA H.T. Cable: NA → NA Make and Place: NA → NA Type (Resistive/Non-resistive): NA → NA Length (mm) (if resistive type): NA → NA Outside dia. (mm) (if resistive type): NA → NA Nominal resistance kilo ohm, (if resistive type): NA → NA Radiator drawing(s): Refer enclosure 4C2 & 4C4 → Refer enclosure 4C2 & 4C4 Radiator drawing(s): Refer enclosurev 4C1 & 4C3 → Refer enclosurev 4C1 & 4C3 Make: Mahle Behr → Mahle Behr Make: Tata Toyo → Tata Toyo Type (s): Cross flow → Cross flow Relief valve pressure setting: 0.9 kg/cm2 → 0.9 kg/cm2 Fan characteristics (Fan power, kW) Enclose the fan power curve corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	No: 9 → 9 Material of blades (metal /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ± 150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150 (AC) Radiator core open area (cm²): 1906 cm2 → 1906 cm2 Cooling system: liquid/air: Liquid → Liquid Nominal setting of the engine temperature control o mechanism (C)::: 85 deg C → 85 deg C Liquid: Yes → Yes Nature of liquid:: Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L → Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L Circulating pump(s): (Yes/No): Yes → Yes Characteristics of Circulating pump:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm → Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm Make(s):: XXXXX		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Fasteners Ltd → XXXXX Fasteners Ltd Type(s):: Impeller Type → Impeller Type Drive ratio(s):: 1: 1.12 → 1: 1.12 Description of the fan and its drive mechanism:: Electric Fan , Motor Driven → Electric Fan , Motor Driven Blower: (Yes/No): No → No Characteristics of blower: NA → NA Make(s):: NA → NA Type(s):: NA → NA Drive ratio(s):: NA → NA Pressure charger: (Yes/No) Turbocharger / Supercharger: Yes, Turbo charger → Yes, Turbo charger Make(s):: M/s. XYZ, India → M/s. XYZ, India Type(s) & Identification Number:: Waste Gate with Vacuum Actuator, X3M1XXXX → Waste Gate with Vacuum Actuator, X3M1XXXX Description of the system (maximum charge pressure:.....(kPa), waste-gate, if applicable): Waste gate turbocharger with vacuum actuator Maximum Charge pressure - 145 kPa → Waste gate turbocharger with vacuum actuator	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Maximum Charge pressure - 145 kPa Inter-cooler: (Yes/No): Yes → Yes Type: air-air / air-water: Air – Air → Air – Air Make :: Mahle Behr → Mahle Behr Make :: Tata Toyo → Tata Toyo		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: B1V42XXX → B1V42XXX Identification Number:: B1V42YYY → B1V42YYY Air pressure drop across the intercooler: 15 kPa → 15 kPa Intake depression at rated engine speed and at 100 per cent load (Specify location of measurement) Minimum allowable:... (kPa) Maximum allowable:... (kPa): Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Description and drawings of inlet pipes and their accessories (plenum chamber, heating device, additional air intakes, etc.):: Intake manifold with Electronic throttle valve Refer Enclosure 4D → Intake manifold with Electronic throttle valve Refer Enclosure 4D Intake manifold description including dimensional drawings and/or	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	photographs):: Aluminium Intake manifold with pressure sensor Refer Enclosure 4D → Aluminium Intake manifold with pressure sensor Refer Enclosure 4D Air filter, Dimensional drawings: Refer Enclosure 4E → Refer Enclosure 4E		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Make(s):: M/s. Fleet guard → M/s. Fleet guard Type(s) & Identification Number:: Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 → Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 Intake silencer, Dimensional drawing: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description and Dimensional drawing of the exhaust manifold:: Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F → Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F Description and drawing with dimensions of the exhaust system:(Description with general arrangement of exhaust system along with its routing	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	indicating the lengths of exhaust pipe, tail pipe and: Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	O2 sensors provided before and after LNT-DPF assembly → Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly Effective volume of exhaust-system (specify the tolerance & range) in liters (from exhaust manifold / TC outlet to tail pipe end), Enclose the exhaust system dimensional drawing and indicate the volume of each parts clearly: 13.05 ± 1 litre Refer Enclosure 4G → 13.05 ± 1 litre Refer Enclosure 4G Description and drawing of the exhaust Muffler/Silencer:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Type(s) & Identification Number:: NA → NA Make: NA → NA Number:……: NA → NA Internal diameter of exhaust pipe (mm)(min): diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G → diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G Minimum distance between exhaust pipe(s) and the fuel		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	line: >150 mm → >150 mm Auxiliary Noise shields for compliance to IS 3028 and / OR AIS-020 (If Provided): NA → NA Material: NA → NA Layout of noise shield / Photographs / Diagram Showing arrangements indicating fitment on vehicle.: NA → NA Maximum allowable exhaust back pressure at rated engine speed and at 100 per cent load (location of measurement): … (kPa): 30.0 kpa (150 mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Minimum cross-sectional areas of inlet and outlet ports:: Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm → Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Inlet (mm): 8.00 → 8.00 Exhaust (mm): 8.00 → 8.00 Opening: 17° bTDC → 17° bTDC Closing: 51° aBDC → 51° aBDC Opening: 66° bBDC → 66° bBDC Closing: 25° aTDC → 25° aTDC Transfer: NA → NA Opening: NA → NA Closing: NA → NA Reference and/or setting ranges (Specify the tolerances):: On Cold Engine → On Cold Engine Inlet: 0.3 to 0.4 mm → 0.3 to 0.4 mm Exhaust: 0.3 to 0.4 mm → 0.3 to 0.4 mm		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Distribution by ports: NA → NA Volume of crank-case cavity with piston at TDC: NA → NA Reed valve fitted (Yes / No): No → No Description of inlet ports, scavenging and exhaust ports with corresponding timing.: NA → NA Device for recycling crankcase gases (description and drawings):: Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J → Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx Catalyst (if any, and if not covered by another	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF / DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No Element – 1 No. → Catalytic Converter – 1No	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	for LNT (periodic / continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:.....cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Bulk Density (GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle and Characteristics:: Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	downstream SCR catalyst for the reduction of NOX during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Type and Identification number:: Type - Passive NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite → Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA Make:: NA → NA Type and Idetification number :: NA → NA NOx Sensor (Before /After of SCR):: No → No Make:: NA → NA Type and Identification number:: NA → NA Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent engine test bench cycles, between two cycles where regenerative phases occur under the conditions equivalent to Type I test (Distance "D" in t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration (Distance “d” in AIS 137): 2 cycles → 2 cycles Description of method employed to determine the number of cycles between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA International standard (where appropriate):: NA → NA Frequency of reagent refill: continuous/maintenance (where appropriate):: NA → NA Normal operating temperature range (°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY → YYYYYYY Air injection: (Yes/No): No → No	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Type (pulse air, air pump, etc.) & Identification Number:: NA → NA Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other): Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY → FEW0XXYY Characteristics (flow rate, etc.):: Flow rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes EGR Cooler Make and Identification Number: M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:.....g: NA → NA Canister working capacity:.....g: NA → NA Canister bed volume:.....litre: NA → NA Canister Purge and vent strategy description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve : (Yes/No): NA → NA Make and Identification Number: NA → NA Operating pressure (kPa): NA → NA Liquid fuel hoses material :: NA → NA Schematic drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	drawing (Periodic / Continuous): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Particulate trap: (Yes/No): Yes → Yes Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dimensions and shape of the particulate trap (capacity):: Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Regeneration Type Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure 4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh: 90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type Material - Silicon Carbide → Structure - Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI Electronic Control Unit (ECU): Yes → Yes Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number: P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA Misfire detection:: NA → NA Oxygen sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K → Refer Enclosure 4K Particulate traps	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method):: Refer Enclosure 4K → Refer Enclosure 4K List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K → Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K A description of the type and number of the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3 EUDC A description of the type of the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	all sensed components with the strategy for fault detection and MI activation (fixed number of driving cycles or statistical method), including a list of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive explanation for the data given in service $05 Test ID $21 to FF and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles – Diagnostics on Controller Area Network (CAN) – Part 4: Requirements for emissions- related systems", a comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA → NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA Systems to ensure the correct operation of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent deactivation of the driver inducement, for use by the rescue services [Note: cross- reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling /disable after parking: NA → NA Lowest		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct operation of NOx control measures: NA → NA Details of any devices designed to influence fuel economy (if not covered by other items):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	LPG fuelling system: (Yes/No): No → No Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to LPG or back:: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.): NA → NA NG fuelling system: (Yes/No): No → No Approval number: NA → NA Electronic engine management control unit for NG fuelling: NA → NA Make(s):: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation:: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NG or back: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.):: NA → NA Hydrogen fuelling system: (Yes/No): No → No Electronic engine management control unit for hydrogen fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to hydrogen or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA H2NG fuelling system: (Yes/No): No → No		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Percentage of hydrogen in the fuel (maximum specified by the manufacturer): NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to H2NG or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA Electric motor(Traction): No → No Type (winding, excitation):: NA → NA Maximum hourly output: kW(manufacturer’s declared value): NA → NA Maximum net power:kW(manufacturer’s declared value): NA → NA Maximum 30 minutes power: kW (manufacturer’s declared value): NA → NA Operating voltage: V: NA → NA Make and Identification Number:: → Battery (Traction): NA → NA Number of cells:: NA → NA Mass: ……(kg): NA → NA Capacity: Ah		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Amp-hours): NA → NA Position:: NA → NA Engines or motor combinations: NA → NA Hybrid Electric Vehicle: (Yes/No): No → No	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Category of Hybrid Electric vehicle Off Vehicle Charging/Not Off Vehicle Charging9: NA → NA Operating mode switch: with/without: NA → NA Selectable modes: NA → NA Pure electric: (Yes/No): NA → NA Pure fuel consuming: (Yes/No): NA → NA Hybrid modes: (Yes/No) (if yes, short description): NA → NA Description of the energy storage device: (traction battery, capacitor, flywheel/generator...): NA → NA Make(s):: NA → NA Type(s) :: NA → NA Identification number:: NA → NA Kind of electrochemical couple:: NA → NA Energy: ..... (for battery: voltage and capacity Ah in 2 h, for capacitor: J): NA → NA Charger: on board/external/without: NA → NA Electric machines (describe each type of electric machine separately): NA → NA Make:: NA → NA Type:: NA → NA Primary use: traction motor/generator: NA → NA When used as traction motor: mono motor / multi motors (number):: NA → NA		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Maximum power: kW: NA → NA Working principle:: NA → NA Direct current/alternating current/number of phases:: NA → NA Separate excitation/series/compound: NA → NA Synchronous/asynchronous: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA Identification number:: NA → NA Power controller: NA → NA Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Vehicle electric range..... km (according to AIS 137):: NA → NA Manufacturer’s recommendation for preconditioning:: NA → NA CO2 (g/km) (Applicable for category M1 with GVW<3.5 T): NA → NA Declared : (rounded to 3 decimal places): NA → NA Fuel consumption (l/100 km) for Petrol, LPG or Diesel and (kg/100km) for CNG and (kWh/100 km) for Electric Driven Vehicles(Applicable for category M1 with GVW<3.5 T) (rounded to 3 decimal places)*: NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA *calculated on the declared CO2: NA → NA CO2 Reducing	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	technologies Available (Yes/No): No → No Regenerative braking: NA → NA Start-Stop System: NA → NA Tyre pressure monitoring system: NA → NA 6 or more Speed Transmission: NA → NA Any other technology (manufacture to enclose detail): NA → NA Electric energy consumption for electric vehicles Wh/km: NA → NA Electric energy consumption for pure electric vehicles Wh/km: NA → NA Electric energy consumption for externally chargeable hybrid electric vehicles: NA → NA Electric energy consumption (condition A, combined) Wh/km: NA → NA Electric energy consumption (condition B, combined) Wh/km: NA → NA Electric energy consumption (weighted combined) ….Wh/km: NA → NA Temperatures permitted by the manufacturer: NA → NA Cooling system: Yes → Yes Liquid cooling: Yes → Yes Maximum temperature at outlet: …….(°C): 100 °C → 100 °C Air cooling: NA → NA Reference point:: NA → NA Maximum temperature at reference point: …….(°C): NA → NA Maximum outlet temperature of the inlet intercooler: .. (°C): 45 °C → 45	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	°C Maximum exhaust temperature at the point in the exhaust pipe(s) adjacent to the outer flange(s) of the exhaust manifold: ...(°C): 740 °C → 740 °C Minimum: .. (°C): - 8 → - 8 Maximum: .. (°C): 42 → 42 Minimum: ...(°C): - 5 → - 5 Maximum: ...(°C): 120 → 120 Lubrication system: Yes → Yes Description of the system: Forced Lubrication system → Forced Lubrication system Position of the lubricant reservoir:: Sump beneath crankcase → Sump beneath crankcase Feed system (by pump/injection into intake/mixing with fuel, etc.): Pump → Pump Lubricating pump: Yes → Yes Make(s):: M/s. ZXWA → M/s. ZXWA Type(s):: XXYY Type → XXYY Type Mixture with fuel: No → No Percentage:: → Oil cooler: (Yes/No): Yes → Yes Drawing(s):..., or: Enclosure 4L → Enclosure 4L Make(s):: M/s. XXYY or Any other equivalent → M/s. XXYY or Any other equivalent Type(s):: DONUT → DONUT Moment of inertia of engine flywheel:: 0.1212 kg/m2 → 0.1212 kg/m2 Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Transmission Control Unit (ECU) (Yes/No):: No → No Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA OBD IUPR Family if any: XX+XYY → XX+XYY exhaust outlet location, indicated in a Schematic dimensional drawing.): Refer Enclosure 4G → Refer Enclosure 4G 1.2.11.2.1.1 4.4 · 3. · Calibration Verification Number:: NA → NA Catalyst · P0420 · Oxygen sensor 1 and 2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Engine speed, engine load, A/F mode, catalyst temperature → Engine speed, engine load, A/F mode, catalyst temperature Catalyst · P0420 · Oxygen sensor 1 and 2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Two Type I cycles → Two Type I cycles Catalyst · P0420 · Oxygen sensor 1 and	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Type I → Type I Type (Manual / power): Manual → Manual No. of wipers: Two → Two Make: XXYY System →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	XXYY System Type: 12V → 12V Identification mark: X5XYZ100 → X5XYZ100 Rated voltage: 12V → 12V Number of sweep Frequencies: 2 → 2 Highest sweep frequency (Cycles / min): 62±8 rpm → 62±8 rpm Lowest sweep frequency (Cycles / min): 40±5 rpm → 40±5 rpm Wiper arm: Provided → Provided Length: 500mm → 500mm Make: M/s XYZ → M/s XYZ Length: 450 mm → 450 mm Make: BCS (SYNDICATE WIPER SYSTEM) → BCS (SYNDICATE WIPER SYSTEM) Identification: X5XY0258 → X5XY0258 Washer tank: BWXY4501 → BWXY4501 Type: Plastic → Plastic Make: XYZ → XYZ Identification No. / Part No.: X5BXYZ168 → X5BXYZ168 Capacity l: 1.5Ltr → 1.5Ltr Material: H.D.P.E → H.D.P.E Washer tank motor / Washer pump: ELECTRIC MOTOR WITH PUMP → ELECTRIC MOTOR WITH PUMP	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make: XYZ → XYZ Model: PEAD64XYZS00 → PEAD64XYZS00 Nozzle(s): XYZ → XYZ No. Of Nozzles: 2 → 2 Defroster: NA → NA Type: - → - Make: - → - Details of Demisting and / or Defrosting system: - → - Schematic Diagram of Engine cooling system: - → - Schematic Diagram of Heating system: - →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	- Cross sectional view of complete air handling system including position of louvers (vents) on dash board: - → - Blower identification: - → - Blower Voltage and capacity (Watts): - → - If defrost system is fitted, specify defrost test temperature (-8 +/- 2 deg C / -18 +/- 3 deg C): → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar		
(continued)	& related dimensions as per related standards.: → Description and sketch showing the detail positions of Tell-Tale and seating layout (For M category vehicles): NA → NA A brief technical description indicating the length of the cleaning period, the consumption of cleaning fluid during the cleaning period and the minimum capacity of the container provided;: -- → -- Description and drawings/photographs of the shapes and constituent materials of the part of the body forming the engine compartment and the part of the passenger compartment nearest to it:: -- → -- Drawings or photographs of the position of the metal components housed in the engine compartment (e.g. heating appliances, spare wheel, air filter, steering mechanism, etc.):: -- → -- List of all subassemblies, which includes an electronic oscillator or switching frequency greater than 9 kHz (like ECU, steering, suspension, Body: -- → -- List of all Electrical components, which include Broadband EMI sources (like HAVC Motor,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Wiper Motor, Window motor and Horn etc.) The list shall include Device name, Make/s and Part No. /Type ID/Drawing No: -- → -- Name of the manufacturer of the system (Steering column shaft / Gear shift control / Transmission) on which the Protective Device functions: XXYY Steering System → XXYY Steering System Type/Part no./Identification number(ID)/Drawing No.of the device on which the Protective Device functions: FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly → FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable only in case of steering column locking systems: Locks the Steering → Locks the Steering SYSTEM SOFTWARE (If fitted): NA → NA Make: -- → -- Version: -- → -- Operating System Details with Version: -- → -- DESCRIPTION OF DEVICE (If fitted): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Description of the VAS: - → - A detailed description of the alarm system and of the vehicle parts related to alarm system installed with Block Diagram: → A list of main components comprising the alarm system: - → - Components placed in Engine compartment: - → - Components placed in Passenger compartment: - → - Availability of radio/wireless transmission Yes/No: - → - Details of radio/wireless transmission if any: - → - frequency: - → - Availability of WPC approval for wireless technology Yes/No: - → - The measures taken against false alarm: - → - Range of protection offered by the device. Details about the detection mechanism (opening of Door, Bonnet, luggage compartment, intrusion etc.): - → - Details & specification of audible alarm: - → - Details & specification of optical alarm: - → - Details & specification of radio alarm if fitted: - → - Method of setting/ unsetting the device: - → - Type of key used for unsetting (Mechanical Key / electronic device): - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Number of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - Exit delay in Sec: - → - Entry Delay in sec: - → - Details of status display (0.5cd): - → - Details of self check , automatic failure indication: - → - Details & specification of audible alarm: - → - Details & specification of optical alarm: - → - Details & specification of radio alarm if fitted: - → - Method of setting/ unsetting the device: - → - Type of key used for unsetting (Mechanical Key / electronic device): - → - Number of interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - Exit delay in Sec: - → - Entry Delay in sec: - → - Details of status display (0.5cd): - → - Details of self check , automatic failure indication: - → - Details about the panic alarm , if provided: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Description of the Immobilizer: - → - A detailed description of the vehicle immobilizer and of the vehicle and the measures		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	taken against inadvertent activation with block diagram.: → The vehicle system(s) on which the immobilizer acts.: - → - Details of radio/wireless transmission if any: - → - Method of setting/ unsetting the device.: - → - Number of interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - A list of main components comprising the device: - → - Description: AL Part No. → AL Part No. Description · AL Part No.: Hella Part No → Hella Part No ECU: FXX01000 → FXX01000 ECU · FXX01000: 015.XYZ-01 → 015.XYZ-01 a. · Make: - → - b. · Type: - → - c. · Serial Number: - → - d. · Name and Address of manufacture: - → - a. · Make: - → - b. · Type: - → - c. · Serial Number: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	d. · Name and Address of manufacture: - → - Example of Engine No: HJEZXYZ891 → HJEZXYZ891		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
Applicable AIS-137 / notified consumption measurement procedure Fuel consumption and CO2 declaration	Base fuel type used for vehicle Type approval including blend level (Ethanol, Bio-diesel etc): Diesel (Maximum 7% bio-diesel blend) → Diesel (Maximum 7% bio-diesel blend) Name and address of engine manufacturing plant (In case of imported vehicles, above details shall be supplied for importer also): Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX → Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX Max. Recommended Gradeability (Stand-start) – in degrees: 10 degs → 10 degs Max. Recommended Gradeability (Stand-start) – in degrees: 8.5 degs → 8.5 degs Vehicle kerb weight kg: Refer Annexure T7 - B Details for Recommended load body Refer Annexure T7 – C → Refer Annexure T7 - B Details for Recommended load body Refer Annexure T7 – C	CEA / report needs review	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Other axle: NA → NA Gross combination weight kg (applicable for articulated / Tractor Trailer combination vehicles): NA → NA Front axle: NA → NA Rear axle: NA → NA Trailer axle / other axles: NA → NA If applicable, maximum mass of trailer whic5h may be coupled: NA →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	NA Unbraked trailer: NA → NA Reference mass (for vehicles with GVW less than or equal to 3.5 ton) (kg.): Refer Annexure T7-B → Refer Annexure T7-B Maximum (Including driver) for completely built vehicles: D+2 → D+2 Maximum (Including driver) for completely built vehicles: → Make: APOLLO → APOLLO Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade description(s): CEAT WINMILE X3-R → CEAT WINMILE X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 Tyre rolling Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer - TC7719XX000 003XXXF → Refer - TC7719XX000 003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022- 2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022- 2023/300002XXX X/TA/XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L- 756XXXX → CM/L- 756XXXX Identification: TAC No. /		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E- Marking: NA → NA Others (For articulated / combination vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00 R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other (for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 325 mm → 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Type: Fully		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	synchronized 5 - forward gears with a Constant mesh reverse gear → Fully synchronized 5 - forward gears with a Constant mesh reverse gear Model name (if any): XXX-XXX → XXX-XXX Model name (if any): YYYY-YY → YYYY-YY	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Gear shifting control system (Sketch showing gear shifting arrangement): → No. of gears: Forward 5 + 1 Rev → Forward 5 + 1 Rev Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Rear axle ratio: 4.56 → 5.13 Type of steering gear (Re-circulating ball / Worm & Roller / Rack & Pinion / Others): Re-circulating ball type → Re-circulating ball type Make: 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd → 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 Type and description (Leaf / Coil / Air / Semi- pneumatic/ Torsion bar): Leaf → Leaf Front: Parabolic spring- Rigid axle suspension → Parabolic spring- Rigid axle suspension Rear: Semi elliptic spring-		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Rigid axle suspension → Semi elliptic spring- Rigid axle suspension Fuel Gauge: LED Bars → LED Bars Engine coolant temperature: Red Color – Thermometer Symbol → Red Color –	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Thermometer Symbol Engine low oil pressure: Red Color – Oil Tank Symbol → Red Color – Oil Tank Symbol Weight of seats: Driver seat : 11.5 Kg & Co-driver seat : 11Kg → Driver seat : 11.5 Kg & Co-driver seat : 11Kg Description of seats: -- → -- Make: -- → -- Number of seats fitted or capable of being fitted with head restraints, adjustable or not adjustable.: -- → -- Description of the adjustment, displacement and locking systems of the seat or of its parts and a description of occupant protection system against displacement of luggage.: -- → -- Description of seat anchorage: -- → -- Longitudinal / Vertical (If applicable) position of the seats during the tests.: -- → -- Drawings, diagrams and plans of the seats, their anchorages on the vehicle, the floor layout, the adjustment and displacement system of the seats and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	their parts, and their locking devices and of additional occupant protection system against displacement of luggage.: -- → -- In the case of seats fitted with head restraints, the head restraint shall be shown on all drawings, diagrams and photographs.: -- → -- Seat Identification No. / Part No/Drawing	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Number.: ID Number/ Part Number/D rawing number Weight Description Make (Seat (kg) Manufactu rer/ Vehicle Manufactu rer): -- → -- Description: Make → Make Description: ID Number/ Part Number/D rawing number (Seat Manufactu rer/ Vehicle Manufactu rer) → ID Number/ Part Number/D rawing number (Seat Manufactu rer/ Vehicle Manufactu rer) Description: Weight (kg) → Weight (kg) Brief description and drawings of the seat type, its attachment fittings and its adjustment, displacement and locking systems including the minimum distance between fitting points.: D+1 → D+1		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Installation drawing showing rear extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	make, unladen pressure) etc., distance between tyre & RUPD RH & LH, frame rear width, P1, P2 points.: - → - Make: M/s. XXXXYYYY → M/s. XXXXYYYY Material (Metallic / Plastic etc.): Metallic → Metallic Nominal thickness mm: 1.0mm → 1.0mm Capacity, Litre: 40 Lts, 30 Lts → 40 Lts, 30 Lts Detailed drawing of the fuel tank assembly with material specifications: FHSXXXXY FHSXXXXY → FHSXXXXY FHSXXXXY Detailed drawing indicating the position / location of the fuel tank(s) in the vehicle.: A5LXXX00 (50 LTS) A5LXXX00 (40 LTS) → A5LXXX00 (50 LTS) A5LXXX00 (40 LTS) Identification No. / Part No.: FHS00544 FfffXXXcdHXXS00744 → FHS00544 FfffXXXcdHXXS00744 Detailed engineering drawing as specified in AIS – 013 or photographs showing the mounting details with dimensions.: NA → NA Cross sectional Drawing of Gear shift	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	lever (Knob): FXA0XXYY (Non AC) / FXA0XXYY (AC) → FXA0XXYY (Non AC) / FXA0XXYY (AC) Drawings and/or photographs of the locations of the statutory plates and inscriptions and of the vehicle		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	identification number: X4W0XYZ4 / VFT0XYZ2 → X4W0XYZ4 / VFT0XYZ2 Drawings and / or photographs of the official part of the plates and inscriptions (completed example with dimensions): X4WXYZ44 → X4WXYZ44 Location of Vehicle Descriptor Section (VDS) in Vehicle Identification Number (VIN): 4thto 9thCharacter of 17 Characters VIN → 4thto 9thCharacter of 17 Characters VIN Brief description of the Provision for emergency exit provided in case of sealed windows: -- → -- Dimensions of the instructions describing each motion necessary to unlatch and open the exit are provided from the release mechanism.: -- → -- In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → -- Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	distance between front extremity & FUPD, width of FUPD, ID		
(continued)	marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Brief description of the vehicle type as regards its structure, dimensions, lines and constituent materials:: -- → -- Description of the protective system installed in the vehicle:: -- → -- Description of the protective system installed in the vehicle:: → Description of interior arrangements or fittings that might affect the tests:: -- → -- A detailed description, including photographs and/or drawings, of the vehicle type with respect to the structure, the dimensions, the design and the constituent materials, the side walls of the passenger compartment (exterior and interior), including specific details of the protection system, where applicable:: -- → -- Drawing / Images of position & arrangement of Engine: -- → -- Drawing/ Images of Under bonnet (Engine house) Components.: -- → -- Construction: Integrated / Attached with Mounts: 4 Mounts, Fixed CABIN → 4 Mounts, Fixed CABIN 1st ·	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st · Overall Ratio / AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69 2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC: 12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56 4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 → 20.44 Description · Make: ID	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number/ Part Number/Drawing number (Seat Manufacturer/ Vehicle Manufacturer) → ID Number/ Part Number/Drawing number (Seat Manufacturer/ Vehicle Manufacturer) Description · Make: Weight (kg) → Weight (kg) B 51.0 B49.0 · Pedestrian protection as per AIS-100: The distance, measured longitudinally on a horizontal plane, between the transverse centerline of the front axle & the R point of the driver’s seat is less than 1100 mm (Yes / No).: NA → NA Brief description of failure warning tell–tale: - → - Brief description and features: - → - Brief description and features: - → - Description: -- → -- Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No Manufacturer’s documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	version as appropriate.: -- → -- Under fault conditions, vehicle response corresponds to that described in the manufacturer’s documents / failure analysis and safety concept is verified.: -- → -- A brief written explanation sufficient to describe the ESC system basic operational characteristics.: -- → -- Logic diagram: -- → -- Understeer information.: -- → -- Static Stability Factor: -- → -- Make and Country of manufacturer(if imported): -- → -- Software Id / version: -- → -- Brief description of failure warning tell–tale: -- → -- Brief description and features: -- → -- Brief description and features: -- → -- System identification including software level identifier: -- → -- List of motor vehicles by description and configuration that are covered by the information document: -- → -- Wheel type (single or twin) and tyre type (e.g. structure, category of use, size): -- → -- Gearbox type (e.g. manual, automated manual, semi-automatic, automatic): -- → -- Management of the engine or any	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	other source(s) of motive power e.g.Torque Management, control of fuel	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	supply etc.: -- → -- Component description:: -- → -- Sensor(s) External to the Controller: -- → -- Function: -- → -- Limitations on the location of the sensors: -- → -- Identification (e.g. part numbers): -- → -- Controller(s): -- → -- General description and function: -- → -- Functionality of internal sensors (if applicable): -- → -- Hardware identification (e.g. part numbers): -- → -- Software identification: -- → -- Limitations on the location of the controller(s): -- → -- Additional features: -- → -- Modulators: -- → -- General description and function: -- → -- Hardware identification (e.g. part numbers): -- → -- Software identification (if applicable): -- → -- Limitations: -- → -- Electrical Equipment: -- → -- Circuit diagrams: -- → -- Powering methods: -- → -- Pneumatic circuits: -- → -- Safety aspects of the electronic system in accordance with Annex S of IS 11852: -- → -- List of complex functions covered::		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	-- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Manufacturer’s documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- A list of input and output variables, including their working range, is provided.: -- → -- Documentation includes an inventory of components, outlines the function of each unit, its interconnection/interaction with other systems and defines signal flow priorities.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software version as appropriate.: -- → -- Safety concept statement verifies that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- System architecture, design methods and tools identified.: -- → -- Information regarding	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	design provisions in case of failure is provided including any error messages, warning signals, partial performance	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	conditions, back-up modes and/or removal of high- level functions.: -- → -- Additional material and analysis (FMEA, FTA) of fault conditions made available for inspection and maintained by the manufacturer.: -- → -- Document reference and date of inspection:: -- → -- Vehicle complies with all other performance requirements of IS 15986:2015 and meets manufacturer’s specifications under non-fault conditions.: -- → -- Under fault conditions, vehicle response corresponds to that described in the manufacturer’s documents / failure analysis and safety concept is verified.: -- → -- Electro-magnetic compatibility: -- → -- Documentation demonstrating compliance with AIS-004(Part 3).: -- → -- Component description: -- → -- Controller(s) (a) General description and function; (b) Identification e.g. part numbers; (c) Limitations on the location of the controller(s); (d) Additional		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	features.: -- → -- Modulators (a) General description and function; (b) Identification; (c) Limitations.: -- → -- Manufacturer’s documents explain the functionality and safety concept of the system and describe how the	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	operational status may be checked.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software version as appropriate.: -- → -- (see annex 4 of this report for test data): -- → -- (see annex 4 of this report for test data): -- → -- Type / Description: Fully built vehicle, D+2, Non AC, 3.39 Ton GVW → Fully built vehicle, D+2, Non AC, 3.39 Ton GVW Make: Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. → Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. Model name / identification: ALXYZABC0 → ALXYZABC0 Type: Compression Ignition → Compression Ignition Bore x stroke (mm): 80 X 98 → 80 X 98 No. of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	cylinders: 3 → 3 Displacement (cm³): 1468 cc → 1468 cc Compression ratio: 16.5±1.0 : 1 → 16.5±1.0 : 1 Max. Engine output (KW @ rpm): 55.1	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Kw @ 3300 rpm → 55.1 Kw @ 3300 rpm Max. Torque (Nm @ rpm): 170Nm @ 1600 to 2400 rpm → 170Nm @ 1600 to 2400 rpm Air cleaner type: Dry type → Dry type Type of Fuel: Diesel → Diesel Make and model: RSB Transmission (T210R) → RSB Transmission (T210R) Make and model: AVTEC (A5M210) → AVTEC (A5M210) Type: Manual gearbox with fully synchro mesh & Rev with sliding mesh → Manual gearbox with fully synchro mesh & Rev with sliding mesh No. of gears: 5 Forward and 1 Reverse → 5 Forward and 1 Reverse 1st: 4.62 → 4.62 1st: 4.60 → 4.60 1st: 21.53 → 21.53 1st: 21.44 → 21.44 2nd: 2.68 → 2.68 2nd: 2.69 → 2.69 2nd: 12.67 → 12.67 2nd: 12.71 → 12.71 3rd: 1.51 → 1.51 3rd: 1.49 → 1.49 3rd: 7.34 → 7.34 3rd: 7.25 → 7.25 4th: 1.0 → 1.0 4th: 1.0 → 1.0 4th: 4.56 → 4.56 4th: 4.56 → 4.56 5th: 0.70 → 0.70 5th: 0.71 → 0.71 5th: 3.64 → 3.64 5th: 3.67 → 3.67		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Reverse: 4.39 → 4.39		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Reverse: 20.36 → 20.36 Reverse: 4.38 → 4.38 Reverse: 20.34 → 20.34 Drive Axle (Front / Rear / All): Rear → Rear Front axle ratio: NA → NA Type / Description: Re-circulating ball type → Re-circulating ball type Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT 12 PR Tyre size designation including ply rating: 215/75 R15 LT → 215/75 R15 LT Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2 → 3.26 kg/cm2 Wiping system (Brief description): Twin blade, Tandem Pattern → Twin blade, Tandem Pattern	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Material: Metallic → Metallic Capacity (l): 50 lts , 40 lts → 50 lts , 40 lts Dimensions: 2490WB → 2490WB		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Dimensions: 2640WB → 2640WB Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Overall width (mm): 1842 → 1842 Overall Length (mm): CC → CC Overall Length (mm): 4750 → 4750 Overall Length (mm): 4988 → 4988 Overall Length (mm): FSD / HSD → FSD / HSD Overall Length (mm): 4925 → 4925 Overall Length (mm): 5225 → 5225 Overall height (mm): 1961 → 1961 Overall height (mm): 1948 → 1948 Maximum gradeability in 1st gear: GVW 2890 kg → GVW 2890 kg Maximum gradeability in 1st gear: GVW 3390 kg → GVW 3390 kg Maximum gradeability in 1st gear: 10 Deg → 10 Deg Maximum gradeability in 1st gear: 8.5 Deg → 8.5 Deg Weights: Base → Base Weights: Variant 1 → Variant 1 Weights: Variant 2 → Variant 2 Weights: Variant 3 → Variant 3 Weights: Variant 4 → Variant 4 Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	spare wheel, tools, etc.) (kg): - → - Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Description of energy converters and power plant. (In the case of a vehicle that can run either on petrol, diesel, etc., or also in combination with another fuel, items shall be repeated.1): Diesel → Diesel Engine Manufacturer:: Ashok Leyland Limited → Ashok Leyland Limited Manufacturer's engine model code (as marked on the engine, or other means of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	identification): Model Code: ALXYZABC0 Enclosure – 4A → Model Code: ALXYZABC0 Enclosure – 4A Location of engine model code and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	engine serial number on block / crankcase (drawing to be attached): Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A → Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A Working principle: positive ignition / compression- ignition, four-stroke / two-stroke / rotary cycle (DI/IDI) (NA/TC/TCIC/Any other): Compression Ignition, four-stroke, DI, TCIC → Compression Ignition, four-stroke, DI, TCIC Number, arrangement of cylinders:: 3, Inline → 3, Inline Bore----------(mm): 80 → 80 Stroke: ----------(mm): 98 → 98 Firing order:: 1-2-3 → 1-2-3 Engine capacity:----------(cm3): 1468 → 1468 Volumetric compression ratio (specify	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	tolerance): 17.5±1.0 : 1 → 17.5±1.0 : 1 Drawings of combustion chamber and piston crown and, in the case of positive ignition engine, piston rings:: Refer Enclosure – 4B → Refer		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure – 4B Vehicle with Defeat Device Yes / No: No → No If yes – Details of the Defeat device: NA → NA Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane equivalent) in PPM content by volume in the exhaust gas with the engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum net power:..... kW at min-1: 55.1 KW @ 3100 ± 50 rpm → 55.1 KW @ 3100 ± 50 rpm Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm Maximum net torque: .....Nm at: -------min-1 ...(manufacturer's declared value): 170 Nm @ 1600 – 2400 rpm → 170 Nm @ 1600 – 2400 rpm Maximum rated speed (Specify	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	tolerance): 3100 rpm ± 50 rpm → 3100 rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Type of fuel used : Mono		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Fuel / Bi-Fuel / Flex Fuel / Dual Fuel: Mono Fuel → Mono Fuel Fuel type: Diesel / Gasoline / LPG / CNG / Biomethane / Bio-H methane / HCNG / Biogas / LNG / Ethanol ((E85) / 2 (E100)) /ED (95) Flex fuel Methanol M15 / M100 / MD95 / Biodiesel up to 100% / Hydrogen.: Diesel → Diesel Maximum amount of biofuel acceptable in fuel (manufacturer's declared value): per cent by volume: 7% → 7% By fuel injection (compression-ignition only): (Yes/No): Yes → Yes System description:: Unit Pump Common Rail Direct Injection system → Unit Pump Common Rail Direct Injection system Working principle: direct-injection/pre-chamber/swirl Chamber: Direct Injection → Direct Injection Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Plunger Type & FHL0XYZ0 → Plunger Type & FHL0XYZ0 Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Injection advance curve (Specify the tolerances):: ECU Controlled → ECU Controlled Governor: ECU Controlled → ECU Controlled Cut-off point under load:…(min-1): 3300 → 3300 Cut-off point without load (Maximum no-load speed)…(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Multi-Hole Injectors (7 holes) FKXXYYZZ → Multi-Hole Injectors (7 holes) FKXXYYZZ Opening pressure (kPa) (specify the tolerances) or characteristic diagram: 15000 to 23000 kPa → 15000 to 23000 kPa Length and internal diameter…(mm): Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm → Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Internal Diameter – 3mm Common rail make & type: M/s. XXXXYYY, Hot Forged Rail Type → M/s. XXXXYYY, Hot Forged Rail Type Cold start system: NA → NA Make(s):: NA → NA Type(s) &		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: NA → NA Description:: NA → NA Auxiliary starting aid: NA → NA Make(s):: NA → NA Type(s):: NA → NA System description:: NA → NA Electronic controlled injection: (Yes/No): Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA → NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s. XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Metering type Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type By fuel injection (positive ignition only): (Yes/No): No → No Working principle: intake manifold (single/multi- point)/direct injection/other (specify): NA → NA Make(s) of Injection Pump :: NA → NA Type(s) & Identification Number of Injection Pump:: NA → NA System description (in the case of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	systems other than continuous injection give equivalent details):: NA → NA Make and type of the control unit:: NA → NA Make and type of the fuel regulator:: NA → NA Make and type of the air-flow sensor:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Make and type of the micro-switch:: NA → NA Make and type of the throttle housing:: NA → NA Make and type of the water temperature sensor:: NA → NA Make and type of the air temperature sensor:: NA → NA Injectors: Opening pressure (Specify the tolerances):,..... (kPa) or characteristic diagram:: NA → NA Make(s):: NA → NA Type(s):: NA → NA Identification Number: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA Cut-off point under load (min-1): NA → NA Injection timing:: NA → NA Cold start system:: NA → NA Operating principle(s):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Operating limits/settings(Specify the tolerances): NA → NA Pressure, (Specify the tolerances).....(kPa) or characteristic diagram:: 320 ± 50 kPa → 320 ± 50 kPa Rated voltage:...V, positive/negative ground: 12V, Negative Ground → 12V, Negative Ground		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Generator/ Alternator: Alternator → Alternator Type:: DC – Alternator with Voltage regulator → DC – Alternator with Voltage regulator Nominal output: ...VA: 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps → 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps Ignition: NA → NA Make(s):: NA → NA Type(s):: NA → NA Working principle:: NA → NA Ignition advance curve or map(Specify the tolerances):: NA → NA Static ignition timing (Specify the tolerances):.....degrees before TDC: NA → NA Spark plugs: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Gap setting----------(mm): NA → NA Nominal resistance (kilo ohm) (if resistive type): NA → NA Ignition coil(s): NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Ignition condenser: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA EMI suppressor cap / Device / Electronic unit: NA → NA Make: NA → NA Type (Resistive/Capacitive): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification number: NA → NA Nominal resistance (kilo ohm): NA → NA Terminology and Drawing of interference Suppression equipment: NA → NA H.T. Cable: NA → NA Make and Place: NA → NA Type (Resistive/Non-resistive): NA → NA Length (mm) (if resistive type): NA → NA Outside dia. (mm) (if resistive type): NA → NA Nominal resistance kilo ohm, (if resistive type): NA → NA Radiator drawing(s): Refer enclosure 4C2 & 4C4 → Refer enclosure 4C2 & 4C4 Radiator drawing(s): Refer enclosurev 4C1 & 4C3 → Refer enclosurev 4C1 & 4C3	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make: Mahle Behr → Mahle Behr Make: Tata Toyo → Tata Toyo Type (s): Cross flow → Cross flow Relief valve pressure setting: 0.9 kg/cm2 → 0.9 kg/cm2 Fan characteristics (Fan power, kW) Enclose the fan power curve corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part No: 9 → 9 Material of blades (metal /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ± 150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150 (AC) Radiator core open area (cm²): 1906 cm2 → 1906 cm2 Cooling system: liquid/air: Liquid → Liquid Nominal setting of the engine temperature control o mechanism (C)::: 85 deg C → 85 deg C	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Liquid: Yes → Yes Nature of liquid:: Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L → Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L Circulating pump(s): (Yes/No): Yes → Yes Characteristics of Circulating pump: Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm → Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm Make(s):: XXXXX		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Fasteners Ltd → XXXXX Fasteners Ltd Type(s):: Impeller Type → Impeller Type Drive ratio(s):: 1: 1.12 → 1: 1.12 Description of the fan and its drive mechanism:: Electric Fan , Motor Driven → Electric Fan , Motor Driven Blower: (Yes/No): No → No Characteristics of blower: NA → NA Make(s):: NA → NA Type(s):: NA → NA Drive ratio(s):: NA → NA Pressure charger: (Yes/No) Turbocharger	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	/ Supercharger: Yes, Turbo charger → Yes, Turbo charger Make(s):: M/s. XYZ, India → M/s. XYZ, India Type(s) & Identification Number:: Waste Gate with Vacuum Actuator, X3M1XXXX → Waste Gate with Vacuum Actuator, X3M1XXXX Description of the system (maximum charge pressure:.....(kPa), waste-gate, if applicable): Waste gate turbocharger with vacuum actuator Maximum Charge pressure - 145 kPa → Waste gate turbocharger with vacuum actuator Maximum Charge pressure - 145 kPa Inter-cooler: (Yes/No): Yes → Yes Type: air-air / air-water: Air – Air → Air – Air Make :: Mahle Behr → Mahle Behr Make :: Tata Toyo → Tata Toyo		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: B1V42XXX → B1V42XXX Identification Number:: B1V42YYY → B1V42YYY Air pressure drop across the intercooler: 15 kPa → 15 kPa Intake depression at rated engine speed and at 100 per cent load (Specify location of measurement) Minimum allowable:... (kPa) Maximum allowable:... (kPa): Minimum : 0 kPa Maximum : 6.0 kPa	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Description and drawings of inlet pipes and their accessories (plenum chamber, heating device, additional air intakes, etc.):: Intake manifold with Electronic throttle valve Refer Enclosure 4D → Intake manifold with Electronic throttle valve Refer Enclosure 4D Intake manifold description including dimensional drawings and/or photographs):: Aluminium Intake manifold with pressure sensor Refer Enclosure 4D → Aluminium Intake manifold with pressure sensor Refer Enclosure 4D Air filter, Dimensional drawings: Refer Enclosure 4E → Refer Enclosure 4E		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Make(s):: M/s. Fleet guard → M/s. Fleet guard Type(s) & Identification Number:: Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 → Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 Intake silencer, Dimensional drawing: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA Description and Dimensional drawing of the exhaust manifold:: Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F → Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F Description and drawing with dimensions of the exhaust system:(Description with general arrangement of exhaust system along with its routing indicating the lengths of exhaust pipe, tail pipe and: Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	O2 sensors provided before and after LNT-DPF assembly → Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly Effective volume of exhaust-system (specify the tolerance & range) in liters (from exhaust manifold / TC outlet to tail pipe end), Enclose the exhaust system dimensional drawing and indicate the volume of each parts clearly: 13.05 ± 1 litre Refer Enclosure 4G → 13.05 ± 1 litre Refer Enclosure 4G Description and drawing of the exhaust Muffler/Silencer:: NA → NA Type(s) & Identification Number:: NA → NA Make: NA → NA Number:.....: NA → NA Internal diameter of exhaust pipe (mm)(min): diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G → diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G Minimum distance between exhaust pipe(s) and the fuel		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	line: >150 mm → >150 mm Auxiliary Noise shields for compliance to IS 3028 and / OR AIS-020 (If Provided): NA → NA Material: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Layout of noise shield / Photographs / Diagram Showing arrangements indicating fitment on vehicle.: NA → NA Maximum allowable exhaust back pressure at rated engine speed and at 100 per cent load (location of measurement): ... (kPa): 30.0 kpa (150 mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Minimum cross-sectional areas of inlet and outlet ports:: Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm → Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm Inlet (mm): 8.00 → 8.00 Exhaust (mm): 8.00 → 8.00 Opening: 17° bTDC → 17° bTDC Closing: 51° aBDC → 51° aBDC Opening: 66° bBDC → 66° bBDC Closing: 25° aTDC → 25° aTDC Transfer: NA → NA Opening: NA → NA Closing: NA → NA Reference and/or setting ranges (Specify the tolerances):: On Cold Engine → On Cold Engine Inlet: 0.3 to 0.4 mm → 0.3 to 0.4 mm Exhaust: 0.3 to 0.4 mm → 0.3 to 0.4 mm		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Distribution by ports: NA → NA Volume of crank-case cavity with piston at TDC: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Reed valve fitted (Yes / No): No → No Description of inlet ports, scavenging and exhaust ports with corresponding timing.: NA → NA Device for recycling crankcase gases (description and drawings):: Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J → Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx Catalyst (if any, and if not covered by another heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF / DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No Element – 1 No. → Catalytic Converter – 1No Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	for LNT (periodic / continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:……cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap Bulk Density (GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle and Characteristics:: Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	downstream SCR catalyst for the reduction of NOX during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY Type and Identification number:: Type - Passive NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite → Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA Make:: NA → NA Type and Idetification number :: NA → NA NOx Sensor (Before /After of SCR):: No → No Make:: NA → NA Type and Identification number:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent engine test bench cycles, between two cycles where regenerative phases occur under the conditions equivalent to Type I test (Distance "D" in t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration (Distance “d” in AIS 137): 2 cycles → 2 cycles Description of method employed to determine the number of cycles	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA International standard (where appropriate):: NA → NA Frequency of reagent refill: continuous/maintenance (where appropriate):: NA → NA Normal operating temperature range (°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY → YYYYYYY Air injection: (Yes/No): No → No Type (pulse air, air pump, etc.) & Identification Number:: NA → NA Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other): Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY → FEW0XXYY Characteristics (flow rate, etc.):: Flow rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes EGR Cooler Make and Identification Number: M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:.....g: NA → NA Canister working capacity:.....g: NA → NA Canister bed volume:.....litre: NA → NA Canister Purge and vent strategy description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve : (Yes/No): NA → NA Make and Identification Number: NA →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA Operating pressure (kPa): NA → NA Liquid fuel hoses material :: NA → NA Schematic drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	drawing (Periodic / Continuous): NA → NA Particulate trap: (Yes/No): Yes → Yes Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Dimensions and shape of the particulate trap (capacity):: Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Regeneration Type Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure 4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh: 90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type Material - Silicon Carbide → Structure - Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI Electronic Control Unit (ECU): Yes → Yes	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number: P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA Misfire detection:: NA → NA Oxygen sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K → Refer Enclosure 4K Particulate traps		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method):: Refer Enclosure 4K → Refer Enclosure 4K List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K → Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K A description of the type and number of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3 EUDC A description of the type of the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	all sensed components with the strategy for fault detection and MI activation (fixed number of driving cycles or statistical method), including a list of relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive explanation for the data given in service $05 Test ID $21 to FF and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles – Diagnostics on Controller Area Network (CAN) – Part 4: Requirements for emissions- related systems", a	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137): Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA → NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA Systems to ensure the correct operation of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent deactivation of the driver inducement, for use by the rescue services [Note: cross- reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	/disable after parking: NA → NA Lowest		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct operation of NOx control measures: NA → NA Details of any devices designed to influence fuel economy (if not covered by other items):: NA → NA LPG fuelling system: (Yes/No): No → No Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to LPG or back:: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.): NA → NA NG fuelling system: (Yes/No): No → No Approval number: NA → NA Electronic engine management control	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	unit for NG fuelling: NA → NA Make(s):: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation:: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to NG or back: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.):: NA → NA Hydrogen fuelling system: (Yes/No): No → No Electronic engine management control unit for hydrogen fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to hydrogen or back:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA H2NG fuelling system: (Yes/No): No → No		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Percentage of hydrogen in the fuel (maximum specified by the manufacturer): NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to H2NG or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA Electric motor(Traction): No → No Type (winding, excitation):: NA → NA Maximum hourly output: kW(manufacturer’s declared value): NA → NA Maximum net power:kW(manufacturer’s declared value): NA → NA Maximum 30 minutes power: kW	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(manufacturer’s declared value): NA → NA Operating voltage: V: NA → NA Make and Identification Number:: → Battery (Traction): NA → NA Number of cells:: NA → NA Mass: .....(kg): NA → NA Capacity: Ah		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Amp-hours): NA → NA Position:: NA → NA Engines or motor combinations: NA → NA Hybrid Electric Vehicle: (Yes/No): No → No Category of Hybrid Electric vehicle Off Vehicle Charging/Not Off Vehicle Charging9: NA → NA Operating mode switch: with/without: NA → NA Selectable modes: NA → NA Pure electric: (Yes/No): NA → NA Pure fuel consuming: (Yes/No): NA → NA Hybrid modes: (Yes/No) (if yes, short description): NA → NA Description of the energy storage device: (traction battery, capacitor, flywheel/generator...): NA → NA Make(s):: NA → NA Type(s) :: NA → NA Identification number:: NA → NA Kind of electrochemical couple:: NA → NA Energy: ..... (for battery: voltage and capacity Ah in 2 h, for capacitor: J): NA →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA Charger: on board/external/without: NA → NA Electric machines (describe each type of electric machine separately): NA → NA Make:: NA → NA Type:: NA → NA Primary use: traction motor/generator: NA → NA When used as traction motor: mono motor / multi motors (number):: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Maximum power: kW: NA → NA Working principle:: NA → NA Direct current/alternating current/number of phases:: NA → NA Separate excitation/series/compound: NA → NA Synchronous/asynchronous: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA Identification number:: NA → NA Power controller: NA → NA Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Vehicle electric range..... km (according to AIS 137):: NA → NA Manufacturer’s recommendation for preconditioning:: NA → NA CO2 (g/km) (Applicable for category M1 with GVW<3.5 T): NA → NA Declared : (rounded to 3 decimal	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	places): NA → NA Fuel consumption (l/100 km) for Petrol, LPG or Diesel and (kg/100km) for CNG and (kWh/100 km) for Electric Driven Vehicles(Applicable for category M1 with GVW<3.5 T) (rounded to 3 decimal places)*: NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA *calculated on the declared CO2: NA → NA CO2 Reducing		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	technologies Available (Yes/No): No → No Regenerative braking: NA → NA Start-Stop System: NA → NA Tyre pressure monitoring system: NA → NA 6 or more Speed Transmission: NA → NA Any other technology (manufacture to enclose detail): NA → NA Electric energy consumption for electric vehicles Wh/km: NA → NA Electric energy consumption for pure electric vehicles Wh/km: NA → NA Electric energy consumption for externally chargeable hybrid electric vehicles: NA → NA Electric energy consumption (condition A, combined) Wh/km: NA → NA Electric energy consumption (condition	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	B, combined) Wh/km: NA → NA Electric energy consumption (weighted combined) ….Wh/km: NA → NA Temperatures permitted by the manufacturer: NA → NA Cooling system: Yes → Yes Liquid cooling: Yes → Yes Maximum temperature at outlet: …….(°C): 100 °C → 100 °C Air cooling: NA → NA Reference point:: NA → NA Maximum temperature at reference point: …….(°C): NA → NA Maximum outlet temperature of the inlet intercooler: .. (°C): 45 °C → 45		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	°C Maximum exhaust temperature at the point in the exhaust pipe(s) adjacent to the outer flange(s) of the exhaust manifold: …(°C): 740 °C → 740 °C Minimum: .. (°C): - 8 → - 8 Maximum: .. (°C): 42 → 42 Minimum: …(°C): - 5 → - 5 Maximum: …(°C): 120 → 120 Lubrication system: Yes → Yes Description of the system: Forced Lubrication system → Forced Lubrication system Position of the lubricant reservoir:: Sump beneath crankcase → Sump beneath crankcase Feed system (by pump/injection into intake/mixing with fuel, etc.): Pump →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Pump Lubricating pump: Yes → Yes Make(s):: M/s. ZXWA → M/s. ZXWA Type(s):: XXYY Type → XXYY Type Mixture with fuel: No → No Percentage:: → Oil cooler: (Yes/No): Yes → Yes Drawing(s):..., or: Enclosure 4L → Enclosure 4L Make(s):: M/s. XXYY or Any other equivalent → M/s. XXYY or Any other equivalent Type(s):: DONUT → DONUT Moment of inertia of engine flywheel:: 0.1212 kg/m2 → 0.1212 kg/m2 Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Transmission Control Unit (ECU) (Yes/No):: No → No Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA OBD IUPR Family if any: XX+XYY → XX+XYY exhaust outlet location, indicated in a Schematic dimensional drawing.): Refer Enclosure 4G → Refer Enclosure 4G 1.2.11.2.1.1 4.4 · 3. · Calibration Verification Number:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Catalyst · P0420 · Oxygen sensor 1 and 2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Engine speed, engine load, A/F mode, catalyst temperature → Engine speed, engine load, A/F mode, catalyst temperature Catalyst · P0420 · Oxygen sensor 1 and 2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Two Type I cycles → Two Type I cycles Catalyst · P0420 · Oxygen sensor 1 and 2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Type I → Type I Type (Manual / power): Manual → Manual No. of wipers: Two → Two Make: XXYY System →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	XXYY System Type: 12V → 12V Identification mark: X5XYZ100 → X5XYZ100 Rated voltage: 12V → 12V Number of sweep Frequencies: 2 → 2 Highest sweep frequency (Cycles / min): 62±8 rpm → 62±8 rpm	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Lowest sweep frequency (Cycles / min): 40±5 rpm → 40±5 rpm Wiper arm: Provided → Provided Length: 500mm → 500mm Make: M/s XYZ → M/s XYZ Length: 450 mm → 450 mm Make: BCS (SYNDICATE WIPER SYSTEM) → BCS (SYNDICATE WIPER SYSTEM) Identification: X5XY0258 → X5XY0258 Washer tank: BWXY4501 → BWXY4501 Type: Plastic → Plastic Make: XYZ → XYZ Identification No. / Part No.: X5BXYZ168 → X5BXYZ168 Capacity l: 1.5Ltr → 1.5Ltr Material: H.D.P.E → H.D.P.E Washer tank motor / Washer pump: ELECTRIC MOTOR WITH PUMP → ELECTRIC MOTOR WITH PUMP Make: XYZ → XYZ Model: PEAD64XYZS00 → PEAD64XYZS00 Nozzle(s): XYZ → XYZ No. Of Nozzles: 2 → 2 Defroster: NA → NA Type: - → - Make: - → - Details of Demisting and / or Defrosting system: - → - Schematic Diagram of Engine cooling system: - → - Schematic Diagram of Heating system: - →		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	- Cross sectional view of complete air handling system including position of louvers (vents) on dash board: - → - Blower identification: - → - Blower Voltage and capacity (Watts): - → - If defrost system is fitted, specify defrost test temperature (-8 +/- 2 deg C / -18 +/- 3 deg C): → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	& related dimensions as per related standards.: → Description and sketch showing the detail positions of Tell-Tale and seating layout (For M category vehicles): NA → NA A brief technical description indicating the length of the cleaning period, the consumption of cleaning fluid during the cleaning period and the minimum capacity of the container provided;: -- → -- Description and drawings/photographs	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	of the shapes and constituent materials of the part of the body forming the engine compartment and the part of the passenger compartment nearest to it:: -- → -- Drawings or photographs of the position of the metal components housed in the engine compartment (e.g. heating appliances, spare wheel, air filter, steering mechanism, etc.):: -- → -- List of all subassemblies, which includes an electronic oscillator or switching frequency greater than 9 kHz (like ECU, steering, suspension, Body: -- → -- List of all Electrical components, which include Broadband EMI sources (like HAVC Motor,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Wiper Motor, Window motor and Horn etc.) The list shall include Device name, Make/s and Part No. /Type ID/Drawing No: -- → -- Name of the manufacturer of the system (Steering column shaft / Gear shift control / Transmission) on which the Protective Device functions: XXYY Steering System → XXYY Steering System Type/Part no./Identification number(ID)/Drawing No.of the device on which the Protective Device functions: FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable only in case of steering column locking systems: Locks the Steering → Locks the Steering SYSTEM SOFTWARE (If fitted): NA → NA Make: -- → -- Version: -- → -- Operating System Details with Version: -- → -- DESCRIPTION OF DEVICE (If fitted): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Description of the VAS: - → - A detailed description of the alarm system and of the vehicle parts related to alarm system installed with Block Diagram: → A list of main components comprising the alarm system: - → - Components placed in Engine compartment: - → - Components placed in Passenger compartment: - → - Availability of radio/wireless transmission Yes/No: - → - Details of radio/wireless transmission if any: - → - frequency: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Availability of WPC approval for wireless technology Yes/No: - → - The measures taken against false alarm: - → - Range of protection offered by the device. Details about the detection mechanism (opening of Door, Bonnet, luggage compartment, intrusion etc.): - → - Details & specification of audible alarm: - → - Details & specification of optical alarm: - → - Details & specification of radio alarm if fitted: - → - Method of setting/ unsetting the device: - → - Type of key used for unsetting (Mechanical Key / electronic device): - → - Number of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - Exit delay in Sec: - → - Entry Delay in sec: - → - Details of status display (0.5cd): - → - Details of self check , automatic failure indication: - → - Details & specification of audible alarm: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Details & specification of optical alarm: - → - Details & specification of radio alarm if fitted: - → - Method of setting/ unsetting the device: - → - Type of key used for unsetting (Mechanical Key / electronic device): - → - Number of interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - Exit delay in Sec: - → - Entry Delay in sec: - → - Details of status display (0.5cd): - → - Details of self check , automatic failure indication: - → - Details about the panic alarm , if provided: - → - Description of the Immobilizer: - → - A detailed description of the vehicle immobilizer and of the vehicle and the measures		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	taken against inadvertent activation with block diagram.: → The vehicle system(s) on which the immobilizer acts.: - → - Details of radio/wireless transmission if any: - → - Method of setting/ unsetting the device.: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Number of interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - A list of main components comprising the device: - → - Description: AL Part No. → AL Part No. Description · AL Part No.: Hella Part No → Hella Part No ECU: FXX01000 → FXX01000 ECU · FXX01000: 015.XYZ-01 → 015.XYZ-01 a. · Make: - → - b. · Type: - → - c. · Serial Number: - → - d. · Name and Address of manufacture: - → - a. · Make: - → - b. · Type: - → - c. · Serial Number: - → - d. · Name and Address of manufacture: - → - Example of Engine No: HJEZXYZ891 → HJEZXYZ891		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
AIS-137 Part 3 OBD interface, fault detection and calibration identification	Base fuel type used for vehicle Type approval including blend level (Ethanol, Bio-diesel etc): Diesel (Maximum 7% bio-diesel blend) → Diesel (Maximum 7% bio-diesel blend) Name and address of engine manufacturing plant (In case of imported vehicles, above details shall be supplied	CEA / report needs review	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	for importer also): Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX → Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX Max. Recommended Gradeability (Stand-start) – in degrees: 10 degs → 10 degs Max. Recommended Gradeability (Stand-start) – in degrees: 8.5 degs → 8.5 degs Vehicle kerb weight kg: Refer Annexure T7 - B Details for Recommended load body Refer Annexure T7 – C → Refer Annexure T7 - B Details for Recommended load body Refer Annexure T7 – C Other axle: NA → NA Gross combination weight kg (applicable for articulated / Tractor Trailer combination vehicles): NA → NA Front axle: NA → NA Rear axle: NA → NA Trailer axle / other axles: NA → NA If applicable, maximum mass of trailer whic5h may be coupled: NA →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	NA Unbraked trailer: NA → NA Reference mass (for vehicles with GVW less than or equal to 3.5 ton) (kg.): Refer	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Annexure T7-B → Refer Annexure T7-B Maximum (Including driver) for completely built vehicles: D+2 → D+2 Maximum (Including driver) for completely built vehicles: → Make: APOLLO → APOLLO Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade description(s): CEAT WINMILE X3-R → CEAT WINMILE X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 Tyre rolling Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer - TC7719XX000 003XXXF → Refer - TC7719XX000 003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022-2023/300002XXX	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	X/TA/XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L- 756XXXX → CM/L- 756XXXX Identification: TAC No. /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E- Marking: NA → NA Others (For articulated / combination	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00 R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other (for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 325 mm → 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	kg/cm2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26 kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Type: Fully		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	synchronized 5 - forward gears with a Constant mesh reverse gear → Fully synchronized 5 - forward gears with a Constant mesh reverse gear Model name (if any): XXX-XXX → XXX-XXX Model name (if any): YYYY-YY → YYYY-YY Gear shifting control system (Sketch showing gear shifting arrangement): → No. of gears: Forward 5 + 1 Rev → Forward 5 + 1 Rev Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Rear axle ratio: 4.56 → 5.13 Type of steering gear (Re-circulating ball / Worm & Roller / Rack & Pinion / Others): Re-circulating ball type → Re-circulating ball type Make: 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd → 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2. YYYY – 18.2 :1	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Type and description (Leaf / Coil / Air / Semi- pneumatic/ Torsion bar): Leaf → Leaf Front: Parabolic spring- Rigid axle suspension → Parabolic spring- Rigid axle suspension Rear: Semi elliptic spring-		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Rigid axle suspension → Semi elliptic spring- Rigid axle suspension Fuel Gauge: LED Bars → LED Bars Engine coolant temperature: Red Color – Thermometer Symbol → Red Color – Thermometer Symbol Engine low oil pressure: Red Color – Oil Tank Symbol → Red Color – Oil Tank Symbol Weight of seats: Driver seat : 11.5 Kg & Co-driver seat : 11Kg → Driver seat : 11.5 Kg & Co-driver seat : 11Kg Description of seats: -- → -- Make: -- → -- Number of seats fitted or capable of being fitted with head restraints, adjustable or not adjustable.: -- → -- Description of the adjustment, displacement and locking systems of the seat or of its parts and a description of occupant protection system against displacement of luggage.: -- → -- Description of seat anchorage: -- → -- Longitudinal / Vertical (If applicable) position of the seats during the tests.: --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	→ -- Drawings, diagrams and plans of the seats, their anchorages on the vehicle, the floor layout, the adjustment and displacement system of the seats and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	their parts, and their locking devices and of additional occupant protection system against displacement of luggage.: -- → -- In the case of seats fitted with head restraints, the head restraint shall be shown on all drawings, diagrams and photographs.: -- → -- Seat Identification No. / Part No/Drawing Number.: ID Number/ Part Number/D rawing number Weight Description Make (Seat (kg) Manufactu rer/ Vehicle Manufactu rer): -- → -- Description: Make → Make Description: ID Number/ Part Number/D rawing number (Seat Manufactu rer/ Vehicle Manufactu rer) → ID Number/ Part Number/D	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	rawing number (Seat Manufactu rer/ Vehicle Manufactu rer) Description: Weight (kg) → Weight (kg) Brief description and drawings of the seat type, its attachment fittings and its adjustment, displacement and locking systems including the minimum distance between fitting points.: D+1 → D+1 Installation drawing showing rear extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	make, unladen pressure) etc., distance between tyre & RUPD RH & LH, frame rear width, P1, P2 points.: - → - Make: M/s. XXXXYYYY → M/s. XXXXYYYY Material (Metallic / Plastic etc.): Metallic → Metallic Nominal thickness mm: 1.0mm → 1.0mm Capacity, Litre: 40 Lts, 30 Lts → 40 Lts, 30 Lts Detailed drawing of the fuel tank assembly with material specifications: FHSXXXXY	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FHSXXXXY → FHSXXXXY FHSXXXXY Detailed drawing indicating the position / location of the fuel tank(s) in the vehicle.: A5LXXX00 (50 LTS) A5LXXX00 (40 LTS) → A5LXXX00 (50 LTS) A5LXXX00 (40 LTS) Identification No. / Part No.: FHS00544 FfffXXXcdHXXS00744 → FHS00544 FfffXXXcdHXXS00744 Detailed engineering drawing as specified in AIS – 013 or photographs showing the mounting details with dimensions.: NA → NA Cross sectional Drawing of Gear shift lever (Knob): FXA0XXYY (Non AC) / FXA0XXYY (AC) → FXA0XXYY (Non AC) / FXA0XXYY (AC) Drawings and/or photographs of the locations of the statutory plates and inscriptions and of the vehicle		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	identification number: X4W0XYZ4 / VFT0XYZ2 → X4W0XYZ4 / VFT0XYZ2 Drawings and / or photographs of the official part of the plates and inscriptions (completed example with dimensions): X4WXYZ44 → X4WXYZ44 Location of Vehicle Descriptor Section (VDS) in Vehicle Identification Number (VIN): 4thto 9thCharacter of 17 Characters VIN → 4thto 9thCharacter of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	17 Characters VIN Brief description of the Provision for emergency exit provided in case of sealed windows: -- → -- Dimensions of the instructions describing each motion necessary to unlatch and open the exit are provided from the release mechanism.: -- → -- In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → -- Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side, distance between front extremity & FUPD, width of FUPD, ID		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Brief description of the vehicle type as regards its structure, dimensions, lines and constituent materials:: -- → -- Description of the protective system installed in the vehicle:: -- → -- Description of the protective system installed in the vehicle:: → Description of interior arrangements or fittings that might affect the tests:: -- → -- A detailed description, including	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	photographs and/or drawings, of the vehicle type with respect to the structure, the dimensions, the design and the constituent materials, the side walls of the passenger compartment (exterior and interior), including specific details of the protection system, where applicable:: -- → -- Drawing / Images of position & arrangement of Engine: -- → -- Drawing/ Images of Under bonnet (Engine house) Components.: -- → -- Construction: Integrated / Attached with Mounts: 4 Mounts, Fixed CABIN → 4 Mounts, Fixed CABIN 1st ·		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st · Overall Ratio / AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69 2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC: 12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 → 20.44 Description · Make: ID		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number/ Part Number/Drawing number (Seat Manufacturer/ Vehicle Manufacturer) → ID Number/ Part Number/Drawing number (Seat Manufacturer/ Vehicle Manufacturer) Description · Make: Weight (kg) → Weight (kg) B 51.0 B49.0 · Pedestrian protection as per AIS-100: The distance, measured longitudinally on a horizontal plane, between the transverse centerline of the front axle & the R point of the driver’s seat is less than 1100 mm (Yes / No).: NA → NA Brief description of failure warning	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	tell–tale: - → - Brief description and features: - → - Brief description and features: - → - Description: -- → -- Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No Manufacturer’s documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	version as appropriate.: -- → -- Under fault conditions, vehicle response corresponds to that described in the manufacturer’s documents / failure analysis and safety concept is verified.: -- → -- A brief written explanation sufficient to describe the ESC system basic operational characteristics.: -- → -- Logic diagram: -- → -- Understeer information.: -- → -- Static Stability Factor: -- → -- Make and Country of manufacturer(if imported): -- → -- Software Id / version: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Brief description of failure warning tell–tale: -- → -- Brief description and features: -- → -- Brief description and features: -- → -- System identification including software level identifier: -- → -- List of motor vehicles by description and configuration that are covered by the information document: -- → -- Wheel type (single or twin) and tyre type (e.g. structure, category of use, size): -- → -- Gearbox type (e.g. manual, automated manual, semi-automatic, automatic): -- → -- Management of the engine or any		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	other source(s) of motive power e.g.Torque Management, control of fuel supply etc.: -- → -- Component description:: -- → -- Sensor(s) External to the Controller: -- → -- Function: -- → -- Limitations on the location of the sensors: -- → -- Identification (e.g. part numbers): -- → -- Controller(s): -- → -- General description and function: -- → -- Functionality of internal sensors (if applicable): -- → -- Hardware identification (e.g. part numbers): -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Software identification: -- → -- Limitations on the location of the controller(s): -- → -- Additional features: -- → -- Modulators: -- → -- General description and function: -- → -- Hardware identification (e.g. part numbers): -- → -- Software identification (if applicable): -- → -- Limitations: -- → -- Electrical Equipment: -- → -- Circuit diagrams: -- → -- Powering methods: -- → -- Pneumatic circuits: -- → -- Safety aspects of the electronic system in accordance with Annex S of IS 11852: -- → -- List of complex functions covered::		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	-- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Manufacturer’s documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- A list of input and output variables, including their working range, is	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	provided.: -- → -- Documentation includes an inventory of components, outlines the function of each unit, its interconnection/interaction with other systems and defines signal flow priorities.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software version as appropriate.: -- → -- Safety concept statement verifies that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- System architecture, design methods and tools identified.: -- → -- Information regarding		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	design provisions in case of failure is provided including any error messages, warning signals, partial performance conditions, back-up modes and/or removal of high- level functions.: -- → -- Additional material and analysis (FMEA, FTA) of fault conditions made available for inspection and maintained by the manufacturer.: -- → -- Document reference and date of inspection:: -- → -- Vehicle complies with all other performance requirements of IS 15986:2015 and meets manufacturer’s specifications under non-fault conditions.: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Under fault conditions, vehicle response corresponds to that described in the manufacturer’s documents / failure analysis and safety concept is verified.: -- → -- Electro-magnetic compatibility: -- → -- Documentation demonstrating compliance with AIS-004(Part 3).: -- → -- Component description: -- → -- Controller(s) (a) General description and function; (b) Identification e.g. part numbers; (c) Limitations on the location of the controller(s); (d) Additional		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	features.: -- → -- Modulators (a) General description and function; (b) Identification; (c) Limitations.: -- → -- Manufacturer’s documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software version as appropriate.: -- → -- (see annex 4 of this report for test data): -- → -- (see annex 4 of this report for test data): -- → -- Type / Description: Fully built vehicle, D+2, Non AC, 3.39 Ton GVW → Fully	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	built vehicle, D+2, Non AC, 3.39 Ton GVW Make: Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. → Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. Model name / identification: ALXYZABC0 → ALXYZABC0 Type: Compression Ignition → Compression Ignition Bore x stroke (mm): 80 X 98 → 80 X 98 No. of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	cylinders: 3 → 3 Displacement (cm³): 1468 cc → 1468 cc Compression ratio: 16.5±1.0 : 1 → 16.5±1.0 : 1 Max. Engine output (KW @ rpm): 55.1 Kw @ 3300 rpm → 55.1 Kw @ 3300 rpm Max. Torque (Nm @ rpm): 170Nm @ 1600 to 2400 rpm → 170Nm @ 1600 to 2400 rpm Air cleaner type: Dry type → Dry type Type of Fuel: Diesel → Diesel Make and model: RSB Transmission (T210R) → RSB Transmission (T210R) Make and model: AVTEC (A5M210) → AVTEC (A5M210) Type: Manual gearbox with fully synchro mesh & Rev with sliding mesh → Manual	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	gearbox with fully synchro mesh & Rev with sliding mesh No. of gears: 5 Forward and 1 Reverse → 5 Forward and 1 Reverse 1st: 4.62 → 4.62 1st: 4.60 → 4.60 1st: 21.53 → 21.53 1st: 21.44 → 21.44 2nd: 2.68 → 2.68 2nd: 2.69 → 2.69 2nd: 12.67 → 12.67 2nd: 12.71 → 12.71 3rd: 1.51 → 1.51 3rd: 1.49 → 1.49 3rd: 7.34 → 7.34 3rd: 7.25 → 7.25 4th: 1.0 → 1.0 4th: 1.0 → 1.0 4th: 4.56 → 4.56 4th: 4.56 → 4.56 5th: 0.70 → 0.70 5th: 0.71 → 0.71 5th: 3.64 → 3.64 5th: 3.67 → 3.67 Reverse: 4.39 → 4.39		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Reverse: 20.36 → 20.36 Reverse: 4.38 → 4.38 Reverse: 20.34 → 20.34 Drive Axle (Front / Rear / All): Rear → Rear Front axle ratio: NA → NA Type / Description: Re-circulating ball type → Re-circulating ball type	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT 12 PR Tyre size designation including ply rating: 215/75 R15 LT → 215/75 R15 LT Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2 → 3.26 kg/cm2 Wiping system (Brief description): Twin blade, Tandem Pattern → Twin blade, Tandem Pattern Material: Metallic → Metallic Capacity (l): 50 lts , 40 lts → 50 lts , 40 lts Dimensions: 2490WB → 2490WB		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Dimensions: 2640WB → 2640WB Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Overall width (mm): 1842 → 1842 Overall Length (mm): CC → CC	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Overall Length (mm): 4750 → 4750 Overall Length (mm): 4988 → 4988 Overall Length (mm): FSD / HSD → FSD / HSD Overall Length (mm): 4925 → 4925 Overall Length (mm): 5225 → 5225 Overall height (mm): 1961 → 1961 Overall height (mm): 1948 → 1948 Maximum gradeability in 1st gear: GVW 2890 kg → GVW 2890 kg Maximum gradeability in 1st gear: GVW 3390 kg → GVW 3390 kg Maximum gradeability in 1st gear: 10 Deg → 10 Deg Maximum gradeability in 1st gear: 8.5 Deg → 8.5 Deg Weights: Base → Base Weights: Variant 1 → Variant 1 Weights: Variant 2 → Variant 2 Weights: Variant 3 → Variant 3 Weights: Variant 4 → Variant 4 Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	spare wheel, tools, etc.) (kg): - → - Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Description of energy converters and power plant. (In the case of a vehicle that can run either on petrol, diesel, etc., or also in combination with another fuel, items shall be repeated.1): Diesel → Diesel Engine Manufacturer:: Ashok Leyland Limited → Ashok Leyland Limited Manufacturer's engine model code (as marked on the engine, or other means of identification): Model Code: ALXYZABC0 Enclosure – 4A → Model Code: ALXYZABC0 Enclosure – 4A Location of engine model code and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	engine serial number on block / crankcase (drawing to be attached): Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Engine Block Refer Enclosure – 4A → Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A Working principle: positive ignition / compression- ignition, four-stroke / two-stroke / rotary cycle (DI/IDI) (NA/TC/TCIC/Any other): Compression Ignition, four-stroke, DI, TCIC → Compression Ignition, four-stroke, DI, TCIC Number, arrangement of cylinders:: 3, Inline → 3, Inline Bore----------(mm): 80 → 80 Stroke: ----------(mm): 98 → 98 Firing order:: 1-2-3 → 1-2-3 Engine capacity:----------(cm3): 1468 → 1468 Volumetric compression ratio (specify tolerance): 17.5±1.0 : 1 → 17.5±1.0 : 1 Drawings of combustion chamber and piston crown and, in the case of positive ignition engine, piston rings:: Refer Enclosure – 4B → Refer		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure – 4B Vehicle with Defeat Device Yes / No: No → No If yes – Details of the Defeat device: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane equivalent) in PPM content by volume in the exhaust gas with the engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum net power:..... kW at min-1: 55.1 KW @ 3100 ± 50 rpm → 55.1 KW @ 3100 ± 50 rpm Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm Maximum net torque: .....Nm at: -------min-1 ...(manufacturer's declared value): 170 Nm @ 1600 – 2400 rpm → 170 Nm @ 1600 – 2400 rpm Maximum rated speed (Specify tolerance): 3100 rpm ± 50 rpm → 3100 rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Type of fuel used : Mono		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Fuel / Bi-Fuel / Flex Fuel / Dual Fuel: Mono Fuel → Mono Fuel	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Fuel type: Diesel / Gasoline / LPG / CNG / Biomethane / Bio-H methane / HCNG / Biogas / LNG / Ethanol ((E85) / 2 (E100)) /ED (95) Flex fuel Methanol M15 / M100 / MD95 / Biodiesel up to 100% / Hydrogen.: Diesel → Diesel Maximum amount of biofuel acceptable in fuel (manufacturer's declared value): per cent by volume: 7% → 7% By fuel injection (compression-ignition only): (Yes/No): Yes → Yes System description:: Unit Pump Common Rail Direct Injection system → Unit Pump Common Rail Direct Injection system Working principle: direct-injection/pre-chamber/swirl Chamber: Direct Injection → Direct Injection Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Plunger Type & FHL0XYZ0 → Plunger Type & FHL0XYZ0 Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Injection advance curve (Specify the	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	tolerances):: ECU Controlled → ECU Controlled Governor: ECU Controlled → ECU Controlled Cut-off point under load:…(min-1): 3300 → 3300 Cut-off point without load (Maximum no-load speed)…(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Multi-Hole Injectors (7 holes) FKXXYYZZ → Multi-Hole Injectors (7 holes) FKXXYYZZ Opening pressure (kPa) (specify the tolerances) or characteristic diagram: 15000 to 23000 kPa → 15000 to 23000 kPa Length and internal diameter…(mm): Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm → Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm Common rail make & type: M/s. XXXXYYY, Hot Forged Rail Type → M/s. XXXXYYY, Hot Forged Rail Type Cold start system: NA → NA Make(s):: NA → NA Type(s) &		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: NA → NA Description:: NA → NA Auxiliary starting aid: NA → NA Make(s):: NA → NA Type(s):: NA → NA System description:: NA → NA Electronic controlled injection: (Yes/No): Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA → NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s. XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Metering type Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type By fuel injection (positive ignition only): (Yes/No): No → No Working principle: intake manifold (single/multi- point)/direct injection/other (specify): NA → NA Make(s) of Injection Pump :: NA → NA Type(s) & Identification Number of Injection Pump:: NA → NA System description (in the case of systems other than continuous injection give equivalent details):: NA → NA Make and type of the control unit:: NA → NA Make and type of the fuel regulator:: NA → NA Make and type of the air-flow sensor:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Make and type of the micro-switch:: NA → NA Make and type of the throttle housing:: NA → NA Make and type of the water temperature sensor:: NA → NA Make and type of the air temperature sensor:: NA → NA Injectors: Opening pressure (Specify the tolerances):,..... (kPa) or characteristic diagram:: NA → NA Make(s):: NA → NA Type(s):: NA → NA Identification Number: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA Cut-off point under load (min-1): NA → NA Injection timing:: NA → NA Cold start system:: NA → NA Operating principle(s):: NA → NA Operating limits/settings(Specify the tolerances): NA → NA Pressure, (Specify the tolerances).....(kPa) or characteristic diagram:: 320 ± 50 kPa → 320 ± 50 kPa Rated voltage:...V, positive/negative ground: 12V, Negative Ground → 12V, Negative Ground	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Generator/ Alternator: Alternator → Alternator Type:: DC – Alternator with Voltage regulator → DC – Alternator with Voltage regulator Nominal output: ...VA: 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps → 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps Ignition: NA → NA Make(s):: NA → NA Type(s):: NA → NA Working principle:: NA → NA Ignition advance curve or map(Specify the tolerances):: NA → NA Static ignition timing (Specify the tolerances):.....degrees before TDC: NA → NA Spark plugs: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Gap setting----------(mm): NA → NA Nominal resistance (kilo ohm) (if resistive type): NA → NA Ignition coil(s): NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Ignition condenser: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA EMI suppressor cap / Device / Electronic	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	unit: NA → NA Make: NA → NA Type (Resistive/Capacitive): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification number: NA → NA Nominal resistance (kilo ohm): NA → NA Terminology and Drawing of interference Suppression equipment: NA → NA H.T. Cable: NA → NA Make and Place: NA → NA Type (Resistive/Non-resistive): NA → NA Length (mm) (if resistive type): NA → NA Outside dia. (mm) (if resistive type): NA → NA Nominal resistance kilo ohm, (if resistive type): NA → NA Radiator drawing(s): Refer enclosure 4C2 & 4C4 → Refer enclosure 4C2 & 4C4 Radiator drawing(s): Refer enclosurev 4C1 & 4C3 → Refer enclosurev 4C1 & 4C3 Make: Mahle Behr → Mahle Behr Make: Tata Toyo → Tata Toyo Type (s): Cross flow → Cross flow Relief valve pressure setting: 0.9 kg/cm2 → 0.9 kg/cm2 Fan characteristics (Fan power, kW) Enclose the fan power curve corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	No: 9 → 9 Material of blades (metal /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ± 150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150 (AC) Radiator core open area (cm²): 1906 cm2 → 1906 cm2 Cooling system: liquid/air: Liquid → Liquid Nominal setting of the engine temperature control o mechanism (C)::: 85 deg C → 85 deg C Liquid: Yes → Yes Nature of liquid:: Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L → Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L Circulating pump(s): (Yes/No): Yes → Yes Characteristics of Circulating pump:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm → Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm Make(s):: XXXXX		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Fasteners Ltd → XXXXX Fasteners Ltd Type(s):: Impeller Type → Impeller Type Drive ratio(s):: 1: 1.12 → 1: 1.12 Description of the fan and its drive mechanism:: Electric Fan , Motor Driven → Electric Fan , Motor Driven Blower: (Yes/No): No → No Characteristics of blower: NA → NA Make(s):: NA → NA Type(s):: NA → NA Drive ratio(s):: NA → NA Pressure charger: (Yes/No) Turbocharger / Supercharger: Yes, Turbo charger → Yes, Turbo charger Make(s):: M/s. XYZ, India → M/s. XYZ, India Type(s) & Identification Number:: Waste Gate with Vacuum Actuator, X3M1XXXX → Waste Gate with Vacuum Actuator, X3M1XXXX Description of the system (maximum charge pressure:.....(kPa), waste-gate, if applicable): Waste gate turbocharger with vacuum actuator Maximum Charge pressure - 145 kPa → Waste gate turbocharger with vacuum actuator	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Maximum Charge pressure - 145 kPa Inter-cooler: (Yes/No): Yes → Yes Type: air-air / air-water: Air – Air → Air – Air Make :: Mahle Behr → Mahle Behr Make :: Tata Toyo → Tata Toyo		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: B1V42XXX → B1V42XXX Identification Number:: B1V42YYY → B1V42YYY Air pressure drop across the intercooler: 15 kPa → 15 kPa Intake depression at rated engine speed and at 100 per cent load (Specify location of measurement) Minimum allowable:... (kPa) Maximum allowable:... (kPa): Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Description and drawings of inlet pipes and their accessories (plenum chamber, heating device, additional air intakes, etc.):: Intake manifold with Electronic throttle valve Refer Enclosure 4D → Intake manifold with Electronic throttle valve Refer Enclosure 4D Intake manifold description including dimensional drawings and/or	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	photographs):: Aluminium Intake manifold with pressure sensor Refer Enclosure 4D → Aluminium Intake manifold with pressure sensor Refer Enclosure 4D Air filter, Dimensional drawings: Refer Enclosure 4E → Refer Enclosure 4E		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Make(s):: M/s. Fleet guard → M/s. Fleet guard Type(s) & Identification Number:: Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 → Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 Intake silencer, Dimensional drawing: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description and Dimensional drawing of the exhaust manifold:: Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F → Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F Description and drawing with dimensions of the exhaust system:(Description with general arrangement of exhaust system along with its routing	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	indicating the lengths of exhaust pipe, tail pipe and: Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	O2 sensors provided before and after LNT-DPF assembly → Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly Effective volume of exhaust-system (specify the tolerance & range) in liters (from exhaust manifold / TC outlet to tail pipe end), Enclose the exhaust system dimensional drawing and indicate the volume of each parts clearly: 13.05 ± 1 litre Refer Enclosure 4G → 13.05 ± 1 litre Refer Enclosure 4G Description and drawing of the exhaust Muffler/Silencer:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Type(s) & Identification Number:: NA → NA Make: NA → NA Number:……: NA → NA Internal diameter of exhaust pipe (mm)(min): diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G → diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G Minimum distance between exhaust pipe(s) and the fuel		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	line: >150 mm → >150 mm Auxiliary Noise shields for compliance to IS 3028 and / OR AIS-020 (If Provided): NA → NA Material: NA → NA Layout of noise shield / Photographs / Diagram Showing arrangements indicating fitment on vehicle.: NA → NA Maximum allowable exhaust back pressure at rated engine speed and at 100 per cent load (location of measurement): … (kPa): 30.0 kpa (150 mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Minimum cross-sectional areas of inlet and outlet ports:: Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm → Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Inlet (mm): 8.00 → 8.00 Exhaust (mm): 8.00 → 8.00 Opening: 17° bTDC → 17° bTDC Closing: 51° aBDC → 51° aBDC Opening: 66° bBDC → 66° bBDC Closing: 25° aTDC → 25° aTDC Transfer: NA → NA Opening: NA → NA Closing: NA → NA Reference and/or setting ranges (Specify the tolerances):: On Cold Engine → On Cold Engine Inlet: 0.3 to 0.4 mm → 0.3 to 0.4 mm Exhaust: 0.3 to 0.4 mm → 0.3 to 0.4 mm		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Distribution by ports: NA → NA Volume of crank-case cavity with piston at TDC: NA → NA Reed valve fitted (Yes / No): No → No Description of inlet ports, scavenging and exhaust ports with corresponding timing.: NA → NA Device for recycling crankcase gases (description and drawings):: Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J → Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx Catalyst (if any, and if not covered by another	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF / DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No Element – 1 No. → Catalytic Converter – 1No	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	for LNT (periodic / continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:.....cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Bulk Density (GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle and Characteristics:: Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	downstream SCR catalyst for the reduction of NOX during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Type and Identification number:: Type - Passive NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite → Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA Make:: NA → NA Type and Idetification number :: NA → NA NOx Sensor (Before /After of SCR):: No → No Make:: NA → NA Type and Identification number:: NA → NA Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent engine test bench cycles, between two cycles where regenerative phases occur under the conditions equivalent to Type I test (Distance "D" in t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration (Distance “d” in AIS 137): 2 cycles → 2 cycles Description of method employed to determine the number of cycles between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA International standard (where appropriate):: NA → NA Frequency of reagent refill: continuous/maintenance (where appropriate):: NA → NA Normal operating temperature range (°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY → YYYYYYY Air injection: (Yes/No): No → No	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Type (pulse air, air pump, etc.) & Identification Number:: NA → NA Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other): Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY → FEW0XXYY Characteristics (flow rate, etc.):: Flow rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes EGR Cooler Make and Identification Number: M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:.....g: NA → NA Canister working capacity:.....g: NA → NA Canister bed volume:.....litre: NA → NA Canister Purge and vent strategy description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve : (Yes/No): NA → NA Make and Identification Number: NA → NA Operating pressure (kPa): NA → NA Liquid fuel hoses material :: NA → NA Schematic drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	drawing (Periodic / Continuous): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Particulate trap: (Yes/No): Yes → Yes Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dimensions and shape of the particulate trap (capacity):: Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Regeneration Type Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure 4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh: 90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type Material - Silicon Carbide → Structure - Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI Electronic Control Unit (ECU): Yes → Yes Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number: P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA Misfire detection:: NA → NA Oxygen sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K → Refer Enclosure 4K Particulate traps	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method):: Refer Enclosure 4K → Refer Enclosure 4K List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K → Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K A description of the type and number of the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3 EUDC A description of the type of the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	all sensed components with the strategy for fault detection and MI activation (fixed number of driving cycles or statistical method), including a list of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive explanation for the data given in service $05 Test ID $21 to FF and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles – Diagnostics on Controller Area Network (CAN) – Part 4: Requirements for emissions- related systems", a comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA → NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA Systems to ensure the correct operation of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent deactivation of the driver inducement, for use by the rescue services [Note: cross- reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling /disable after parking: NA → NA Lowest		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct operation of NOx control measures: NA → NA Details of any devices designed to influence fuel economy (if not covered by other items):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	LPG fuelling system: (Yes/No): No → No Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to LPG or back:: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.): NA → NA NG fuelling system: (Yes/No): No → No Approval number: NA → NA Electronic engine management control unit for NG fuelling: NA → NA Make(s):: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation:: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NG or back: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.):: NA → NA Hydrogen fuelling system: (Yes/No): No → No Electronic engine management control unit for hydrogen fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to hydrogen or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA H2NG fuelling system: (Yes/No): No → No		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Percentage of hydrogen in the fuel (maximum specified by the manufacturer): NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to H2NG or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA Electric motor(Traction): No → No Type (winding, excitation):: NA → NA Maximum hourly output: kW(manufacturer’s declared value): NA → NA Maximum net power:kW(manufacturer’s declared value): NA → NA Maximum 30 minutes power: kW (manufacturer’s declared value): NA → NA Operating voltage: V: NA → NA Make and Identification Number:: → Battery (Traction): NA → NA Number of cells:: NA → NA Mass: ……(kg): NA → NA Capacity: Ah		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Amp-hours): NA → NA Position:: NA → NA Engines or motor combinations: NA → NA Hybrid Electric Vehicle: (Yes/No): No → No	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Category of Hybrid Electric vehicle Off Vehicle Charging/Not Off Vehicle Charging9: NA → NA Operating mode switch: with/without: NA → NA Selectable modes: NA → NA Pure electric: (Yes/No): NA → NA Pure fuel consuming: (Yes/No): NA → NA Hybrid modes: (Yes/No) (if yes, short description): NA → NA Description of the energy storage device: (traction battery, capacitor, flywheel/generator...): NA → NA Make(s):: NA → NA Type(s) :: NA → NA Identification number:: NA → NA Kind of electrochemical couple:: NA → NA Energy: ..... (for battery: voltage and capacity Ah in 2 h, for capacitor: J): NA → NA Charger: on board/external/without: NA → NA Electric machines (describe each type of electric machine separately): NA → NA Make:: NA → NA Type:: NA → NA Primary use: traction motor/generator: NA → NA When used as traction motor: mono motor / multi motors (number):: NA → NA		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Maximum power: kW: NA → NA Working principle:: NA → NA Direct current/alternating current/number of phases:: NA → NA Separate excitation/series/compound: NA → NA Synchronous/asynchronous: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA Identification number:: NA → NA Power controller: NA → NA Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Vehicle electric range..... km (according to AIS 137):: NA → NA Manufacturer’s recommendation for preconditioning:: NA → NA CO2 (g/km) (Applicable for category M1 with GVW<3.5 T): NA → NA Declared : (rounded to 3 decimal places): NA → NA Fuel consumption (l/100 km) for Petrol, LPG or Diesel and (kg/100km) for CNG and (kWh/100 km) for Electric Driven Vehicles(Applicable for category M1 with GVW<3.5 T) (rounded to 3 decimal places)*: NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA *calculated on the declared CO2: NA → NA CO2 Reducing	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	technologies Available (Yes/No): No → No Regenerative braking: NA → NA Start-Stop System: NA → NA Tyre pressure monitoring system: NA → NA 6 or more Speed Transmission: NA → NA Any other technology (manufacture to enclose detail): NA → NA Electric energy consumption for electric vehicles Wh/km: NA → NA Electric energy consumption for pure electric vehicles Wh/km: NA → NA Electric energy consumption for externally chargeable hybrid electric vehicles: NA → NA Electric energy consumption (condition A, combined) Wh/km: NA → NA Electric energy consumption (condition B, combined) Wh/km: NA → NA Electric energy consumption (weighted combined) ….Wh/km: NA → NA Temperatures permitted by the manufacturer: NA → NA Cooling system: Yes → Yes Liquid cooling: Yes → Yes Maximum temperature at outlet: …….(°C): 100 °C → 100 °C Air cooling: NA → NA Reference point:: NA → NA Maximum temperature at reference point: …….(°C): NA → NA Maximum outlet temperature of the inlet intercooler: .. (°C): 45 °C → 45	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	°C Maximum exhaust temperature at the point in the exhaust pipe(s) adjacent to the outer flange(s) of the exhaust manifold: ...(°C): 740 °C → 740 °C Minimum: .. (°C): - 8 → - 8 Maximum: .. (°C): 42 → 42 Minimum: ...(°C): - 5 → - 5 Maximum: ...(°C): 120 → 120 Lubrication system: Yes → Yes Description of the system: Forced Lubrication system → Forced Lubrication system Position of the lubricant reservoir:: Sump beneath crankcase → Sump beneath crankcase Feed system (by pump/injection into intake/mixing with fuel, etc.): Pump → Pump Lubricating pump: Yes → Yes Make(s):: M/s. ZXWA → M/s. ZXWA Type(s):: XXYY Type → XXYY Type Mixture with fuel: No → No Percentage:: → Oil cooler: (Yes/No): Yes → Yes Drawing(s):..., or: Enclosure 4L → Enclosure 4L Make(s):: M/s. XXYY or Any other equivalent → M/s. XXYY or Any other equivalent Type(s):: DONUT → DONUT Moment of inertia of engine flywheel:: 0.1212 kg/m2 → 0.1212 kg/m2 Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Transmission Control Unit (ECU) (Yes/No):: No → No Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA OBD IUPR Family if any: XX+XYY → XX+XYY exhaust outlet location, indicated in a Schematic dimensional drawing.): Refer Enclosure 4G → Refer Enclosure 4G 1.2.11.2.1.1 4.4 · 3. · Calibration Verification Number:: NA → NA Catalyst · P0420 · Oxygen sensor 1 and 2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Engine speed, engine load, A/F mode, catalyst temperature → Engine speed, engine load, A/F mode, catalyst temperature Catalyst · P0420 · Oxygen sensor 1 and 2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Two Type I cycles → Two Type I cycles Catalyst · P0420 · Oxygen sensor 1 and	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Type I → Type I Type (Manual / power): Manual → Manual No. of wipers: Two → Two Make: XXYY System →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	XXYY System Type: 12V → 12V Identification mark: X5XYZ100 → X5XYZ100 Rated voltage: 12V → 12V Number of sweep Frequencies: 2 → 2 Highest sweep frequency (Cycles / min): 62±8 rpm → 62±8 rpm Lowest sweep frequency (Cycles / min): 40±5 rpm → 40±5 rpm Wiper arm: Provided → Provided Length: 500mm → 500mm Make: M/s XYZ → M/s XYZ Length: 450 mm → 450 mm Make: BCS (SYNDICATE WIPER SYSTEM) → BCS (SYNDICATE WIPER SYSTEM) Identification: X5XY0258 → X5XY0258 Washer tank: BWXY4501 → BWXY4501 Type: Plastic → Plastic Make: XYZ → XYZ Identification No. / Part No.: X5BXYZ168 → X5BXYZ168 Capacity l: 1.5Ltr → 1.5Ltr Material: H.D.P.E → H.D.P.E Washer tank motor / Washer pump: ELECTRIC MOTOR WITH PUMP → ELECTRIC MOTOR WITH PUMP	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make: XYZ → XYZ Model: PEAD64XYZS00 → PEAD64XYZS00 Nozzle(s): XYZ → XYZ No. Of Nozzles: 2 → 2 Defroster: NA → NA Type: - → - Make: - → - Details of Demisting and / or Defrosting system: - → - Schematic Diagram of Engine cooling system: - → - Schematic Diagram of Heating system: - →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	- Cross sectional view of complete air handling system including position of louvers (vents) on dash board: - → - Blower identification: - → - Blower Voltage and capacity (Watts): - → - If defrost system is fitted, specify defrost test temperature (-8 +/- 2 deg C / -18 +/- 3 deg C): → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar		
(continued)	& related dimensions as per related standards.: → Description and sketch showing the detail positions of Tell-Tale and seating layout (For M category vehicles): NA → NA A brief technical description indicating the length of the cleaning period, the consumption of cleaning fluid during the cleaning period and the minimum capacity of the container provided;: -- → -- Description and drawings/photographs of the shapes and constituent materials of the part of the body forming the engine compartment and the part of the passenger compartment nearest to it:: -- → -- Drawings or photographs of the position of the metal components housed in the engine compartment (e.g. heating appliances, spare wheel, air filter, steering mechanism, etc.):: -- → -- List of all subassemblies, which includes an electronic oscillator or switching frequency greater than 9 kHz (like ECU, steering, suspension, Body: -- → -- List of all Electrical components, which include Broadband EMI sources (like HAVC Motor,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Wiper Motor, Window motor and Horn etc.) The list shall include Device name, Make/s and Part No. /Type ID/Drawing No: -- → -- Name of the manufacturer of the system (Steering column shaft / Gear shift control / Transmission) on which the Protective Device functions: XXYY Steering System → XXYY Steering System Type/Part no./Identification number(ID)/Drawing No.of the device on which the Protective Device functions: FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly → FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable only in case of steering column locking systems: Locks the Steering → Locks the Steering SYSTEM SOFTWARE (If fitted): NA → NA Make: -- → -- Version: -- → -- Operating System Details with Version: -- → -- DESCRIPTION OF DEVICE (If fitted): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Description of the VAS: - → - A detailed description of the alarm system and of the vehicle parts related to alarm system installed with Block Diagram: → A list of main components comprising the alarm system: - → - Components placed in Engine compartment: - → - Components placed in Passenger compartment: - → - Availability of radio/wireless transmission Yes/No: - → - Details of radio/wireless transmission if any: - → - frequency: - → - Availability of WPC approval for wireless technology Yes/No: - → - The measures taken against false alarm: - → - Range of protection offered by the device. Details about the detection mechanism (opening of Door, Bonnet, luggage compartment, intrusion etc.): - → - Details & specification of audible alarm: - → - Details & specification of optical alarm: - → - Details & specification of radio alarm if fitted: - → - Method of setting/ unsetting the device: - → - Type of key used for unsetting (Mechanical Key / electronic device): - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Number of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - Exit delay in Sec: - → - Entry Delay in sec: - → - Details of status display (0.5cd): - → - Details of self check , automatic failure indication: - → - Details & specification of audible alarm: - → - Details & specification of optical alarm: - → - Details & specification of radio alarm if fitted: - → - Method of setting/ unsetting the device: - → - Type of key used for unsetting (Mechanical Key / electronic device): - → - Number of interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - Exit delay in Sec: - → - Entry Delay in sec: - → - Details of status display (0.5cd): - → - Details of self check , automatic failure indication: - → - Details about the panic alarm , if provided: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Description of the Immobilizer: - → - A detailed description of the vehicle immobilizer and of the vehicle and the measures		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	taken against inadvertent activation with block diagram.: → The vehicle system(s) on which the immobilizer acts.: - → - Details of radio/wireless transmission if any: - → - Method of setting/ unsetting the device.: - → - Number of interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - A list of main components comprising the device: - → - Description: AL Part No. → AL Part No. Description · AL Part No.: Hella Part No → Hella Part No ECU: FXX01000 → FXX01000 ECU · FXX01000: 015.XYZ-01 → 015.XYZ-01 a. · Make: - → - b. · Type: - → - c. · Serial Number: - → - d. · Name and Address of manufacture: - → - a. · Make: - → - b. · Type: - → - c. · Serial Number: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	d. · Name and Address of manufacture: - → - Example of Engine No: HJEZXYZ891 → HJEZXYZ891		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
AIS-137 Part 3; applicable diesel provisions SCR reagent / inducement and dosing system	Base fuel type used for vehicle Type approval including blend level (Ethanol, Bio-diesel etc): Diesel (Maximum 7% bio-diesel blend) → Diesel (Maximum 7% bio-diesel blend) Name and address of engine manufacturing plant (In case of imported vehicles, above details shall be supplied for importer also): Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX → Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX Max. Recommended Gradeability (Stand-start) – in degrees: 10 degs → 10 degs Max. Recommended Gradeability (Stand-start) – in degrees: 8.5 degs → 8.5 degs Vehicle kerb weight kg: Refer Annexure T7 - B Details for Recommended load body Refer Annexure T7 – C → Refer Annexure T7 - B Details for Recommended load body Refer Annexure T7 – C	CEA / report needs review	Confirm SCR fitted. Unknown fitment / scope is retained for review.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Other axle: NA → NA Gross combination weight kg (applicable for articulated / Tractor Trailer combination vehicles): NA → NA Front axle: NA → NA Rear axle: NA → NA Trailer axle / other axles: NA → NA If applicable, maximum mass of trailer whic5h may be coupled: NA →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	NA Unbraked trailer: NA → NA Reference mass (for vehicles with GVW less than or equal to 3.5 ton) (kg.): Refer Annexure T7-B → Refer Annexure T7-B Maximum (Including driver) for completely built vehicles: D+2 → D+2 Maximum (Including driver) for completely built vehicles: → Make: APOLLO → APOLLO Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade description(s): CEAT WINMILE X3-R → CEAT WINMILE X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 Tyre rolling Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer - TC7719XX000 003XXXF → Refer - TC7719XX000 003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022- 2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022- 2023/300002XXX X/TA/XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L- 756XXXX → CM/L- 756XXXX Identification: TAC No. /		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E- Marking: NA → NA Others (For articulated / combination vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00 R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other (for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 325 mm → 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Type: Fully		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	synchronized 5 - forward gears with a Constant mesh reverse gear → Fully synchronized 5 - forward gears with a Constant mesh reverse gear Model name (if any): XXX-XXX → XXX-XXX Model name (if any): YYYY-YY → YYYY-YY	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Gear shifting control system (Sketch showing gear shifting arrangement): → No. of gears: Forward 5 + 1 Rev → Forward 5 + 1 Rev Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Rear axle ratio: 4.56 → 5.13 Type of steering gear (Re-circulating ball / Worm & Roller / Rack & Pinion / Others): Re-circulating ball type → Re-circulating ball type Make: 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd → 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 Type and description (Leaf / Coil / Air / Semi- pneumatic/ Torsion bar): Leaf → Leaf Front: Parabolic spring- Rigid axle suspension → Parabolic spring- Rigid axle suspension Rear: Semi elliptic spring-		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Rigid axle suspension → Semi elliptic spring- Rigid axle suspension Fuel Gauge: LED Bars → LED Bars Engine coolant temperature: Red Color – Thermometer Symbol → Red Color –	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Thermometer Symbol Engine low oil pressure: Red Color – Oil Tank Symbol → Red Color – Oil Tank Symbol Weight of seats: Driver seat : 11.5 Kg & Co-driver seat : 11Kg → Driver seat : 11.5 Kg & Co-driver seat : 11Kg Description of seats: -- → -- Make: -- → -- Number of seats fitted or capable of being fitted with head restraints, adjustable or not adjustable.: -- → -- Description of the adjustment, displacement and locking systems of the seat or of its parts and a description of occupant protection system against displacement of luggage.: -- → -- Description of seat anchorage: -- → -- Longitudinal / Vertical (If applicable) position of the seats during the tests.: -- → -- Drawings, diagrams and plans of the seats, their anchorages on the vehicle, the floor layout, the adjustment and displacement system of the seats and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	their parts, and their locking devices and of additional occupant protection system against displacement of luggage.: -- → -- In the case of seats fitted with head restraints, the head restraint shall be shown on all drawings, diagrams and photographs.: -- → -- Seat Identification No. / Part No/Drawing	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Number.: ID Number/ Part Number/D rawing number Weight Description Make (Seat (kg) Manufactu rer/ Vehicle Manufactu rer): -- → -- Description: Make → Make Description: ID Number/ Part Number/D rawing number (Seat Manufactu rer/ Vehicle Manufactu rer) → ID Number/ Part Number/D rawing number (Seat Manufactu rer/ Vehicle Manufactu rer) Description: Weight (kg) → Weight (kg) Brief description and drawings of the seat type, its attachment fittings and its adjustment, displacement and locking systems including the minimum distance between fitting points.: D+1 → D+1		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Installation drawing showing rear extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	make, unladen pressure) etc., distance between tyre & RUPD RH & LH, frame rear width, P1, P2 points.: - → - Make: M/s. XXXXYYYY → M/s. XXXXYYYY Material (Metallic / Plastic etc.): Metallic → Metallic Nominal thickness mm: 1.0mm → 1.0mm Capacity, Litre: 40 Lts, 30 Lts → 40 Lts, 30 Lts Detailed drawing of the fuel tank assembly with material specifications: FHSXXXXY FHSXXXXY → FHSXXXXY FHSXXXXY Detailed drawing indicating the position / location of the fuel tank(s) in the vehicle.: A5LXXX00 (50 LTS) A5LXXX00 (40 LTS) → A5LXXX00 (50 LTS) A5LXXX00 (40 LTS) Identification No. / Part No.: FHS00544 FfffXXXcdHXXS00744 → FHS00544 FfffXXXcdHXXS00744 Detailed engineering drawing as specified in AIS – 013 or photographs showing the mounting details with dimensions.: NA → NA Cross sectional Drawing of Gear shift	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	lever (Knob): FXA0XXYY (Non AC) / FXA0XXYY (AC) → FXA0XXYY (Non AC) / FXA0XXYY (AC) Drawings and/or photographs of the locations of the statutory plates and inscriptions and of the vehicle		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	identification number: X4W0XYZ4 / VFT0XYZ2 → X4W0XYZ4 / VFT0XYZ2 Drawings and / or photographs of the official part of the plates and inscriptions (completed example with dimensions): X4WXYZ44 → X4WXYZ44 Location of Vehicle Descriptor Section (VDS) in Vehicle Identification Number (VIN): 4thto 9thCharacter of 17 Characters VIN → 4thto 9thCharacter of 17 Characters VIN Brief description of the Provision for emergency exit provided in case of sealed windows: -- → -- Dimensions of the instructions describing each motion necessary to unlatch and open the exit are provided from the release mechanism.: -- → -- In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → -- Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	distance between front extremity & FUPD, width of FUPD, ID		
(continued)	marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Brief description of the vehicle type as regards its structure, dimensions, lines and constituent materials:: -- → -- Description of the protective system installed in the vehicle:: -- → -- Description of the protective system installed in the vehicle:: → Description of interior arrangements or fittings that might affect the tests:: -- → -- A detailed description, including photographs and/or drawings, of the vehicle type with respect to the structure, the dimensions, the design and the constituent materials, the side walls of the passenger compartment (exterior and interior), including specific details of the protection system, where applicable:: -- → -- Drawing / Images of position & arrangement of Engine: -- → -- Drawing/ Images of Under bonnet (Engine house) Components.: -- → -- Construction: Integrated / Attached with Mounts: 4 Mounts, Fixed CABIN → 4 Mounts, Fixed CABIN 1st ·	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st · Overall Ratio / AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69 2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC: 12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56 4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 → 20.44 Description · Make: ID	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number/ Part Number/Drawing number (Seat Manufacturer/ Vehicle Manufacturer) → ID Number/ Part Number/Drawing number (Seat Manufacturer/ Vehicle Manufacturer) Description · Make: Weight (kg) → Weight (kg) B 51.0 B49.0 · Pedestrian protection as per AIS-100: The distance, measured longitudinally on a horizontal plane, between the transverse centerline of the front axle & the R point of the driver’s seat is less than 1100 mm (Yes / No).: NA → NA Brief description of failure warning tell–tale: - → - Brief description and features: - → - Brief description and features: - → - Description: -- → -- Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No Manufacturer’s documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	version as appropriate.: -- → -- Under fault conditions, vehicle response corresponds to that described in the manufacturer’s documents / failure analysis and safety concept is verified.: -- → -- A brief written explanation sufficient to describe the ESC system basic operational characteristics.: -- → -- Logic diagram: -- → -- Understeer information.: -- → -- Static Stability Factor: -- → -- Make and Country of manufacturer(if imported): -- → -- Software Id / version: -- → -- Brief description of failure warning tell–tale: -- → -- Brief description and features: -- → -- Brief description and features: -- → -- System identification including software level identifier: -- → -- List of motor vehicles by description and configuration that are covered by the information document: -- → -- Wheel type (single or twin) and tyre type (e.g. structure, category of use, size): -- → -- Gearbox type (e.g. manual, automated manual, semi-automatic, automatic): -- → -- Management of the engine or any	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	other source(s) of motive power e.g.Torque Management, control of fuel	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	supply etc.: -- → -- Component description:: -- → -- Sensor(s) External to the Controller: -- → -- Function: -- → -- Limitations on the location of the sensors: -- → -- Identification (e.g. part numbers): -- → -- Controller(s): -- → -- General description and function: -- → -- Functionality of internal sensors (if applicable): -- → -- Hardware identification (e.g. part numbers): -- → -- Software identification: -- → -- Limitations on the location of the controller(s): -- → -- Additional features: -- → -- Modulators: -- → -- General description and function: -- → -- Hardware identification (e.g. part numbers): -- → -- Software identification (if applicable): -- → -- Limitations: -- → -- Electrical Equipment: -- → -- Circuit diagrams: -- → -- Powering methods: -- → -- Pneumatic circuits: -- → -- Safety aspects of the electronic system in accordance with Annex S of IS 11852: -- → -- List of complex functions covered::		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	-- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Manufacturer’s documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- A list of input and output variables, including their working range, is provided.: -- → -- Documentation includes an inventory of components, outlines the function of each unit, its interconnection/interaction with other systems and defines signal flow priorities.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software version as appropriate.: -- → -- Safety concept statement verifies that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- System architecture, design methods and tools identified.: -- → -- Information regarding	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	design provisions in case of failure is provided including any error messages, warning signals, partial performance	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	conditions, back-up modes and/or removal of high- level functions.: -- → -- Additional material and analysis (FMEA, FTA) of fault conditions made available for inspection and maintained by the manufacturer.: -- → -- Document reference and date of inspection:: -- → -- Vehicle complies with all other performance requirements of IS 15986:2015 and meets manufacturer’s specifications under non-fault conditions.: -- → -- Under fault conditions, vehicle response corresponds to that described in the manufacturer’s documents / failure analysis and safety concept is verified.: -- → -- Electro-magnetic compatibility: -- → -- Documentation demonstrating compliance with AIS-004(Part 3).: -- → -- Component description: -- → -- Controller(s) (a) General description and function; (b) Identification e.g. part numbers; (c) Limitations on the location of the controller(s); (d) Additional		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	features.: -- → -- Modulators (a) General description and function; (b) Identification; (c) Limitations.: -- → -- Manufacturer’s documents explain the functionality and safety concept of the system and describe how the	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	operational status may be checked.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software version as appropriate.: -- → -- (see annex 4 of this report for test data): -- → -- (see annex 4 of this report for test data): -- → -- Type / Description: Fully built vehicle, D+2, Non AC, 3.39 Ton GVW → Fully built vehicle, D+2, Non AC, 3.39 Ton GVW Make: Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. → Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. Model name / identification: ALXYZABC0 → ALXYZABC0 Type: Compression Ignition → Compression Ignition Bore x stroke (mm): 80 X 98 → 80 X 98 No. of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	cylinders: 3 → 3 Displacement (cm³): 1468 cc → 1468 cc Compression ratio: 16.5±1.0 : 1 → 16.5±1.0 : 1 Max. Engine output (KW @ rpm): 55.1	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Kw @ 3300 rpm → 55.1 Kw @ 3300 rpm Max. Torque (Nm @ rpm): 170Nm @ 1600 to 2400 rpm → 170Nm @ 1600 to 2400 rpm Air cleaner type: Dry type → Dry type Type of Fuel: Diesel → Diesel Make and model: RSB Transmission (T210R) → RSB Transmission (T210R) Make and model: AVTEC (A5M210) → AVTEC (A5M210) Type: Manual gearbox with fully synchro mesh & Rev with sliding mesh → Manual gearbox with fully synchro mesh & Rev with sliding mesh No. of gears: 5 Forward and 1 Reverse → 5 Forward and 1 Reverse 1st: 4.62 → 4.62 1st: 4.60 → 4.60 1st: 21.53 → 21.53 1st: 21.44 → 21.44 2nd: 2.68 → 2.68 2nd: 2.69 → 2.69 2nd: 12.67 → 12.67 2nd: 12.71 → 12.71 3rd: 1.51 → 1.51 3rd: 1.49 → 1.49 3rd: 7.34 → 7.34 3rd: 7.25 → 7.25 4th: 1.0 → 1.0 4th: 1.0 → 1.0 4th: 4.56 → 4.56 4th: 4.56 → 4.56 5th: 0.70 → 0.70 5th: 0.71 → 0.71 5th: 3.64 → 3.64 5th: 3.67 → 3.67		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Reverse: 4.39 → 4.39		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Reverse: 20.36 → 20.36 Reverse: 4.38 → 4.38 Reverse: 20.34 → 20.34 Drive Axle (Front / Rear / All): Rear → Rear Front axle ratio: NA → NA Type / Description: Re-circulating ball type → Re-circulating ball type Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT 12 PR Tyre size designation including ply rating: 215/75 R15 LT → 215/75 R15 LT Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2 → 3.26 kg/cm2 Wiping system (Brief description): Twin blade, Tandem Pattern → Twin blade, Tandem Pattern	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Material: Metallic → Metallic Capacity (l): 50 lts , 40 lts → 50 lts , 40 lts Dimensions: 2490WB → 2490WB		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Dimensions: 2640WB → 2640WB Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Overall width (mm): 1842 → 1842 Overall Length (mm): CC → CC Overall Length (mm): 4750 → 4750 Overall Length (mm): 4988 → 4988 Overall Length (mm): FSD / HSD → FSD / HSD Overall Length (mm): 4925 → 4925 Overall Length (mm): 5225 → 5225 Overall height (mm): 1961 → 1961 Overall height (mm): 1948 → 1948 Maximum gradeability in 1st gear: GVW 2890 kg → GVW 2890 kg Maximum gradeability in 1st gear: GVW 3390 kg → GVW 3390 kg Maximum gradeability in 1st gear: 10 Deg → 10 Deg Maximum gradeability in 1st gear: 8.5 Deg → 8.5 Deg Weights: Base → Base Weights: Variant 1 → Variant 1 Weights: Variant 2 → Variant 2 Weights: Variant 3 → Variant 3 Weights: Variant 4 → Variant 4 Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	spare wheel, tools, etc.) (kg): - → - Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Description of energy converters and power plant. (In the case of a vehicle that can run either on petrol, diesel, etc., or also in combination with another fuel, items shall be repeated.1): Diesel → Diesel Engine Manufacturer:: Ashok Leyland Limited → Ashok Leyland Limited Manufacturer's engine model code (as marked on the engine, or other means of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	identification): Model Code: ALXYZABC0 Enclosure – 4A → Model Code: ALXYZABC0 Enclosure – 4A Location of engine model code and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	engine serial number on block / crankcase (drawing to be attached): Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A → Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A Working principle: positive ignition / compression- ignition, four-stroke / two-stroke / rotary cycle (DI/IDI) (NA/TC/TCIC/Any other): Compression Ignition, four-stroke, DI, TCIC → Compression Ignition, four-stroke, DI, TCIC Number, arrangement of cylinders:: 3, Inline → 3, Inline Bore----------(mm): 80 → 80 Stroke: ----------(mm): 98 → 98 Firing order:: 1-2-3 → 1-2-3 Engine capacity:----------(cm3): 1468 → 1468 Volumetric compression ratio (specify	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	tolerance): 17.5±1.0 : 1 → 17.5±1.0 : 1 Drawings of combustion chamber and piston crown and, in the case of positive ignition engine, piston rings:: Refer Enclosure – 4B → Refer		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure – 4B Vehicle with Defeat Device Yes / No: No → No If yes – Details of the Defeat device: NA → NA Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane equivalent) in PPM content by volume in the exhaust gas with the engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum net power:..... kW at min-1: 55.1 KW @ 3100 ± 50 rpm → 55.1 KW @ 3100 ± 50 rpm Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm Maximum net torque: .....Nm at: -------min-1 ...(manufacturer's declared value): 170 Nm @ 1600 – 2400 rpm → 170 Nm @ 1600 – 2400 rpm Maximum rated speed (Specify	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	tolerance): 3100 rpm ± 50 rpm → 3100 rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Type of fuel used : Mono		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Fuel / Bi-Fuel / Flex Fuel / Dual Fuel: Mono Fuel → Mono Fuel Fuel type: Diesel / Gasoline / LPG / CNG / Biomethane / Bio-H methane / HCNG / Biogas / LNG / Ethanol ((E85) / 2 (E100)) /ED (95) Flex fuel Methanol M15 / M100 / MD95 / Biodiesel up to 100% / Hydrogen.: Diesel → Diesel Maximum amount of biofuel acceptable in fuel (manufacturer's declared value): per cent by volume: 7% → 7% By fuel injection (compression-ignition only): (Yes/No): Yes → Yes System description:: Unit Pump Common Rail Direct Injection system → Unit Pump Common Rail Direct Injection system Working principle: direct-injection/pre-chamber/swirl Chamber: Direct Injection → Direct Injection Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Plunger Type & FHL0XYZ0 → Plunger Type & FHL0XYZ0 Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Injection advance curve (Specify the tolerances):: ECU Controlled → ECU Controlled Governor: ECU Controlled → ECU Controlled Cut-off point under load:…(min-1): 3300 → 3300 Cut-off point without load (Maximum no-load speed)…(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Multi-Hole Injectors (7 holes) FKXXYYZZ → Multi-Hole Injectors (7 holes) FKXXYYZZ Opening pressure (kPa) (specify the tolerances) or characteristic diagram: 15000 to 23000 kPa → 15000 to 23000 kPa Length and internal diameter…(mm): Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm → Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Internal Diameter – 3mm Common rail make & type: M/s. XXXXYYY, Hot Forged Rail Type → M/s. XXXXYYY, Hot Forged Rail Type Cold start system: NA → NA Make(s):: NA → NA Type(s) &		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: NA → NA Description:: NA → NA Auxiliary starting aid: NA → NA Make(s):: NA → NA Type(s):: NA → NA System description:: NA → NA Electronic controlled injection: (Yes/No): Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA → NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s. XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Metering type Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type By fuel injection (positive ignition only): (Yes/No): No → No Working principle: intake manifold (single/multi- point)/direct injection/other (specify): NA → NA Make(s) of Injection Pump :: NA → NA Type(s) & Identification Number of Injection Pump:: NA → NA System description (in the case of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	systems other than continuous injection give equivalent details):: NA → NA Make and type of the control unit:: NA → NA Make and type of the fuel regulator:: NA → NA Make and type of the air-flow sensor:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Make and type of the micro-switch:: NA → NA Make and type of the throttle housing:: NA → NA Make and type of the water temperature sensor:: NA → NA Make and type of the air temperature sensor:: NA → NA Injectors: Opening pressure (Specify the tolerances):,..... (kPa) or characteristic diagram:: NA → NA Make(s):: NA → NA Type(s):: NA → NA Identification Number: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA Cut-off point under load (min-1): NA → NA Injection timing:: NA → NA Cold start system:: NA → NA Operating principle(s):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Operating limits/settings(Specify the tolerances): NA → NA Pressure, (Specify the tolerances).....(kPa) or characteristic diagram:: 320 ± 50 kPa → 320 ± 50 kPa Rated voltage:...V, positive/negative ground: 12V, Negative Ground → 12V, Negative Ground		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Generator/ Alternator: Alternator → Alternator Type:: DC – Alternator with Voltage regulator → DC – Alternator with Voltage regulator Nominal output: ...VA: 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps → 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps Ignition: NA → NA Make(s):: NA → NA Type(s):: NA → NA Working principle:: NA → NA Ignition advance curve or map(Specify the tolerances):: NA → NA Static ignition timing (Specify the tolerances):.....degrees before TDC: NA → NA Spark plugs: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Gap setting----------(mm): NA → NA Nominal resistance (kilo ohm) (if resistive type): NA → NA Ignition coil(s): NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Ignition condenser: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA EMI suppressor cap / Device / Electronic unit: NA → NA Make: NA → NA Type (Resistive/Capacitive): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification number: NA → NA Nominal resistance (kilo ohm): NA → NA Terminology and Drawing of interference Suppression equipment: NA → NA H.T. Cable: NA → NA Make and Place: NA → NA Type (Resistive/Non-resistive): NA → NA Length (mm) (if resistive type): NA → NA Outside dia. (mm) (if resistive type): NA → NA Nominal resistance kilo ohm, (if resistive type): NA → NA Radiator drawing(s): Refer enclosure 4C2 & 4C4 → Refer enclosure 4C2 & 4C4 Radiator drawing(s): Refer enclosurev 4C1 & 4C3 → Refer enclosurev 4C1 & 4C3	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make: Mahle Behr → Mahle Behr Make: Tata Toyo → Tata Toyo Type (s): Cross flow → Cross flow Relief valve pressure setting: 0.9 kg/cm2 → 0.9 kg/cm2 Fan characteristics (Fan power, kW) Enclose the fan power curve corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part No: 9 → 9 Material of blades (metal /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ± 150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150 (AC) Radiator core open area (cm²): 1906 cm2 → 1906 cm2 Cooling system: liquid/air: Liquid → Liquid Nominal setting of the engine temperature control o mechanism (C)::: 85 deg C → 85 deg C	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Liquid: Yes → Yes Nature of liquid:: Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L → Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L Circulating pump(s): (Yes/No): Yes → Yes Characteristics of Circulating pump: Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm → Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm Make(s):: XXXXX		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Fasteners Ltd → XXXXX Fasteners Ltd Type(s):: Impeller Type → Impeller Type Drive ratio(s):: 1: 1.12 → 1: 1.12 Description of the fan and its drive mechanism:: Electric Fan , Motor Driven → Electric Fan , Motor Driven Blower: (Yes/No): No → No Characteristics of blower: NA → NA Make(s):: NA → NA Type(s):: NA → NA Drive ratio(s):: NA → NA Pressure charger: (Yes/No) Turbocharger	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	/ Supercharger: Yes, Turbo charger → Yes, Turbo charger Make(s):: M/s. XYZ, India → M/s. XYZ, India Type(s) & Identification Number:: Waste Gate with Vacuum Actuator, X3M1XXXX → Waste Gate with Vacuum Actuator, X3M1XXXX Description of the system (maximum charge pressure:.....(kPa), waste-gate, if applicable): Waste gate turbocharger with vacuum actuator Maximum Charge pressure - 145 kPa → Waste gate turbocharger with vacuum actuator Maximum Charge pressure - 145 kPa Inter-cooler: (Yes/No): Yes → Yes Type: air-air / air-water: Air – Air → Air – Air Make :: Mahle Behr → Mahle Behr Make :: Tata Toyo → Tata Toyo		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: B1V42XXX → B1V42XXX Identification Number:: B1V42YYY → B1V42YYY Air pressure drop across the intercooler: 15 kPa → 15 kPa Intake depression at rated engine speed and at 100 per cent load (Specify location of measurement) Minimum allowable:... (kPa) Maximum allowable:... (kPa): Minimum : 0 kPa Maximum : 6.0 kPa	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Description and drawings of inlet pipes and their accessories (plenum chamber, heating device, additional air intakes, etc.):: Intake manifold with Electronic throttle valve Refer Enclosure 4D → Intake manifold with Electronic throttle valve Refer Enclosure 4D Intake manifold description including dimensional drawings and/or photographs):: Aluminium Intake manifold with pressure sensor Refer Enclosure 4D → Aluminium Intake manifold with pressure sensor Refer Enclosure 4D Air filter, Dimensional drawings: Refer Enclosure 4E → Refer Enclosure 4E		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Make(s):: M/s. Fleet guard → M/s. Fleet guard Type(s) & Identification Number:: Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 → Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 Intake silencer, Dimensional drawing: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA Description and Dimensional drawing of the exhaust manifold:: Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F → Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F Description and drawing with dimensions of the exhaust system:(Description with general arrangement of exhaust system along with its routing indicating the lengths of exhaust pipe, tail pipe and: Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	O2 sensors provided before and after LNT-DPF assembly → Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly Effective volume of exhaust-system (specify the tolerance & range) in liters (from exhaust manifold / TC outlet to tail pipe end), Enclose the exhaust system dimensional drawing and indicate the volume of each parts clearly: 13.05 ± 1 litre Refer Enclosure 4G → 13.05 ± 1 litre Refer Enclosure 4G Description and drawing of the exhaust Muffler/Silencer:: NA → NA Type(s) & Identification Number:: NA → NA Make: NA → NA Number:.....: NA → NA Internal diameter of exhaust pipe (mm)(min): diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G → diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G Minimum distance between exhaust pipe(s) and the fuel		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	line: >150 mm → >150 mm Auxiliary Noise shields for compliance to IS 3028 and / OR AIS-020 (If Provided): NA → NA Material: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Layout of noise shield / Photographs / Diagram Showing arrangements indicating fitment on vehicle.: NA → NA Maximum allowable exhaust back pressure at rated engine speed and at 100 per cent load (location of measurement): ... (kPa): 30.0 kpa (150 mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Minimum cross-sectional areas of inlet and outlet ports:: Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm → Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm Inlet (mm): 8.00 → 8.00 Exhaust (mm): 8.00 → 8.00 Opening: 17° bTDC → 17° bTDC Closing: 51° aBDC → 51° aBDC Opening: 66° bBDC → 66° bBDC Closing: 25° aTDC → 25° aTDC Transfer: NA → NA Opening: NA → NA Closing: NA → NA Reference and/or setting ranges (Specify the tolerances):: On Cold Engine → On Cold Engine Inlet: 0.3 to 0.4 mm → 0.3 to 0.4 mm Exhaust: 0.3 to 0.4 mm → 0.3 to 0.4 mm		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Distribution by ports: NA → NA Volume of crank-case cavity with piston at TDC: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Reed valve fitted (Yes / No): No → No Description of inlet ports, scavenging and exhaust ports with corresponding timing.: NA → NA Device for recycling crankcase gases (description and drawings):: Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J → Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx Catalyst (if any, and if not covered by another heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF / DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No Element – 1 No. → Catalytic Converter – 1No Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	for LNT (periodic / continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:……cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap Bulk Density (GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle and Characteristics:: Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	downstream SCR catalyst for the reduction of NOX during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY Type and Identification number:: Type - Passive NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite → Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA Make:: NA → NA Type and Idetification number :: NA → NA NOx Sensor (Before /After of SCR):: No → No Make:: NA → NA Type and Identification number:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent engine test bench cycles, between two cycles where regenerative phases occur under the conditions equivalent to Type I test (Distance "D" in t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration (Distance “d” in AIS 137): 2 cycles → 2 cycles Description of method employed to determine the number of cycles	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA International standard (where appropriate):: NA → NA Frequency of reagent refill: continuous/maintenance (where appropriate):: NA → NA Normal operating temperature range (°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY → YYYYYYY Air injection: (Yes/No): No → No Type (pulse air, air pump, etc.) & Identification Number:: NA → NA Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other): Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY → FEW0XXYY Characteristics (flow rate, etc.):: Flow rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes EGR Cooler Make and Identification Number: M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:.....g: NA → NA Canister working capacity:.....g: NA → NA Canister bed volume:.....litre: NA → NA Canister Purge and vent strategy description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve : (Yes/No): NA → NA Make and Identification Number: NA →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA Operating pressure (kPa): NA → NA Liquid fuel hoses material :: NA → NA Schematic drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	drawing (Periodic / Continuous): NA → NA Particulate trap: (Yes/No): Yes → Yes Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Dimensions and shape of the particulate trap (capacity):: Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Regeneration Type Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure 4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh: 90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type Material - Silicon Carbide → Structure - Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI Electronic Control Unit (ECU): Yes → Yes	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number: P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA Misfire detection:: NA → NA Oxygen sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K → Refer Enclosure 4K Particulate traps		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method):: Refer Enclosure 4K → Refer Enclosure 4K List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K → Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K A description of the type and number of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3 EUDC A description of the type of the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	all sensed components with the strategy for fault detection and MI activation (fixed number of driving cycles or statistical method), including a list of relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive explanation for the data given in service $05 Test ID $21 to FF and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles – Diagnostics on Controller Area Network (CAN) – Part 4: Requirements for emissions- related systems", a	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137): Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA → NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA Systems to ensure the correct operation of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent deactivation of the driver inducement, for use by the rescue services [Note: cross- reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	/disable after parking: NA → NA Lowest		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct operation of NOx control measures: NA → NA Details of any devices designed to influence fuel economy (if not covered by other items):: NA → NA LPG fuelling system: (Yes/No): No → No Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to LPG or back:: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.): NA → NA NG fuelling system: (Yes/No): No → No Approval number: NA → NA Electronic engine management control	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	unit for NG fuelling: NA → NA Make(s):: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation:: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to NG or back: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.):: NA → NA Hydrogen fuelling system: (Yes/No): No → No Electronic engine management control unit for hydrogen fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to hydrogen or back:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA H2NG fuelling system: (Yes/No): No → No		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Percentage of hydrogen in the fuel (maximum specified by the manufacturer): NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to H2NG or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA Electric motor(Traction): No → No Type (winding, excitation):: NA → NA Maximum hourly output: kW(manufacturer’s declared value): NA → NA Maximum net power:kW(manufacturer’s declared value): NA → NA Maximum 30 minutes power: kW	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(manufacturer’s declared value): NA → NA Operating voltage: V: NA → NA Make and Identification Number:: → Battery (Traction): NA → NA Number of cells:: NA → NA Mass: .....(kg): NA → NA Capacity: Ah		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Amp-hours): NA → NA Position:: NA → NA Engines or motor combinations: NA → NA Hybrid Electric Vehicle: (Yes/No): No → No Category of Hybrid Electric vehicle Off Vehicle Charging/Not Off Vehicle Charging9: NA → NA Operating mode switch: with/without: NA → NA Selectable modes: NA → NA Pure electric: (Yes/No): NA → NA Pure fuel consuming: (Yes/No): NA → NA Hybrid modes: (Yes/No) (if yes, short description): NA → NA Description of the energy storage device: (traction battery, capacitor, flywheel/generator...): NA → NA Make(s):: NA → NA Type(s) :: NA → NA Identification number:: NA → NA Kind of electrochemical couple:: NA → NA Energy: ..... (for battery: voltage and capacity Ah in 2 h, for capacitor: J): NA →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA Charger: on board/external/without: NA → NA Electric machines (describe each type of electric machine separately): NA → NA Make:: NA → NA Type:: NA → NA Primary use: traction motor/generator: NA → NA When used as traction motor: mono motor / multi motors (number):: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Maximum power: kW: NA → NA Working principle:: NA → NA Direct current/alternating current/number of phases:: NA → NA Separate excitation/series/compound: NA → NA Synchronous/asynchronous: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA Identification number:: NA → NA Power controller: NA → NA Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Vehicle electric range..... km (according to AIS 137):: NA → NA Manufacturer’s recommendation for preconditioning:: NA → NA CO2 (g/km) (Applicable for category M1 with GVW<3.5 T): NA → NA Declared : (rounded to 3 decimal	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	places): NA → NA Fuel consumption (l/100 km) for Petrol, LPG or Diesel and (kg/100km) for CNG and (kWh/100 km) for Electric Driven Vehicles(Applicable for category M1 with GVW<3.5 T) (rounded to 3 decimal places)*: NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA *calculated on the declared CO2: NA → NA CO2 Reducing		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	technologies Available (Yes/No): No → No Regenerative braking: NA → NA Start-Stop System: NA → NA Tyre pressure monitoring system: NA → NA 6 or more Speed Transmission: NA → NA Any other technology (manufacture to enclose detail): NA → NA Electric energy consumption for electric vehicles Wh/km: NA → NA Electric energy consumption for pure electric vehicles Wh/km: NA → NA Electric energy consumption for externally chargeable hybrid electric vehicles: NA → NA Electric energy consumption (condition A, combined) Wh/km: NA → NA Electric energy consumption (condition	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	B, combined) Wh/km: NA → NA Electric energy consumption (weighted combined) ….Wh/km: NA → NA Temperatures permitted by the manufacturer: NA → NA Cooling system: Yes → Yes Liquid cooling: Yes → Yes Maximum temperature at outlet: …….(°C): 100 °C → 100 °C Air cooling: NA → NA Reference point:: NA → NA Maximum temperature at reference point: …….(°C): NA → NA Maximum outlet temperature of the inlet intercooler: .. (°C): 45 °C → 45		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	°C Maximum exhaust temperature at the point in the exhaust pipe(s) adjacent to the outer flange(s) of the exhaust manifold: …(°C): 740 °C → 740 °C Minimum: .. (°C): - 8 → - 8 Maximum: .. (°C): 42 → 42 Minimum: …(°C): - 5 → - 5 Maximum: …(°C): 120 → 120 Lubrication system: Yes → Yes Description of the system: Forced Lubrication system → Forced Lubrication system Position of the lubricant reservoir:: Sump beneath crankcase → Sump beneath crankcase Feed system (by pump/injection into intake/mixing with fuel, etc.): Pump →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Pump Lubricating pump: Yes → Yes Make(s):: M/s. ZXWA → M/s. ZXWA Type(s):: XXYY Type → XXYY Type Mixture with fuel: No → No Percentage:: → Oil cooler: (Yes/No): Yes → Yes Drawing(s):..., or: Enclosure 4L → Enclosure 4L Make(s):: M/s. XXYY or Any other equivalent → M/s. XXYY or Any other equivalent Type(s):: DONUT → DONUT Moment of inertia of engine flywheel:: 0.1212 kg/m2 → 0.1212 kg/m2 Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Transmission Control Unit (ECU) (Yes/No):: No → No Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA OBD IUPR Family if any: XX+XYY → XX+XYY exhaust outlet location, indicated in a Schematic dimensional drawing.): Refer Enclosure 4G → Refer Enclosure 4G 1.2.11.2.1.1 4.4 · 3. · Calibration Verification Number:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Catalyst · P0420 · Oxygen sensor 1 and 2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Engine speed, engine load, A/F mode, catalyst temperature → Engine speed, engine load, A/F mode, catalyst temperature Catalyst · P0420 · Oxygen sensor 1 and 2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Two Type I cycles → Two Type I cycles Catalyst · P0420 · Oxygen sensor 1 and 2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Type I → Type I Type (Manual / power): Manual → Manual No. of wipers: Two → Two Make: XXYY System →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	XXYY System Type: 12V → 12V Identification mark: X5XYZ100 → X5XYZ100 Rated voltage: 12V → 12V Number of sweep Frequencies: 2 → 2 Highest sweep frequency (Cycles / min): 62±8 rpm → 62±8 rpm	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Lowest sweep frequency (Cycles / min): 40±5 rpm → 40±5 rpm Wiper arm: Provided → Provided Length: 500mm → 500mm Make: M/s XYZ → M/s XYZ Length: 450 mm → 450 mm Make: BCS (SYNDICATE WIPER SYSTEM) → BCS (SYNDICATE WIPER SYSTEM) Identification: X5XY0258 → X5XY0258 Washer tank: BWXY4501 → BWXY4501 Type: Plastic → Plastic Make: XYZ → XYZ Identification No. / Part No.: X5BXYZ168 → X5BXYZ168 Capacity l: 1.5Ltr → 1.5Ltr Material: H.D.P.E → H.D.P.E Washer tank motor / Washer pump: ELECTRIC MOTOR WITH PUMP → ELECTRIC MOTOR WITH PUMP Make: XYZ → XYZ Model: PEAD64XYZS00 → PEAD64XYZS00 Nozzle(s): XYZ → XYZ No. Of Nozzles: 2 → 2 Defroster: NA → NA Type: - → - Make: - → - Details of Demisting and / or Defrosting system: - → - Schematic Diagram of Engine cooling system: - → - Schematic Diagram of Heating system: - →		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	- Cross sectional view of complete air handling system including position of louvers (vents) on dash board: - → - Blower identification: - → - Blower Voltage and capacity (Watts): - → - If defrost system is fitted, specify defrost test temperature (-8 +/- 2 deg C / -18 +/- 3 deg C): → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	& related dimensions as per related standards.: → Description and sketch showing the detail positions of Tell-Tale and seating layout (For M category vehicles): NA → NA A brief technical description indicating the length of the cleaning period, the consumption of cleaning fluid during the cleaning period and the minimum capacity of the container provided;: -- → -- Description and drawings/photographs	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	of the shapes and constituent materials of the part of the body forming the engine compartment and the part of the passenger compartment nearest to it:: -- → -- Drawings or photographs of the position of the metal components housed in the engine compartment (e.g. heating appliances, spare wheel, air filter, steering mechanism, etc.):: -- → -- List of all subassemblies, which includes an electronic oscillator or switching frequency greater than 9 kHz (like ECU, steering, suspension, Body: -- → -- List of all Electrical components, which include Broadband EMI sources (like HAVC Motor,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Wiper Motor, Window motor and Horn etc.) The list shall include Device name, Make/s and Part No. /Type ID/Drawing No: -- → -- Name of the manufacturer of the system (Steering column shaft / Gear shift control / Transmission) on which the Protective Device functions: XXYY Steering System → XXYY Steering System Type/Part no./Identification number(ID)/Drawing No.of the device on which the Protective Device functions: FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable only in case of steering column locking systems: Locks the Steering → Locks the Steering SYSTEM SOFTWARE (If fitted): NA → NA Make: -- → -- Version: -- → -- Operating System Details with Version: -- → -- DESCRIPTION OF DEVICE (If fitted): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Description of the VAS: - → - A detailed description of the alarm system and of the vehicle parts related to alarm system installed with Block Diagram: → A list of main components comprising the alarm system: - → - Components placed in Engine compartment: - → - Components placed in Passenger compartment: - → - Availability of radio/wireless transmission Yes/No: - → - Details of radio/wireless transmission if any: - → - frequency: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Availability of WPC approval for wireless technology Yes/No: - → - The measures taken against false alarm: - → - Range of protection offered by the device. Details about the detection mechanism (opening of Door, Bonnet, luggage compartment, intrusion etc.): - → - Details & specification of audible alarm: - → - Details & specification of optical alarm: - → - Details & specification of radio alarm if fitted: - → - Method of setting/ unsetting the device: - → - Type of key used for unsetting (Mechanical Key / electronic device): - → - Number of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - Exit delay in Sec: - → - Entry Delay in sec: - → - Details of status display (0.5cd): - → - Details of self check , automatic failure indication: - → - Details & specification of audible alarm: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Details & specification of optical alarm: - → - Details & specification of radio alarm if fitted: - → - Method of setting/ unsetting the device: - → - Type of key used for unsetting (Mechanical Key / electronic device): - → - Number of interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - Exit delay in Sec: - → - Entry Delay in sec: - → - Details of status display (0.5cd): - → - Details of self check , automatic failure indication: - → - Details about the panic alarm , if provided: - → - Description of the Immobilizer: - → - A detailed description of the vehicle immobilizer and of the vehicle and the measures		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	taken against inadvertent activation with block diagram.: → The vehicle system(s) on which the immobilizer acts.: - → - Details of radio/wireless transmission if any: - → - Method of setting/ unsetting the device.: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Number of interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - A list of main components comprising the device: - → - Description: AL Part No. → AL Part No. Description · AL Part No.: Hella Part No → Hella Part No ECU: FXX01000 → FXX01000 ECU · FXX01000: 015.XYZ-01 → 015.XYZ-01 a. · Make: - → - b. · Type: - → - c. · Serial Number: - → - d. · Name and Address of manufacture: - → - a. · Make: - → - b. · Type: - → - c. · Serial Number: - → - d. · Name and Address of manufacture: - → - Example of Engine No: HJEZXYZ891 → HJEZXYZ891		